Monetary-Financial Model:
A global projection and scenario tool
User Manual and Methodological Documentation[1]

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Source code: https://github.com/matyasfarkas/BVARScenarioTool

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Abstract

This document describes the Monetary-Financial Model, a global projection and scenario tool: a browser-based application for conditional forecasting, scenario analysis, and policy counterfactual evaluation. The tool combines a large Bayesian vector autoregression (BVAR) estimated with hierarchical Minnesota priors (Giannone et al., 2015) with structural impulse responses drawn from the Macroeconomic Model Data Base (MMB; Wieland et al., 2012, 2016). Users condition the BVAR forecast on arbitrary variable paths via a Durbin–Koopman simulation smoother (Durbin and Koopman, 2002, 2012), and evaluate monetary and fiscal policy counterfactuals using DSGE-identified impulse response functions. We describe the methodology employed, the implementation of the conditional forecasting algorithms, and the three-layer architecture that separates the unconditional baseline, the user's scenario, and the policy counterfactual. All computation runs client-side in the browser; the tool requires no server infrastructure and can be deployed as a static website. The manual is in two parts: Part I explains how to *use* the tool — what you do, how you do it, and what you find on the screen — in plain language, assuming no econometrics; Part II documents the methodology in full technical detail.

Keywords: BVARs, Conditional Forecasting, Scenario Analysis, Kalman Filter, Durbin–Koopman Smoother, Policy Counterfactuals, DSGE Models, MMB

JEL classification: C32, C53, E37, E52, E58

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Contents

1 What this tool does

This tool shows you quarterly forecasts for about fifty economies — growth, inflation, interest rates, public finances, and more — and lets you bend them.

Click on a chart where you think inflation will be, and every other chart updates so the whole forecast stays consistent with the estimated economics: if you push United States inflation up, the policy rate rises, growth cools, and — because the economies are linked through trade and financial channels — the euro area, Japan, and China move too. You can set where a variable should settle in the long run, ask what a different interest-rate path would have done, and put uncertainty bands around any of it. Everything runs in your browser; nothing to install.

Three honest rules govern everything you see:

- **Data are data.** Solid history is observed. Where a country's latest quarter has not been published yet, the model fills the gap using everything already observed elsewhere — and those filled-in points are always drawn in amber, never disguised as data.
- **The model is estimated once, then frozen.** Within the year, new data shift the starting point of the forecast but never silently change the model itself. Re-estimation happens once a year, on a complete final year of data.
- **Judgment is visible.** If you move a forecast away from what the model says — a scenario, a long-run target, a policy counterfactual — the chart says so.

How to use this manual. Part I is the manual proper: each section tells you *what you can do*, *how to do it* (step by step, naming the exact buttons), and *what you will find* on the screen when you have done it. You can read any section on its own. If you have five minutes, read Section 2 and skip the rest until you need it.

Part II explains how the machinery works — the estimation, the filtering, the cross-country system — with the mathematics. You do not need any of it to use the tool. It is there so that every number on the screen can be traced to a method, and every method to code and data.

Part I
Using the tool

2 Your first scenario in five minutes

This chapter is one continuous walk-through. You open the tool, look at one chart, bend the United States policy rate with your mouse, watch the rest of the economy respond, put uncertainty bands around the result, and clean up. Nothing else. Every button and label named below appears on screen exactly as written. The full tour of every control comes later, in §3.

2.1 Open the tool and pick a country

What you do. Load the page and make sure you are looking at the United States.

How.

1. Open the tool's web address in a modern browser. There is nothing to install; the model files load into the page itself, so give the first visit a few seconds.
2. Look at the top bar. The **Country** dropdown already reads **United States** — that is the default. If someone changed it, open the dropdown and pick **United States** from the **GVAR** group. That group lists the economies linked into the cross-country system; every other economy sits under **Other countries** and moves on its own.
3. Leave everything else as it is: **Units** on **q/q ann.** and **Uncertainty** on **None**.

What you find. A grid of charts fills the screen — growth, inflation, the policy rate, public finances, credit, and more. In the left control panel a scenario named **Scenario 1** is already active, and at the bottom of the screen it has a tab of its own. You have not touched anything yet, so every chart still shows the model's own view.

2.2 Read your first chart

What you do. Learn to tell history from forecast on one chart before you change anything.

How. Find the chart titled **Inflation**, at the top left.

What you find. Three things to read:

- *History sits on a lightly shaded background.* The solid line there is observed data. The x-axis is labelled with calendar years; hover over any point to see its exact quarter and value.
- *The forecast is the dotted line* on the white part of the chart, labelled **Baseline** in the legend. The faint grey fan around it is the model's own uncertainty about that path — it is there from the start, before you compute anything. The first forecast quarter is the latest quarter whose data have not been published yet; the sidebar and exports label it **NC** — short for nowcast, the model's estimate of the current, not-yet-published quarter.
- *Amber means model-filled.* If a series' latest quarters have not been published, the tail of its history is drawn as an amber dashed tail with open circles — values the model filled in from everything observed elsewhere, never disguised as data (§10).

The unit is % q/q ann.: the change from the previous quarter, stated at an annual rate. If you prefer the change over the same quarter a year earlier, click **y/y** in the **Units** toggle — then click **q/q ann.** to come back.

2.3 Bend the policy rate

What you do. Tell the tool: "suppose the policy rate is higher than the model expects one year from now — say half a percentage point higher."

How.

1. Find the chart titled **Policy Rate**, next to Inflation.
2. Click the small **condition** button in its header. The button turns blue and now reads **CONDITIONED**, the panel gets a blue border, and a hint badge appears: **drag to adjust policy**.
3. Press the mouse down on the forecast part of the chart, about four quarters in, and drag upward — half a percentage point above the dotted baseline. Release.

A shortcut for later: right-clicking an unconditioned chart does steps 2 and 3 at once — it conditions the variable and plants the point exactly where you clicked.

What you find. A dot marks the quarter you pinned, and the scenario line — solid, in Scenario 1's colour — now passes through it. While you hold the mouse the line follows a fast preview; the full answer settles in about a second after you release. Only the quarter you dragged is imposed; every other quarter of the rate path is left free, and the model fills it in with the most plausible route through your point. In the left control panel, a new entry for the rate appears in the conditioning list, with one row per forecast quarter from **NC** to **t+19** — you will use those rows in §4.

2.4 Watch the rest of the economy respond

What you do. See what a tighter policy path means for everything else.

How. Scroll through the grid. Compare each chart's solid scenario line with its dotted **Baseline**.

What you find. Every chart has updated, not just the one you touched. The estimated model knows how United States variables have moved together in the data, and it re-draws the whole forecast so it stays consistent with your pinned rate: with a tighter path, growth typically cools, inflation eases with a lag, and the financial variables shift too. How this works is spelled out in Part II, §14.1. Two notes. First, the response is the model's estimate of typical historical co-movement, not a law. Read it as "what usually accompanies a higher rate," not "what the rate causes." For a true cause-and-effect experiment, use the policy counterfactual (§7). Second, other countries have not moved: cross-border spillovers are off by default, so the United States moves alone until you switch them on (§5).

2.5 Put uncertainty bands around it

What you do. Ask how sure the model is about the scenario you just drew.

How.

1. In the top bar, set **Uncertainty** to **Quick**.
2. Press **Compute**. An overlay appears: *Computing scenario uncertainty (50 draws)...* — the model redraws its own forecast fifty times, each time with a different plausible version of its estimated parameters and shocks, and centres that range on your scenario.

What you find. A shaded band in the scenario's colour now wraps the solid line on every chart: the middle two-thirds of the plausible outcomes. Notice the band on the Policy Rate chart pinches shut at the quarter you dragged — you *imposed* that value, so there is no uncertainty about it there. The band is widest where the model is least sure, usually far out in the forecast. **Full** does the same with 200 draws — smoother edges, longer wait. What the bands do and do not cover is in §9, with the machinery in Part II, §19.6.

2.6 Clean up

What you do. Put everything back the way you found it.

How.

1. In the left control panel, at the top of the **Scenario** section, click **Reset**. This clears the pins of the active scenario for the current country only — other scenarios and other countries are untouched.
2. Set **Uncertainty** back to **None**.

What you find. The dot is gone, the scenario line lies back on the dotted **Baseline**, and the conditioning list in the left control panel is empty again. You are back at the model's own view.

2.7 Where to go next

You have now done the core loop: pin a point, read the response, band it, reset. The rest of Part I builds outward from here.

- §3 — the full tour: every control on the screen, top bar to bottom bar.
- §4 — richer scenarios: several variables and quarters at once, quarter-by-quarter editing, ready-made cookbook recipes, saving and loading.

- §5 — switch on cross-border transmission and let a United States scenario reach the euro area, Japan, and China.
- §6 — set where a variable settles in the long run.
- §7 — the true policy experiment: what a different interest-rate path would have *caused*.
- §8 — revenue, spending, deficits, and debt.
- §9 — what the bands mean, Quick versus Full, and what they leave out.
- §10 — where the data come from, what the amber points are, and how to export everything you see.

3 A tour of the screen

The screen has four parts. A *top bar* across the top: country choice, display units, uncertainty, and links. A *control panel* down the left side: scenarios, long-run targets, cross-border switches, and the policy counterfactual. The *chart grid* in the middle: one chart per variable, and this is where you click and drag. A *scenario bar* along the bottom: scenario tabs and export buttons. This chapter walks through each part, and — most importantly — teaches you to read one chart correctly.

3.1 The top bar

3.1.1 Picking a country

What you do. You choose which economy fills the chart grid.

How.

1. Find the dropdown labelled **Country** in the top bar.
2. Open it. The list has two groups: **GVAR** and **Other countries**.
3. Click an economy. The grid redraws for that economy within a moment.

What you find. The two groups matter. Economies under **GVAR** belong to the linked cross-country system — the global model that ties economies together through trade and financial channels (§16) — so a scenario you build there can spill over to partners, and their scenarios can reach it (see §5). Economies under **Other countries** are forecast on their own. Switching country never deletes your work: each economy keeps its own scenario settings until you reset them.

3.1.2 The rest of the bar

Left to right, the remaining controls are:

- **The app title** (*Monetary-Financial Model: A global projection and scenario tool*), with a small "i" button that opens an **About** note.
- **Data & coverage** — opens the data-provenance panel, telling you for every variable where the numbers come from and how far the observations reach (§10).
- **Three links**: **User Manual** (this document as a PDF), **Docs** (the full online documentation), and **Monitor** (a page showing how the latest data update moved every economy's forecast).
- **Units** — the display-units toggle, covered below.
- **Uncertainty** — a toggle with the choices **None**, **Quick**, and **Full** (§9).
- **Compute** — runs the uncertainty and long-run calculations (§9, §6).

Compute is greyed out until you either pick **Quick** or **Full** uncertainty or set at least one long-run target.

3.2 How to read one chart

What you do. Before you touch anything, learn what the lines and shading on a single chart mean. Every chart in the grid follows the same grammar.

How. Pick any chart — say **Inflation** — and look at it from left to right.

What you find.

- **The shaded region on the left is history.** A light background tint covers all quarters up to the last historical quarter. Everything to its right is forecast.
- **The solid coloured line is the data, then your scenario.** Inside the shaded region the solid line is observed history. To the right it continues as the active scenario's forecast. If you have not changed anything, it lies exactly on top of the baseline.
- **The amber dashed tail is model-filled, not observed.** Quarters not yet published are drawn amber and dashed at the end of the history line, with small open circles marking exactly the filled-in quarters. Hover over one and the label reads *Model-implied nowcast (imputed).* Solid means observed; amber dashed with open circles means the model's best guess. What the filling uses is explained in §10; see also §19.5.
- **The dotted grey line is the baseline.** This is what the model expects with no intervention from you. It is labelled **Baseline** in the legend. When you move a forecast, your scenario line departs from it, so the gap between solid and dotted is exactly what your judgment added.

- **The grey fan is the model's own uncertainty.** A pale grey band around the baseline covers the central two-thirds of the outcomes the model considers plausible — roughly, "the future usually lands inside this band." It is there from the start; the coloured band around your *scenario* appears only after you press **Compute** (§9; construction in §19.6).
- **The date axis.** Tick labels mark calendar years. For the exact quarter, hover anywhere on a line: the label shows the date and value, for example *2027 Q3 · 2.10*. The first forecast quarter is the one right after the last historical quarter.
- **The header.** Each chart's title sits top-left with a **condition** button next to it — click it to start steering this variable (§4); it turns into **CONDITIONED** and gives the panel a blue border once active. Charts computed from other charts by an accounting identity — the change in debt-to-GDP (Δ Debt-to-GDP) or the external-sector charts, for example — show a **derived** badge instead and cannot be steered directly. If you have moved a variable's long-run target off the model's estimate, its title turns amber and shows a small **judgement** badge with a pencil mark (§6). While a chart is steerable, a hint badge appears top-right: **drag to set path** — or, on the policy-rate chart, **drag to adjust policy**.
- **The footer.** Two small lines under each chart give the variable's definition and its unit, so you never have to guess what a series measures. The full variable catalogue is in §19.

One more thing you may see *above* the grid on GVAR economies: a badge reporting whether the cross-country solve you are looking at is a fast preview or the full solve, sometimes with a **Solve full GVAR** button. Ignore it for now; §5 explains it.

3.3 Three ways to display the numbers

What you do. You choose the units every growth-rate chart is drawn in. The numbers are the same forecast; only the lens changes.

How.

1. In the top bar, find the toggle labelled **Units**.
2. Click one of **q/q ann.**, **y/y**, or **% vs base**.

What you find.

- **q/q ann.** shows each quarter's change from the quarter before, scaled up to a full year — the "annualised" pace economists quote. It is the model's native unit (§12.3) and the most responsive view: a one-quarter dip shows up as one sharp notch.
- **y/y** shows the change against the same quarter one year earlier. It is smoother and matches how inflation is usually reported in the press. The same forecast therefore looks calmer here; nothing has changed but the lens.

- **% vs base** shows the *difference* between your scenario and the baseline, centred at zero — an impact view. A flat line at zero means your scenario changed nothing for that variable. This view is read-only: you cannot drag on charts while it is active. A second pair of buttons appears in this view only: **Level** (the default; the cumulative difference in the level of the series) and **Rate (IRF)** (the quarter-by-quarter difference, matching the convention used to compare against the structural policy models of §7; details in §15).

Two kinds of charts opt out of this toggle. Charts that are already rates — the policy rate, unemployment — are always shown in % and ignore the growth-rate lenses. And a handful of price and index series — the oil price, exchange-rate indices, equity and house prices — are always drawn as *levels* (an index number, or dollars per barrel for oil), because a level is how you think about them. On those charts, dragging sets the level itself.

3.4 The left control panel

The panel down the left side holds everything that is not a chart. Top to bottom:

- **Scenario** — buttons to **Reset**, **Reset all**, **Save**, and **Load** a scenario, plus **Cookbook**, a menu of ready-made, documented scenarios you can load in one click. Below them, a text box to name the scenario. All of it is covered in §4.
- **Long-run targets** — a collapsible panel where you set where each variable settles in the long run. Covered in §6.
- **Cross-border spillovers** — two on/off switches controlling whether your scenario and your long-run targets propagate to other economies. Covered in §5.
- **Policy counterfactual** — a mode for asking what a different interest-rate or spending path would do, judged through a structural model of your choice. Covered in §7.
- **The conditioning list** — once you steer a variable, it appears here with a slider and an on/off button for each forecast quarter, and a **Show deviations / Show levels** toggle for how you type the numbers. Covered in §4.
- **Acknowledgments** and the disclaimer sit at the very bottom.

3.5 The bottom bar

What you do. You manage up to four scenarios side by side and export your results.

How. The bar along the bottom shows one tab per scenario — a colour dot, the name, and the structural model and policy rule it uses for counterfactuals. Click a tab to make that scenario active; click its **x** to remove it. **+ Add Scenario** creates a new one (up to four). To rename a scenario, use the name box in the left control panel, not the tab. On the right sit four export buttons: **CSV**, **Excel**, **PNG**, and **PDF**.

What you find. Every scenario draws on every chart in its own colour, so you can compare, say, a soft-landing path against a stagflation path on the same axes. The exports carry the numbers behind what you see — baseline, scenarios, and uncertainty ranges — and are described in §**??**.

4 Building scenarios

A scenario is a forecast you have bent. You pin one or more variables to a path you believe in — say, inflation half a point higher for the next year — and the model redraws every other chart so the whole picture stays consistent with the estimated economics. This "pin some, let the model fill in the rest" calculation is called a conditional forecast; the mechanics live in §14.1. This chapter shows you every way to set pins, how to run several scenarios side by side, and how to start over.

4.1 Pin a variable with the mouse

What you do. You tell the tool where you think one variable is heading, directly on its chart.

How.

1. Find the chart you want to steer. In its header, click **condition**. The button changes to **CONDITIONED** and the panel gets a blue border. Nothing moves yet — you have only declared the variable steerable.
2. Drag on the plot. A hint badge — **drag to set path** — appears while the chart is draggable. As you drag, the forecast quarter nearest your cursor is pinned at the height of the cursor.
3. Or skip step 1 entirely: *right-click* anywhere on the forecast region of an unconditioned chart. That single click conditions the variable *and* plants a pin exactly at the quarter and value you clicked. On a chart that is already conditioned, right-click toggles that one quarter's pin on or off, placing the dot at the cursor value.

What you find. The pinned quarter shows a solid dot at your value — the dot snaps to the click immediately, and the full redrawn path follows a moment later. Every other quarter of that variable, and every other chart, adjusts to stay consistent: pin growth down and you will typically see inflation soften and the policy rate follow. Two things cannot be dragged. Charts with a **derived** badge are computed by an accounting identity from other variables, so you steer their drivers instead — except the fiscal primary-balance and debt charts, where a drag sets a target trajectory directly (see §8). And nothing is draggable in the **% vs base** view, which is a read-only impact display.

If cross-border spillovers are switched on for a linked economy, the line follows a quick local preview while you drag, and the cross-country solve — the fast direct preview, flagged by the amber **GVAR DIRECT preview** badge — fires once when you release the mouse. The full cross-country solve still runs only on demand, via the **Solve full GVAR** button or **Compute** (§5).

4.2 Fine-tune quarters in the conditioning list

What you do. You set exact numbers, quarter by quarter, instead of eyeballing a drag.

How.

1. Look at the conditioning list at the bottom of the left control panel. Every conditioned variable appears there with twenty quarter rows, labelled **NC**, **t+1** through **t+19**. **NC** is short for nowcast, the model's estimate of the current, not-yet-published quarter — its read of where the economy stands right now.
2. Click a quarter label to switch that quarter's pin on or off. Click **All** to switch all twenty at once.
3. Move a quarter's slider to set its value. The slider covers roughly ±3 in steps of 0.1, and the number you set is shown next to it.
4. By default you type absolute values (say, a policy rate of 3.5). Click **Show deviations** to enter distances from the baseline forecast instead (+0.5); click **Show levels** to switch back. The same pins are shown either way.
5. Click **remove** next to a variable's name to take it out of the scenario entirely.

What you find. Only the quarters you switched on are held fixed; the quarters that are off are filled in by the model. A single pinned quarter far out — say **t+8** — gives a smooth model-chosen path to that point. Pins are what-you-set-is-what-you-see: a pinned quarter sits exactly at your number on the chart, always. If a pinned path looks kinked or implausible elsewhere, that is the model telling you what your assumption costs; loosen a quarter and watch it relax.

4.3 Condition several variables at once

What you do. You build a richer story — for example, weaker growth *and* a lower policy rate together, to describe a downturn the central bank leans against.

How. Condition each variable as above; they stack. Every conditioned variable gets its own block in the conditioning list, and you can pin different quarters on each.

What you find. All pins are imposed jointly, not one after the other: the model finds the single most plausible forecast that passes through every pin at once. Beware of fighting yourself — pinning inflation up while pinning the policy rate down asks for an unusual combination, and the unconditioned charts will show you how strained it is. Fewer, well-chosen pins usually tell a cleaner story than many.

4.4 Run scenarios side by side

What you do. You keep up to four alternative scenarios and compare them on the same charts.

How.

1. In the bar along the bottom of the screen, click **+ Add Scenario**. A new tab appears with its own colour dot and becomes active; the button is offered while you have fewer than four scenarios.

2. Click a tab to make that scenario active. All conditioning you do — charts, the conditioning list, cookbook loads — edits the active scenario only.

3. Rename the active scenario by typing in the **Scenario name** box at the top of the left control panel. The tabs have no rename control of their own.

4. Click the small **x** on a tab to delete that scenario. The **x** is shown only while you have more than one.

What you find. Each scenario is drawn on every chart in its own colour — blue, red, green, purple, matching the dot on its tab — with the active scenario's line thicker. There is no hide button: all scenarios stay visible until you delete them, so if a chart gets crowded, remove the tabs you no longer need. Each tab also carries a small caption naming the scenario's policy-model choice, which matters only in counterfactual mode (§7).

4.5 Start over: Reset vs. Reset all

What you do. You clear your work — either just the current scenario in the current country, or everything everywhere.

How. Both buttons sit in the **Scenario** header of the left control panel. **Reset** clears the *active* scenario's pins and paths for the *current country only*. **Reset all** (in red) clears every scenario and all cross-country conditioning in all countries; because that is drastic, it first shows "Clear all scenarios & conditioning?" and waits for you to click **Yes, reset all** or **Cancel**.

What you find. After **Reset**, the active scenario's line falls back onto the baseline forecast here, but pins you set in other countries survive. After **Yes, reset all**, you are back to a clean baseline everywhere. If a scenario is worth keeping, click **Save** first — it writes the scenario to an Excel file that **Load** restores later (§10).

4.6 When to press Compute

What you do. Nothing, most of the time. The central scenario path updates live with every pin — **Compute** is never needed to see it.

How. Press **Compute** in the top bar only when you want uncertainty bands around the scenario or when you have set a long-run target; the button's enable rule and the **Uncertainty** switch are described in the tour (§3).

What you find. Bands appear around your scenario line, showing the range of outcomes consistent with your pins — how to read them is §9; long-run targets are §6. Bands do not refresh on their own: change a pin afterwards and press **Compute** again.

4.7 Load a ready-made scenario from the Cookbook

What you do. Instead of building pins from scratch, you load a named, documented recipe — an oil-price shock, an inflation surge — as a worked example, then adjust it.

How.

1. In the **Scenario** header of the left control panel, click **Cookbook ▼**. A panel titled **Scenario cookbook** opens, listing the recipes available for the current economy.
2. Click **Why** on a recipe to read its rationale — what the shock is and why the pins are set where they are.
3. Click one of the two load buttons. **Load naive** pins only the headline variable and lets the model fill in everything else — the model's own read of the shock. **Load + restriction** adds short-run pins on related variables that shape the shock into its economically meaningful form.
4. Adjust from there: everything a recipe loads is ordinary conditioning, so the sliders in the conditioning list, the quarter toggles, and chart drags all work on it.

What you find. Loading *replaces* the active scenario's conditioning — your existing pins are wiped, not merged, so switch to a fresh tab first if you want to keep them. The panel closes and the charts redraw under the recipe's pins. Comparing **Load naive** against **Load + restriction** for the same recipe, on two tabs, is the fastest way to see what the extra pins buy. If a recipe names a variable this economy does not carry, an amber note lists what was skipped; click **Dismiss** to clear it.

5 Cross-country spillovers

Economies trade with each other, so a shock to one moves the others. This chapter shows you how to make your scenario travel: you condition inflation or growth in one country, and the forecasts for its trading partners move too. The machinery behind this is the linked cross-country system: country models tied together by trade-weighted links, called a GVAR (§16). On screen it comes down to two switches, one button, and a small status badge.

A quick map of who is linked. Open the **Country** dropdown in the top bar: economies listed under the **GVAR** heading are full members of the linked system — conditioning any of them can reach all the others. Economies under **Other countries** include the individual euro-area members (Germany, France, Italy, ...), which are a special case: they receive shocks through the euro-area aggregate rather than through their own link (see the last subsection).

By default both spillover switches are *off*. Until you turn something on or press a solve button, a scenario moves only the country you are looking at, and its partners stay frozen. That is deliberate: local-only is instant, it is explicit about what has and has not been computed, and it is often all you need.

5.1 Turning on scenario spillovers

What you do. You let your conditioning — the points you have pinned on the charts — propagate across borders, so that, say, a United States growth scenario also moves the euro area, Japan, and China.

How.

1. In the left control panel, find the section headed **Cross-border spillovers**.
2. The first switch is **Scenario spillovers (GVAR solve)**. Click its On/Off pill to turn it **On**.
3. Condition a variable as usual (§4): click **condition** on a chart and drag the forecast, or right-click to plant a point directly.

What you find. With the switch on, every edit to a pinned point triggers a live cross-country computation. A status line appears above the chart grid. While the computation runs you see a small **computing...** pill with a spinner — the heavier recomputations can take several seconds, so the pill tells you the app is working, not frozen. When it finishes, a badge (marked with a small double-arrow icon) reports what was computed, for example **GVAR DIRECT preview** with a short technical read-out (how many passes it took, a convergence number, and how long it ran). Switch to another country in the **Country** dropdown and you will see its forecasts shifted by your scenario. If the switch is *off* instead, the status line says *local forecast (partners frozen)* whenever you have active pins — a plain reminder that what you see has not travelled anywhere.

5.2 Preview versus the complete solve

The live result you get while editing is a *preview*, and it is worth understanding what that means.

What you do. You read the badge above the charts to know how much of the cross-country feedback is in the numbers on screen.

How. Look at the badge colour and text:

- An *amber* badge reading **GVAR DIRECT preview** means you are seeing the fast first-pass answer: your shock has travelled *one hop* to each partner, but the partners' responses have not yet fed back — the echo of the echo is missing. This keeps editing responsive.
- A *blue* badge reading **GVAR full solve** means the complete answer is on screen: every feedback loop between every linked economy, including the feedback from the rest of the world onto the United States, has been solved through to consistency (see §16.6).

What you find. For small shocks the preview and the full solve are usually close, because the second-round echoes are smaller than the first hop. For large shocks, or when you care about the country *receiving* the spillover, run the full solve before you quote numbers. The hint text next to the badge tells you which situation you are in: *live direct-channel preview — click "Solve full GVAR" for all feedback loops* when a preview is showing, and *conditioning one bloc propagates to its trade partners* once the full answer is up.

5.3 The "Solve full GVAR" button

What you do. You compute the complete cross-country answer on demand — whether or not the scenario-spillover switch is on.

How.

1. Set up your scenario first: pin the paths you want.
2. Click **Solve full GVAR** (the button is marked with a filled play-triangle icon). The button appears in two places — in the **Cross-border spillovers** section of the control panel, and above the chart grid whenever a fuller answer than the one on screen is available. Pressing **Compute** in the top bar runs the same complete solve as part of its work.
3. Wait for the **computing. . .** pill to clear. The complete solve is the heavy option; expect it to take roughly ten seconds rather than an instant.

What you find. The badge turns blue and reads **GVAR full solve**, and the button above the grid disappears — there is nothing more to compute for this scenario. Every linked country's charts now show the fully consistent answer. This is the workflow to prefer if you leave the spillover switch off: work locally and fast while you shape the scenario, then press **Solve full GVAR** once to propagate it everywhere. Be aware that pressing the button also turns the **Scenario spillovers (GVAR solve)** switch **On**, so later pin edits run the live cross-country preview; switch it back **Off** if you want to return to local-only editing. Note also that the full solve reflects the pins as they are when you press it; if you keep editing afterwards, press it again.

5.4 Long-run target spillovers

What you do. You let a *long-run target* — a value you have set for where a variable settles far out (§6) — shift the long-run forecasts of the trading partners too, not just the economy you set it on.

How.

1. Set a long-run target as described in §6.
2. In **Cross-border spillovers**, turn **On** the second switch, **Long-run target spillovers (GVAR constant shifts)**.

What you find. This channel is instant — no solve button and no computing pill. The cross-country pass-through of long-run shifts is worked out once, ahead of time, so flipping the switch just applies it (the construction is in §16.6 and §18.2). The receiving country tells you what happened: the banner above its chart grid, which normally reads **Unconditional forecast:** *model-based*, turns amber and adds **+ spillover from** followed by the names of the countries whose targets are feeding in. Hover it for the full explanation. This keeps judgment visible: you can always tell a partner's baseline has been tilted by targets you set elsewhere. Note the two switches are independent — one governs scenario pins, the other long-run targets, and either can be on without the other.

5.5 Euro-area members: one interest rate, one doorway

The individual euro-area members — Germany, France, Italy, Spain, and the smaller members — do not set their own interest rate. They share the European Central Bank's policy rate, the area-wide long yield, and the area corporate bond yield (each member also carries its own sovereign spread over the Bund). The tool respects this: a member's model does not have its own policy channel to move. Instead, shocks reach a member through one doorway — the euro-area aggregate. A United States or euro-area scenario first moves the area-wide model. The solved area response — the common interest rate, area growth, area inflation — is then fed into the member's own model, which translates it into that member's GDP, inflation, and yields.

What you do. You run, say, a United States scenario and check what it does to Germany.

How.

1. Condition your scenario on the United States (or euro-area) charts.
2. Switch the **Country** dropdown to Germany.
3. Look above the chart grid. If the area shock has not been passed through yet, you see an amber note: **Cross-border spillover from the US/EA scenario not yet computed**, next to a **Solve full GVAR** button.
4. Click **Solve full GVAR** (or press **Compute** in the top bar). This solves the euro-area aggregate and pushes the area shock into the member.

What you find. After the solve, Germany's growth, inflation, and interest-rate charts show the scenario's imprint, transmitted through the common euro-area block. Members never update silently in the background — the amber note is the tool telling you plainly that the answer on screen is still the pre-scenario one, and the solve runs only when you ask for it. Check the note is gone before reading member charts under a US or euro-area scenario. If you edit the scenario again, the note reappears and you re-solve. On a member you will not see the **GVAR** badge itself; that badge belongs to the full members of the linked system, and the member's answer arrives through the euro-area doorway instead.

Two caveats to close. First, spillovers only flow between economies in the linked system — for the exact variable-by-variable channels, see §16.2. Second, on a linked economy the **Long-run targets** panel always shows an amber **GVAR bloc** note warning that, with the target-spillover switch off, partners are *not* re-solved — the realised joint outcome may differ from the single-country picture on screen.

6 Long-run targets

Every forecast in this tool eventually settles down: growth returns to a trend rate, inflation to a typical level, a fiscal ratio to its long-run average. Where each variable settles comes from the estimated model. A *long-run target* lets you overrule that: you say where a variable should end up — inflation at 2%, say, or trend growth at 1.5% — and the tool bends the whole forecast so it glides to your number instead of the model's. How that bending works is explained in Part II (§18 and §18.2); this chapter shows you the controls.

Two things to hold on to. First, every control starts at the model's own estimate, so an untouched panel changes nothing — the default view is always the raw model. Second, targets apply *instantly*: the moment you change a value, the baseline and every scenario update. You never press **Compute** for this; that button is for uncertainty bands and is covered in §9.

6.1 Set a target

What you do. Pick one variable and fix where it settles at the far end of the forecast.

How.

1. In the left control panel, click the **Long-run targets** header (just below the scenario name). The panel unfolds; the chevron turns from ► to open.

2. Find the row for your variable. There is one row for every chart that can be targeted, in the same order as the chart grid, labelled with the chart title and its unit — for example **Inflation · % q/q ann.** (quarter-on-quarter at an annual rate; units are explained in §12.3).

3. Set the value. Either drag the slider, or type in the number box, or use its arrows — each arrow click steps the target by 0.1. The small grey **est** figure next to the slider is the model's own estimate, your reference point.

4. To take a target back, click the row's reset-to-estimate arrow button — it resets that variable to the model estimate. Setting the value exactly back to the estimate does the same thing: the target is removed, not stored as a coincidence.

What you find. The row highlights amber as soon as the value leaves the estimate, the panel header gains a count such as **(1 set)**, and a green badge appears at the foot of the panel reading **1 target · exact**. "Exact" means what it says: the forecast is built to hit your number at the end of the horizon by construction, not approximately. On the chart, the forecast path for that variable now drifts toward your value instead of the model's. A **reset all to estimate** link next to the badge clears every target at once.

6.2 See which numbers are yours

What you do. Keep a visible record, on the charts themselves, of where you have overruled the model.

How. Nothing to click — the flags appear on their own. Look in two places: the chart title, and the status line above the grid.

What you find. Two flags appear automatically. On the affected chart, the title turns amber and shows a small **judgement** badge with a pencil mark; hover it and the tooltip states your target and the model estimate side by side, in the chart's units. Above the whole grid, the status line **Unconditional forecast:** switches from **model-based** to **with judgement (1 variable anchored)**, counting up as you add targets. When every target is cleared, both flags disappear and the line reads **model-based** again. This is the tool's third honest rule at work: judgment is visible, always.

6.3 Load the consensus anchors

What you do. Instead of setting values by hand, load a ready-made set of publicly defensible long-run numbers for the country you are viewing — one click.

How.

1. Open **Long-run targets**. If a preset exists for this country, a **Consensus anchors** note sits near the top of the panel, saying how many variables it covers.
2. Click **Apply consensus anchors**. The button turns green and reads ✓ **Consensus applied** while your targets exactly match the preset.
3. Click the adjacent **Clear** to drop all targets and return every variable to its model estimate.

What you find. The preset is deliberately minimal — and audited. A comprehensive review (July 2026) tested every candidate anchor against the forecast it produces and kept only those that fix a genuine long-run bias *without* distorting the near-term path: 32 anchors across 23 economies, drawn from long-run growth, inflation, and — for some public-finance economies — revenue, spending, or the effective interest rate on debt. It never sets policy rates, unemployment, or asset prices. Where a preset ships it is *on* by default (the affected charts carry the judgement flag, §6); **Clear** returns the country to the raw model. Most economies ship *no* preset at all — their long run is left entirely model-driven, so the block simply does not appear there (Japan is one; so are the economies whose long-run biases are too large to fix by a target and are queued for re-estimation instead).

6.4 Bring in an outside forecast

What you do. Feed the model someone else's forecast — IMF WEO, a consensus survey, a market curve, your own desk numbers — as an *ingredient* rather than as a hard fact. You decide how much to trust it; the model reconciles it with everything else.

How.

1. Prepare a small JSON file listing your forecast, in the units the charts show (% q/q annualised for growth and inflation, the level in % for a rate). A template ships at `data/sample_external_forecast.json`

```
{ "source": "WEO", "bloc": "US",
  "forecasts": { "GDPH": { "2026Q1": 2.4, "2026Q2": 2.2 } } }
```

2. In the left panel, open **External forecasts** and click **Import forecasts. . .**, then pick your file. The source appears in a list with a **trust** slider.
3. Set the trust. At **100 %** the forecast is pinned exactly (identical to conditioning it by hand); lower it and the model is allowed to pull the number toward its own view. Half-way means the band around that point is about half the model's own.

What you find. Each imported point shows as a **teal diamond** with a small error bar — taller when you trust it less — and every other chart moves in response, because the outside number enters through the same estimated dynamics as any pin: an imported growth path cools the rate, shifts inflation, and so on. A cell you have pinned by hand always wins over an imported one. Turning the trust to zero, or pressing the source's , removes it. This release applies imported forecasts to the *viewed* economy; the cross-country version (a WEO path for the United States moving the euro area) is planned.

Under the hood. An imported forecast is a *soft* conditioning: the Durbin–Koopman smoother (Part II, §14) is run with a measurement variance on that cell, set from the trust dial relative to the model's own predictive band — trust t gives variance $\mathrm{predVar} \cdot (1-t)/t$, so $t \to 1$ is an exact observation and $t \to 0$ is no information at all. The band therefore shrinks only *partially* at a soft pin, which is the honest thing to draw.

Saving the whole picture. Because an outside forecast, your conditioning, the long-run targets and the cross-country pins together define your view, the scenario header carries **Export session** / **Import session**: one JSON that bundles all of it across every country — so a colleague opens exactly what you see, or you pick the work up tomorrow. (The older per-scenario **Save** exports one country's conditioning to Excel.)

6.5 Several targets at once: Per-variable or Joint

What you do. When you set two or more targets, they can interfere: pushing inflation up also moves the interest rate, which moves growth, and so on. You choose how the tool handles that interference.

How. As soon as a second target is set, a **Multiple-target solve** toggle appears at the top of the panel with two buttons: **Per-variable** and **Joint**. **Per-variable** is the default.

What you find. The trade-off, in plain words:

- **Joint** solves all your targets together, so each one lands exactly on its value, knock-on effects included. This is the correct answer — but if two targets are strongly related (say two interest rates the model cannot move apart in the long run), the solve can become unstable and the paths can blow up. If the charts go wild after you add a target, switch to **Per-variable**.

- **Per-variable** sets each target on its own, ignoring what one target's shift does to the others. It is stable and bounded — but approximate: it understates the knock-on effects, so with several targets set, each one lands slightly off its stated value.

A footnote line under the toggle restates which behaviour is active. With a single target the two modes coincide, so the toggle is hidden. The mathematics behind both modes is in §18.2.

6.6 Greyed-out sliders

A few variables cannot be retargeted reliably: the model's long-run response to them is too weak for the calculation to be trustworthy. Their rows are disabled, and hovering shows *"Near-singular response — this variable cannot be retargeted reliably."* This is a safety guard, not a bug — forcing a target there would produce an explosive or meaningless path. Target a closely related variable instead.

6.7 Growth targets move the debt path

Rows for real-growth variables carry an extra tag, · **sets debt path**. Setting long-run growth changes the gap between the interest rate and growth, and that gap compounds on the stock of public debt — so a growth target quietly reshapes the debt-to-GDP forecast too. It is a consequence of arithmetic, not a structural fiscal fix. The public-finance charts are covered in §8, and the accounting behind them in §17.2.

6.8 Targets and other countries

For an economy inside the linked cross-country system (§16), the panel shows an amber note beginning **GVAR bloc.** — it means that, by default, your target moves only the country you set it on; trading partners are not re-solved. To let a long-run target travel across borders, turn on **Long-run target spillovers (GVAR constant shifts)** — the switch, and how imported judgment is flagged on the receiving side, are covered in §5.

7 Policy counterfactuals

A scenario answers "what if the data come in differently?" A policy counterfactual answers a sharper question: *what if the central bank had set interest rates differently?* You choose a theory model of how the economy responds to policy, push the policy rate up or down quarter by quarter, and the tool draws a second forecast — the counterfactual — as a dashed line next to your scenario on every chart. The scenario line stays put; the dashed line shows what that same scenario would have looked like under the alternative rate path. The machinery behind this is described in §15 and §15.1.

One term needs defining up front. The theory models used here are DSGE models — economist shorthand for models where households, firms, and the central bank are written down as decision-makers, so the model has an internal story about *why* a rate change moves inflation and output. Each model ships with pre-computed IRFs — impulse response functions, a ledger of how a one-time rate move ripples through every variable over the following quarters. The counterfactual is built by adding that ledger, scaled to your rate adjustment, on top of your scenario.

7.1 Turning counterfactual mode on

What you do. You switch the active scenario into counterfactual mode so the policy controls appear.

How.

1. In the left control panel, find the **Policy Counterfactual** section.
2. Click **Policy counterfactual**. The button turns orange and its label changes to **Exit counterfactual mode**; click it again at any time to leave.

What you find. The section expands to show four blocks: **DSGE Model**, **Policy Rule**, **Macro response**, and the adjustment sliders. Nothing on the charts changes yet — the counterfactual line only appears once you actually move the rate.

7.2 Picking a model and a policy rule

What you do. You choose which theory model, and which interest-rate rule inside it, translates your rate move into effects on the rest of the economy. This choice matters: the same one-quarter rate move can have a much larger or smaller effect under one model-and-rule pair than under another, so treat the pass-through as model-dependent, not as a fact.

How.

1. Under **DSGE Model**, the current model name is shown (the default is **US_SW07**, a widely used estimated model of the US economy). Click **Browse...** to open the model list, grouped by category with a count per group, or type in **Search models...** to filter by name or author. Click a model to select it; the list closes.
2. Under **Policy rule**, pick a rule from the dropdown. Each entry shows the rule name and a short description. Rules without pre-computed responses for the selected model are greyed out and marked **(no IRFs)** — you cannot select them.

What you find. In the default **Rule-based** mode a note below the buttons reads "IRFs pre-computed with the ... rule embedded in the DSGE equilibrium" — meaning the rule is baked into the model's ledger of responses, not applied afterwards. Only the model list is filtered to models that have at least one computed rule.

7.3 Setting the rate adjustment

What you do. You specify how much higher or lower the policy rate should be than in your scenario, quarter by quarter, in percentage points.

How. There are two equivalent ways.

1. *Sliders.* Under **Interest rate adjustment (pp)** you find twelve sliders labelled **Q1** through **Q12**, each running from -2 to $+2$ percentage points in steps of 0.05, with a signed readout on the right. Moving a slider sets that quarter. **Undo** steps back through your recent slider moves; **Reset** clears the whole adjustment.

2. *Drag on the chart.* With counterfactual mode on, the policy-rate chart shows a **drag to adjust policy** badge. Drag the forecast part of the rate line to where you want it; the nearest quarter takes the difference between your cursor and the scenario's rate as its adjustment.

What you find. Every quarter you touch — by slider or by drag — becomes *pinned*: the counterfactual rate holds your value there. Quarters you never touched are left to the model: under the rule-consistent mode below, the model's own rule decides how the rate behaves in them, responding to the inflation and output your pinned quarters have caused. Right-click a quarter on the policy-rate chart to toggle it between pinned and free; releasing a quarter also zeroes its adjustment so a later re-pin starts clean. So a single $+0.5$ in Q1 is not a five-year rate promise — it is a one-quarter surprise, after which policy follows the rule.

7.4 Two ways the economy can respond

What you do. You choose who works out the consequences of the rate move: the theory model or the forecasting model.

How. Use the **Macro response** dropdown:

- **DSGE/Taylor-rule (IRF-consistent)** — the default. The theory model's ledger of responses dictates how inflation, output, and the rest react, and the unpinned rate quarters follow the model's policy rule. This is the mode that answers "what does *this theory* say the rate move would have done?"

- **BVAR-endogenous (let adjust)** — no theory model. The rate path is handed to the estimated forecasting model as extra conditioning information, and it works out everything else's response from the historical correlations, exactly as in a scenario (§4, method in §15.3). This answers "what does *the data* say usually accompanies such a rate path?"

What you find. The two modes can disagree, and that disagreement is informative: theory attributes the move to a deliberate policy surprise; the historical pattern cannot tell a surprise from a response to something else. If the selected model-and-rule pair has no computed responses, an amber warning appears — "No IRFs for ... — using BVAR-only re-conditioning" — telling you the tool has silently fallen back to the second mode. Pick a different rule if you wanted the theory answer.

7.5 Reading the result, and the Level | Rate (IRF) toggle

What you do. You read the counterfactual off the charts and, if you want, switch to a view that shows only the policy effect.

How.

1. Look at any chart: the counterfactual is the dashed line labelled **Counterfactual** on hover, drawn in a distinct rust colour next to your scenario line.
2. To isolate the effect, set the **Units** toggle in the top bar to **% vs base**. Charts now plot deviations from baseline, and the counterfactual line becomes the pure policy effect. This view is read-only — you cannot drag in it.
3. In that view only, a second toggle appears: **Level** versus **Rate (IRF)**. **Level** (the default) shows the cumulative effect on the *level* of each variable — where GDP ends up relative to baseline, the "% chg." reporting convention. **Rate (IRF)** shows the quarter-by-quarter effect in each variable's native units — the way responses are plotted in the model database the DSGE models come from, so you can compare the chart directly against published model responses.

What you find. Under a rate hike of, say, +1 percentage point for one quarter, expect the dashed line to show the rate above the scenario at the start, inflation and growth drifting below it over the following quarters, and everything closing back towards the scenario as the effect dies out. How large and how persistent is a property of the model and rule you chose — switch models and watch the dashed line change while your scenario does not. Uncertainty bands, if computed, belong to the scenario; the counterfactual is a single model-implied path (§14.6).

7.6 Letting the tool choose the rate path: Optimal (B&M)

What you do. Instead of hand-crafting a rate path, you state what the central bank should care about — price stability versus employment — and the tool computes the rate path that best achieves it, using the Barnichon–Mesters method — it reads the best rate path directly off the model's ledger of responses, no extra estimation (§15.4).

How.

1. Under **Policy Rule**, click **Optimal (B&M)** instead of **Rule-based**.
2. Move the weight slider between **Price stability** and **Full employment**; the readout shows the two weights, written $\lambda\pi$ and λy on screen.
3. Set the goals with the π*** target** slider (inflation, 0 to 5, step 0.1) and the **y* target** slider (output, −3 to 3, step 0.1).

4. Click **Compute optimal rate path**.

What you find. The rate-adjustment sliders fill in with the computed path, and — unlike a hand-set path — *every* quarter is pinned: the optimal path is a full plan, not a one-off surprise. The dashed counterfactual line updates as usual. If the selected model and rule have no computed responses, an alert says so and nothing is loaded; pick another rule first. The caption "Barnichon & Mesters (2023) sufficient statistics" names the method.

7.7 A fiscal lever too

What you do. You add a government-spending move to the counterfactual, alongside (or instead of) the rate move.

How. Below the rate sliders, find **Fiscal adjustment — gov spending (pp)**: twelve more sliders (**Q1**–**Q12**, ±3, step 0.1) with their own **Undo** and **Reset**. They work exactly like the rate sliders, scaling the model's government-spending responses on top of your scenario.

What you find. As the caption notes, this needs the selected model to include fiscal responses — fewer than half of the models do — so if the dashed line does not react, that is why. For projecting the public finances themselves (debt, deficits), use the fiscal block instead (§8).

Everything here belongs to the *active* scenario. Each scenario tab carries its own model, rule, and adjustments — shown as a small "model/rule" caption on the tab — so you can keep, say, a hawkish and a dovish counterfactual side by side and flip between them in the bottom bar.

8 Public finances and the external accounts

For the economies that carry public-finance data, the tool adds a government block to the chart grid: what the government collects, what it spends, what it pays in interest, and where its debt goes. For three economies it also adds an external block: trade with the rest of the world and the country's net position toward it. This chapter shows you how to read both blocks, how to bend the debt path, and which numbers are data and which are stated modelling assumptions.

8.1 Reading the fiscal charts

What you do. You inspect a country's public finances: the forecast for revenue, spending, interest costs, the budget balance, and the debt ratio.

How.

1. Pick a fiscal economy in the **Country** dropdown. The block exists for the United States, the euro area and each euro-area member, Japan, the United Kingdom, Canada, and China. Other countries show no fiscal charts.

2. In the main grid, find the four estimated fiscal charts: **Revenue** and **Primary Expenditure** (both in % of GDP), **Effective Interest Rate** (in % per year), and **Stock-Flow Adjustment** (in % of GDP).
3. Below them, find the row headed **Fiscal block — derived by accounting identity**. It holds three more charts: **Primary Balance**, **Δ Debt-to-GDP**, and **Debt-to-GDP**.

What you find. The four charts in the main grid are *estimated*: the model forecasts them directly, like inflation or growth. In plain words:

- **Revenue** and **Primary Expenditure** are what the government takes in and what it spends, excluding interest, both as a share of the economy.
- The **Effective Interest Rate** is the average rate the government actually pays: interest costs divided by the debt stock. It moves slowly, because old debt was issued at old rates.
- The **Stock-Flow Adjustment** is the bookkeeping gap between the deficit and the change in debt — items such as asset sales or valuation effects that move debt without passing through the budget.

The three charts in the derived row are *not* estimated. They are computed from the four drivers by budget arithmetic. The **Primary Balance** is simply revenue minus primary expenditure. The debt line is an accounting identity rolled forward: each quarter, debt grows with the interest the government pays, shrinks as the economy grows in money terms, and falls one-for-one with the primary surplus. The tool applies that rule quarter by quarter, starting from the last observed debt ratio. **Δ Debt-to-GDP** shows the resulting change per quarter. The exact arithmetic is in Part II, §17.2.

Three reading notes. First, the derived charts show the forecast horizon only — the history sits in the driver charts above. Second, the **Units** toggle in the top bar does not change them; they are always ratios to GDP or rates. Third, **Δ Debt-to-GDP** carries a grey **derived** badge and cannot be conditioned — it is implied by the others.

8.2 Move the fiscal forecast through its drivers

What you do. You impose a view on a fiscal driver — say, revenue one point of GDP weaker — and read off what it does to the balance and the debt.

How. Condition the driver exactly as you condition any variable (§4): click **condition** on the **Revenue**, **Primary Expenditure**, **Effective Interest Rate**, or **Stock-Flow Adjustment** chart, then drag the line or use the sliders in the left panel.

What you find. The whole model updates, and the derived row updates with it. A note appears next to the block header — **derived variables implicitly conditioned** — reminding you that once you pin any fiscal variable, the primary balance and the debt path are pinned too, through the identity. Growth and inflation matter as much as the budget itself: push growth down and the debt ratio rises even with unchanged fiscal numbers, because the denominator shrinks and the gap between the interest rate and growth widens.

8.3 Set a debt path or a primary-balance path directly

What you do. Instead of working through the drivers, you draw the outcome you want: "debt stabilises at, say, 100% of GDP" or "the primary balance reaches, say, plus one percent of GDP by 2028" — and let the tool find the budget that delivers it.

How.

1. On the **Debt-to-GDP** or **Primary Balance** chart, click **condition**, or simply right-click where you want the path to pass — that plants a dot exactly at the clicked quarter and value.
2. Drag the line to shape the trajectory. The badge reads **drag to set path**. On these two charts, the value you drag to is the *absolute* target — 100 means 100% of GDP, not a deviation.
3. Right-click again on a quarter to pin or release it. Released quarters follow the model.
4. To remove the target, click the **CONDITIONED** button. Note that these two trajectory targets live on the chart itself; they do not appear in the conditioning list in the left panel.

What you find. A primary-balance target is hit *exactly*: it is a straight linear condition on revenue minus expenditure, and the model finds the most plausible mix. A debt target is *approximate*, and says so. The tool inverts the debt arithmetic: it computes the primary balance that would deliver your debt path *given the model's forecast* interest rate and growth, and pins that. But the interest rate and growth themselves respond a little to the implied consolidation, and the tool does not iterate on that feedback. In practice the realised debt path can miss your target — by, say, a point or so of GDP over the horizon. If you need the debt line exact to the decimal, this is not the instrument; if you want a coherent consolidation scenario, it is. Details in Part II, §17.4.

8.4 Long-run fiscal targets

Every fiscal driver can also be given a long-run destination in the **Long-run targets** panel (§6). Two things are specific to the fiscal block. First, the real growth target carries the note **sets debt path**: where growth settles decides whether the debt ratio drifts up or down, because it sets the gap between the interest rate and growth — it is not a structural budget fix. Second, for the main fiscal economies the **Apply consensus anchors** button loads public long-run values for revenue, primary expenditure, and the effective interest rate alongside growth and inflation; it is off by default, so the raw model is what you see first.

8.5 The external accounts

What you do. For the United States, the euro area, and China, you read the country's balance with the rest of the world: exports minus imports, the broader current account, and the accumulated net position.

How. Select one of those three economies. Below the fiscal block, find the row headed **External-sector block — derived by balance-of-payments identity** with three charts: **Trade Balance**, **Current Account**, and **NFA / GDP**, all in % of GDP.

What you find. **Trade Balance** is exports minus imports. **Current Account** adds the income the country earns on its net foreign assets. **NFA / GDP** is the stock of net foreign assets — what the country owns abroad minus what it owes abroad — rolled forward, quarter by quarter, from the latest published net position: each surplus adds to it, each deficit subtracts, and exchange-rate moves revalue it. All three charts are read-only and carry a **derived** note: to move them, condition the drivers — the export, import, exchange-rate, or growth charts in the country grid — and the external row follows. A weaker currency, for instance, revalues foreign-currency assets upward and shifts the net position even before trade responds.

8.6 Which numbers are modelling defaults

Some inputs are data and some are assumptions, and the distinction matters. Data: the estimated drivers, the starting debt ratio, and the starting net foreign asset position — the latter two come from published official statistics for each economy. Stated assumptions, shipped flagged in the model files:

- **The structural stock-flow adjustment.** The debt roll-forward uses a fixed per-quarter constant for the deficit–debt gap, not the forecast from the **Stock-Flow Adjustment** chart, because that forecast can be dragged to extremes by other conditions and would distort the debt line. The chart shows the model's forecast; the debt arithmetic deliberately uses the stable constant.

- **The return on net foreign assets**, set to about 3% per year, which generates the income line inside the current account.

- **The share of net foreign assets exposed to the exchange rate**, set to about 0.4, which scales the revaluation effect.

Neither of the last two is estimated; treat the level of the income and revaluation terms as indicative and the direction of movement as the reliable part.

Finally, uncertainty: after you press **Compute** with **Quick** or **Full** selected (§9), shaded bands appear on the derived fiscal and external charts too. Each band is built consistently: every one of the model's alternative re-solves is pushed through the same accounting identity, so the debt band widens exactly as fast as the underlying budget and growth uncertainty says it should. Where you have pinned a quarter of a debt or primary-balance target, the band closes onto the line at that quarter.

9 Uncertainty bands

A forecast line is a best guess, not a promise. This chapter shows you how to read the range around that guess, and how to put the same kind of range around a scenario you have built yourself. There are two kinds of bands on the charts: a grey fan that is always there, and a coloured scenario band that you compute on demand.

9.1 The grey fan: the model's own range

What you do. You read how sure the model is about its own baseline forecast, before you have touched anything.

How.

1. Nothing. The grey fan is always on. Open any chart and look at the shaded region around the dotted **Baseline** line in the forecast area.

What you find. The fan covers roughly the middle two-thirds of the outcomes the model considers plausible: it runs from the 16th to the 84th percentile of the model's simulated futures, meaning about one outcome in six falls above the fan and one in six below. Two things to check:

- The fan *widens* as you look further out. Next quarter is fairly well pinned down; five years out much less so. That is a feature, not a flaw.
- The fan belongs to the *baseline*. It was computed when the model was estimated and does not react to your clicks. When you condition a chart, the grey fan stays where it is — your scenario gets its own band, described next.

How the range is built — from repeated simulations of the estimated model, including both coefficient and shock uncertainty — is in §19.6.

9.2 Putting a band around your scenario

What you do. You have pinned a path — say, inflation half a point higher for four quarters (§4) — and you want to know how uncertain the rest of the system's response is.

How.

1. In the top bar, find the **Uncertainty** toggle. It has three settings: **None**, **Quick**, and **Full**.

2. Pick **Quick** or **Full**. Quick uses 50 alternative versions of the model; Full uses 200. An "alternative version" here is one set of model coefficients that is also consistent with the historical data — the estimation does not produce a single model but a whole family of plausible ones, and the band samples from that family (§14.6).

3. Press **Compute**. Every chart greys out behind a message — **Computing scenario uncertainty (50 draws)...** or **(200 draws)...** — and comes back with the band drawn.

Expect a wait. Quick usually takes a few seconds. Full takes proportionally longer. The very first Compute for a country is the slowest, because the browser first has to download that country's file of sampled coefficients (several megabytes); after that the file is cached and only the computation remains.

What you find. A band tinted in your scenario's colour appears around the scenario line on every chart — not just the ones you conditioned. It answers the question: *given* the path I pinned, how much could everything else still vary? By default this band keeps the full width of the model's own forecast range and is re-centred on your scenario path, so it is directly comparable to the grey fan.

Three caveats:

- The band is a *snapshot*. If you move a pin, add a variable, or change the conditioning after computing, the displayed band no longer matches — press **Compute** again to refresh it.

- After switching country or switching the active scenario, treat any band still on screen as stale — press **Compute** again before reading it.

- Setting **Uncertainty** back to **None** removes the scenario band. The grey baseline fan never goes away.

The **CSV** and **Excel** exports include the scenario's band columns only after you have pressed **Compute**; until then those cells are empty (§10).

One convenience: **Compute** also lights up when you have a long-run target set, even with uncertainty on **None** — pressing it then switches the setting to **Quick** automatically, because the long-run tilt needs the sampled model versions to work with (§18.1).

9.3 Why the band pinches shut at your pins

What you find. On a chart you have conditioned, the scenario band collapses to zero width at every quarter you pinned, then opens up again at the quarters in between and beyond.

This is correct behaviour, not a rendering glitch. A pinned quarter is something you have *asserted*: in your scenario, inflation in that quarter *is* the value you set, by construction. There is no uncertainty about an assumption. The uncertainty lives in everything the assumption does not fix — the same variable at the quarters you left free, and every other variable at every quarter. So read a pinched band as a reminder of what you have imposed, and the open parts as what the model still has to say. If a band stays wide where you expected it to pinch, check the conditioning list: the quarter may be toggled off (**NC** — short for nowcast, the model's estimate of the current, not-yet-published quarter — or an unlit quarter label, §4).

9.4 Long-run targets move the bands instantly

What you do. You adjust a value in the **Long-run targets** panel — say, potential growth a quarter-point higher — after having already computed a scenario band.

What you find. The forecast line *and* its bands glide to the new path together, immediately, with no recomputation and no wait. A long-run target is a deterministic shift of the whole forecast path (§6, §18.2): it moves where the forecast is headed, not how uncertain the model is about getting there. So both the grey fan and any computed scenario band are simply carried along rigidly — their widths are preserved exactly. You can drag a long-run slider back and forth and watch line and band move as one; you never need to press **Compute** for this. Check it yourself: note the band width at some horizon, move a target, and confirm the width is unchanged while the centre has shifted.

9.5 Experimental: the per-draw cross-country band

What you do. You switch to an alternative, slower way of building the scenario band on countries that are part of the linked cross-country system (§5).

How.

1. Find the small **Experimental** switch in the bottom-left corner of the screen.
2. Turn it on.
3. Press **Compute**.

What you find. With the switch on, **Compute** re-solves the whole cross-country system separately for each of the 50 or 200 alternative model versions, producing a true conditional cloud. That cloud pinches naturally at your pins without any overlay, but it can collapse *too* tightly when only a point or two is pinned. It is also much slower than the default. With the switch off — the default, and the recommended setting — the band is the model's own forecast range shifted onto the scenario path, as described above: full width, fast. Flipping the switch clears any band on screen, because a band built one way must not be read as if it were built the other; press **Compute** again after toggling. The construction details are in §14.6 and §16.6.

10 The data behind the charts

Every chart starts with a solid line: history. This chapter tells you exactly what that line is made of — which quarters are published data, which are filled in by the model, where each series comes from, and how the whole panel is kept up to date. The rule throughout is simple: *data are data*. Anything the model filled in is drawn in amber and labelled, never disguised as an observation.

10.1 What vintage you are looking at

What you do. You establish the "as of" date of everything on screen: the last quarter of history, and the quarter where the forecast begins.

How.

1. Look at any chart. The solid history line ends at the *panel origin* — the last quarter of history, shared by every economy; the forecast lines continue from there.
2. Hover near the boundary — the hover box shows the calendar quarter of each point.
3. For the exact per-series dates, open **Data & coverage** in the top bar (next subsection but one).

What you find. At the time of writing, the panel origin is 2026 Q1: that is the last quarter of history for every economy, and the first forecast quarter — labelled **NC** (short for nowcast, the model's estimate of the current, not-yet-published quarter) in the conditioning list and in the exports — is 2026 Q2. All economies share this single origin. That is deliberate: the linked cross-country system (labelled GVAR in the app, short for global vector autoregression — see §16) couples economies row by row, so every country's history must end on the same line. One exception: Israel's statistics arrive with a long lag, so its charts run on Israel's own calendar, about a year behind the rest of the panel. Details of the dating convention are in §19.5.

10.2 Observed history versus model-filled history

What you do. You check whether the most recent quarters of a series are published data or values the model filled in because the statistical office has not published them yet.

How.

1. Look at the end of the history line on any chart. If the last few quarters are drawn as an *amber dashed* segment with small open circles, those quarters are model-filled, not published.
2. Hover over one of the open circles. The hover box reads **Model-implied nowcast (imputed)**.

3. Compare with the solid line just before it: the dashed segment starts at the last genuinely observed point, so you can see exactly where data end and filling begins.

What you find. The amber dashed tail marks quarters that have not been published for that series. The model fills them using everything already observed elsewhere — the same economy's other series and, through the trade and financial links, every other economy's published data. (A value estimated this way for the most recent past is called a *nowcast*: a model estimate of a quarter that has already happened but has not been reported yet.) A series with no amber tail is fully observed through the panel origin. At this vintage only the United States is genuinely observed at the origin; every other economy's latest quarter is model-filled, so its series carry a short amber tail. The filling method is described in §19.5; how the linked system shares information across borders is in §16.

10.3 Check where every number comes from

What you do. For the economy you are viewing, you list every series in the model: its source, the span of quarters it covers, its last observed quarter, and any data caveats.

How.

1. In the top bar, click **Data & coverage**.
2. A dialog opens, titled **Data & coverage** followed by the country code. The header line states the last historical quarter, the last quarter at which *all* series are observed together, and the variable count.
3. Scan the table. The columns are **Variable**, **Source**, **Coverage**, **Last obs**, and **Note**.
4. Click anywhere outside the dialog, or the × button, to close it.

What you find. A last-observed quarter drawn in amber means the series ends before the panel origin — exactly the series that carry an amber dashed tail on their charts. The **Note** column carries plain-language flags: quarters around the pandemic treated as missing during estimation (see §19.4), an interior gap in the raw series, or a source still under review. Sources are the genuine national, FRED, Eurostat, BIS, ECB, and IMF providers; the full source list is in §19.3 and the cross-economy coverage matrix in §20.59. The view is read-only: opening it never re-pulls data or changes anything on the charts.

10.4 See how the latest data release moved the world

What you do. After a data update, you ask: how did the new numbers change each economy's estimate of the current quarter — and which economies moved only because *someone else's* data came in?

How.

1. In the top bar, click **Monitor**. The **Release-impact monitor** opens in the same tab, with a ← **back to the tool** link at the top.

2. Read the intro line: it names the update quarter and lists the economies with their own new releases this round.

3. Scan the headline table. Columns: **Economy**, **GDP nowcast revision (pp, q/q ann)**, **Inflation revision (pp)**, **Policy rate (pp)**, and **Channel**. Use the two buttons next to **sort by** to rank economies by the absolute size of the GDP or inflation revision.

4. Click **detail** on any row to expand a per-variable table for that economy; click **hide** to collapse it.

What you find. Each number is a *revision*: the economy's estimate of the update quarter after the new data, minus what the previous vintage had already predicted for that same quarter. The **Channel** badge is the key column. **observed** (green) means the economy's own new data drove the move — its genuine surprise against what was expected. **cross-block** (blue) means the economy published nothing itself: its revision is pure cross-border transmission of other economies' releases through the trade and financial links. Watching, say, the euro area or Canada shift on a United States release is the linked system at work on real data — the same mechanism you steer by hand in §5. In the per-economy detail, each variable shows its before and after values, the revision, and how that revision carries into the first and the final forecast quarter; an **obs** or **nowcast** badge says whether that particular cell was published or model-filled. If a source restated its own history, the intro also lists the *accepted source revisions* for that update.

10.5 The update cycle: quarterly shifts, annual re-estimation

What you do. You understand what changes — and, just as important, what does *not* change — each time the panel is updated.

How. Nothing to click: the cycle is an operating rule, and every update leaves visible traces you can inspect with the two views above.

1. *Within the year*, the panel gets quarterly *data shifts*. Newly published quarters are pulled in for the economies with wired sources; every other economy's new quarter is filled by the linked system from everything just observed, and flagged as model-implied (the amber tails). The estimated model itself is untouched — frozen exactly as estimated.

2. *Once a year*, after the previous calendar year is fully published for every economy, the whole system is re-estimated on that complete year of data. Only then do the model's estimated relationships change.

3. *Revisions are accepted and logged.* When a statistical office restates its history, the revised figures replace the old ones as the current best knowledge, and the change is recorded — the Monitor page lists the accepted revisions of each update.

What you find. After a quarterly shift, the history line is one quarter longer, the forecast starts one quarter later, and the amber tails move with the new origin — but the model's behaviour, and therefore the way your scenarios respond, is unchanged. Any movement in the forecast comes from new data, not from a silently different model. After the annual re-estimation, the model itself is new, and the manual's technical chapters (§19.5, §22) describe how that vintage is documented. If a chart surprises you after an update, the reading order is: hover the tail (observed or model-filled?), open **Data & coverage** (did the series' last observation move?), then open **Monitor** (was it your economy's own release, or transmission from abroad?).

Part II

How the machinery works

This part documents the methods behind the screen — the estimation, the filtering, the cross-country system, the accounting — with the mathematics and the references. You do not need it to use the tool. It is here so that every number on the screen can be traced to a method, every method to code, and every series to a source.

11 System Overview

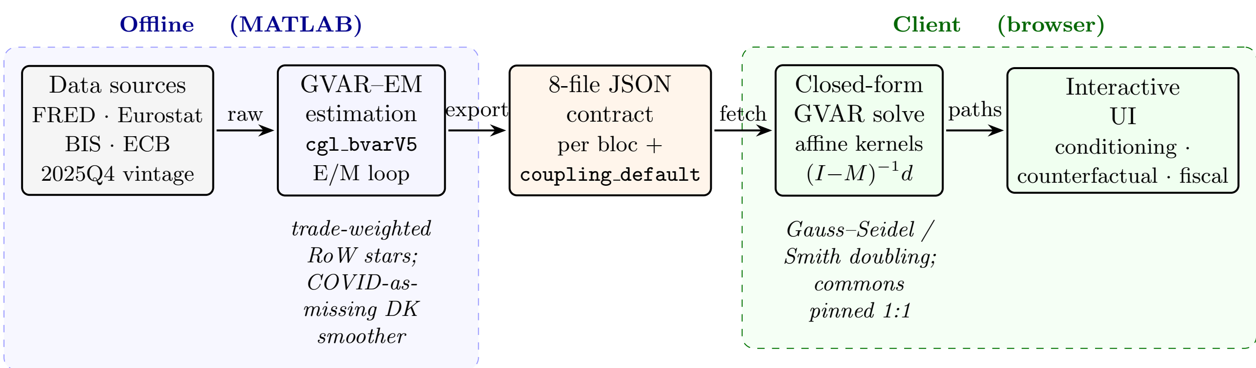

Figure 1: End-to-end system architecture of the EM-Bayesian Global VAR platform. Harmonized macro-financial data (FRED, Eurostat, BIS, ECB; common 2025Q4 *estimation* vintage, with the observed panel subsequently refreshed to 2026Q1 with no re-estimation, §19.5) feed the *offline* MATLAB GVAR–EM estimator: per-bloc Bayesian VARs (`cgl_bvarV5`) are re-fit in an EM loop that, at each sweep, rebuilds the trade-weighted rest-of-world "star" variables (ROWDEM/ROWINF/ROWFIN) from partners' Durbin–Koopman-completed panels and re-estimates every bloc with the 2020Q2–Q4 *macro* columns masked (the foreign-rate star ROWFIN and all domestic financials kept observed; COVID-as-missing), iterating to $\max|\Delta\text{star}| < 0.03$ (cf. §16). Each converged bloc is serialized to the eight-file JSON contract (`spec`, `companion_mean`, `companion_draws`, `baseline`, `baseline_levels`, `historical`, `historical_levels`, `transformations`; plus `fiscal_meta` and the precomputed `coupling_default.json.gz` coupling artifact). In the *client* (browser), the cross-country system is solved in closed form: the scenario delta obeys the affine fixed point $x = Mx + d$, evaluated by Gauss–Seidel sweeps or directly via $(I - M)^{-1}d$ (Smith doubling), with global commons (US WTI oil, US 10Y, US BAA, US equity) pinned 1:1 onto every RoW bloc. The interactive UI exposes conditioning, counterfactuals, and the stock-flow fiscal block (§17).

Conditional forecasting with Bayesian vector autoregressions (BVARs) has become a standard tool in central bank forecasting and policy analysis. The seminal contributions of Doan et al. (1984) and Waggoner and Zha (1999) established the methodological foundations, while Bańbura et al. (2015) extended these techniques to large cross-sections using state-space methods. The Monetary-Financial Model described in this document builds on this literature to provide an interactive, browser-based platform that makes conditional forecasting and policy counterfactual analysis accessible without specialized software.

The tool implements a three-layer analytical framework:

(i) **Baseline forecast:** An unconditional forecast from a large BVAR estimated on quarterly macroeconomic and financial data, following the hierarchical prior framework of Giannone et al. (2015).

(ii) **Conditional scenario:** The user conditions any subset of variables on desired paths; a Durbin–Koopman smoother (Durbin and Koopman, 2002, 2012) propagates these constraints to all remaining variables, respecting the full covariance structure of the VAR.

(iii) **Policy counterfactual:** An alternative monetary or fiscal policy path is overlaid on the scenario using impulse response functions from a DSGE model drawn from the Macroeconomic Model Data Base (MMB; Wieland et al., 2012), and the resulting macroeconomic responses are re-conditioned through the BVAR to ensure cross-variable consistency.

The tool is designed for economists at central banks and international institutions who need to construct and communicate conditional projections and policy scenarios. It runs entirely in the browser, requiring no server-side computation, and can be deployed as a static website on any hosting platform.

Part II is organized as follows. Section 12 describes the BVAR estimation methodology, and Section 13 derives the companion-form state-space representation. Section 14 presents the Kalman filter and Durbin–Koopman smoother used for conditional forecasting. Section 15 details the policy counterfactual framework, and Section 16 the cross-country system. Section 17 covers the fiscal block and Section 18 the long-run targets. Section 19 documents the data, coverage, and vintage policy. Section 21 discusses technical implementation details, Section 22 reproducibility, and Section 24 concludes with limitations and directions for future development. The guide to the user interface is Part I (Sections 2–10).

11.1 Notation

Table 1 collects the symbols used throughout the manual, grouped by the layer in which they appear (BVAR estimation, the companion-form state space and smoother, the GVAR coupling, the fiscal recursion, and the long-run drift setter). A plain-language *Glossary* of the named concepts—BVAR, GVAR, the trade-weighted stars, COVID-as-missing, the stock-flow adjustment, the EM loop, the endogenous dominant unit, the long-run drift setter, and the evaluation statistics (CRPS, Giacomini–White, . . .)—is provided in Appendix A.

Table 1: Notation used across the BVAR, GVAR, fiscal, and long-run layers.

Symbol	**Meaning**
BVAR estimation (Sec. 12)	
y_t	$n \times 1$ vector of endogenous variables of a bloc at quarter t (in estimation space).
n	number of variables in a bloc ($n = 41$ for the endogenous-US bloc).
p	VAR lag order ($p = 3$ throughout).
B_ℓ	$n \times n$ reduced-form autoregressive coefficient matrix on lag ℓ.
ε_t	reduced-form innovation, $\varepsilon_t \sim \mathcal{N}(0, \Sigma)$.
Σ	$n \times n$ innovation covariance; posterior-mean $\hat{\Sigma}$.
β, $\hat{\beta}$	stacked VAR coefficients; posterior mean.
λ	overall Minnesota/GLP shrinkage (tightness) hyperparameter.
μ	sum-of-coefficients prior hyperparameter (Giannone et al., 2015).
ψ	residual-variance scale of the prior (per variable).
δ_i	prior mean on variable i's own first lag: $\delta_i = 1$ (random walk, `RW`) or $\delta_i = 0$ (stationary white noise, `WN`).
`pos`	index set of the `WN` variables; sets $\texttt{diagb}(\texttt{pos}) = 0$ (Sec. 12.5).
D	number of retained posterior draws $\{(\hat{\beta}^{(d)}, \hat{\Sigma}^{(d)})\}$.
State space & smoother (Sec. 13–14)	
α_t	companion-form state vector (stacked current and lagged y).
T, Z, Q, H	transition, observation, state-noise, and measurement-noise matrices.
$Z_t^{(o)}$	rows of Z corresponding to the *observed* variables at t (partial observation).
$\alpha_{t\|t-1}, P_{t\|t-1}$	one-step predicted state mean and covariance.
v_t, F_t, K_t	innovation, innovation variance, Kalman gain.
r_t, L_t	Durbin–Koopman backward recursion vector and transition.
$\alpha_{t\|H}$	smoothed (full-interval) state.
$\mathcal{W}$	COVID-as-missing window (the fixed three quarters 2020Q2–Q4).
GVAR coupling (Sec. 16)	
c, j	bloc (economy) indices; c receiver, j partner.
w_{cj}	trade weight of partner j in bloc c's stars, $\sum_j w_{cj} = 1$.
`ROWDEM`, `ROWINF`, `ROWFIN`	trade-weighted foreign demand, inflation, and financial "star" variables.
$s_{c,v,t}$	scenario pin-delta of bloc c's star v at horizon t.
$g\Delta_{p,t}$	deviation of US-determined global column p (oil, US10Y, USBAA, USeq).
x	stacked deviation state of the coupling fixed point.
M	solve-time coupling operator; $x = Mx + d$.
d	exogenous forcing (own-scenario pins) of the fixed point.
$\rho(M)$	spectral radius of M (multiplier strength; $< 1 \Rightarrow$ contractive).
Fiscal recursion (Sec. 17)	
d_t	gross public debt as a percent of GDP.
$rev_t, pexp_t$	revenue and primary expenditure (% of GDP); `FREV_GDP`, `FPEXP_GDP`.
i_t^{eff}	effective interest rate (`EFFI`, % p.a.).
sfa_t	stock-flow adjustment (`SFA`, % of GDP).

continued on next page

Table 1 continued from previous page

Symbol	Meaning
g_t^n	nominal GDP growth.
pb_t	primary balance $= rev_t - pexp_t$.
Long-run targets & evaluation (Sec. 18, 23)	
$\psi_j^\star, \psi_j^0$	user long-run target and current terminal display forecast of variable j.
c, δc_j	companion drift (constant row); shift of variable j's own equation.
S_h	partial companion-power sum $\sum_{k<h} T^k$; $S_h[:, j]\, \delta c_j$ is the response.
sens_j	display-unit sensitivity of j's terminal forecast to a unit δc_j.
h	forecast horizon (quarters); $h \in \{1, 2, 4, 8\}$ in the horse-race.
t_0	forecast origin in the recursive out-of-sample scheme.
d_t (loss)	loss differential between two models at t (overloads d_t; context disambiguates).
CRPS	continuous ranked probability score (primary density metric).

12 BVAR Estimation

12.1 Reduced-Form VAR

Consider an n-variable VAR(p) in reduced form:

$$y_t = c + \sum_{\ell=1}^{p} B_\ell\, y_{t-\ell} + u_t, \qquad u_t \sim \mathcal{N}(0, \Sigma), \tag{1}$$

where $y_t \in \mathbb{R}^n$ is the vector of observables, $c \in \mathbb{R}^n$ is a vector of intercepts, $B_1, \ldots, B_p \in \mathbb{R}^{n\times n}$ are the autoregressive coefficient matrices, and $\Sigma \in \mathbb{R}^{n\times n}$ is the positive-definite innovation covariance matrix. In the US specification, $n = 41$ and $p = 3$, yielding a coefficient matrix $\hat{\beta} \in \mathbb{R}^{124\times 41}$ (with row 0 containing the intercept). The 41 US variables include the four US-determined *global commons* (`GLB_OIL`, `GLB_USLR`, `GLB_USBAA`, `GLB_USEQ`) that the GVAR layer of Section 16 carries into every other bloc, the four-variable fiscal block (Section 17), and the three trade-weighted `ROW*_US` star variables through which the US, now an endogenous GVAR bloc, receives Rest-of-World feedback.

12.2 Data Transformations

Variables enter the VAR in one of two forms:

- **Log-transformed** (`log`): the variable is modeled in log levels. Growth rates displayed to the user are quarter-over-quarter annualized:

$$g_t^{\text{q/q ann.}} = 400 \times \big(\ln Y_t - \ln Y_{t-1}\big). \tag{2}$$

  Year-over-year growth rates are computed as:

$$g_t^{\text{y/y}} = 100 \times \big(\ln Y_t - \ln Y_{t-4}\big). \tag{3}$$

- **Linear** (`lin`): the variable enters in levels (e.g., interest rates in percent, unemployment rate). No growth-rate conversion is applied.

12.3 Reporting Units

The display unit is chosen per variable for interpretability, independently of the estimation transformation:

- **National-accounts / activity flows** (GDP, consumption, investment, trade volumes) are shown as growth (q/q annualized or y/y).
- **Prices** (CPI/HICP/deflators) are shown as inflation.
- **Rates and ratios** (policy and market rates, unemployment, fiscal ratios) are shown as levels in percent / percentage points. Unemployment-rate uncertainty bands are floored at 0 (a rate cannot be negative); policy and market rates are *not* floored, because negative nominal rates do occur.
- **Index / FX / equity / commodity series** (the trade-weighted dollar `FXTWM`, US equity `GLB_USEQ`, house prices `HP`, oil `PZTEXP`/`GLB_OIL`, the non-fuel commodity index `GSNE`, and the other blocs' `NEER_EA`, `NEER_UK`, `NEER_CA`, `REER_CN`, `EQ_CN`) are reported as the *index level* (e.g. $2010 = 100$) rather than as q/q growth: a trade-weighted exchange rate or equity index q/q-annualized is not an informative number, whereas its level is. These series remain `log` for estimation and are conditioned in the stored $100\ln(\text{index})$ space; only the display is exponentiated back to the level, $\exp(\text{stored}/100)$, with a level-space uncertainty fan.

12.4 Hierarchical Minnesota Prior

The model is estimated following Giannone et al. (2015), who cast the Minnesota prior (Litterman, 1986; Doan et al., 1984) in a hierarchical framework with data-driven hyperparameter selection. The prior specification includes:

(a) The **Minnesota shrinkage** prior on B_ℓ, centered on the random-walk or white-noise prior for each variable (depending on the variable's stationarity properties). The tightness parameter λ controls the overall degree of shrinkage toward the prior mean.

(b) The **sum-of-coefficients** prior (Doan et al., 1984), which shrinks toward the belief that a unit increase in every variable's level produces a unit increase in its forecast, imposing a form of cointegration.

(c) The **single-unit-root** prior, which constrains the system to have at most one common stochastic trend.

The hyperparameters governing the tightness of each prior component are treated as additional parameters and selected by maximizing the marginal likelihood, as advocated by Giannone et al. (2015).

Posterior draws are obtained via Markov chain Monte Carlo (MCMC). The estimator runs a long chain with burn-in, retaining the posterior mean $\hat{\beta}$ and $\hat{\Sigma}$ for the deterministic conditional forecasting engine. For the browser the chain is *thinned* to $D = 500$ retained draws $\{(\hat{\beta}^{(d)}, \hat{\Sigma}^{(d)})\}_{d=1}^{D}$, the count shipped in every bloc's `companion_draws.json`, used for uncertainty quantification via the simulation smoother (Section 14.6). The thinned export keeps the per-bloc draw artifact small enough to load on demand (the US file is $\approx 34\,\mathrm{MB}$ at $D = 500$, $n = 41$, $p = 3$).

12.5 Per-variable own-lag prior mean: random walk vs. stationary

The Minnesota/GLP prior is centered, for each equation, on a univariate model in which the variable's own first lag carries a prior mean of δ_i and every other coefficient a prior mean of zero (Litterman, 1986; Doan et al., 1984; Giannone et al., 2015). The *default* choice $\delta_i = 1$ encodes a random-walk (`RW`) prior belief—"the best forecast of the level is the current level"—which is appropriate for variables that are (near-)integrated in the estimation space (log real activity, log price levels, nominal yields). For a variable that is genuinely *stationary*—a mean-reverting ratio or rate that fluctuates around a constant level—the random-walk center is the wrong null: it shrinks the dynamics toward a unit root and lets the variable *drift* at long horizons rather than return to its historical mean. For such variables the tool sets the own-lag prior mean to $\delta_i = 0$, a stationary white-noise (`WN`) prior whose null model is "the best forecast is the unconditional mean (the equation constant), not the last observation."

The mechanism (`pos`/`stationary`). The choice is implemented exactly once, at the point where the prior mean of the coefficient matrix is built. In `BVAR/MFBVAR/GLPreplicationWeb/bvarGLPmf.m` the own-first-lag prior block is the diagonal matrix $\texttt{diag}(\texttt{diagb})$ with $\texttt{diagb} = \mathbf{1}_n$ by default; the line

$$\texttt{diagb}(\texttt{pos}) = 0$$

zeroes the own-lag prior mean for every variable index in the vector `pos`, so those equations are shrunk toward white noise instead of a random walk. The index set is threaded into the sampler as the third positional argument `stationary` of `BVAR/MFBVAR/cgl_bvarV5.m` (internally $\texttt{pos} = \texttt{stationary}$), which forwards it as the `'pos'` option to `bvarGLPmf`. The estimators build the set directly from the per-variable `prior` tag in each bloc's specification,

$$\texttt{pos} \;=\; \texttt{find}(\texttt{prior} == \text{`}\texttt{WN}\text{'}),$$

so a variable opts in to the stationary prior simply by carrying `prior:"WN"` in its spec (see `pipeline/export_bvar_gvar24_loop.m` and `pipeline/export_bvar_euro_fsap.m`). For the 44 non-fiscal blocs the set is empty and the call is byte-identical to the historical all-`RW` behaviour.

Prior, not constraint. Crucially $\delta_i = 0$ sets only the prior *mean*; it does not *impose* stationarity. The posterior own-lag coefficient is the usual GLP balance of prior and likelihood, governed by the estimated tightness λ (itself selected by the marginal-likelihood maximization, Section 12.4). A genuinely persistent series whose data favour a near-unit root will still produce a large posterior own-lag coefficient despite the white-noise center; the `WN` prior simply removes the *built-in* drift that the random-walk center would otherwise inject at the long end. This is the mechanism that makes the fiscal GDP-ratios mean-revert (Section 17.1).

13 Companion-Form State Space

To apply the Kalman filter, the VAR(p) in (1) is cast in companion form as a first-order state-space model (see, e.g., Durbin and Koopman, 2012, ch. 3).

13.1 State Vector

Define the np-dimensional state vector:

$$\alpha_t = \begin{pmatrix} y_t \\ y_{t-1} \\ \vdots \\ y_{t-p+1} \end{pmatrix} \in \mathbb{R}^{np}. \tag{4}$$

13.2 Transition Equation

The state evolves according to:

$$\alpha_t = T\,\alpha_{t-1} + d + R\,\eta_t, \qquad \eta_t \sim \mathcal{N}(0, \Sigma), \tag{5}$$

where the $np \times np$ transition matrix T, the np-dimensional intercept vector d, and the $np \times n$ selection matrix R are:

$$T = \begin{pmatrix} B_1' & B_2' & \cdots & B_{p-1}' & B_p' \\ I_n & 0 & \cdots & 0 & 0 \\ 0 & I_n & \cdots & 0 & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & I_n & 0 \end{pmatrix}, \quad d = \begin{pmatrix} c \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad R = \begin{pmatrix} I_n \\ 0 \\ \vdots \\ 0 \end{pmatrix}. \tag{6}$$

Remark 1. The B_ℓ' blocks in T are transposed relative to (1) because the coefficient matrix $\hat{\beta}$ is stored as $(1 + np) \times n$, with row 0 containing the intercept c and rows $1, \ldots, np$ containing the stacked lag coefficients. Specifically, $T(i, j) = \hat{\beta}(j + 1, i)$ for $0 \le i < n$ and $0 \le j < np$.

13.3 Observation Equation

The observation equation extracts the contemporaneous variables:

$$y_t = Z\,\alpha_t + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, H), \tag{7}$$

where $Z = \big(I_n \mid 0_{n\times n(p-1)}\big)$ and $H = \sigma_\varepsilon^2 I_n$ with $\sigma_\varepsilon^2 = 10^{-12}$ (a negligible measurement error included for numerical stability of the Kalman filter recursions).

13.4 State-Noise Covariance

The $np \times np$ state-noise covariance matrix is:

$$Q = R\,\Sigma\,R' = \begin{pmatrix} \Sigma & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & & \ddots & \\ 0 & 0 & \cdots & 0 \end{pmatrix}. \tag{8}$$

14 Kalman Filter and Smoother

14.1 Conditioning via Partial Observations

The key insight underlying conditional forecasting in a state-space framework, following Bańbura et al. (2015), is that conditioning on variable i at horizon h is equivalent to observing $y_{T+h,i}$ in the Kalman filter. For each forecast period $t = T+1, \dots, T+H$, the observation vector is partitioned into:

- **Conditioned variables:** $y_t^{(o)}$, with known target values specified by the user.
- **Free variables:** $y_t^{(m)}$, treated as missing observations (coded as `NaN`).

The Kalman filter processes only the observed subset at each time step, adapting Z_t, H_t, and the innovation v_t accordingly.

14.2 Forward Pass: Kalman Filter

Algorithm: Kalman Filter with Partial Observations. **Input:** T, Z, Q, H, d; initial state α_0, covariance P_0; partial observations $\{y_t^{(o)}\}_{t=1}^H$.
For $t = 1, \dots, H$:

$$\alpha_{t|t-1} = T\,\alpha_{t-1|t-1} + d \qquad \text{(state prediction)} \tag{9}$$

$$P_{t|t-1} = T\,P_{t-1|t-1}\,T' + Q \qquad \text{(covariance prediction)} \tag{10}$$

If observations exist at time t, let $Z_t^{(o)}$ be the rows of Z for observed variables, then:

$$
\begin{aligned}
v_t &= y_t^{(o)} - Z_t^{(o)}\,\alpha_{t|t-1} && \text{(innovation)} && (11)\\
F_t &= Z_t^{(o)}\,P_{t|t-1}\,(Z_t^{(o)})' + H_t^{(o)} && \text{(innovation variance)} && (12)\\
K_t &= P_{t|t-1}\,(Z_t^{(o)})'\,F_t^{-1} && \text{(Kalman gain)} && (13)\\
\alpha_{t|t} &= \alpha_{t|t-1} + K_t\,v_t && \text{(state update)} && (14)\\
P_{t|t} &= P_{t|t-1} - K_t\,Z_t^{(o)}\,P_{t|t-1} && \text{(covariance update)} && (15)
\end{aligned}
$$

If no observations exist at time t: $\alpha_{t|t} = \alpha_{t|t-1}$ and $P_{t|t} = P_{t|t-1}$.

The matrix inversion F_t^{-1} is computed via LU decomposition with partial pivoting; singularity is flagged when the pivot falls below 10^{-15}.

14.3 Backward Pass: Durbin–Koopman Smoother

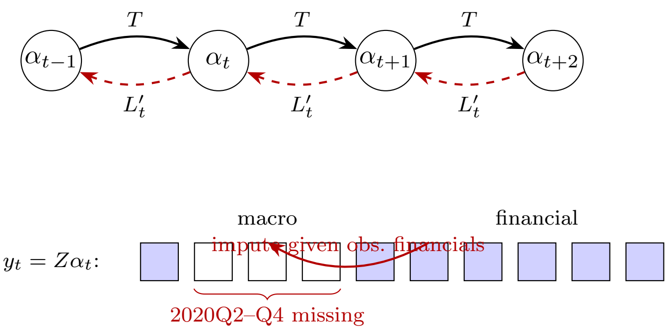

Figure 2: Companion-form state space with the Durbin–Koopman filter/smoother under the COVID-as-missing treatment. States α_t (Eq. (4)) evolve forward through the transition matrix T (solid, Kalman filter, Eq. (9)); the backward smoother propagates $r_{t-1} = (Z_t^{(o)})'F_t^{-1}v_t + L_t'r_t$ (dashed, Eq. (17)). In the observation row $y_t = Z\alpha_t$ (Eq. (7)), over the fixed window 2020Q2–Q4 (three quarters) the *macro* cells (real activity, prices, and the macro stars ROWDEM/ROWINF) are set missing (hollow) while the *financial* cells (rates, yields, equity, house prices, and the foreign-rate star ROWFIN) remain observed (filled). Because the filter processes only the observed rows $Z_t^{(o)}$ of each partially-observed vector, the macro holes are imputed conditional on the observed financials.

The smoother refines the filtered estimates using all future information, yielding $\alpha_{t|H}$ for all t (Durbin and Koopman, 2002, 2012).

Algorithm: Durbin–Koopman Fixed-Interval Smoother. **Input:** Filtered quantities $\{\alpha_{t|t-1}, P_{t|t-1}, v_t, Z_t^{(o)}$
Initialize $r_H = 0$. For $t = H, H-1, \ldots, 1$:

If observations exist at time t:

$$
L_t = T'\left(I - P_{t|t-1}\,(Z_t^{(o)})'\,F_t^{-1}\,Z_t^{(o)}\right), \tag{16}
$$

$$
r_{t-1} = (Z_t^{(o)})'\,F_t^{-1}\,v_t + L_t'\,r_t. \tag{17}
$$

Otherwise: $r_{t-1} = T' \, r_t$.
The smoothed state is:

$$\alpha_{t|H} = \alpha_{t|t-1} + P_{t|t-1} \, r_{t-1}. \tag{18}$$

Simulation smoother (draws, not just means). For uncertainty quantification and for the COVID-as-missing data augmentation (Section 19.4) the tool needs *draws* of the missing/conditioned states, not only their smoothed mean. The Durbin and Koopman (2002) simulation smoother obtains a draw $\tilde{\alpha}$ from the conditional distribution $p(\alpha \mid y^{(o)})$ by the elegant "mean-correction" device: simulate an unconditional path, smooth it, and add the smoothed *discrepancy* back to the unconditional draw. Algorithm 1 states the form used inside each `cgl_bvarV5` sweep, with partial (ragged + COVID-masked) observations handled by processing only the observed rows $Z_t^{(o)}$ at each date.

Algorithm 1 Durbin–Koopman simulation smoother with partial observations (one draw inside the data-augmentation Gibbs sweep).

Input: system matrices T, Z, Q, H; partial observations $\{y_t^{(o)}\}_{t=1}^{H}$ (COVID-macro and ragged rows absent); current draw of (β, Σ).
Output: one draw $\tilde{\alpha}_{1:H}$ from $p(\alpha_{1:H} \mid y^{(o)}, \beta, \Sigma)$, hence imputed values for the missing y cells.

1: ▷ (1) Forward simulation of an unconditional path.
2: Draw α_0^+ from the initial state law; **for** $t = 1, \ldots, H$: draw η_t, ε_t and set $\alpha_t^+ = T\alpha_{t-1}^+ + \eta_t$, $y_t^+ = Z\alpha_t^+ + \varepsilon_t$.
3: Form the synthetic partial observations $y_t^{+(o)}$ on the *same* observed pattern as the data (drop the cells that are `NaN` in $y_t^{(o)}$).
4: ▷ (2) Kalman filter (Algorithm of Sec. 14.2) on the artificial series $y_t^{(o)} - y_t^{+(o)}$.
5: **for** $t = 1, \ldots, H$: run the prediction/update recursions on $Z_t^{(o)}$, storing $\{\alpha_{t|t-1}, P_{t|t-1}, v_t, F_t\}$. ▷ at fully-missing t, no update
6: ▷ (3) Backward Durbin–Koopman smoothing recursion.
7: Set $r_H = 0$; **for** $t = H, \ldots, 1$:
8: **if** observed at t: $L_t = T'\big(I - P_{t|t-1}(Z_t^{(o)})'F_t^{-1}Z_t^{(o)}\big)$, $r_{t-1} = (Z_t^{(o)})'F_t^{-1}v_t + L_t'r_t$;
9: **else**: $r_{t-1} = T'r_t$.
10: smoothed discrepancy $\hat{\alpha}_t = \alpha_{t|t-1} + P_{t|t-1}r_{t-1}$.
11: ▷ (4) Mean-correction: add the discrepancy back to the unconditional draw.
12: $\tilde{\alpha}_t \leftarrow \alpha_t^+ + \hat{\alpha}_t$; read the imputed $\tilde{y}_t = Z\tilde{\alpha}_t$ at the missing cells.

Averaging $\tilde{\alpha}$ across Gibbs sweeps recovers the posterior-mean completed panel $\widehat{X}_c$ that Algorithm 4 feeds back into the stars; retaining the per-draw paths yields the predictive draw cloud `Dforecast` used for the uncertainty bands (Section 19.6) and for density scoring (Section 23.2).

14.4 Conditional Forecast Computation

The conditional forecast is computed as follows:

1. Initialize the state from the last p historical observations: $\alpha_0 = (y_T', y_{T-1}', \ldots, y_{T-p+1}')'$.

2. Construct the conditioning matrix $Y^* \in \mathbb{R}^{H \times n}$, where $Y^*_{t,j}$ is the target value for variable j at horizon t if conditioned, and `NaN` otherwise.

3. Run the Kalman filter (Algorithm 14.2) and smoother (Algorithm 14.3) with partial observations given by the non-`NaN` entries of Y^*.

4. Extract the observation block (first n elements) from each smoothed state $\alpha_{t|H}$.

14.5 Delta-Based Display

To ensure that zero user deviations produce exactly the stored baseline forecast, the tool computes both an unconditional smoother pass (with all entries of Y^* set to `NaN`) and a conditional pass. The displayed forecast is:

$$\hat{y}^{\text{display}}_{T+h} = \bar{y}^{\text{baseline}}_{T+h} + \underbrace{\left(\hat{y}^{\text{cond}}_{T+h} - \hat{y}^{\text{uncond}}_{T+h}\right)}_{\text{smoother delta}}, \tag{19}$$

where $\bar{y}^{\text{baseline}}$ is the pre-computed baseline forecast (posterior mean). This construction guarantees that the displayed forecast matches the stored baseline exactly when no conditioning constraints are imposed, avoiding small numerical discrepancies between the smoother's unconditional output and the stored baseline. The unconditional smoother result is cached and reused across conditioning changes.

For log-transformed variables, the smoother operates in log-level space, and the delta is scaled to annualized growth-rate space:

$$\Delta g_h = 4 \times \Delta \ell_h, \qquad \Delta \ell_h = \ell^{\text{cond}}_h - \ell^{\text{uncond}}_h, \tag{20}$$

where ℓ_h denotes the log-level forecast at horizon h.

14.6 Uncertainty Quantification

Forecast uncertainty bands are computed via the Durbin–Koopman *simulation smoother* (Durbin and Koopman, 2002), which generates draws from the posterior predictive distribution conditional on the user's constraints.

For each posterior draw $d = 1, \dots, D$:

1. **Forward simulation:** Draw shocks $\eta^+_t \sim \mathcal{N}(0, I_n)$ and compute a simulated path using the d-th posterior draw of the VAR coefficients:

$$\alpha^+_t = T^{(d)}\, \alpha^+_{t-1} + d^{(d)} + R\, L^{(d)}\, \eta^+_t, \tag{21}$$

   where $L^{(d)}$ is the lower Cholesky factor of $\Sigma^{(d)}$. The simulated observations are $y^+_t = Z\, \alpha^+_t$.

2. **Deviation computation:** At conditioned entries, compute $y^*_t = Y^{\text{target}}_t - y^+_t$; at free entries, set $y^*_t =$ `NaN`.

3. **Smoother on deviations:** Run the Kalman smoother on $\{y_t^*\}$ with initial state $\alpha_0^* = 0$ and small initial covariance $P_0^* = 10^{-7} I_{np}$, producing adjustment paths $\hat{\alpha}_t^*$.

4. **Combine:** The conditional draw is $\tilde{\alpha}_t = \hat{\alpha}_t^* + \alpha_t^+$.

The 16th and 84th percentiles across D draws yield the reported 68% credible bands. The tool uses $D = 50$ in "Quick" mode and $D = 200$ in "Full" mode. The simulation smoother correctly integrates over both parameter uncertainty (through the posterior draws) and forecast uncertainty (through the simulated shocks).

15 Policy Counterfactuals

The counterfactual layer evaluates how macroeconomic outcomes change under an alternative policy stance. It operates *on top of* the conditional scenario described in Section 14: starting from the DK-smoothed scenario paths, the user specifies deviations in the policy rate or government spending, and the tool traces the implied effects through a DSGE model's impulse response functions before re-conditioning the full system through the BVAR.

15.1 DSGE Impulse Response Functions

Impulse response functions (IRFs) are drawn from the Macroeconomic Model Data Base (MMB; Wieland et al., 2012, 2016). The MMB provides a library of pre-solved DSGE models with standardized variable names and multiple policy rule specifications. The tool includes IRFs from 169 models under 9 monetary policy rules:

Rule	**Reference**
Taylor	Taylor (1993)
SW	Smets and Wouters (2007)
CEE	Christiano et al. (2005)
GR	Gerdesmeier & Roffia (2004)
LWW	Levin, Wieland & Williams (2003)
OW08	Orphanides & Wieland (2008)
OW13	Orphanides & Wieland (2013)
Coenen	Coenen et al. (2012)
CMR	Christiano, Motto & Rostagno (2014)

Each IRF file provides the dynamic response of key macroeconomic variables—inflation, output gap, interest rate, consumption, investment, and others—to a one-standard-deviation monetary policy shock under the specified rule. Fiscal policy IRFs are available for 60 models that include a fiscal sector.

15.2 Monetary Policy Shock Inversion

Given a user-specified interest-rate adjustment path $\Delta r = (\Delta r_1, \ldots, \Delta r_H)'$, the tool solves for the structural monetary policy shock sequence $\varepsilon = (\varepsilon_1, \ldots, \varepsilon_H)'$ via the lower-triangular Toeplitz system:

$$\underbrace{\begin{pmatrix} h_r(0) & 0 & \cdots & 0 \\ h_r(1) & h_r(0) & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ h_r(H-1) & h_r(H-2) & \cdots & h_r(0) \end{pmatrix}}_{\mathcal{H}_r \in \mathbb{R}^{H \times H}} \varepsilon = \Delta r, \tag{22}$$

where $h_r(s)$ is the impulse response of the interest rate at horizon s. Since $\mathcal{H}_r$ is lower-triangular, the system is solved by forward substitution:

$$\varepsilon_t = \frac{\Delta r_t - \sum_{j=0}^{t-1} h_r(t-j)\,\varepsilon_j}{h_r(0)}, \qquad t = 0, 1, \ldots, H-1. \tag{23}$$

The implied responses of inflation and output are:

$$\Delta\pi_t^{\mathrm{IRF}} = \sum_{j=0}^{t} h_\pi(t-j)\,\varepsilon_j, \qquad \Delta y_t^{\mathrm{IRF}} = \sum_{j=0}^{t} h_y(t-j)\,\varepsilon_j. \tag{24}$$

An analogous procedure applies to fiscal policy shocks, using the government-spending IRFs (h_g, h_π^g, h_y^g, h_r^g) from the DSGE models in the MMB that include a fiscal sector.

15.3 Three Conditioning Modes

After computing the DSGE-implied deltas, the counterfactual is re-conditioned through the BVAR to enforce cross-variable consistency. Three modes govern what is held fixed versus free in the second-stage DK smoother pass:

(a) Hold-targets: All user-conditioned variables are pinned at their scenario values; the interest rate is pinned at the scenario rate plus the adjustment. Remaining variables are free and determined by the DK smoother:

$$Y_t^{\mathrm{cf}}(j) = \hat{y}_{T+t,j}^{\mathrm{cond}}, \qquad \text{for } j \in \mathcal{C}, \tag{25}$$

$$Y_t^{\mathrm{cf}}(r) = \hat{y}_{T+t,r}^{\mathrm{cond}} + \Delta r_t, \tag{26}$$

$$Y_t^{\mathrm{cf}}(j) = \texttt{NaN}, \qquad \text{otherwise.} \tag{27}$$

(b) Let-adjust: Only the policy instrument(s) are conditioned; all other variables, including inflation and output, are free and adjust endogenously through the BVAR covariance structure.

(c) IRF-consistent (default): The interest rate is pinned at the scenario rate plus the adjustment, and inflation and output are pinned at the scenario values plus the DSGE-implied deltas from (24):

$$Y_t^{\mathrm{cf}}(\pi) = \hat{y}_{T+t,\pi}^{\mathrm{cond}} + \Delta\pi_t^{\mathrm{IRF}}, \tag{28}$$
$$Y_t^{\mathrm{cf}}(y) = \hat{y}_{T+t,y}^{\mathrm{cond}} + \Delta y_t^{\mathrm{IRF}}, \tag{29}$$
$$Y_t^{\mathrm{cf}}(r) = \hat{y}_{T+t,r}^{\mathrm{cond}} + \Delta r_t. \tag{30}$$

The remaining variables are determined by the DK smoother, producing responses consistent with both the DSGE-identified monetary transmission and the BVAR covariance structure.

In all modes, the counterfactual display uses the same delta construction as (19):

$$\hat{y}_{T+h}^{\mathrm{cf,display}} = \hat{y}_{T+h}^{\mathrm{scen,display}} + f\big(\hat{y}_{T+h}^{\mathrm{cf,smooth}} - \hat{y}_{T+h}^{\mathrm{scen,smooth}}\big), \tag{31}$$

where $f(\cdot)$ applies the $\times 4$ scaling from (20) for log-transformed variables.

15.4 Optimal Policy: Barnichon–Mesters

Following Barnichon and Mesters (2023), the tool solves for the optimal monetary policy shock sequence that minimizes a quadratic loss function over deviations from target:

$$\mathcal{L} = \sum_{t=1}^{H} \Big[\lambda_\pi \big(\pi_t - \pi^*\big)^2 + \lambda_y \big(y_t - y^*\big)^2\Big], \tag{32}$$

subject to the linear mapping from shocks to outcomes given by the DSGE impulse responses.

Define the Toeplitz matrices $\mathcal{H}_\pi, \mathcal{H}_y \in \mathbb{R}^{H\times H}$ analogously to (22) using the inflation and output-gap IRFs. Stacking the weighted system:

$$\underbrace{\begin{pmatrix} \sqrt{\lambda_\pi}\, \mathcal{H}_\pi \\ \sqrt{\lambda_y}\, \mathcal{H}_y \end{pmatrix}}_{\tilde{\mathcal{H}} \in \mathbb{R}^{2H\times H}} \varepsilon = \underbrace{\begin{pmatrix} \sqrt{\lambda_\pi}\, \Delta\pi^{\mathrm{target}} \\ \sqrt{\lambda_y}\, \Delta y^{\mathrm{target}} \end{pmatrix}}_{\tilde{b} \in \mathbb{R}^{2H}}, \tag{33}$$

the optimal shock sequence is obtained via Tikhonov-regularized least squares:

$$\varepsilon^* = \big(\tilde{\mathcal{H}}'\tilde{\mathcal{H}} + \alpha I_H\big)^{-1} \tilde{\mathcal{H}}'\, \tilde{b}, \tag{34}$$

with regularization parameter $\alpha = 10^{-6}$. The implied policy rate path is $\Delta r_t^* = \sum_{j=0}^{t} h_r(t-j)\, \varepsilon_j^*$, which is then passed to the IRF-consistent conditioning mode (Section 15.3c) to obtain the full counterfactual.

The user controls the relative weight on price stability versus full employment via a slider that sets $\lambda_\pi/(\lambda_\pi + \lambda_y)$.

16 Global VAR (GVAR) and Global Commons

The single-country engine of Sections 12–15 generalizes to a multi-country setting in which a conditioning imposed on one economy spills over to all others. The tool implements a *Global VAR* (GVAR) layer in the tradition of Pesaran et al. (2004) and Dees et al. (2007), comprising the United States together with the euro area as a single aggregate bloc and 31 further Rest-of-World (RoW) blocs—33 coupled blocs in all—that span approximately 90% of global GDP. The decisive implementation choice is that *cross-country coupling is imposed at solve time, not at estimation time*: each bloc is an independently estimated single-country BVAR, and the spillovers enter only through the scenario *deviation*, which propagates as a linear fixed point. Two further channels distinguish the current version: the United States is now a *fully endogenous* coupled bloc—it drives the rest of the world *and* receives feedback from it, rather than acting as a frozen anchor—and four US-determined *global commons* (oil, the long US rate, US corporate credit, and US equity) are carried by every bloc.

16.1 Architecture: an endogenous US and 32 partner blocs

The bloc set is `US`, `EA`, `JP`, `UK`, `CA`, `CN`, `CH`, `KR`, `MX`, `AU`, `TR`, `SE`, `PL`, `NO`, `DK`, `IN`, `BR`, `RU`, `ID`, `SA`, `AR`, `ZA`, `TH`, `MY`, `CL`, `CO`, `CR`, `CZ`, `HU`, `IL`, `IS`, `NZ`, and `TW` (33 blocs: the US, the euro-area aggregate `EA`, and 31 further blocs). The euro area enters as a *single aggregate* bloc rather than as its individual members: a fully disaggregated euro GVAR is non-stationary ($\rho(M) \approx 1.089 > 1$), and aggregating the euro area into one bloc restores stationarity. The 17 euro-area member states are therefore retained only as *standalone* single-country models outside the coupled system (Section 19.2), not as GVAR blocs.

The US is now endogenous, not a frozen anchor. In earlier versions the US was treated as a *closed anchor*, exogenous in the Dees–PSW sense (Dees et al., 2007): it carried no foreign variables, was solved once, and never iterated in the cross-country fixed point. The current US model instead carries its own three trade-weighted RoW *star* variables (`ROWDEM_US`, `ROWINF_US`, `ROWFIN_US`) built from a US trade-weights row, so the US is a *fully coupled* bloc with $n = 41$ (its domestic and fiscal variables, the global commons of Section 16.7 as its own endogenous variables, and the three US star variables). It both drives every partner—through its weight in their trade-weighted aggregates and through the global commons—*and* receives partner feedback through its own stars. The US is the first entry of the coupled set and iterates inside the cross-country fixed point alongside everyone else; there is no longer a privileged frozen bloc. This RoW→US feedback is the qualitative gain of the endogenous treatment, and it is also what makes the complete solve more expensive (Section 16.6).

The topology is **not hardcoded**: it self-configures from the deployed model specifications. The US is treated as endogenous if and only if its model carries the three `ROW*_US` star variables *and* it has a row in the trade-weights file; absent either, the engine falls back to the legacy frozen-anchor solve, which remains byte-identical. More generally, a bloc is an active GVAR member if and only if its `spec.json` carries variables tagged `block: 'row'` *and* it has a row in the trade-weights file (`weights_24bloc.json`; the filename is historical—the file now lists all 33 coupled blocs, the US

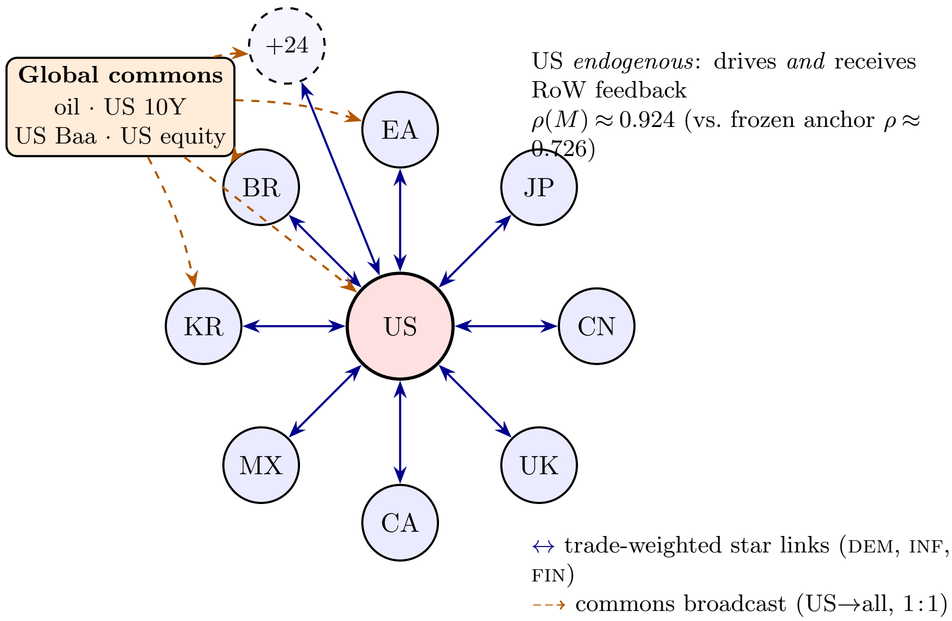

Figure 3: GVAR bloc topology. A central, dominant United States bloc is coupled to a ring of 32 Rest-of-World partner blocs (eight labelled—EA, JP, UK, CA, CN, MX, KR, BR—plus 24 further blocs) through *bidirectional*, trade-weighted "RoW star" links carrying foreign demand (DEM), inflation (INF), and the short rate (FIN). A small global-commons node (oil, the long US rate, US corporate credit, US equity) broadcasts *one-directionally* and identically (1 : 1) to every bloc. The US is now a *fully endogenous* coupled bloc rather than the classical frozen anchor: it feeds the rest of the world *and* receives feedback, raising the coupling operator's spectral radius from $\rho(M) \approx 0.726$ (frozen-US, RoW-only) to $\rho(M) \approx 0.924$ at the full 33-bloc endogenous-US configuration (the value stored in the shipped `coupling_default.json.gz`; it moves slightly with each refit), both contractive so the fixed point $x = Mx + d$ is well defined (Sections 16.1–16.5).

row included). At load, the engine derives each bloc's GDP, price, and short-rate columns from its specification (mirroring the estimator's column-resolution order) and reads its RoW-star indices, so that bloc indices are never assumed by position. Adding an economy to the GVAR therefore requires only shipping its estimated model with the appropriate variable tags and a weights row.

16.2 Trade-weighted star variables

Every coupled bloc—now including the United States—consists of its domestic VAR augmented by three trade-weighted *star* variables that summarize the foreign environment, following the weak-exogeneity device of Pesaran et al. (2004):

- `ROWDEM`: foreign demand, a trade-weighted index of partner real-GDP growth;
- `ROWINF`: foreign inflation, the analogous index on partner price levels;
- `ROWFIN`: foreign financial conditions, a trade-weighted average of partner short-term policy rates.

Let $\mathcal{W}_c \subseteq \{\text{partners of } c\}$ collect the partners that contribute to a given star at quarter t, with w_{cj} the (renormalized) trade weight of partner j in receiver c. For the activity and price stars, the partner level is read in $100\ln$ space and the star is the annualized growth index

$$\mathrm{ROWDEM}_{c,t} = 4\sum_{j\in\mathcal{W}_c} \tilde{w}_{cj}\left(\ell^{\text{gdp}}_{j,t} - \ell^{\text{gdp}}_{j,t-1}\right), \qquad \mathrm{ROWINF}_{c,t} = 4\sum_{j\in\mathcal{W}_c} \tilde{w}_{cj}\left(\ell^{\text{price}}_{j,t} - \ell^{\text{price}}_{j,t-1}\right), \tag{35}$$

where the scaling $4\,\Delta(100\ln\cdot)$ is the quarter-over-quarter annualized convention of (2), and the financial star is the level aggregate

$$\mathrm{ROWFIN}_{c,t} = \sum_{j\in\mathcal{W}_c} \tilde{w}_{cj}\, r_{j,t}. \tag{36}$$

The weights are *per-component* renormalized over the partners that are present and eligible at each t, $\tilde{w}_{cj} = w_{cj}\left(\sum_k w_{ck}\right)/\left(\sum_{k\in\mathcal{W}_c} w_{ck}\right)$, so that the mass of any absent or excluded partner is reabsorbed and the active weights still sum to one. Three eligibility rules mirror the estimator exactly: rate-less blocs (Switzerland, Norway) carry no own short rate and are dropped from partners' `ROWFIN` (but retained in `ROWDEM`/`ROWINF`); the hyperinflation bloc (Argentina) is dropped from partners' `ROWINF` and `ROWFIN` (retained in `ROWDEM`); and China's GDP is reported in *nominal* terms, so its activity contribution is deflated by *subtraction* in log space, $\ell^{\text{gdp}}_{\text{CN}} - \ell^{\text{price}}_{\text{CN}} = 100\ln(\text{NGDP}/P)$, rather than by division.

16.3 Trade weights

The bilateral weights are constructed from IMF International Merchandise Trade Statistics (IMTS) and shipped as `weights_24bloc.json` (a historical filename: the file now holds the full roster), which lists the 33 blocs together with 33 weight rows—one per bloc, the US included, now that the US is endogenous—each summing to one (zero diagonal). At load, the live weight matrix W is renormalized over the *active* bloc set, so a deployment that activates a subset of the candidate blocs remains internally consistent.

16.4 EM estimation of the coupled system: iterative completion of the star variables

The solve-time coupling of Section 16.5 takes each bloc's estimated model as given. This subsection describes how those models are *produced*—the step that is the methodological core of the system and is easy to mistake for an off-the-shelf GVAR estimation. It is not. Classical fixed-weight GVAR estimation builds the foreign-variable ("star") regressors *once*, from observed partner data, on a balanced panel, and runs a single pass of country-by-country regressions. Our panels are deeply unbalanced and the stars cannot be formed directly; we therefore estimate the blocs by an *EM-flavoured fixed-point iteration* that alternates a Durbin–Koopman data-augmentation completion of the partner panels (the E-step) with a Bayesian re-estimation of each bloc given the recomputed stars (the M-step), iterating to a fixed point in the *completed stars*. This estimation-time loop is a distinct object from the solve-time deviation system $x = Mx + d$ of Section 16.5: the former is run *offline*, once, to produce the shipped posterior-mean coefficients; the latter is run *online*, per scenario, with those coefficients frozen.

Why the stars cannot be formed directly. For receiver bloc c the star at quarter t is the trade-weighted partner aggregate $x^*_{c,t} = \sum_j w_{cj}\, x_{j,t}$ of equations (35)–(36), evaluated on the partner *levels* $\ell^{\text{gdp}}_{j,t}$, $\ell^{\text{price}}_{j,t}$, $r_{j,t}$. In a balanced classical GVAR every $x_{j,t}$ is observed for every t in a common span and the sum is immediate. Here it is not:

- **Ragged starts and ends.** Blocs enter the sample at different dates (some emerging-market series begin a decade after the US), so at early quarters only a subset of c's partners exist; partner panels also end on different quarters.
- **COVID macro masking.** Following the data-augmentation treatment of Section 19.4 and Lenza and Primiceri (2022) in spirit, the three pandemic quarters 2020Q2–Q4 are treated as *missing* in every bloc's macro (activity, price) columns—and, for a receiver, in the incoming *macro* stars `ROWDEM`/`ROWINF`—while the financial columns and `ROWFIN` stay observed. The partner GDP and price levels that would feed `ROWDEM`/`ROWINF` are therefore `NaN` during the pandemic.
- **Short credit/financial series.** Several blocs' credit or rate series begin late, leaving leading holes in the columns that feed `ROWFIN`.

A star built naïvely over the observed cells would drop every quarter in which *any* contributing partner is missing, truncating the usable sample to the common balanced span—exactly the information loss the data-augmentation design is meant to avoid. The resolution is to *complete* each partner panel with a model-based imputation and build the stars from the completed panels; but the completion of bloc j depends on j's own model, which in turn was estimated conditional on j's stars, which depend on *its* partners' completions. The completed stars are thus the fixed point of a coupled system, and we solve it by iteration.

Relation to EM and data augmentation. The loop is best read as an *EM-flavoured block-coordinate iteration* rather than a textbook EM on a single global likelihood. Within each bloc, the per-bloc step is a genuine data-augmentation Gibbs/Metropolis sampler in the sense of **?**: the missing cells (the latent "complete data") are imputed by the Durbin–Koopman simulation smoother (Durbin and Koopman, 2002, 2012) and the VAR is re-estimated given the augmented panel, exactly the expectation/maximization split of **?**. What differs from the canonical EM is the *cross-bloc* structure: there is no one likelihood being maximized across all blocs simultaneously; instead the shared latent object iterated to a fixed point is the set of *completed partner panels* (equivalently the stars they generate), and the per-bloc E- and M-steps are fused inside a single call to the COVID-aware sampler `cgl_bvarV5`, which performs both the imputation and the hyperparameter/coefficient draws in one sweep. We are explicit about this so the construction is not over-sold as a convergence-guaranteed EM: it is a contraction in the stars (verified empirically by the convergence metric below), EM in flavour and motivation, with the data-augmentation and Bayesian-estimation machinery of Sections 12–14 doing the per-bloc work.

Notation. Index the iterations by $m = 0, 1, 2, \ldots$. Let $D_j^{(m)} \in \mathbb{R}^{T_j \times n_j}$ denote the *completed* domestic panel of bloc j after iteration m, in $100 \ln$ level space (the `comp` object in code; n_j counts j's domestic variables). Write $\mathcal{S}_c(\{D_j\})$ for the star-construction map of equations (35)–(36), which reads each partner's three components from its completed panel and returns receiver c's three star columns $x_c^{*(m)} = (\texttt{ROWDEM}_c, \texttt{ROWINF}_c, \texttt{ROWFIN}_c)$, including the per-component per-quarter renormalization and the deflation/exclusion rules already stated. Let $\mathcal{E}_c$ denote one call to `cgl_bvarV5` on the augmented matrix $X_c = \left[D_c^{\text{obs}} \,\middle|\, x_c^*\right]$ with the COVID macro mask applied; it returns the draw stack $X_c^{\text{cond}} \in \mathbb{R}^{T_c \times (n_c+3) \times G}$ of G Durbin–Koopman–completed panels and the post-burn coefficient draws $\{(\beta_c^{(g)}, \Sigma_c^{(g)})\}$.

E-step (completion). Given the current stars $x_c^{*(m)}$, the E-step runs $\mathcal{E}_c$ and forms the *posterior-mean* completed panel

$$\widehat{X}_c^{(m+1)} \;=\; \frac{1}{G}\sum_{g=1}^{G} X_c^{\text{cond}}[\,\cdot,\cdot,g\,] \;=\; \overline{X_c^{\text{cond}}} \qquad (\texttt{mean(Xcond,3)}), \tag{37}$$

trimmed to the estimation rows. The simulation smoother conditions each missing cell on the *observed* cells of the same and surrounding periods: its missing-data step drops only the `NaN` *elements* of each observation vector ($y \leftarrow y(\text{ix})$, $C \leftarrow C(\text{ix},:)$, $R \leftarrow R(\text{ix},\text{ix})$ with $\text{ix} = \neg\,\text{isnan}(y)$), so a partially observed row still informs the imputation of its own holes. During the COVID window this means the masked macro cells—domestic activity and price, *and* the two incoming macro stars `ROWDEM`/`ROWINF`—are filled conditional on the financial columns and `ROWFIN`, which remain observed. The domestic block of $\widehat{X}_c^{(m+1)}$ defines the next completed panel $D_c^{(m+1)} = \texttt{panelfrom}(\widehat{X}_c^{(m+1)})$ that the *other* blocs' stars will read.

M-step (re-estimation). The same call $\mathcal{E}_c$ returns the Bayesian posterior over the bloc's VAR coefficients given the augmented panel: `cgl_bvarV5` maximizes the Giannone–Lenza–Primiceri marginal likelihood over the Minnesota hyperparameters by `csminwel`, then runs a random-walk Metropolis sampler over the tightness/sum-of-coefficients hyperparameters (λ, μ), recomputing (β_c, Σ_c) at each accepted draw (Giannone et al., 2015; Bańbura et al., 2015). The three star columns are appended *after* the domestic block and always keep the random-walk (own-lag-one) Minnesota prior; the prior mean on the first own lag is set to zero only for the variables tagged white-noise/stationary (e.g. the fiscal ratios of Section 17). The deployed coefficients are the posterior means $\hat{\beta}_c = \frac{1}{G}\sum_g \beta_c^{(g)}$, $\hat{\Sigma}_c = \frac{1}{G}\sum_g \Sigma_c^{(g)}$; the draws themselves furnish the predictive bands. Because the imputation (37) and this re-estimation happen inside one $\mathcal{E}_c$ call, the E- and M-steps are *fused* per bloc per iteration.

Outer fixed-point iteration. The blocs are swept *Jacobi-style*: all stars for iteration $m+1$ are built from the *frozen* start-of-iteration completions $\{D_j^{(m)}\}$, and the blocs are then re-estimated in parallel (`parfor`). Only after the parallel sweep are the completions overwritten, $D_j^{(m)} \leftarrow D_j^{(m+1)}$, so the next sweep sees the updated panels. A Gauss–Seidel ordering (update each bloc's completion immediately and let later blocs in the same sweep see it) reaches the same fixed point; the Jacobi choice is order-independent and parallelizes across cores, at the cost of at most one extra iteration. The recursion is

$$x_c^{*(m+1)} \;=\; \mathcal{S}_c\Big(\big\{\, \texttt{panelfrom}\big(\overline{\mathcal{E}_j\big[D_j^{\mathrm{obs}} \,\big|\, x_j^{*(m)}\big]}\big)\big\}_{j\in\mathrm{partners}(c)}\Big), \qquad c=1,\ldots,N_{\mathrm{bloc}}, \tag{38}$$

i.e. each iteration re-completes every bloc given its current stars and rebuilds every receiver's stars from the refreshed completions. The shipped models are the $(\hat{\beta}_c, \hat{\Sigma}_c)$ at the fixed point x_c^* of (38).

Convergence criterion. Convergence is measured on the stars actually used in estimation. Let $x_c^{*(m)}$ be receiver c's star matrix at iteration m, with the COVID rows of its two *macro* stars set to `NaN` (cross-iteration churn in the pandemic-imputed macro-star cells is Monte-Carlo noise of the imputation, not signal, so it is excluded; `ROWFIN` stays observed through COVID and remains in the metric). The sup-norm change is

$$\mathrm{mc}^{(m)} \;=\; \max_c \,\big\|\, x_c^{*(m)} - x_c^{*(m-1)} \,\big\|_{\infty,\,\mathrm{omitnan}}, \tag{39}$$

and the loop stops when $\mathrm{mc}^{(m)} < \textsc{tol}$ (default $\textsc{tol} = 0.03$, in annualized-percent units for the macro stars and rate-level units for `ROWFIN`), or after a hard cap of $\textsc{maxit} = 12$ iterations. A Monte-Carlo noise floor of order 0.1 from the imputation means the criterion is a practical contraction check rather than an asymptotic guarantee; setting $\textsc{maxit} = 1$ recovers the single-pass, one-shot historical-star mode (no coupling iteration) used for standalone blocs.

Numerical specifics. Five implementation details are load-bearing and match the estimator exactly:

(i) **The $4\times$ (not $400\times$) scaling.** The completed panels are already in $100\ln$ space, so a one-quarter difference $\Delta(100\ln\cdot)$ is a percent-scale quarterly change; the activity and price stars annualize it by $\times 4$ (equations (35)). A $400\times$ would double-apply the $\times 100$ and inflate the foreign-demand/inflation stars hundred-fold. The financial star is a *level* average with no $\times 4$ (equation (36)).

(ii) **Per-quarter, per-component renormalization.** The weight applied to a present partner is rescaled so the contributing weights sum to one *separately for each component and each quarter* ($\tilde{w}_{cj}$ of Section 16.2); the mass of absent (ragged-edge) or excluded (rate-less, hyperinflation) partners is reabsorbed rather than treated as zero.

(iii) **Leading-`NaN` back-fill.** A star column's leading `NaN` prefix—early quarters before any partner exists—is back-filled with the first valid value (or zero if the whole column is empty), because a `NaN`-prefixed foreign-demand regressor yields a singular state covariance; *internal* and COVID `NaN`s are left for the smoother to fill. The same guard lead-fills short domestic credit series.

(iv) **COVID macro mask.** The masked columns on X_c are the domestic macro columns *plus* the two macro stars (indices n_c+1, n_c+2); the financial star `ROWFIN` (index n_c+3) and the domestic financial columns stay observed, so the smoother fills the macro holes conditional on observed financials.

(v) **Dominant-unit / `_US` handling.** The US closed-anchor bootstrap is run *once before* the loop: the US domestic panel (COVID macro cells `NaN`'d) is completed with *no* stars, and its posterior-mean completion seeds the first-iteration partner stars. Under the legacy frozen-anchor design the US is then held out of the loop (`rowB = blocs(2:end)`); under `US_ENDOGENOUS=1` the US is folded into the sweep like any other bloc, carrying its own `ROW*_US` stars built from a `US` weights row, and iterates to the fixed point alongside the partners.

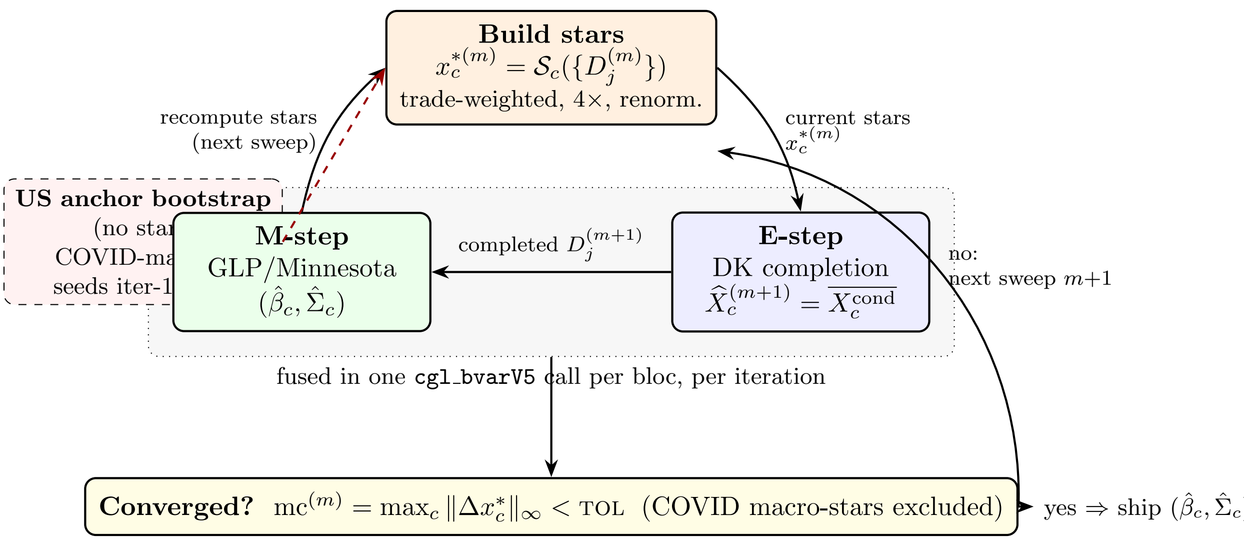

Figure 4: The estimation-time EM fixed-point loop over the completed star variables. A one-shot US anchor bootstrap (red, dashed) seeds the first iteration's stars. Each iteration then (i) builds every receiver's trade-weighted stars from the frozen start-of-iteration completed partner panels, (ii) runs the *fused* E-step (Durbin–Koopman posterior-mean completion of the bloc's missing cells, `mean(Xcond,3)`) and M-step (GLP/Minnesota Bayesian re-estimation) inside one `cgl_bvarV5` call per bloc, and (iii) recomputes the stars from the refreshed completions for the next Jacobi sweep. The loop stops when the sup-norm change in the estimation stars falls below $\text{TOL} = 0.03$ (with the COVID-imputed macro-star cells excluded as Monte-Carlo noise), capped at $\text{MAXIT} = 12$. This is distinct from the solve-time deviation fixed point $x = Mx + d$ (Fig. 5), which runs online with these coefficients frozen.

Algorithm 2 states the loop (`pipeline/export_bvar_gvar24_loop.m`; the credit-validation driver `credit_validate_loop.m` carries an identical loop).

Algorithm 2 EM fixed-point estimation of the coupled GVAR (completion of the star variables; `export_bvar_gvar24_loop.m`).

Input: per-bloc observed domestic panels with `NaN` holes (ragged edges, COVID macro mask, short credit series); trade weights W; eligibility rules (rate-less, hyperinflation, China nominal-GDP deflation); `US_ENDOGENOUS` flag; $\textsc{maxit} = 12$; $\textsc{tol} = 0.03$.

Output: per-bloc posterior-mean models $\{(\hat\beta_c, \hat\Sigma_c)\}$ and coefficient draws; converged completed stars $\{x_c^*\}$; iteration count.

$\triangleright$ US anchor bootstrap (once). Complete the US domestic panel with *no* stars (COVID macro cells `NaN`'d), $\widehat{X}_{\text{US}} = \overline{X_{\text{US}}^{\text{cond}}}$; set $D_{\text{US}}^{(0)} = \texttt{panelfrom}(\widehat{X}_{\text{US}})$. **if** `US_ENDOGENOUS` then US joins the swept set; **else** rowB = blocs(2:end) (US frozen).

for $m = 1$ **to** $\textsc{maxit}$:

  $\triangleright$ Jacobi sweep: stars built from the FROZEN $\{D_j^{(m-1)}\}$.

  parfor each swept bloc c:

    $x_c^* \leftarrow \mathcal{S}_c(\{D_j^{(m-1)}\})$ $\triangleright$ `buildstars`: trade-weighted, $4\times$ macro / level fin, per-quarter renorm, lead-`NaN` back-fill.

    $X_c \leftarrow [\, D_c^{\text{obs}} \mid x_c^* \,]$; apply COVID macro mask (domestic macro cols + `ROWDEM`,`ROWINF`; `ROWFIN` observed).

    $X_c^{\text{cond}}, \{\beta_c^{(g)}, \Sigma_c^{(g)}\} \leftarrow \texttt{cgl\_bvarV5}(X_c, \dots)$ $\triangleright$ E+M fused: DK completion + GLP/MH draws.

    $\widehat{X}_c \leftarrow \overline{X_c^{\text{cond}}} = \texttt{mean(Xcond,3)}$; $D_c^{(m)} \leftarrow \texttt{panelfrom}(\widehat{X}_c)$; store $(\hat\beta_c, \hat\Sigma_c)$ and draws.

  $\triangleright$ after the parfor: commit $\{D_c^{(m)}\}$ so the next sweep's `buildstars` sees them.

  $\triangleright$ Convergence. mc $\leftarrow \max_c \|x_c^{*(m)} - x_c^{*(m-1)}\|_\infty$ with COVID macro-star cells `NaN`'d.

  if $m > 1$ **and** mc $< \textsc{tol}$: **break** ($\textsc{converged}$).

return $\{(\hat\beta_c, \hat\Sigma_c)\}$, draws, $\{x_c^*\}$, m.

Contrast with classical fixed-weight OLS GVAR. The textbook GVAR of Pesaran et al. (2004) and Dees et al. (2007) (surveyed by Chudik and Pesaran, 2016) differs from this estimator on every axis, and the differences are exactly the contribution.

(1) **Iteration vs. single pass.** Classical GVAR builds each foreign star $x_{it}^* = \sum_j w_{ij} x_{jt}$ *once* from observed partner data, treats it as weakly exogenous, and estimates each country VARX* in a single OLS (or reduced-rank ML) pass—no re-aggregation, no feedback into the stars. Here the stars are recomputed from model-completed panels every sweep and iterated to the fixed point (38).

(2) **Missing-data completion vs. truncation.** Classical GVAR requires a *balanced* panel: ragged edges are dropped and the sample is truncated to the common span, with no imputation. Here the panel is unbalanced by design, and the Durbin–Koopman simulation smoother imputes the missing cells (ragged starts/ends, COVID macro mask, short credit) conditional on the observed cells—and crucially the imputed posterior means *flow into the stars*, which is precisely what makes the iteration necessary.

(3) **Per-quarter renormalized weights.** Trade weights are fixed (IMF IMTS, 2021–2023), as

in classical GVAR, *but* they are renormalized per quarter and per component over the partners actually present and eligible, with hyperinflation and rate-less exclusions and China's nominal-GDP deflation—an unbalanced-panel device absent from the fixed-weight formula.

(4) **Bayesian posterior vs. OLS point estimates.** Each bloc is a hierarchical GLP/Minnesota BVAR with Metropolis-sampled hyperparameters, yielding a posterior (draws $\rightarrow$ predictive bands) and a per-variable random-walk-vs-white-noise prior choice, not OLS point estimates on a single balanced sample.

(5) **Jointly solved linkages.** The GVAR linkages and the estimation are solved *jointly* here—the stars are a fixed point of the completion loop—whereas classical GVAR fixes the stars pre-estimation and stacks the country models into a single global linear system only afterwards.

In one line: the novelty is a *Durbin–Koopman data-augmentation completion of unbalanced, COVID-masked partner panels, feeding a per-quarter-renormalized trade-weighted star aggregation, inside a Bayesian per-bloc re-estimation, iterated to a star fixed point*—so the cross-country linkages are an estimated object rather than a fixed pre-estimation input. The frozen posterior means it produces are what the solve-time coupling of Section 16.5 then propagates.

16.5 Coupling at solve time, not estimation

Because each bloc is estimated independently, there is no joint, block-exogenous GVAR system; the cross-country linkage is imposed when the scenario is solved. The key observation is that, with coefficients fixed at their posterior mean $(\hat{\beta}, \hat{\Sigma})$, the conditional-forecast operator of Section 14 is *affine* in the conditioning values (the Kalman filter and Durbin–Koopman smoother are linear in the pinned data). The *baseline* forecast of each bloc is therefore data-driven and computed exactly once, via an all-`NaN` (unconditional) smoother pass; only the scenario *deviation* couples across blocs.

Because the US is now endogenous it is no longer held outside the fixed point, and the compact deviation state is correspondingly *augmented*. It collects (i) the star pin-deltas $s_{c,v,t}$ of *every* coupled bloc c—the US included—for star variable $v \in \{\mathrm{DEM}, \mathrm{INF}, \mathrm{FIN}\}$ at horizon t, and (ii) the per-period *deviation of each US global column* $g\Delta_{p,t}$, which under the frozen-anchor design was a constant and is now itself part of the unknown. Stacking these into a vector x, the deviation system is the linear fixed point

$$x = M\,x + d, \tag{40}$$

where d is the exogenous forcing (each bloc's own scenario pins moving its linked series at zero partner and global delta) and M now carries three coupling families: star $\leftarrow$ star (receiver c's star responds to a partner j's free-variable path—with the US participating both as partner and as receiver), the US-global definition rows (a global column's deviation as the US's own response to its star pins), and star $\leftarrow$ global (a partner's star responds to its own global pin). The US-determined globals therefore live *inside* M rather than in d, which is precisely the new RoW$\rightarrow$US feedback loop. The augmented state has dimension ≈ 2060 at the full 33-bloc roster.

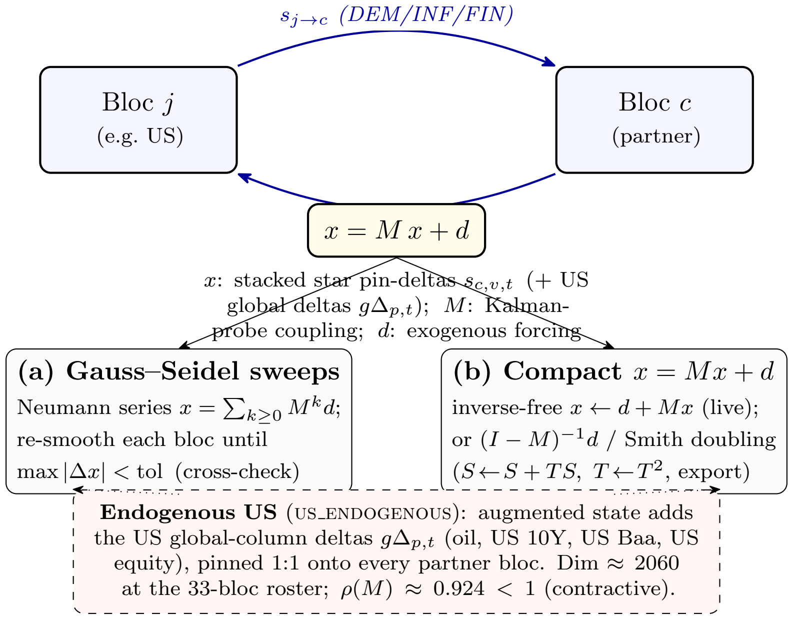

Figure 5: Solve-time cross-country coupling as the affine fixed point $x = M\,x+d$. Two coupled blocs exchange trade-weighted star pin-deltas (DEM/INF/FIN), forming the feedback cycle; the deviation state x stacks these pins (plus the US global-column deltas under an endogenous US). The fixed point admits two equivalent solutions—(a) Gauss–Seidel/Neumann sweeps through the Durbin–Koopman smoother, used as the cross-check, and (b) the compact M-state recursion, solved inverse-free by the Neumann fixed-point $x \leftarrow d + Mx$ on the production path (the explicit $(I - M)^{-1}d$ / Smith doubling are reserved for the small frozen-anchor case and the export-time precompute, since the in-browser $O(\dim^3)$ inverse is prohibitive at dim ≈ 2060). With the US endogenous the augmented state has dimension ≈ 2060 and spectral radius $\rho(M) \approx 0.924 < 1$, so the iteration contracts and the two routes agree to $\sim 10^{-9}$.

The fixed point (40) is solved two equivalent ways:

(a) **Gauss–Seidel / Neumann sweeps** (the cross-check) sum the Neumann series $x = \sum_{k\geq 0} M^k d$ by re-conditioning each bloc in turn through the Durbin–Koopman smoother (Durbin and Koopman, 2002, 2012) until the maximum level change falls below tolerance.

(b) **Direct closed form**

$$x = (I - M)^{-1}d, \tag{41}$$

after which one final conditional-forecast pass per bloc reconstructs the full set of free variables consistent with the converged pin-deltas.

At the full 33-bloc endogenous-US configuration the spectral radius is $\rho(M) \approx 0.924 < 1$, so the Neumann series converges and $(I - M)$ is nonsingular; the two solvers agree to $\sim 10^{-9}$. (The legacy frozen-anchor RoW-only operator had a smaller radius, $\rho(M) \approx 0.726$; closing the RoW→US loop raises the coupling but leaves the system comfortably contractive.)

Algorithm 3 states both solve paths (`app/src/lib/compute/gvarSolve.ts`). The coupling operator M is never formed by writing down a joint VAR; each of its columns is *probed* by perturbing one pinned cell and reading the affine conditional-forecast response of every linked bloc through the Durbin–Koopman smoother—so M encodes the same per-bloc Kalman operator that produces the baseline, restricted to the deviation.

Algorithm 3 Solve-time coupling fixed point $x = Mx + d$ (Gauss–Seidel/Neumann and direct $(I - M)^{-1}$; `gvarSolve.ts`).

Input: per-bloc posterior-mean models $\{(\hat{\beta}_c, \hat{\Sigma}_c)\}$; scenario pins; trade weights W; `US_ENDOGENOUS` flag; tolerance TOL.
Output: converged deviation state x and per-bloc scenario deltas `delta`[c]; sweep count; $\rho(M)$.

▷ Forcing. Build d: each bloc's own scenario pins, evaluated at *zero* partner-star and global deltas (one DK conditional-forecast pass per pinned bloc).
▷ Coupling families in M (endogenous-US augmented state). star←star (a receiver's star responds to a partner's free path; the US is both partner and receiver); US-global rows ($g\Delta$ as the US's own response to its star pins); star←global (a partner star responds to its own global pin).
Path A — Gauss–Seidel / Neumann sweeps (inverse-free).
  $x \leftarrow d$
  repeat (sweep): **for each** coupled bloc c (US included **if** `US_ENDOGENOUS`): re-condition c through the DK smoother on the current partner-star and global deltas ⇒ update x's block for c.
    ▷ this sums the Neumann series $x = \sum_{k\geq 0} M^k d$ term by term.
  until $\max|\Delta x| <$ TOL; record ρ_{emp} from the geometric residual decay.
Path B — direct closed form (cached).
  assemble M by probing columns; **if** a precomputed $(I - M)^{-1}$ matches the model signature & pin structure, hydrate it; **else** fall back to Path A (the dense in-browser inverse is never built live).
  $x \leftarrow (I - M)^{-1}d$; one final per-bloc DK pass reconstructs all free variables consistent with the converged pin-deltas.
▷ Cross-check. The two paths agree to the verified tolerance ($\max|\Delta| < 10^{-4}$ across all blocs in the closed-form vs. iterative regression tests, with the iterative reference itself converged to 10^{-7}), confirming $\rho(M) < 1$; report $\rho(M)$ via power iteration on M (in practice on M^2, see Sec. 16.6).
return x, `delta`[c], sweep count, $\rho(M)$.

A practical consequence drives the deployment. The operator M depends on the estimated models and on the conditioning *structure* (which cells are pinned) but *not* on the pinned *values*, so it can be assembled once and reused for every value change during interactive dragging. Its dense inverse is, however, expensive: at the augmented endogenous dimension (≈ 2060) the naive in-browser inversion would take tens of minutes (the cubic inverse is of order three-quarters of an hour), making a cold build infeasible even in a Web Worker. The tool therefore **ships a precomputed coupling artifact** (`coupling_default.json.gz`) containing M, $(I - M)^{-1}$, and $\rho(M)$ for the default (empty) pin structure, with the inverse computed offline by a high-performance BLAS routine. The artifact is large (the dense 2060×2060 inverse), so it is shipped *gzipped* (about 30 MB compressed) and decompressed in the browser via the native `DecompressionStream` API at load. This artifact is mandatory, not an optimization: hydrating it reduces the first complete solve from the tens-of-minutes cold build to a few seconds. A model-signature gate guards against using a stale inverse after any re-estimation; a genuinely novel pin structure misses the cache and falls back to an inverse-free Neumann solve (the in-browser cubic inverse is never built live).

16.6 Two-tier solve: live preview vs. the full fixed point

The endogenous US makes the *complete* cross-country solve materially more expensive than the legacy frozen-anchor solve, because placing or moving a US pin re-probes the (large) US bloc, the heavy step. To keep the interface reactive the tool exposes **two solve tiers**:

- **Direct preview** (the default while dragging). A fast, first-order approximation that propagates the scenario through the pre-warmed local per-bloc elasticities—one cross-country hop, without the full multiplier feedback—so the propagated paths update live as the user drags a pin. The per-bloc response kernels are cached by model signature and own-pin structure, so a value-only re-drag (same pinned cells) reuses them and re-solves cheaply; a *structure* change (pinning a new cell) re-probes the US bloc and is the slower direct step (a few seconds), during which a small "computing..." pill is shown so the wait does not look like a freeze.

- **Full GVAR** (on demand). The complete fixed point of (40)—all feedback loops, including RoW→US—computed when the user clicks the **Solve full GVAR** button. This consumes the hydrated $(I-M)^{-1}$ artifact and takes a few seconds.

A badge above the charts reports which tier produced the current result—an amber "GVAR DIRECT preview" versus a blue "GVAR full solve"—together with the sweep count, the spectral radius ρ, and the solve time. When a DIRECT preview is on screen, the **Solve full GVAR** button is offered to upgrade it to the complete fixed point. Drag-release dispatch and debouncing keep the preview channel from firing mid-drag, and the heavy coupling artifact is stripped from the worker messages on the direct path (it is only needed for the full solve). For the legacy frozen-anchor topology—or any deployment where the US is not endogenous—both tiers resolve to the same already-fast cached closed form, so the two-tier distinction is invisible there.

16.7 Global commons: US-determined common shocks

The headline addition of version 2.0 is a set of four *global common* variables that are determined by the United States but carried by *every* bloc:

- `GLB_OIL` — the spot oil price (WTI);

- `GLB_USLR` — the US 10-year Treasury yield;

- `GLB_USBAA` — the US Baa corporate-bond yield (a credit-spread proxy);

- `GLB_USEQ` — US equity prices.

In the US bloc these enter as ordinary *endogenous* variables; in each partner bloc the same four identifiers (matched one-to-one by *name*, never by position) are carried as weakly-exogenous commons that the solver *pins*.

The pinning is by *deviation*, not by level. Each partner bloc c sets each global column to its own baseline plus the US deviation of the matching global,

$$\widehat{\mathrm{GLB}}_{c,t} = \bar{\mathrm{GLB}}_{c,t} + \big(\mathrm{GLB}^{\mathrm{scen}}_{\mathrm{US},t} - \bar{\mathrm{GLB}}_{\mathrm{US},t}\big), \tag{42}$$

where $\bar{(\cdot)}$ denotes the bloc's own unconditional baseline. This is a load-bearing design choice. Each RoW bloc estimated its *own* baseline path for the globals as weakly-exogenous regressors, and those baseline levels differ from the US's by tens of $100 \ln$ units (for example, of order +17, +70, and +28 for the euro area, Japan, and China on oil). Pinning the US *level* would inject that entire baseline gap as a spurious deviation the instant a bloc is selected, even under a zero scenario. Pinning the US *deviation* on top of each bloc's own baseline instead yields exactly zero spillover at a zero scenario and an *identical* common shock under conditioning—precisely mirroring the star convention $\bar{\mathrm{RoW}}$ + partner-delta of (35)–(36). A US oil pin therefore propagates the same oil deviation to all 32 partner blocs (verified to within 10^{-9}), while each bloc's domestic variables respond through its *own* estimated coefficients.

Mechanically, with the US now endogenous the US global *deviation* is itself an unknown of the fixed point rather than a frozen constant. The global columns therefore enter the *augmented* coupling operator M of Section 16.5: a US-global definition row writes each global's deviation as the US's own response to its star pins, and a star $\leftarrow$ global block routes each partner's global pin back into its stars. This is exactly the channel by which a partner shock now feeds back to the US. Indirect effects—a common shock moving a bloc's macroeconomy, which moves its stars, which spill to other blocs—flow through the same operator and are not double-counted. A bloc that carries no globals reduces the pin to a no-op, leaving the legacy globals-free solve byte-identical.

On the user interface the globals are *conditionable* on the US screen (where they are endogenous) and *read-only*, carried over from the US path, on every other bloc.

16.8 Relation to the fiscal and counterfactual layers

The GVAR layer composes with the single-country machinery rather than replacing it. The displayed scenario line for an active GVAR bloc is the cross-country propagated path of this section; for a standalone (non-GVAR) economy it is the single-country conditional forecast of Section 14. The US derived fiscal block (the debt-to-GDP accounting identity, whose empirical motivation is the fiscal reaction function of Bohn, 1998) and the DSGE policy counterfactual—including the Barnichon and Mesters (2023) optimal-policy problem—continue to operate on the US bloc's propagated path. Conditional forecasts throughout use the large-cross-section state-space method of Bańbura et al. (2015) with the hierarchical prior of Giannone et al. (2015), applied independently to each bloc and stitched together only through the linear deviation system (40).

17 Fiscal Blocks: Estimated Drivers and Debt Accounting

Six of the economies in the tool—the United States, the Euro Area, Japan, the United Kingdom, Canada, and China—carry a dedicated *fiscal block*. The block follows the stock-flow-consistent division of labour shared by the U.S. Congressional Budget Office's conditional-BVAR budget mechanics and the IMF Financial Programming (FPP) / debt-sustainability frameworks: the BVAR is used as a *macro-fiscal scenario generator* that jointly forecasts the macroeconomy together with a small number of fiscal *flow drivers*, while the public-debt stock and the headline reporting variables (primary and overall balance, interest, debt-to-GDP, gross financing needs) are *derived outside the VAR* by the government budget-constraint identity. No reduced-form equation is ever allowed to forecast the debt stock freely, because an unconstrained debt VAR is explosive: the debt-dynamics identity has a near-unit autoregressive root $(1 + i)/(1 + g^n)$, and small parameter errors compound into divergent debt paths. Imposing the identity by construction is exactly the discipline that the CBO and FPP/DSA frameworks demand, and it is the structure implemented in `app/src/lib/compute/fiscalAccounting.ts`.

17.1 Estimated Fiscal Drivers

The BVAR jointly estimates the macro block of Section 19 together with *four* fiscal flow drivers, all entering as linear (`lin`) variables:

- **Revenue** (`FREV_GDP`): general-government receipts as a percentage of GDP, an annual ratio.
- **Primary expenditure** (`FPEXP_GDP`): non-interest outlays as a percentage of GDP, an annual ratio.
- **Effective interest rate** (`EFFI`, % p.a.): the average rate the government actually pays on its outstanding stock, constructed as net interest divided by the lagged gross debt and annualized. Because the stock turns over only as maturing securities are refinanced, this effective rate lags market yields and is therefore distinct from—and estimated separately from—the policy rate and the market yields already in the macro block.
- **Stock-flow adjustment** (`SFA`, % of GDP): the deficit–debt discrepancy described below.

The effective rate `EFFI` is the estimated interest driver of the deployed block. (The parametrisation has changed over time: an early version carried Δ(debt/GDP) directly, and a later one carried debt service to GDP `DSVC`; the current block reverts to the effective rate.) The interest bill is then formed inside the accounting recursion as $int_t = i^{\text{eff}}_t\, d_{t-1}/100$.

The stock-flow adjustment as an estimated variable. The novelty of the current block is that the stock-flow adjustment (`SFA`) is now an *estimated* BVAR variable rather than a constant plug. It is the exact residual of the tool's own debt recursion—the part of the observed change in gross debt that the flow side (deficit plus the interest–growth "snowball") does *not* explain. Economically this seam arises because the headline deficit is measured on a federal-flows perimeter while the debt stock is gross debt (so intragovernmental holdings, valuation effects, and other below-the-line "other means of financing" operations move the two apart). It is constructed so that it closes the recursion to machine precision on history, and the pandemic quarters are set to missing (`NaN`) so the COVID-as-missing Gibbs sampler imputes them (Section 19.4) rather than letting the 2020 swing contaminate the estimate. Carrying `SFA` as a forecast variable means the debt-to-GDP *projection* now propagates its own `SFA` path forward (with dynamics and uncertainty bands) instead of defaulting the discrepancy to zero, which materially affects the high-debt blocs whose historical SFA is large and persistent. The historical SFA means (in % of GDP), which each bloc also ships as its `sfaDefault` fallback, are: US 1.75, EA 0.33, JP 2.23, UK 3.15, CA 3.19, CN 4.02.

Because all four drivers are ordinary BVAR variables, they inherit the full estimation machinery of Sections 12–14: the hierarchical Minnesota prior (Giannone et al., 2015), the COVID-as-missing data-augmentation Gibbs treatment, the companion-form Durbin–Koopman conditional smoother (Durbin and Koopman, 2002, 2012), and the Bańbura et al. (2015) conditioning-via-partial-observations construction. In the five Global-VAR fiscal blocs (EA, JP, UK, CA, CN)—and now in the endogenous US bloc itself—the fiscal drivers are part of the domestic block and propagate to the other economies through the shared Rest-of-World stars, exactly as the macro variables do (Pesaran et al., 2004; Dees et al., 2007).

A stationary prior for the fiscal ratios. The three GDP-ratio drivers—revenue `FREV_GDP`, primary expenditure `FPEXP_GDP`, and the stock-flow adjustment `SFA`—are mean-reverting shares that fluctuate around a roughly constant in-sample level; they are *not* integrated. Carrying them under the default random-walk own-lag prior ($\delta = 1$) is therefore the wrong null: it injects a built-in drift that lets the projected ratio wander away from its historical level at the long end of the horizon. These three drivers are accordingly tagged `prior:"WN"` in every fiscal bloc's specification and estimated with the *stationary* white-noise own-lag prior ($\delta = 0$) of Section 12.5: the estimator collects them via `pos` = `find`(`prior` == 'WN') and passes the set as the `stationary` argument of `cgl_bvarV5`, so `diagb`(`pos`) = 0 centers their own-lag dynamics on mean reversion to the equation constant. The effective interest rate `EFFI`, by contrast, *retains* the random-walk prior: it is a slow-moving stock-average rate (net interest over lagged debt) whose level is genuinely persistent—it turns over only as maturing securities refinance—so a near-unit own-lag is the right prior null for it.

The empirical effect is exactly the intended one. Under the random-walk prior the U.S. revenue ratio was pinned near its end-of-sample value ($\sim 19\%$) and drifted from there; under the white-noise prior it now *reverts* to its $\sim 17.4\%$ in-sample mean over the projection, and the expenditure ratio and `SFA` likewise return to their historical means rather than carrying the last observation forward indefinitely. We emphasise (per Section 12.5) that this is a shift of the prior *center*, not a hard stationarity constraint: the posterior own-lag coefficient is still the GLP prior–likelihood balance at the estimated tightness λ, so a fiscal ratio whose data genuinely trend will still be allowed to do so—the `WN` prior only removes the mechanical drift that the random-walk center would otherwise build in. Because `pos` is empty for the 44 non-fiscal blocs, this change is a no-op (byte-identical output) everywhere except the six fiscal blocs.

Each fiscal bloc ships a `fiscal_meta.json` file that pins the column layout used by the accounting recursion. It carries the integer indices `idx` $= \{$`gdp`, `deflator`, `revenue`, `primaryExp`, `effRate`, `sfa`$\}$ into the level-space forecast matrix—the first two locate the real-GDP and deflator log-levels that supply nominal growth, the next two locate the revenue and primary-expenditure drivers, `effRate` locates the estimated effective-rate column, and `sfa` locates the estimated stock-flow-adjustment column—together with the seed debt ratio d_0 and a fallback `sfaDefault` (e.g. the U.S. block has $d_0 = 122.57$, `idx.effRate` $= 35$, `idx.sfa` $= 36$, and `sfaDefault` $= 1.75$; the U.K. block has $d_0 = 96.24$, `idx.effRate` $= 18$, `idx.sfa` $= 19$). The presence of `idx.sfa` is the switch that selects the per-quarter estimated SFA: when it is set, `computeFiscalBlock` reads the forecast SFA off that column so the debt projection carries the discrepancy forward with its own dynamics and bands; when it is unset, the recursion falls back to the constant `sfaDefault` (or zero). A legacy block that instead carried an `idx.debtService` column would read that column directly as the period interest flow and back out the effective rate; the current effective-rate blocks do not.

17.2 The Budget-Identity Recursion

Let d_t denote the debt-to-GDP ratio (in percent), i_t^{eff} the effective rate (percent per annum), and g_t^n nominal GDP growth (percent per quarter). All flows below are annual ratios in percent of GDP. Nominal growth is read directly off the level-space forecast as the sum of the real-GDP and deflator log-differences, in the $100 \cdot \ln$ space of Section 12.3:

$$g_t^n \;=\; \left(\ell_t^{\text{gdp}} - \ell_{t-1}^{\text{gdp}}\right) \;+\; \left(\ell_t^{\text{defl}} - \ell_{t-1}^{\text{defl}}\right), \tag{43}$$

where ℓ is the stored $100 \cdot \ln$ level and the right-hand side is already in percent per quarter (it is not annualized by the factor of four used for display in (2)). The primary balance is the revenue–expenditure gap (positive = surplus),

$$pb_t \;=\; rev_t \;-\; primexp_t, \tag{44}$$

the interest bill (debt service) is the estimated effective rate applied to the lagged debt stock,

$$int_t \;=\; i_t^{\text{eff}} \,\frac{d_{t-1}}{100}, \tag{45}$$

i.e. the BVAR forecast of the effective rate `EFFI` drives the interest flow directly, with the lagged debt d_{t-1} converting the rate into a percent-of-GDP flow. (A legacy debt-service block would instead

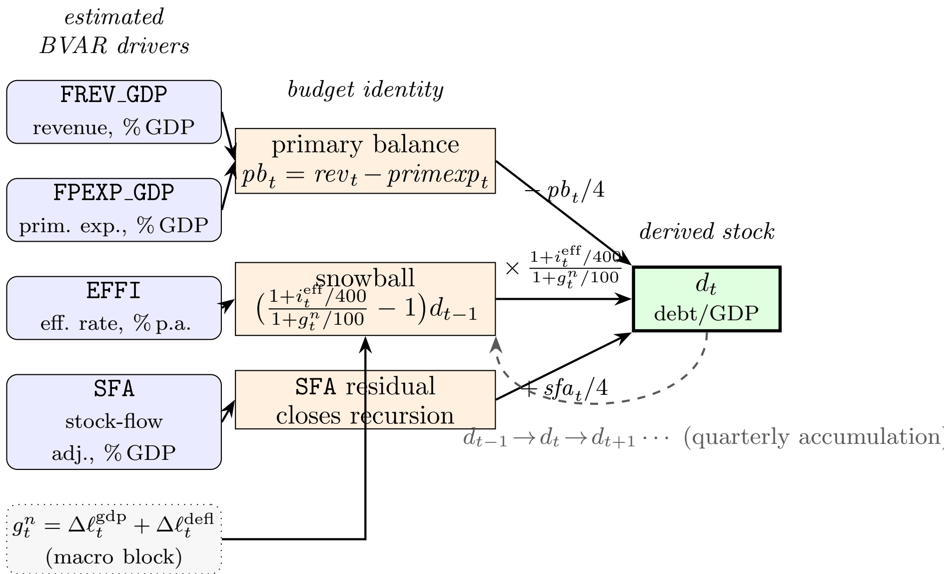

Figure 6: Fiscal stock-flow schematic. The four estimated BVAR drivers (`FREV_GDP`, `FPEXP_GDP`, `EFFI`, `SFA`; Section 17.1) feed the budget identity: revenue and primary expenditure form the primary balance pb_t (44); the effective rate `EFFI` and nominal growth g_t^n (43) form the interest–growth *snowball* (49); and the estimated `SFA` enters as the residual that closes the recursion. These flows accumulate into the derived debt-to-GDP stock d_t by the quarterized roll-forward $d_t = d_{t-1}\,(1+i_t^{\text{eff}}/400)/(1+g_t^n/100) - pb_t/4 + sfa_t/4$ (47), with the lagged stock d_{t-1} fed back each quarter (dashed). Debt is *derived*, never estimated (`computeFiscalBlock`, Section 17.2).

supply int_t as the estimated `DSVC` ratio and back out the effective rate as $i_t^{\text{eff}} = 100\,int_t/d_{t-1}$.) The overall balance and deficit follow as

$$ob_t \;=\; pb_t - int_t, \qquad def_t \;=\; -\,ob_t \;=\; int_t - pb_t. \tag{46}$$

The debt stock is then rolled forward one quarter by the FPP/DSA debt-dynamics equation,

$$d_t \;=\; d_{t-1}\,\frac{1+i_t^{\text{eff}}/400}{1+g_t^n/100} \;-\; \frac{pb_t}{4} \;+\; \frac{sfa_t}{4}, \qquad d_{-1} \equiv d_0, \tag{47}$$

seeded at d_0. The rate enters quarterized as $i_t^{\text{eff}}/400$ and growth as $g_t^n/100$, while the annual flow ratios pb_t and sfa_t are scaled by 1/4 so that one quarter's worth of the flow accrues per step; sfa_t is the stock-flow adjustment (the deficit–debt discrepancy of Section 17.1). The `SFA` column is estimated as its own BVAR variable and is forecast with its own dynamics and bands; the deployed debt roll-forward, however, deliberately uses the *structural constant sfaDefault* (the long-run SFA mean) rather than the per-quarter forecast column, because the forecast column is pulled to spurious extremes by the long-run consensus anchors' cross-response and would collapse the projected debt path (see §17.3). Gross financing needs add amortization to the deficit,

$$gfn_t \;=\; def_t \;+\; amort_t. \tag{48}$$

The recursion in (44)–(48) is the function `computeFiscalBlock`, which is the *only* fiscal recursion wired to the displayed debt-to-GDP charts and bands (it is called from `ChartGrid.tsx` for the baseline, scenario, and counterfactual paths, and from `runGvarBandSolve.ts` for the predictive bands). This is a **passive accounting recursion**: the debt ratio is rolled forward purely from the four BVAR-forecast drivers (*rev*, *primexp*, the effective rate, and SFA) with *no* endogenous response of the primary balance to the debt level and *no* debt-driven risk premium on the rate (§17.3 explains the dormant lever variant). The exact quarterly convention is pinned by a historical back-cast test that reconstructs observed debt-to-GDP from the realized drivers.

Snowball decomposition. Subtracting d_{t-1} from (47) isolates the *automatic* debt dynamics from the discretionary primary balance. The change in the debt ratio is

$$\Delta d_t = \underbrace{\left(\frac{1 + i_t^{\text{eff}}/400}{1 + g_t^n/100} - 1\right) d_{t-1}}_{\text{snowball (interest–growth)}} - \frac{pb_t}{4} + \frac{sfa_t}{4} \approx \frac{i_t^{\text{eff}} - g_t^n}{1 + g_t^n} d_{t-1} - pb_t + sfa_t, \tag{49}$$

where the right-hand approximation is the familiar Fiscal-Monitor $(i - g)$ form (here in annualized flow units). The snowball term is positive—debt rises mechanically—whenever the effective rate exceeds nominal growth; a primary surplus and a favourable growth-interest differential are the two forces that stabilize the ratio. The tool returns the per-quarter snowball contribution $\big((1 + i_t^{\text{eff}}/400)/(1 + g_t^n/100) - 1\big)d_{t-1}$ alongside the derived balances so the user can read off the source of any debt drift.

Remark 2. Because the debt stock is *not* a BVAR variable, the estimated effective rate $\hat{i}_t^{\text{eff}}$ and primary balance $\widehat{pb}_t$ are debt-agnostic baselines: the recursion uses *lagged* debt d_{t-1} throughout, so it stays explicit (no within-period fixed point). This is the design that lets a single forward sweep produce a consistent debt path, and it is also what would let the dormant feedback levers of Section 17.3 layer on without double-counting were they ever activated in the live path.

17.3 Debt-Feedback Channels (dormant; possible future extension)

Not active in the live tool. The two debt-feedback channels described in this subsection—the Bohn fiscal-reaction function (γ) and the sovereign risk-premium channel (β)—are *not* part of the displayed debt path. They exist only in a separate code path (`runFiscalRecursion`) that is exercised by the unit-test suite and has *no* production or UI caller; the deployed recursion that produces every debt-to-GDP chart and band (`computeFiscalBlock`, §17) runs with $\gamma = \beta = 0$ and exposes no `feedbackOn` control to the user. We document the channels here because they are validated and ready, and because they show how the passive accounting layer could be extended to an *endogenous* debt feedback, but the debt path the user sees today has **no** debt→primary-balance reaction (Bohn) and **no** debt→yield risk premium. The only user-reachable fiscal lever is the SFA / conditioning closure described below and the scenario conditioning of the four drivers themselves.

The would-be channels modify the estimated baselines in terms of the lagged debt gap $d_{t-1} - d^*$, where the anchor d^* defaults to d_0.

Bohn fiscal-reaction function (γ) — not wired in. Following Bohn (1998), the primary balance *could* be allowed to respond to the lagged debt gap, so that fiscal policy consolidates as debt rises above its anchor:

$$pb_t \;=\; \big(rev_t - primexp_t\big) \;+\; \gamma\,\big(d_{t-1} - d^*\big), \qquad \gamma \geq 0. \tag{50}$$

A value $\gamma = 0.03$ (three basis points of primary surplus per percentage point of excess debt) is the canonical Bohn (1998) estimate and the threshold above which a no-Ponzi debt process is mean-reverting. In the live tool this term is *absent*: the displayed primary balance is exactly $rev_t - primexp_t$ (§17, eq. (44)), with no γ term. The Bohn (1998) reaction function is the *empirical motivation* for caring about a debt anchor, not an implemented feedback in the deployed path.

Risk-premium channel (β) — not wired in. The effective rate *could* be allowed to rise with debt above the anchor, capturing a sovereign risk premium:

$$i_t^{\text{eff}} \;=\; \widehat{\imath}_t^{\,\text{eff}} \;+\; \beta\,\max\!\big(0,\; d_{t-1} - d^*\big), \qquad \beta \geq 0, \tag{51}$$

where the premium would bite only *above* the anchor (the $\max(0, \cdot)$ rectifier). Were it active, β would feed back into both the interest bill (45) and the snowball term (49), producing the nonlinear debt-spiral dynamics that motivate the FPP/DSA realism literature. In the live tool this term is *absent*: the displayed effective rate is the BVAR-forecast $\widehat{\imath}_t^{\,\text{eff}}$ alone, with no debt-level premium. This channel is the natural candidate for a future endogenous extension, but it is not current behaviour.

Stock-flow-adjustment closure (the genuine SFA term). The SFA term sfa_t in (47) is a genuine, non-optional part of the debt-dynamics identity (the deficit–debt discrepancy / stock-flow adjustment), *not* a behavioural feedback. It is estimated as its own BVAR variable (Section 17.1) and forecast with its own dynamics and bands. Note, however, a deliberate implementation choice: the deployed debt roll-forward uses a *structural constant* (*sfaDefault*, the long-run mean of the SFA) rather than the per-quarter forecast SFA column, because the forecast column is dragged to spurious extremes by the long-run consensus anchors' cross-response (which would collapse the projected debt path). The structural SFA is the economically-correct long-run residual. This closure can also be used as a scenario lever—set to zero for pure flow accounting, held at the historical average, or solved (via the debt-targeting inversion of §17.4) to reconcile a target debt path with the flow drivers.

17.4 Debt-Targeting Inversion

The recursion can also be run backwards. Given a desired debt-to-GDP path $\{d_t^{\text{target}}\}$, the effective rate, nominal growth, and SFA, the annual primary balance required at each quarter is obtained by solving (47) for pb_t along the target trajectory:

$$pb_t \;=\; 4\left[d_{t-1}^{\text{target}}\,\frac{1+i_t^{\text{eff}}/400}{1+g_t^n/100} \;-\; d_t^{\text{target}}\right] \;+\; sfa_t, \tag{52}$$

with $d_{-1}^{\text{target}} \equiv d_0$. This is the IMF DDT/DSA "required primary balance / adjustment to the debt anchor" (`requiredPrimaryBalance`). The caller then conditions the BVAR on this primary-balance path through the linear constraint $rev_t - primexp_t = pb_t$, propagated by the DK smoother of Section 14. Because the effective rate and growth themselves respond to the primary balance through the BVAR covariance structure, an exact hit is obtained by iterating invert → condition → re-read (i, g) → re-invert; one or two passes converge.

17.5 Display Layout

The estimated and derived fiscal objects are presented in different chart rows, reflecting their different epistemic status. The four *estimated* fiscal drivers are shown as inputs alongside the macro block: revenue and primary expenditure as % of GDP, and the two new charts

- **Effective interest rate** (`EFFI`, % p.a.): the average rate the government pays on its outstanding stock. It is plotted as a level in percent per annum next to the policy rate and the market yields. Read it as the slow-moving cost of the debt: it lags the policy rate and the 10-year yield, because only maturing securities reprice, so a rate hike shows up here gradually rather than at once. This is the rate that enters the interest bill (45) and the snowball (49).

- **Stock-flow adjustment** (`SFA`, % of GDP): the deficit–debt discrepancy. It is plotted as a level in percent of GDP. A positive value means gross debt rose by more than the deficit plus interest–growth dynamics imply (debt-increasing other-means-of-financing operations); a value near zero means the flow accounting closes cleanly. Its history is informative—it is large and persistent for the high-debt blocs (Table values of order +2 to +4% of GDP for JP, UK, CA, CN)—and, crucially, its *forecast* now feeds the debt projection, so a persistently positive SFA pushes the projected debt path *above* what the deficit alone would deliver.

A further row holds the *derived* reporting variables produced by the accounting recursion, in the order

(i) **Primary Balance** (pb_t, % of GDP), which is itself conditionable: the user can drag a primary-balance path and the tool maps it back to the $rev - primexp$ linear combination;

(ii) **Δ Debt-to-GDP** (Δd_t, percentage points per quarter), the per-quarter debt change of (49), which is not directly conditionable because it is a pure identity output;

(iii) **Debt-to-GDP** (d_t, % of GDP), the rolled-forward stock, which can be targeted via the inversion of Section 17.4. Note that this projection now *carries the forecast SFA forward* (with dynamics and bands): a bloc with a large positive estimated SFA will show a debt path that drifts up faster than its primary balance and snowball alone explain, which is the intended, honest behaviour.

Separating the estimated drivers (effective rate, SFA, revenue, expenditure) from the derived balances and debt makes the stock-flow-consistent structure visible on screen: what the data and the prior speak to is shown as an input, and what the budget identity implies is shown as an output.

17.6 Data Treatment: the U.K. Index-Linked-Gilt Spike

Interest series constructed from accrued interest can contain mechanical, non-recurring spikes that are not informative about the underlying refinancing cost. In the United Kingdom, the large stock of *index-linked gilts* produced an interest spike in 2008–09: the inflation uplift on the principal of index-linked debt is recorded as accrued interest, so the 2008 inflation spike (and its subsequent unwinding) translated into an outsized swing in the measured effective rate `EFFI` that would otherwise contaminate the BVAR's autoregressive dynamics. These quarters are handled with the same machinery used for the COVID-19 observations: the affected effective-rate entries are set to missing (`NaN`) and imputed by the Kalman filter inside the data-augmentation Gibbs sampler, rather than being allowed to enter the likelihood as if they were ordinary observations. This keeps the estimated effective-rate process—and hence the derived interest bill and debt path—free of a one-off accounting artefact while preserving the genuine information in the surrounding quarters.

18 Long-Run Targets: In-Prior Prior vs the Forward-Pass Drift Setter

The forecasts of Section 14 inherit their long-run behaviour entirely from the estimated VAR. Because the real-activity and price blocks are estimated on $100 \ln$ levels of $I(1)$ series, the companion matrix (6) carries roots on or near the unit circle, and the implied long-horizon central path can drift away from any value the user regards as a credible steady state (e.g. a 2% inflation objective, a neutral policy rate, or a balanced great ratio). The tool exposes a *long-run target* control that lets the user pin the long-horizon forecast of a chosen variable to a target. We support two conceptually distinct mechanisms, summarized in Table 2: an *in-prior* long-run prior applied *before* estimation (computed offline), and an *after-estimation forward-pass drift setter* that maps each target to a constant shift of the companion drift and is applied instantly in the browser. The latter is the user-facing control and *replaces* the earlier entropic-tilt mechanism (Remark 6).

18.1 In-Prior Long-Run Prior (PLR)

The first mechanism is the "prior for the long run" (PLR) of Giannone et al. (2019), which augments the hierarchical Minnesota prior of Section 12.4 with additional dummy observations that shrink a small set of linear combinations Hy toward stationarity (no-cointegration / random-walk behaviour). Let $H \in \mathbb{R}^{q\times n}$ collect q long-run combinations and let y_0 be the prior pre-sample mean. Following the authors' construction, H is rotated into a full-rank basis $\bar{H} = \big(H'; \, \text{null}(H)'\big)'$ with inverse $\bar{H}^{-1}$, and for each active long-run row j a single dummy observation

$$y_j^d = \frac{(y_0 \bar{H}')_j}{\phi_j} \left(\bar{H}^{-1}\right)'_{\cdot j}, \qquad x_j^d = \Big(0, \; \tfrac{(y_0 \bar{H}')_j}{\phi_j} \, \mathbf{1}_p \otimes \left(\bar{H}^{-1}\right)'_{\cdot j}\Big), \tag{53}$$

is appended *to the regression system* (both the Gibbs draw and the marginal-likelihood normalizing constant T_d), with ϕ_j controlling the tightness of the j-th long-run restriction. Crucially, (53) combines the *equations* through the columns of $\bar{H}^{-1}$; combining the regressors instead produces a dummy whose residual is identically zero at the sum-of-coefficients mode and hence carries no tightness information. The tightness hyperparameters ϕ_j are held fixed so that the Metropolis–Hastings search over the Minnesota λ and the sum-of-coefficients μ remains two-dimensional.

The PLR is a *proper Bayesian prior*: it modifies the posterior and requires re-estimation. It is therefore computed offline as part of the data-export pipeline, not in the browser.

Remark 3 (What the PLR actually does in the US sample). A US head-to-head ($n = 36$, $p = 3$, COVID treated as missing per Section 14.6, $q = 13$ long-run combinations comprising six great ratios, three nominal anchors, and four rate/credit spreads) shows that the PLR is a *separate dummy block* rather than a change to the Minnesota tightness: the maximizing λ is essentially unchanged (0.563 baseline vs 0.562 with the PLR at either $\phi = 1.0$ or $\phi = 0.1$). Because the sum-of-coefficients-style dummy in (53) shrinks toward a *random walk* (no cointegration), and the US sample treats most of these ratios as genuinely $I(1)$, the PLR does *not* pull the long-run central forecast toward a historical mean: the mean absolute drift of the $H{=}40$ forecast from a recent historical anchor moves from 8.49 (baseline) to 8.41 at $\phi = 1.0$ (-1%), and is marginally *worse* at $\phi = 0.1$. The visible effect is on the *bands*, not the path: the GDP-growth 68% band tightens from 4.24 to 3.95 percentage points ($\approx 7\%$) at $\phi = 1.0$. The PLR is thus the appropriate instrument for principled band tightening and a soft no-cointegration restriction; it is *not* a device for pinning a long-run combination to a chosen number.

18.2 The Forward-Pass Drift Setter (δc)

The user-facing mechanism, exposed in the interface as the **Long-run targets** panel (Section 3), does *not* re-estimate the model and does *not* reweight the draws. It is a deterministic *counterfactual on the companion drift* in the spirit of Villani (2009): the user states where a variable's long-run forecast should land, and the tool solves for the single constant shift of that variable's own equation that places it there, then applies the resulting system response as a cheap, exact overlay on top of the precomputed paths and bands. It is instant (no Compute, no re-solve), exact (the terminal forecast hits the target by construction), and fully reversible (an unmoved slider is a byte-exact no-op).

The finite-horizon response operator. Write the companion recursion (5) in level space as $Y_t = c + T\,Y_{t-1} + \varepsilon_t$, with T the companion matrix and c the companion drift (the constant row of the estimated coefficients). Switching innovations off, the h-step deterministic forecast is

$$Y_{T+h} = T^h\,Y_T + S_h\,c, \qquad S_h = \sum_{k=0}^{h-1} T^k, \tag{54}$$

where S_h is the partial sum of companion powers. A shift δc of the drift moves the entire forecast path by *exactly* $\Delta Y_{T+h} = S_h\,\delta c$; this is an algebraic identity, not an approximation (verified to 10^{-11} against the iterated recursion). We deliberately do *not* use the closed-form steady state $Y^\star = (I - \sum_\ell B_\ell)^{-1} c$. For the near-unit-root level VARs the matrix $(I - \sum_\ell B_\ell)$ is near-singular, so $Y^\star$ is a meaningless quantity of order 10^5, whereas the finite-horizon operator S_H in (54) is well-posed. This is precisely the object that makes the construction work on the random-walk-in-levels BVAR, whose exact steady state does not exist.

One slider, one equation. A long-run target on variable j is imposed by shifting *only* j's own equation, $\delta c = e_j\,\delta c_j$. The coherent system-wide response is then the j-th column of the response operator,

$$\Delta Y_{T+h} = S_h[:, j]\,\delta c_j \qquad \text{for } \textit{all} \text{ variables.} \tag{55}$$

The scalar δc_j is chosen so that variable j's *displayed* terminal forecast (Section 12.3) equals the user's target $\psi_j^\star$. Let sens_j be the display-unit sensitivity of j's terminal forecast to a unit δc_j:

$$\mathrm{sens}_j = \begin{cases} 4\big(S_H[j,j] - S_{H-1}[j,j]\big), & \texttt{log} \text{ var (display} = 4\,\Delta\text{level)}, \\ S_H[j,j], & \texttt{lin} \text{ var (display} = \text{level)}, \end{cases} \qquad \delta c_j = \frac{\psi_j^\star - \psi_j^0}{\mathrm{sens}_j}, \tag{56}$$

where ψ_j^0 is the variable's *current* terminal display forecast — the slider's default thumb. Setting $\psi_j^\star = \psi_j^0$ gives $\delta c_j = 0$ and hence $\Delta Y \equiv 0$: an unmoved slider is an exact, byte-identical no-op.

Joint long-run restriction. When *several* sliders are moved at once a naive per-anchor superposition — solving each δc_j from (56) in isolation and adding the columns of (55) — is *incoherent*: a δc_j on equation j also shifts every other anchored variable's terminal through the off-diagonal cross-response $S_H[k, j]$, so the displayed terminals no longer land on their targets. A stationary, low-sensitivity anchor (e.g. a fiscal-block ratio with a near-zero own-sens_j, hence a large δc_j) can drag a macro target arbitrarily far off. The tool therefore imposes all enabled targets *jointly*, as a single linear long-run restriction. Let $\mathcal{K}$ be the anchored equations that pass the sensitivity guard, and define the display-unit response matrix M by

$$M_{kj} = \begin{cases} 4\big(S_H[k,j] - S_{H-1}[k,j]\big), & k \in \mathcal{K}\ \texttt{log}, \\ S_H[k,j], & k \in \mathcal{K}\ \texttt{lin}, \end{cases} \qquad M\,\delta c = \psi^\star - \psi^0, \quad k,j \in \mathcal{K}. \tag{57}$$

The diagonal M_{kk} is exactly the own-sensitivity sens_k of (56), so a single anchor ($|\mathcal{K}| = 1$) reduces (57) to the scalar $\delta c_k = (\psi^\star_k - \psi^0_k)/\text{sens}_k$, and $\psi^\star \equiv \psi^0$ (all targets at their estimates) gives $\delta c \equiv 0$ exactly — the byte-identical no-op survives. The system is solved by Gaussian elimination with partial pivoting; the δc are then superposed back through (55) as before, so *every* anchored terminal hits its target *simultaneously*.

Per-variable vs. joint (user toggle). The exact joint solve is well-posed only when the anchored variables' terminal responses are linearly well separated. For a *near-collinear* anchor set — e.g. two anchored interest rates that the estimated dynamics cannot move independently in the long run — M^{-1} produces huge *offsetting* δc's that hit the $|\mathcal{K}|$ terminals exactly while wrecking the path everywhere else (on one estimated bloc the exact solution required $|\delta c| \approx 459$ where the per-anchor diagonal needed ≈ 11). The tool therefore exposes a *solve-mode toggle* (shown once two or more targets are set):

- **Per-variable** — each $\delta c_k = (\psi^\star_k - \psi^0_k)/\text{sens}_k$ set independently from its own diagonal sensitivity. *Stable* and bounded (the drift can never blow up), but *approximate*: it ignores the off-diagonal cross-responses, so it *underestimates the spillovers* and the other anchored terminals drift off their set values by the cross-drag.

- **Joint** — the full $M\,\delta c = \psi^\star - \psi^0$ system, so *every* anchored terminal lands exactly on its value. *Correct at the terminal*, but a strongly-correlated target set makes M ill-conditioned and can *destabilise* the path; the tool flags that case and the user switches to per-variable. A genuinely singular M falls back to the per-variable δc numerically (never a NaN path).

Both modes reduce to the same scalar for a single target ($|\mathcal{K}| = 1$), and $\psi^\star \equiv \psi^0$ is a byte-exact no-op in either. The shipped default anchor sets are deliberately kept to a *bare minimum* — since the July-2026 anchor review, 32 anchors across 23 economies, each individually verified to fix a genuine terminal-bias pathology while keeping the anchored path inside the economy's own (COVID-masked) historical envelope, with no origin discontinuity and no >1 pp cross-drag onto unanchored headline variables (`docs/anchor-review-2026-07.md`) — so the joint system stays well conditioned out of the box. Well-conditioned is not the same as well-shaped, however: because δc is a *rigid translation* of the whole path, joint exactness at the terminal is bought with a front-loaded near-term shift, and when the model's own terminal sits far from the anchor the first forecast years can be pushed to implausible values even under the bare-minimum set (at the 2026 Q1 origin the US deflator — model terminal 4.3 against the 2.0 anchor — displayed *negative* inflation over the first two forecast years under joint, against +1.7 under per-variable). **Per-variable is therefore the shipped**

default — stable and near-term-plausible, at the cost of terminals that land only approximately — and joint is the deliberate opt-in for users who need every anchor met exactly and accept the near-term reshaping.

Rigid band translation. The deviation ΔY of (55) is added *identically* to the mean level path and to *every* shipped quantile $(q_{05}, q_{16}, q_{50}, q_{84}, q_{95})$ of both the baseline and the scenario fans. The forecast distribution is therefore translated rigidly: band widths are provably invariant and the quantile ordering cannot be reordered or rescaled by an additive shift. This is what makes the overlay safe to apply on top of the precomputed bands of Section 14.6 without recomputing them, and it is the qualitative difference from a reweighting scheme, which deforms the density. The same ΔY, in level space, is forwarded into the fiscal recursion (Section 17) so the debt path responds coherently to the new macro long run.

Responsiveness. Building $\{S_h\}_{h=1}^{H}$ costs $O(H)$ companion mat-muls and is done once per bloc on load (a few milliseconds, then cached). Every subsequent slider move is a single cached mat-vec add (55) and a rigid translation of the bands — no draw loop, no Kalman re-smooth, no GVAR re-solve. The sliders are therefore as responsive as the drag interaction: the chart updates on every pointer move.

The policy rate. Because the policy rate is an ordinary `lin` variable of the VAR, it carries its own long-run-target slider like any other series: the user sets the terminal rate directly through the drift setter. There is no separate Taylor-rule rate overlay — the rate's long run is governed solely by its own slider, consistent with every other variable.

Remark 4 (Honest caveats, surfaced in the panel). The drift setter shifts the long-run *drift*, not the model's *persistence*; the sliders are accordingly labelled "long-run target" rather than "mean reversion", and they do not add a new attractor to the dynamics. The cross-variable response $S_h[:, j]$ in (55) drags the other variables along the *estimated* dynamics — a genuine model side effect, not a bug. Most consequentially, the *real-growth* target owns the direction of the debt path through the $(i - g)$ snowball, so it is labelled distinctly ("long-run real growth target") and the panel states explicitly that it is not a structural debt fix. Finally, a variable whose sensitivity (56) is near-singular ($|\text{sens}_j| < 0.3$, the `SENS_GUARD`) yields an ill-conditioned δc_j and its slider is disabled.

Remark 5 (On a GVAR bloc the target is marginal). The drift setter operates on a *single* bloc's companion. On a GVAR-linked economy it shifts *that bloc's own* long-run forecast; the trade partners and the US-determined global commons (Section 16.7) are not re-solved, so the realised joint long-run value can differ from the marginal target. The panel states this caveat when a GVAR bloc is selected. A fully joint re-anchoring of the cross-country fixed point is left for future work.

Remark 6 (Why the entropic tilt was retired). An earlier release pinned the long run by an after-estimation *entropic tilt* — a relative-entropy reweighting of the posterior draws (Robertson et al., 2005) so that the weighted H-step moment matched the target. It was correct in principle but degenerated in practice: as a target was pushed away from the draw-cloud mean the importance weights concentrated and the effective sample size ESS $= 1/\sum_d w_d^2$ collapsed toward 1 (with $D =$ 500 draws, a *joint* 13-constraint tilt fell to ESS $\to$ 1 because the joint target lay outside the convex hull of the draws), so the pinned band rested on a handful of draws and was unreliable exactly where the user most wanted a firm anchor. It was also slow: it required a draw cloud and a Newton solve in the browser. The forward-pass drift setter replaces it wholesale — it is deterministic (no ESS to collapse), exact at any target, instant, and band-coherent by construction — and the tilt control has been removed from the interface.

Remark 7 (Model-based vs. with-judgement, and a consensus preset). The drift setter induces a natural per-variable flag. With every slider at its model estimate the displayed forecast is purely *model-based* ($\delta c = 0$, an exact no-op); the moment a target is set the variable becomes *with-judgement* and the panel marks it as such (a highlighted target, a reset-to-estimate control, and an N `targets · exact` badge). The flag is per-country and per-variable — a single chart grid may mix purely model-based series with judgemental ones — and is cleared on a country switch. To make a transparent judgemental starting point one click away, blocs with publicly-defensible long-run values carry a *consensus-anchors* preset, deliberately kept to a *bare minimum*: potential growth and the central-bank inflation objective per economy, plus the effective-rate/revenue/expenditure fiscal trio for the six fiscal-block economies (the values of §18.2). No policy or market rates, unemployment, house prices or spreads are anchored by default — collinear anchored rates make the joint long-run system ill-conditioned (see the tameness guard of §18.2) and those long runs stay model-driven. The preset seeds enabled so the big-6 debt trajectories display their realistic slope out of the box, and it is fully reversible to model-based (clearing the anchors is a byte-exact no-op).

18.3 When to Use Which

The two surviving mechanisms are complementary. The in-prior PLR is the gold standard — a proper offline posterior — but its data-decides property means that, at the tightnesses examined, it leaves the US central long-run forecast essentially unchanged and acts mainly on the bands. The forward-pass drift setter is in-browser, requires no re-estimation, and *forces* the displayed long-run forecast onto an explicit target by construction, with rigidly translated bands and no degeneracy. As a rule of thumb: to *pin* a variable's long run to a number interactively, use the drift setter; for principled band tightening and a soft no-cointegration restriction, use the PLR (offline).

Table 2: Long-run targets: in-prior PLR versus the forward-pass drift setter.

	In-prior PLR	**Drift setter (δc)**
Source	Giannone et al. (2019)	Villani (2009)
Acts on	prior (dummy obs.)	companion drift c (forward pass)
Re-estimation	yes (offline)	no (in-browser, instant)
Object	proper posterior	deterministic counterfactual
Target	Hy stationarity	terminal display forecast $\psi_j^\star$
Bands	tightened	rigidly translated (widths invariant)
Main effect	tightens bands	shifts central path to the target
Failure mode	none (proper)	slider disabled if $\lvert\mathrm{sens}_j\rvert < 0.3$

19 Economy Coverage, Data Sources, and Estimation

The tool ships estimated BVARs for roughly fifty economies. Thirty-three of them are integrated into a single *Global VAR* (GVAR) system (Pesaran et al., 2004; Dees et al., 2007) covering approximately 90% of world GDP; the remaining blocs—including the 17 euro-area member states, which are deliberately kept out of the coupled system because a disaggregated euro GVAR is non-stationary—are standalone single-country models that share the same estimation machinery but are not wired into the cross-country fixed point. This section documents the roster, the underlying data, the COVID-as-missing estimator, the common-vintage harmonization, and the construction of the predictive bands. Throughout, variables enter the VAR in the $100\ln(\cdot)$ space for `log` series and in levels for `lin` series (Section 12.3); annualized quarter-over-quarter growth and inflation are reported as $4 \times \Delta(100\ln)$, consistent with (2) and (20).

19.1 The GVAR System: 33 Integrated Blocs

The GVAR comprises the United States—now a fully endogenous coupled bloc, not a frozen anchor (Section 16.1)—the euro area as a single aggregate bloc, plus 31 further Rest-of-World (RoW) blocs:

> US; EA (euro-area aggregate), JP, UK, CA, CN (the core majors); CH, KR, MX, AU, TR, SE, PL, NO, DK, IN, BR, RU, ID, SA, AR, ZA, TH, MY; and the nine further non-euro economies CL, CO, CR, CZ, HU, IL, IS, NZ, TW added in the OECD expansion.

The euro area enters as one aggregate bloc by necessity: a fully disaggregated euro GVAR is non-convergent ($\rho(M) \approx 1.089 > 1$), and treating the euro area as a single bloc restores stationarity. The 17 euro-area members are consequently *standalone* models (Section 19.2), not coupled GVAR blocs. Each bloc—the US included—is an independently estimated single-country BVAR in which the foreign sector is summarized not by a raw foreign-variable block but by three trade-weighted RoW "star" variables, following the GVAR tradition of Pesaran et al. (2004) and Dees et al. (2007):

- `ROWDEM` — foreign demand: the trade-weighted average of partners' real-GDP growth;

- `ROWINF` — foreign inflation: the trade-weighted average of partners' price inflation;
- `ROWFIN` — foreign financial conditions: the trade-weighted average of partners' short rates.

For bloc c with partner set $\mathcal{P}_c$ and weights w_{cj}, the demand and inflation stars are built in annualized growth space from the partners' stored $100\ln$ levels,

$$\mathrm{ROWDEM}_{c,t} = 4 \times \frac{\sum_{j \in \mathcal{P}_c} w_{cj}\left(x^g_{j,t} - x^g_{j,t-1}\right)}{\sum_{j \in \mathcal{P}_c} w_{cj}\,\mathbb{1}\{x^g_{j,t}, x^g_{j,t-1} \text{ observed}\}}, \tag{58}$$

where $x^g_{j,t}$ is partner j's real-activity level (analogously for `ROWINF` from prices, and `ROWFIN` from the level of the short rate). Weights are *renormalized per component* over the partners actually observed in each quarter, so a partner that is missing a given series simply drops out of the corresponding star and the remaining weights are rescaled. Two implementation details matter for the econometrician: China's reported GDP is nominal, so its real-activity level is formed by subtracting the price level in $100\ln$ space, $x^g_{\mathrm{CN}} = x^{\mathrm{NGDP}}_{\mathrm{CN}} - x^P_{\mathrm{CN}}$ (a log-space deflation); and Argentina's 2023–24 hyperinflation episode is excluded from the `ROWINF` and `ROWFIN` aggregates of its partners (its genuine real demand is retained in `ROWDEM`), with renormalization absorbing the dropped weight.

Trade weights. Bilateral weights w_{cj} are constructed from goods-trade flows in the IMF International Trade in Goods statistics (IMTS, the renamed bilateral DOTS), queried from the SDMX 2.1 endpoint `api.imf.org` over the 2021–2023 average of exports (`XG_FOB_USD`) plus imports (`MG_CIF_USD`). The Euro Area is treated as a single counterpart (IMTS aggregate code `G163`). The weight matrix is row-normalized over the 33 blocs with a zero diagonal; now that the US is endogenous it carries its own trade-weights row, so all 33 blocs have a row.

Rate-optional blocs. The short rate is *optional*: a GDP series and a price series are required of every bloc, but two blocs (Switzerland and Norway) carry no domestic short rate. Their rate column is `NaN` throughout, they are excluded from their partners' `ROWFIN` aggregate, and the `ROWFIN` weights of any bloc having them as a partner are renormalized accordingly. They still contribute to `ROWDEM` and `ROWINF`.

Cross-country coupling. As in Section 19's original treatment, the baseline forecast of each bloc is computed independently; cross-country coupling enters only through the scenario layer. A shock conditioned in one bloc perturbs its star variables, which propagate to its trade partners, whose own stars then feed back. Stacking the per-bloc deviations Δ yields the linear fixed point

$$\Delta = M\,\Delta + d, \tag{59}$$

where M encodes the trade-weighted star couplings together with the four US-determined global commons (the spot oil price, the US 10-year yield, the US Baa corporate yield, and US equity; Section 16.7) and d collects the user's direct conditioning forcing. With the US endogenous the deviation state is *augmented* to carry the US global-column deviations as unknowns (Section 16.5), so the global commons close the RoW→US feedback loop inside M rather than entering d as a frozen forcing. The system is solved at display time both by a direct $(I - M)^{-1}d$ evaluation and by Neumann sweeps (summing $\sum_k M^k d$); the two agree to $\sim 10^{-9}$. The coupling is contractive, with spectral radius $\rho(M) \approx 0.924$, so the series converges. Because the full 2060×2060 $(I - M)^{-1}$ inverse is expensive in-browser (a cold build is on the order of three-quarters of an hour), it is precomputed offline and shipped as a *gzipped* coupling cache (`coupling_default.json.gz`, about 30 MB compressed), decompressed in the browser via the native `DecompressionStream` API. The interactive interface uses a two-tier solve—a fast direct preview on drag and an explicit "Solve full GVAR" button for the complete fixed point—to keep the endogenous-US solve reactive (Section 16.6).

19.2 Standalone Single-Country Blocs

Beyond the 33 GVAR blocs, the tool ships additional single-country BVARs estimated by the same procedure but not coupled into (59). Together with the GVAR members these complete coverage of all 38 OECD members and add several large non-OECD emerging markets. The standalone remainder is dominated by the 17 *euro-area member states* (DE, FR, IT, ES, NL, AT, BE, FI, GR, IE, LU, PT, SI, SK, EE, LT, LV), which are kept out of the coupled GVAR *by construction*: the euro area is represented in the GVAR by the aggregate `EA` bloc, because a disaggregated euro GVAR is non-stationary ($\rho(M) \approx 1.089 > 1$) whereas aggregating the euro area restores convergence. Each euro member is nonetheless estimated as a full single-country model—its domestic block augmented by the four global commons and by trade-weighted foreign *star* variables—but in *separation* (with the GVAR iteration count fixed to `MAXIT` $= 1$, so its foreign stars are read off data rather than from a cross-country solve) and it carries no trade-weights row. The country selector groups the roster into two sections, a *GVAR* group and an *Other countries* group; a bloc is GVAR-linked, and hence placed in the *GVAR* group, if and only if it has a trade-weights row and its specification carries the three `ROW*` star variables.

19.3 Data Sources

No single provider supplies current quarterly real GDP for the full roster, so the data are assembled bloc-by-bloc from a small number of authoritative source families. Each bloc's variables are wired to a specific provider series by the per-bloc build scripts (`pipeline/build_*.py`; the US source registry lives in `app/scripts/refresh-quarter.mjs`), and the exact provider series behind every shipped variable is recorded in the machine-readable *coverage manifest* (`app/public/data/coverage_manifest.json`), from which the coverage matrix of Section 20.59 is generated.

Source families. The classification of each bloc to its primary source family is anchored on its real-GDP series; the resulting split is **2 FRED** (US, CN), **25 Eurostat** (EU members plus the EFTA economies CH/IS/NO), and **23 OECD/FRED** (the remaining OECD members and FRED-sourced emerging markets, including CA, UK, JP).

- **FRED (St. Louis Fed).** The US block ($n = 41$, Table 3) is pulled in full from FRED with exact series IDs (e.g. `GDPC1` real GDP, `PCEPI/CPIAUCSL` prices, `FEDFUNDS` policy rate, `DGS10` 10-year yield, `UNRATE` unemployment, `CSUSHPISA` house prices). The China bloc is built entirely from verified real FRED series, with quarterly real activity anchored by industrial production since clean quarterly *real* GDP is not available on FRED for China (only annual constant-price GDP and nominal quarterly GDP that ends early — the documented China real-GDP trap). Cross-bloc, FRED also supplies the US-determined global commons (`GLB_OIL` from `WTISPLC`, `GLB_USLR` from `DGS10`, `GLB_USBAA` from `BAA`, `GLB_USEQ` from the US equity index) and the BIS credit-to-GDP and residential-property-price series re-distributed through FRED (`Q<ISO2>NAM770A`, BIS RPP).

- **Eurostat.** The IMF's thematic SDMX API exposes no per-country quarterly chain-volume GDP (its national-accounts flow carries current-price values only, for a limited country set), so the EU/EFTA economies are sourced from Eurostat: `namq_10_gdp` chain-linked volumes (GDP and the expenditure components), `prc_hicp_midx` HICP, `une_rt_q` unemployment, `irt_st_q/irt_lt_mcby_m` short and long interest rates, `ert_eff_ic_q` NEER, and `prc_hpi_q` house prices. Euro-area members carry the ECB policy rate (deposit facility) and ECB MIR lending rate rather than the generic BIS series.

- **OECD QNA and national sources.** The non-EU OECD members and FRED-sourced emerging markets (e.g. CA, UK, JP, AU, NZ, KR, MX, BR, ZA, Türkiye, Chile) draw real GDP and national accounts from the OECD Quarterly National Accounts (largely re-distributed on FRED), short/policy rates from national-bank or BIS `CBPOL` series (e.g. RBA/RBNZ cash rates), CPI and unemployment from national/OECD flows, and NEER where available.

- **IMF WEO (back-fill only).** Where genuine quarterly real activity or prices end short of the common estimation window, the Step-1 harmonization of Section 19.5 temporally disaggregates IMF World Economic Outlook annual projections (`NGDP_RPCH`, `PCPIPCH`) to fill the gap. WEO is a *back-fill of last resort*, used only for the macro tail of blocs whose national source lags; it never overrides observed data and is flagged wherever it is the live tail (CN/IN; see below).

- **Derived series.** The three rest-of-world "star" variables (`ROWDEM/ROWINF/ROWFIN`), the fiscal effective-interest rate (`EFFI`) and stock-flow adjustment (`SFA`), and the rest-of-euro-area aggregates (`ROWEA_*`) are not pulled from a provider but *computed* from other series (trade-weighted partner aggregates; the debt recursion residual of Section 17.2).

Coverage and last-observation conventions. Coverage start varies by bloc and series (typically the mid-1990s to mid-2010s; the US sample is the longest); coverage end is the bloc's last historical quarter, governed by the ragged edge of Section 19.5. The *default* last observation is **2026Q1**; `CN` and `IN` are held at **2025Q4** and `IL` at **2024Q4**. Within a bloc, all core macro variables share the same trailing endpoint (the only interior gaps are the COVID 2020Q2–2021Q1 macro mask of Section 19.4 and the credit/SFA seed gaps), so the last *balanced* observation equals the last historical quarter for every bloc. Real-versus-nominal GDP is handled per bloc: every fiscal bloc now carries a chain-volume real GDP series used directly — China was re-sourced to a real chain-linked SA volume index (base 2011Q1=100) in the June-2026 endogenous-US 33-bloc re-estimate, replacing the earlier nominal proxy; and thin or volatile samples are estimated with deliberately wide, honest predictive bands rather than spuriously tight ones.

Provenance flags surfaced for the reader. Two classes of tail are flagged explicitly, both consistent with the no-fabrication rule:

- **WEO back-fill (confirmed) — `CN`, `IN`.** The domestic macro tail to 2025Q4 is the WEO temporal-disaggregation of Section 19.5, held ragged and never observed. CN's GDP series is the nominal-GDP / industrial-production proxy (the China real-GDP trap above).
- **Source-review (flagged, not rebuilt) — `KR`, `RU`, `ID`, `TH`.** For these blocs the GVAR pipeline's genuine national/OECD-FRED source ended earlier (KR/RU ≈2024Q1, ID/TH ≈2025Q1) and was WEO-extended in the 2025Q4 build, yet the shipped panels carry a 2026Q1 endpoint; June credit/policy-rate rebuilds also touched these blocs. Whether the shipped trailing quarters are a genuine refresh or the synthetic extension is *not verifiable from the shipped data alone*, so these are surfaced as a *source-review* flag for human verification rather than asserted or rebuilt. `KR`'s genuine domestic macro additionally derives from a frozen OECD-MEI-on-FRED mirror (≈2023Q3 vintage) before any extension.

These flags appear in the last column of the coverage matrix (Section 20.59) and as `blocFlags` in the coverage manifest.

19.4 COVID-as-Missing Estimation

The 2020 pandemic quarters are an extreme outlier that, left untreated, would contaminate the autoregressive estimates and the innovation covariance. Rather than scaling them out, the tool treats the pandemic window as *missing data* and imputes it within the estimation itself, via a data-augmentation Gibbs sampler that nests a Durbin–Koopman simulation smoother (Durbin and Koopman, 2002, 2012). Each Gibbs sweep alternates:

(i) a Durbin–Koopman simulation-smoother draw of the missing pandemic rows given the current coefficients, treating those rows as `NaN` observations in the companion-form state space of Section 13;

(ii) a Metropolis–Hastings update of the hierarchical-prior hyperparameters on the *completed-data* marginal likelihood, so the overall tightness λ and the sum-of-coefficients parameter μ (Giannone et al., 2015) are selected by empirical Bayes without any fixed-tightness restriction; and

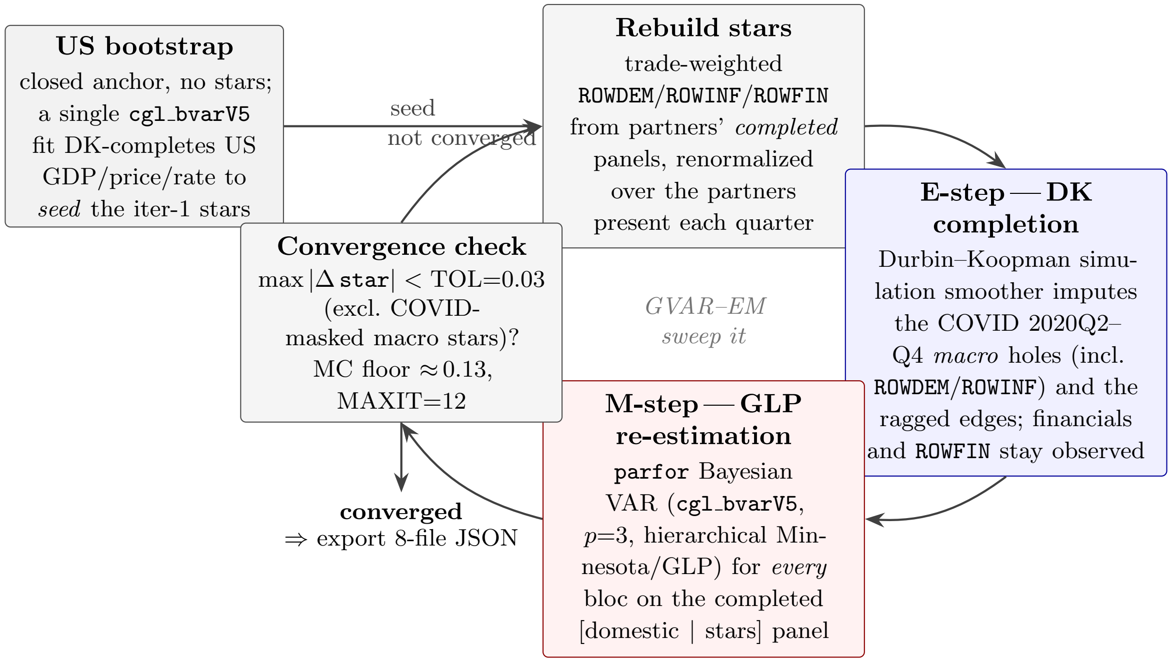

Figure 7: The GVAR-EM estimation loop under pervasive missingness (Giannone et al., 2019; Durbin and Koopman, 2002). A closed US bootstrap (no stars) is estimated once to seed the first trade-weighted foreign variables. Each sweep then (i) rebuilds the `ROWDEM`/`ROWINF`/`ROWFIN` stars from partners' completed panels, renormalized over the partners present each quarter, and (ii) re-estimates every bloc by hierarchical-prior Bayesian VAR (Gibbs, run in `parfor`). Within each bloc's single `cgl_bvarV5` pass the **E-step**—a Durbin–Koopman simulation-smoother completion that imputes the COVID macro holes (the fixed 2020Q2–Q4 window, with financial series and `ROWFIN` held observed) and the ragged, unbalanced-panel edges—and the **M-step**—the posterior re-estimation of the VAR coefficients on the completed [domestic | stars] panel—are carried out jointly by the same sampler. The sweep repeats until the foreign components stop moving, $\max|\Delta\, \texttt{star}| <$ TOL $= 0.03$ (evaluated after the second sweep and excluding the COVID-masked macro stars, whose cross-iteration churn is Monte-Carlo noise with floor ≈ 0.13), capped at MAXIT $= 12$ iterations.

(iii) a draw of the VAR coefficients (β, Σ) from the completed data.

The pandemic history is thus *preserved* (the surrounding quarters remain in the sample and inform the imputation) while the outlier itself does not drive the dynamics.

Window selection. The COVID window self-calibrates to each bloc's pandemic trough rather than being hard-coded. It is the *argmin* of the quarter-on-quarter GDP change *restricted to the window 2019Q4–2021Q1*:

$$k^\star = \operatorname*{arg\,min}_{t\,\in\,\{2019\text{Q4},\ldots,2021\text{Q1}\}} \big(x^g_t - x^g_{t-1}\big), \qquad \mathcal{W} = \{\, k^\star - 1,\ k^\star,\ k^\star + 1,\ k^\star + 2\,\}, \tag{60}$$

and the four rows of $\mathcal{W}$ (intersected with the sample) are NaN'd across *all* variables before estimation. Restricting the search to 2019Q4–2021Q1 is deliberate: an unrestricted global argmin misfires for economies whose deepest contraction is a different crisis (for example Korea's 2008Q4 Global-

Financial-Crisis drop exceeds its 2020Q2 pandemic drop).

Fixed window and macro-only masking in the production GVAR loop. The auto-detected window above is the legacy single-bloc convention; the production GVAR estimator (`pipeline/export_bvar_gvar24` instead uses a *fixed* three-quarter window $\mathcal{W} = \{$2020Q2, 2020Q3, 2020Q4$\}$ for every bloc, which avoids the ∼seven-quarter spans the unrestricted trough search could produce. Equally important, the production loop masks *only the macro columns*: over $\mathcal{W}$ the domestic macro variables (`isFinancial` = false) and the two macro stars `ROWDEM`/`ROWINF` are set `NaN`, while the domestic financial series (rates, yields, equity, house prices, corporate spreads) and the financial star `ROWFIN` are *kept observed*. The Durbin–Koopman smoother then imputes the pandemic macro holes *conditional on* the observed financials, which carry genuine signal through 2020—the masking is thus "macro-masked / financials-observed," the same convention the out-of-sample evaluation re-applies in-sample at every origin (Section 23.2).

Algorithm 4 states the full estimation loop. It is the operational form of the cycle sketched in Figure 7: a no-stars US bootstrap seeds the first foreign variables, after which each sweep rebuilds every bloc's trade-weighted stars from partners' completed panels (the *E-step* regressor), re-estimates all blocs in parallel under macro masking (the per-bloc `cgl_bvarV5` call, whose own Durbin–Koopman completion is the within-bloc *E-step* and whose Gibbs re-estimation is the *M-step*), and recomputes the stars from the freshly completed data, iterating to a fixed point in the completed stars.

The sweep is a *Jacobi* update—all stars are rebuilt from the common start-of-sweep state *before* the parallel re-estimation—so the parallel blocs are mutually independent given the current stars and the sweep has the same fixed point as a sequential (Gauss–Seidel) pass, at a wall-clock set by the slowest single bloc (the $n = 41$ US bloc) rather than the sum. The convergence metric excludes the COVID-masked macro-star cells, whose cross-sweep churn is pure imputation Monte-Carlo noise (floor ≈ 0.13); the financial star `ROWFIN`, kept observed through the pandemic, stays in the metric. Each `cgl_bvarV5` call is itself the data-augmentation Gibbs sampler of Algorithm 1—its DK completion is the bloc-level E-step and its hyperparameter/coefficient draws are the M-step—so the whole loop is a block-coordinate EM over (foreign stars, completed-data states, VAR coefficients).

19.5 Common-Vintage Harmonization and the 2026Q1 Observed Refresh

The shipped panels are the product of two distinct, clearly-separated steps: an *estimation-build harmonization* to a common 2025Q4 endpoint (the vintage every bloc's coefficients were estimated on), and a later *no-re-estimation observed-data refresh* that advances the *observed* panel one quarter to 2026Q1 without touching the frozen estimates. The two are described in turn.

Step 1 — estimation-build harmonization to 2025Q4. So that estimation runs on a uniform window, every bloc's domestic series was first extended to a common 2025Q4 endpoint by the GVAR build (`pipeline/build_gvar_2025q4.py`, `TARGET_END` = 2025Q4). For blocs whose quarterly data

Algorithm 4 EM-style GVAR estimation loop (Jacobi star re-convergence; `export_bvar_gvar24_loop.m`).

Input: per-bloc panels $\{X_c\}$ ($X_c = [\text{domestic} \mid \text{stars}]$); trade weights W; lags $p = 3$; `WN`/stationary index sets $\texttt{pos}_c$; tolerance $\textsc{tol} = 0.03$; cap $\textsc{maxit} = 12$; fixed COVID window $\mathcal{W} = \{2020\text{Q2..Q4}\}$.
Output: per-bloc posterior means $(\hat{\beta}_c, \hat{\Sigma}_c)$, completed panels, predictive draws $\texttt{Dforecast}_c$, exported 8-file JSON.

1: ▷ Bootstrap (seed the first stars).
2: Mask the US macro columns over $\mathcal{W}$ and fit the closed, no-stars US bloc with `cgl_bvarV5`; the posterior-mean DK completion gives the initial US real-GDP/price/rate panel (this seeds partners' iteration-1 stars).
3: **for** $it = 1, \ldots, \textsc{maxit}$ **do** ▷ Jacobi sweep
4:     ▷ Build all stars from the FROZEN start-of-sweep completed panels.
5:     **for each** re-estimated bloc c: $\texttt{star}_c \leftarrow$ trade-weighted `ROWDEM`/`ROWINF`/`ROWFIN` from partners' completed panels (renormalize over partners present each quarter).
6:     **parfor each** bloc c **do** ▷ independent given the frozen stars
7:         $x_c \leftarrow [\,\text{domestic}_c \mid \texttt{star}_c\,]$
8:         over $\mathcal{W}$, set `NaN` the macro columns of x_c *and* `ROWDEM`,`ROWINF` (keep financials, `ROWFIN`)
9:         $(\texttt{Xcond}_c, \beta_c, \Sigma_c, \texttt{Dforecast}_c) \leftarrow \texttt{cgl\_bvarV5}(x_c, p, \texttt{pos}_c, \ldots)$ ▷ E-step DK completion + M-step Gibbs
10:         completed panel $\widehat{X}_c \leftarrow \text{mean}_{\text{draws}}\, \texttt{Xcond}_c$
11:     **end parfor**
12:     $mc \leftarrow \max_c \max \left| \texttt{star}_c^{(it)} - \texttt{star}_c^{(it-1)} \right|$ ▷ exclude COVID-masked `ROWDEM`/`ROWINF` cells
13:     **if** $it > 1$ **and** $mc < \textsc{tol}$ **then break** ▷ converged
14: **end for**
15: Export each bloc's $(\hat{\beta}_c, \hat{\Sigma}_c)$, completed history, stars, and predictive draws to the 8-file JSON contract.

ended short of 2025Q4:

- **Real-activity and price series** (GDP, consumption, investment, government, exports, imports, CPI/HICP, all `log`) are benchmarked to IMF World Economic Outlook annual projections (`NGDP_RPCH` real growth, `PCPIPCH` inflation) via temporal disaggregation: a within-year quarterly profile is laid down at the WEO-implied annual growth and then proportional-Denton corrected so that each fully appended year's four-quarter average reproduces the WEO annual level (already-observed quarters in a partial year are held fixed). The realized annual growth of each appended year matches WEO to within 0.2pp.

- **Financial series with no WEO analogue** (NEER, REER, equity indices, and all `lin` rates — short rates, 3-month rates, 10-year yields, unemployment) are carried forward by *hold-last* and flagged as such.

The original raw vintages are left untouched; the extended panels are written separately and consumed by the estimation drivers. The estimated companion, posterior draws, coupling artifact, and trade weights are all frozen as of this 2025Q4 build.

Step 2 — the 2026Q1 observed-data refresh (no re-estimation). The live panel then advances one quarter to a default last-historical of **2026Q1**, making 2026Q2 the first projected quarter for the refreshed blocs. This is a pure data extension performed by `app/scripts/refresh-quarter.mjs`: for each refreshed bloc the single new 2026Q1 row is pulled from the *same* live sources, in the *same* transform, that produced the prior 2025Q4 row, and the forecast simply re-anchors from the *unchanged* companion. No estimation runs — `cgl_bvarV5.m` is never invoked, and `companion_mean`, `companion_draws`, `spec`, `transformations`, `coupling`, and `weights` stay byte-frozen. The script is fail-closed: a bloc is advanced only once a verified live-source registry (`SOURCES[<CC>]`) has been authored and gated for it, so the rollout is per-bloc rather than a single sweep.

The ragged edge (panel default 2026Q1; three exceptions). The default last-historical quarter is **2026Q1**; three blocs are held earlier because their genuine source data have not advanced and are *never* fabricated forward (the no-fabrication rule of Section 19.4):

- **China (`CN`) and India (`IN`)** — held at **2025Q4**. Their real source data still end ≈2025Q2, so their 2025Q4 tail is the WEO temporal-disaggregation back-fill of Step 1 and they are deliberately *not* advanced to 2026Q1.
- **Israel (`IL`)** — held at **2024Q4**, its quarterly source not yet having advanced.

These per-bloc endpoints are pinned in `app/src/lib/dates.ts` (`DEFAULT_LAST_HIST_DATE` = 2026Q1 with `CN`/`IN`/`IL` overrides) so the ragged edge plots correctly: each bloc's first forecast quarter is the one immediately after its own last observation. A small number of emerging-market blocs (`KR`, `RU`, `ID`, `TH`) carry a short trailing tail whose genuine-versus-synthetic provenance is flagged for review in Section 19.3.

19.6 Predictive Uncertainty Bands

For every bloc the exported baseline carries *predictive* bands, not parameter-only bands. The COVID-missing Gibbs sampler emits a forecast density that integrates over three sources of uncertainty simultaneously — posterior parameter uncertainty, future innovation uncertainty, and missing-data (imputation) uncertainty. The reported quantiles are the 5th, 16th, 84th, and 95th percentiles of this predictive density (with the 50th retained in level space),

$$\big(q_{05},\, q_{16},\, q_{84},\, q_{95}\big)_{T+h} = \text{Quantile}_{\{0.05, 0.16, 0.84, 0.95\}}\big\{\, \tilde{y}^{(d)}_{T+h} \,\big\}_{d=1}^{D}, \tag{61}$$

yielding the 68% band (q_{16}, q_{84}) shaded in the charts and the 90% band (q_{05}, q_{95}). Crucially, the central path $\bar{y}^{\text{baseline}}$ remains the *deterministic* forecast from the posterior-mean coefficients (so the unconditional smoother of Section 14.5 reproduces it exactly, and zero user deviations return the stored baseline); only the bands are predictive. This is the designed fix for the previously over-tight bands that resulted from spreading posterior-mean paths under a tight prior — wide bands now reflect genuine macro volatility (e.g. CN, JP, UK), not the COVID outlier. On-the-fly conditional bands computed in the browser use the same predictive construction, drawing innovations through the simulation smoother of Section 14.6.

19.7 US Specification

The US is the largest bloc, with $n = 41$ quarterly variables and $p = 3$ lags. Its full variable list is given in Table 3; the four-variable fiscal block (`FREV_GDP`, `FPEXP_GDP`, `EFFI`, `SFA`) supports scenario conditioning on the fiscal drivers and the debt-targeting inversion (the debt-anchor motivation being the fiscal reaction function of Bohn, 1998, though no γ feedback is active in the live recursion; see §17.3), and the conditional-forecast and optimal-policy machinery of Bańbura et al. (2015) and Barnichon and Mesters (2023) applies uniformly across all blocs. The count of 41 comprises the domestic and fiscal variables, the four global commons, and the three trade-weighted `ROW*_US` star variables that make the US a fully coupled GVAR bloc (Section 16.1).

Table 3: Variable coverage for the US BVAR model. "Trans." denotes the data transformation: `log` = log levels (growth rates displayed), `lin` = levels. "Prior" denotes the prior mean: RW = random walk, WN = white noise.

ID	**Description**	**Trans.**	**Prior**
GDPH	Real Gross Domestic Product	log	RW
CH	Real Personal Consumption Expenditures	log	RW
FRH	Real Private Residential Fixed Investment (nom/deflator)	log	RW
FNH	Real Private Nonresidential Fixed Investment (nom/deflator)	log	RW
XH	Real Exports of Goods & Services	log	RW
MH	Real Imports of Goods & Services	log	RW
GH	Real Government Consumption Expenditures & Gross Investment	log	RW
JGDP	Gross Domestic Product: Chain Price Index	log	RW
JC	Personal Consumption Expenditures: Chain Price Index	log	RW
JCXFE	PCE less Food & Energy: Chain Price Index	log	RW
UI	CPI-U: All Items	log	RW
UIXFDG	CPI-U: All Items Less Food & Energy	log	RW
LXBC	Business Sector: Compensation Per Hour (nominal)	log	RW

ID	Description	Trans.	Prior
LANAGRA	All Employees: Total Nonfarm	log	RW
LR	Civilian Unemployment Rate: 16 yr +	lin	RW
IP	Industrial Production Index	log	RW
CUMFG	Capacity Utilization: Manufacturing [SIC]	lin	RW
HST	Housing Starts	log	RW
YPDH	Real Disposable Personal Income	log	RW
CSENT	University of Michigan: Consumer Sentiment	lin	RW
POLRATE	Effective Federal Funds Rate (policy rate)	lin	RW
FCM2	2-Year Treasury Note Yield at Constant Maturity	lin	RW
FCM5	5-Year Treasury Note Yield at Constant Maturity	lin	RW
GLB_USLR	US 10-Year Treasury Yield (global common)	lin	RW
FAAA	Moodys Seasoned Aaa Corporate Bond Yield	lin	RW
GLB_USBAA	US Baa Corporate Bond Yield, credit proxy (global common)	lin	RW
FXTWM	Nominal Trade-Weighted Exch Value of US$ vs AFE Currencies	log	RW
GLB_USEQ	US Equity Price Index (global common)	log	RW
PZTEXP	Spot Oil Price: West Texas Intermediate	log	RW
GSNE	Non-Fuel Commodity Price Index (IMF, non-energy proxy)	log	RW
SP500VOL	CBOE Volatility Index for S&P 500 Stock Price Index	lin	RW
HP	House Price	log	RW
LEV	Real nonfinancial business debt	log	RW
FREV_GDP	Federal Receipts (% of GDP)	lin	WN
FPEXP_GDP	Federal Primary Expenditure (% of GDP)	lin	WN
EFFI	Federal Effective Interest Rate (% p.a.)	lin	RW
SFA	Stock-Flow Adjustment (% of GDP)	lin	WN

ID	Description	Trans.	Prior
GLB_OIL	Spot Oil Price, WTI (global common)	log	RW
ROWDEM_US	US RoW demand (trade-weighted partner GDP growth)	lin	RW
ROWINF_US	US RoW inflation (trade-weighted partner inflation)	lin	RW
ROWFIN_US	US RoW financial conditions (trade-weighted partner rates)	lin	RW

The other coupled blocs follow the same estimation methodology with country-appropriate variable selections, replacing the US fiscal and detailed financial block with the three `ROW*` stars of Section 19.1. Across the deployed roster the partner blocs range from $n = 14$ (Costa Rica) to $n = 25$ (UK and Canada), with the euro-area aggregate at $n = 24$ and the core majors at $n \in \{18, \ldots, 25\}$ (e.g. CH, NO = 18, CN = 20, JP = 23); the US is the largest at $n = 41$. The China bloc uses a tighter Minnesota tightness to discipline its shorter, noisier sample.

19.8 Variable Transformations, Priors, and Financial Tags by Economy

Section 19.7 tabulates the US bloc in full; this subsection extends that detail to *every* deployed economy. For all 50 blocs (the 33 GVAR members of Section 19.1 and the 17 euro-area standalones of Section 19.2) each variable's *transformation* (`log` = log level, growth displayed; `lin` = level), its *financial* tag (which governs whether a series is held observed through the COVID window, Section 19.4), and its *prior* mean (RW = random walk in levels; WN = white noise / stationary) are read directly from the deployed `public/data/bvar/<CC>/spec.json` specifications, so the tables below mirror the current re-estimated state of the tool exactly.

Cross-economy coherence matrix. Table 4 collects the common concepts that recur across blocs. Because the transformation, financial tag, and prior for each concept are assigned *identically* in every economy that carries it, a single row characterises the concept tool-wide. Every variable is given a random-walk (RW) prior *except* the three fiscal GDP-ratios (`FREV_GDP`, `FPEXP_GDP`, `SFA`), which are bounded mean-reverting shares and therefore carry a white-noise / stationary (WN) prior; the effective-interest series `EFFI` remains RW.

The 33 GVAR blocs. Each carries a domestic block, the four global commons (`GLB_*`), and the three trade-weighted `ROW*_<CC>` stars. Tables follow in alphabetical order.

Table 5: Argentina (`AR`, $n = 18$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
`GDP_AR`	`log`	–	RW

Argentina (AR), continued

ID	Trans.	Fin.	Prior
CONS_AR	log	–	RW
INV_AR	log	–	RW
GOV_AR	log	–	RW
EXP_AR	log	–	RW
IMP_AR	log	–	RW
CPI_AR	log	–	RW
LR_AR	lin	–	RW
RATE_AR	lin	✓	RW
NEER_AR	log	✓	RW
POLRATE_AR	lin	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_AR	lin	✓	RW
ROWINF_AR	lin	✓	RW
ROWFIN_AR	lin	✓	RW

Table 6: Australia (AU, $n = 19$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_AU	log	–	RW
CONS_AU	log	–	RW
GOV_AU	log	–	RW
INV_AU	log	–	RW
EXP_AU	log	–	RW
IMP_AU	log	–	RW
CPI_AU	log	–	RW
LR_AU	lin	–	RW
NEER_AU	log	✓	RW
RATE_AU	lin	✓	RW
YIELD10_AU	lin	✓	RW
HP_AU	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_AU	lin	✓	RW
ROWINF_AU	lin	✓	RW
ROWFIN_AU	lin	✓	RW

Table 4: Cross-economy coherence matrix for the common concepts. "Trans." is `log/lin`; "Fin." marks financial series (✓); "Prior" is RW (random walk) or WN (white noise / stationary). Each concept takes the same triple in every bloc that carries it.

Concept	**Trans.**	**Fin.**	**Prior**
Real GDP (level)	`log`	–	RW
Price index (CPI/HICP/deflator)	`log`	–	RW
Policy rate	`lin`	✓	RW
Short rate (3M/money-mkt)	`lin`	✓	RW
10Y government yield	`lin`	✓	RW
Corporate / Baa spread	`lin`	✓	RW
House prices	`log`	✓	RW
Equity price index	`log`	✓	RW
Oil price (WTI, global)	`log`	✓	RW
FX / NEER	`log`	✓	RW
Fiscal revenue ratio (FREV_GDP)	`lin`	–	WN
Fiscal primary-exp. ratio (FPEXP_GDP)	`lin`	–	WN
Stock-flow adjustment (SFA)	`lin`	–	WN
Effective interest (EFFI)	`lin`	✓	RW
RoW demand star (ROWDEM)	`lin`	✓	RW
RoW inflation star (ROWINF)	`lin`	✓	RW
RoW financial star (ROWFIN)	`lin`	✓	RW

Table 7: Brazil (`BR`, $n = 18$): variable transformations, financial tags, and priors.

ID	**Trans.**	**Fin.**	**Prior**
`GDP_BR`	`log`	–	RW
`CONS_BR`	`log`	–	RW
`INV_BR`	`log`	–	RW
`GOV_BR`	`log`	–	RW
`EXP_BR`	`log`	–	RW
`IMP_BR`	`log`	–	RW
`CPI_BR`	`log`	–	RW
`LR_BR`	`lin`	–	RW
`NEER_BR`	`log`	✓	RW
`POLRATE_BR`	`lin`	✓	RW
`HP_BR`	`log`	✓	RW
`GLB_OIL`	`log`	✓	RW
`GLB_USLR`	`lin`	✓	RW
`GLB_USBAA`	`lin`	✓	RW
`GLB_USEQ`	`log`	✓	RW
`ROWDEM_BR`	`lin`	✓	RW
`ROWINF_BR`	`lin`	✓	RW
`ROWFIN_BR`	`lin`	✓	RW

Table 8: Canada (CA, $n = 25$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_CA	log	–	RW
CONS_CA	log	–	RW
INV_CA	log	–	RW
GOV_CA	log	–	RW
EXP_CA	log	–	RW
IMP_CA	log	–	RW
IP_CA	log	–	RW
LR_CA	lin	–	RW
CPI_CA	log	–	RW
RATE_CA	lin	✓	RW
POLRATE_CA	lin	✓	RW
YIELD10_CA	lin	✓	RW
HP_CA	log	✓	RW
NEER_CA	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
FREV_GDP	lin	–	WN
FPEXP_GDP	lin	–	WN
EFFI	lin	✓	RW
SFA	lin	–	WN
ROWDEM_CA	lin	✓	RW
ROWINF_CA	lin	✓	RW
ROWFIN_CA	lin	✓	RW

Table 9: Switzerland (CH, $n = 18$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_CH	log	–	RW
CONS_CH	log	–	RW
INV_CH	log	–	RW
GOV_CH	log	–	RW
EXP_CH	log	–	RW
IMP_CH	log	–	RW
HICP_CH	log	–	RW
LR_CH	lin	–	RW
NEER_CH	log	✓	RW
POLRATE_CH	lin	✓	RW
HP_CH	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW

Switzerland (CH), continued

ID	Trans.	Fin.	Prior
GLB_USEQ	log	✓	RW
ROWDEM_CH	lin	✓	RW
ROWINF_CH	lin	✓	RW
ROWFIN_CH	lin	✓	RW

Table 10: Chile (CL, $n = 20$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_CL	log	–	RW
CONS_CL	log	–	RW
INV_CL	log	–	RW
GOV_CL	log	–	RW
EXP_CL	log	–	RW
IMP_CL	log	–	RW
CPI_CL	log	–	RW
LR_CL	lin	–	RW
RATE_CL	lin	✓	RW
POLRATE_CL	lin	✓	RW
YIELD10_CL	lin	✓	RW
HP_CL	log	✓	RW
NEER_CL	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_CL	lin	✓	RW
ROWINF_CL	lin	✓	RW
ROWFIN_CL	lin	✓	RW

Table 11: China (CN, $n = 20$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_CN	log	–	RW
CPI_CN	log	–	RW
RATE_CN	lin	✓	RW
EXP_CN	log	–	RW
IMP_CN	log	–	RW
REER_CN	log	✓	RW
EQ_CN	log	✓	RW
POLRATE_CN	lin	✓	RW
HP_CN	log	✓	RW
GLB_OIL	log	✓	RW

China (CN), continued

ID	Trans.	Fin.	Prior
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
FREV_GDP	lin	–	WN
FPEXP_GDP	lin	–	WN
EFFI	lin	✓	RW
SFA	lin	–	WN
ROWDEM_CN	lin	✓	RW
ROWINF_CN	lin	✓	RW
ROWFIN_CN	lin	✓	RW

Table 12: Colombia (CO, $n = 20$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_CO	log	–	RW
CONS_CO	log	–	RW
INV_CO	log	–	RW
GOV_CO	log	–	RW
EXP_CO	log	–	RW
IMP_CO	log	–	RW
CPI_CO	log	–	RW
LR_CO	lin	–	RW
RATE_CO	lin	✓	RW
POLRATE_CO	lin	✓	RW
YIELD10_CO	lin	✓	RW
HP_CO	log	✓	RW
NEER_CO	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_CO	lin	✓	RW
ROWINF_CO	lin	✓	RW
ROWFIN_CO	lin	✓	RW

Table 13: Costa Rica (CR, $n = 14$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_CR	log	–	RW
EXP_CR	log	–	RW
IMP_CR	log	–	RW
CPI_CR	log	–	RW

Costa Rica (CR), continued

ID	Trans.	Fin.	Prior
LR_CR	lin	–	RW
RATE_CR	lin	✓	RW
NEER_CR	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_CR	lin	✓	RW
ROWINF_CR	lin	✓	RW
ROWFIN_CR	lin	✓	RW

Table 14: Czechia (CZ, $n = 19$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_CZ	log	–	RW
CONS_CZ	log	–	RW
INV_CZ	log	–	RW
GOV_CZ	log	–	RW
EXP_CZ	log	–	RW
IMP_CZ	log	–	RW
HICP_CZ	log	–	RW
LR_CZ	lin	–	RW
YIELD10_CZ	lin	✓	RW
NEER_CZ	log	✓	RW
POLRATE_CZ	lin	✓	RW
HP_CZ	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_CZ	lin	✓	RW
ROWINF_CZ	lin	✓	RW
ROWFIN_CZ	lin	✓	RW

Table 15: Denmark (DK, $n = 20$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_DK	log	–	RW
CONS_DK	log	–	RW
INV_DK	log	–	RW
GOV_DK	log	–	RW
EXP_DK	log	–	RW

Denmark (DK), continued

ID	Trans.	Fin.	Prior
IMP_DK	log	–	RW
HICP_DK	log	–	RW
LR_DK	lin	–	RW
RATE_DK	lin	✓	RW
YIELD10_DK	lin	✓	RW
NEER_DK	log	✓	RW
POLRATE_DK	lin	✓	RW
HP_DK	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_DK	lin	✓	RW
ROWINF_DK	lin	✓	RW
ROWFIN_DK	lin	✓	RW

Table 16: Euro Area (EA, $n = 24$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_EA	log	–	RW
CONS_EA	log	–	RW
INV_EA	log	–	RW
GOV_EA	log	–	RW
EXP_EA	log	–	RW
IMP_EA	log	–	RW
HICP_EA	log	–	RW
RATE_EA	lin	✓	RW
YIELD10_EA	lin	✓	RW
NEER_EA	log	✓	RW
POLRATE_EA	lin	✓	RW
HPI_EA	log	✓	RW
EACORP	lin	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
FREV_GDP	lin	–	WN
FPEXP_GDP	lin	–	WN
EFFI	lin	✓	RW
SFA	lin	–	WN
ROWDEM_EA	lin	✓	RW
ROWINF_EA	lin	✓	RW
ROWFIN_EA	lin	✓	RW

Table 17: Hungary (HU, $n = 19$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_HU	log	–	RW
CONS_HU	log	–	RW
INV_HU	log	–	RW
GOV_HU	log	–	RW
EXP_HU	log	–	RW
IMP_HU	log	–	RW
HICP_HU	log	–	RW
LR_HU	lin	–	RW
YIELD10_HU	lin	✓	RW
NEER_HU	log	✓	RW
POLRATE_HU	lin	✓	RW
HP_HU	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_HU	lin	✓	RW
ROWINF_HU	lin	✓	RW
ROWFIN_HU	lin	✓	RW

Table 18: Indonesia (ID, $n = 18$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_ID	log	–	RW
CONS_ID	log	–	RW
INV_ID	log	–	RW
GOV_ID	log	–	RW
EXP_ID	log	–	RW
IMP_ID	log	–	RW
CPI_ID	log	–	RW
RATE_ID	lin	✓	RW
NEER_ID	log	✓	RW
POLRATE_ID	lin	✓	RW
HP_ID	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_ID	lin	✓	RW
ROWINF_ID	lin	✓	RW
ROWFIN_ID	lin	✓	RW

Table 19: Israel (IL, $n = 19$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_IL	log	–	RW
CONS_IL	log	–	RW
INV_IL	log	–	RW
GOV_IL	log	–	RW
EXP_IL	log	–	RW
IMP_IL	log	–	RW
CPI_IL	log	–	RW
LR_IL	lin	–	RW
RATE_IL	lin	✓	RW
POLRATE_IL	lin	✓	RW
NEER_IL	log	✓	RW
HP_IL	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_IL	lin	✓	RW
ROWINF_IL	lin	✓	RW
ROWFIN_IL	lin	✓	RW

Table 20: India (IN, $n = 20$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_IN	log	–	RW
CONS_IN	log	–	RW
INV_IN	log	–	RW
GOV_IN	log	–	RW
EXP_IN	log	–	RW
IMP_IN	log	–	RW
CPI_IN	log	–	RW
RATE_IN	lin	✓	RW
RATE3M_IN	lin	✓	RW
YIELD10_IN	lin	✓	RW
NEER_IN	log	✓	RW
POLRATE_IN	lin	✓	RW
HP_IN	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_IN	lin	✓	RW
ROWINF_IN	lin	✓	RW
ROWFIN_IN	lin	✓	RW

Table 21: Iceland (IS, $n = 19$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_IS	log	–	RW
CONS_IS	log	–	RW
INV_IS	log	–	RW
GOV_IS	log	–	RW
EXP_IS	log	–	RW
IMP_IS	log	–	RW
CPI_IS	log	–	RW
LR_IS	lin	–	RW
RATE_IS	lin	✓	RW
NEER_IS	log	✓	RW
POLRATE_IS	lin	✓	RW
HP_IS	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_IS	lin	✓	RW
ROWINF_IS	lin	✓	RW
ROWFIN_IS	lin	✓	RW

Table 22: Japan (JP, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_JP	log	–	RW
CONS_JP	log	–	RW
INV_JP	log	–	RW
GOV_JP	log	–	RW
EXP_JP	log	–	RW
IMP_JP	log	–	RW
LR_JP	lin	–	RW
CPI_JP	log	–	RW
RATE_JP	lin	✓	RW
POLRATE_JP	lin	✓	RW
YIELD10_JP	lin	✓	RW
HP_JP	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
FREV_GDP	lin	–	WN
FPEXP_GDP	lin	–	WN
EFFI	lin	✓	RW
SFA	lin	–	WN

Japan (JP), continued

ID	Trans.	Fin.	Prior
ROWDEM_JP	lin	✓	RW
ROWINF_JP	lin	✓	RW
ROWFIN_JP	lin	✓	RW

Table 23: Korea (KR, $n = 19$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_KR	log	–	RW
CONS_KR	log	–	RW
INV_KR	log	–	RW
GOV_KR	log	–	RW
EXP_KR	log	–	RW
IMP_KR	log	–	RW
CPI_KR	log	–	RW
LR_KR	lin	–	RW
RATE_KR	lin	✓	RW
POLRATE_KR	lin	✓	RW
YIELD10_KR	lin	✓	RW
HP_KR	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_KR	lin	✓	RW
ROWINF_KR	lin	✓	RW
ROWFIN_KR	lin	✓	RW

Table 24: Mexico (MX, $n = 20$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_MX	log	–	RW
CONS_MX	log	–	RW
INV_MX	log	–	RW
GOV_MX	log	–	RW
EXP_MX	log	–	RW
IMP_MX	log	–	RW
CPI_MX	log	–	RW
LR_MX	lin	–	RW
RATE_MX	lin	✓	RW
POLRATE_MX	lin	✓	RW
YIELD10_MX	lin	✓	RW
HP_MX	log	✓	RW

Mexico (MX), continued

ID	Trans.	Fin.	Prior
NEER_MX	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_MX	lin	✓	RW
ROWINF_MX	lin	✓	RW
ROWFIN_MX	lin	✓	RW

Table 25: Malaysia (MY, $n = 19$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_MY	log	–	RW
CONS_MY	log	–	RW
INV_MY	log	–	RW
GOV_MY	log	–	RW
EXP_MY	log	–	RW
IMP_MY	log	–	RW
CPI_MY	log	–	RW
LR_MY	lin	–	RW
RATE_MY	lin	✓	RW
NEER_MY	log	✓	RW
POLRATE_MY	lin	✓	RW
HP_MY	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_MY	lin	✓	RW
ROWINF_MY	lin	✓	RW
ROWFIN_MY	lin	✓	RW

Table 26: Norway (NO, $n = 18$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_NO	log	–	RW
CONS_NO	log	–	RW
INV_NO	log	–	RW
GOV_NO	log	–	RW
EXP_NO	log	–	RW
IMP_NO	log	–	RW
HICP_NO	log	–	RW

Norway (NO), continued

ID	Trans.	Fin.	Prior
LR_NO	lin	–	RW
NEER_NO	log	✓	RW
POLRATE_NO	lin	✓	RW
HP_NO	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_NO	lin	✓	RW
ROWINF_NO	lin	✓	RW
ROWFIN_NO	lin	✓	RW

Table 27: New Zealand (NZ, $n = 19$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_NZ	log	–	RW
CONS_NZ	log	–	RW
INV_NZ	log	–	RW
GOV_NZ	log	–	RW
EXP_NZ	log	–	RW
IMP_NZ	log	–	RW
CPI_NZ	log	–	RW
LR_NZ	lin	–	RW
RATE_NZ	lin	✓	RW
YIELD10_NZ	lin	✓	RW
HP_NZ	log	✓	RW
NEER_NZ	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_NZ	lin	✓	RW
ROWINF_NZ	lin	✓	RW
ROWFIN_NZ	lin	✓	RW

Table 28: Poland (PL, $n = 19$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_PL	log	–	RW
CONS_PL	log	–	RW
INV_PL	log	–	RW
GOV_PL	log	–	RW

Poland (PL), continued

ID	Trans.	Fin.	Prior
EXP_PL	log	–	RW
IMP_PL	log	–	RW
HICP_PL	log	–	RW
LR_PL	lin	–	RW
YIELD10_PL	lin	✓	RW
NEER_PL	log	✓	RW
POLRATE_PL	lin	✓	RW
HP_PL	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_PL	lin	✓	RW
ROWINF_PL	lin	✓	RW
ROWFIN_PL	lin	✓	RW

Table 29: Russia (RU, $n = 15$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_RU	log	–	RW
CPI_RU	log	–	RW
EXP_RU	log	–	RW
LR_RU	lin	–	RW
RATE_RU	lin	✓	RW
NEER_RU	log	✓	RW
POLRATE_RU	lin	✓	RW
HP_RU	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_RU	lin	✓	RW
ROWINF_RU	lin	✓	RW
ROWFIN_RU	lin	✓	RW

Table 30: Saudi Arabia (SA, $n = 18$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_SA	log	–	RW
CONS_SA	log	–	RW
INV_SA	log	–	RW
GOV_SA	log	–	RW

Saudi Arabia (SA), continued

ID	Trans.	Fin.	Prior
EXP_SA	log	–	RW
IMP_SA	log	–	RW
CPI_SA	log	–	RW
LR_SA	lin	–	RW
RATE_SA	lin	✓	RW
NEER_SA	log	✓	RW
POLRATE_SA	lin	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_SA	lin	✓	RW
ROWINF_SA	lin	✓	RW
ROWFIN_SA	lin	✓	RW

Table 31: Sweden (SE, $n = 20$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_SE	log	–	RW
CONS_SE	log	–	RW
INV_SE	log	–	RW
GOV_SE	log	–	RW
EXP_SE	log	–	RW
IMP_SE	log	–	RW
HICP_SE	log	–	RW
LR_SE	lin	–	RW
RATE_SE	lin	✓	RW
YIELD10_SE	lin	✓	RW
NEER_SE	log	✓	RW
POLRATE_SE	lin	✓	RW
HP_SE	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_SE	lin	✓	RW
ROWINF_SE	lin	✓	RW
ROWFIN_SE	lin	✓	RW

Table 32: Thailand (TH, $n = 20$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_TH	log	–	RW
CONS_TH	log	–	RW
INV_TH	log	–	RW
GOV_TH	log	–	RW
EXP_TH	log	–	RW
IMP_TH	log	–	RW
CPI_TH	log	–	RW
LR_TH	lin	–	RW
RATE_TH	lin	✓	RW
YIELD10_TH	lin	✓	RW
NEER_TH	log	✓	RW
POLRATE_TH	lin	✓	RW
HP_TH	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_TH	lin	✓	RW
ROWINF_TH	lin	✓	RW
ROWFIN_TH	lin	✓	RW

Table 33: Türkiye (TR, $n = 18$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_TR	log	–	RW
CONS_TR	log	–	RW
INV_TR	log	–	RW
GOV_TR	log	–	RW
EXP_TR	log	–	RW
IMP_TR	log	–	RW
CPI_TR	log	–	RW
LR_TR	lin	–	RW
RATE_TR	lin	✓	RW
POLRATE_TR	lin	✓	RW
HP_TR	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_TR	lin	✓	RW
ROWINF_TR	lin	✓	RW
ROWFIN_TR	lin	✓	RW

Table 34: Taiwan (TW, $n = 18$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_TW	log	–	RW
CONS_TW	log	–	RW
GOV_TW	log	–	RW
INV_TW	log	–	RW
EXP_TW	log	–	RW
IMP_TW	log	–	RW
CPI_TW	log	–	RW
LR_TW	lin	–	RW
RATE_TW	lin	✓	RW
MMRATE_TW	lin	✓	RW
YIELD10_TW	lin	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_TW	lin	✓	RW
ROWINF_TW	lin	✓	RW
ROWFIN_TW	lin	✓	RW

Table 35: United Kingdom (UK, $n = 25$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_UK	log	–	RW
CONS_UK	log	–	RW
INV_UK	log	–	RW
GOV_UK	log	–	RW
EXP_UK	log	–	RW
IMP_UK	log	–	RW
IP_UK	log	–	RW
LR_UK	lin	–	RW
CPI_UK	log	–	RW
RATE_UK	lin	✓	RW
POLRATE_UK	lin	✓	RW
YIELD10_UK	lin	✓	RW
HP_UK	log	✓	RW
NEER_UK	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
FREV_GDP	lin	–	WN
FPEXP_GDP	lin	–	WN
EFFI	lin	✓	RW

United Kingdom (UK), continued

ID	Trans.	Fin.	Prior
SFA	lin	–	WN
ROWDEM_UK	lin	✓	RW
ROWINF_UK	lin	✓	RW
ROWFIN_UK	lin	✓	RW

Table 36: United States (US, $n = 41$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDPH	log	–	RW
CH	log	–	RW
FRH	log	–	RW
FNH	log	–	RW
XH	log	–	RW
MH	log	–	RW
GH	log	–	RW
JGDP	log	–	RW
JC	log	–	RW
JCXFE	log	–	RW
UI	log	–	RW
UIXFDG	log	–	RW
LXBC	log	–	RW
LANAGRA	log	–	RW
LR	lin	–	RW
IP	log	–	RW
CUMFG	lin	–	RW
HST	log	–	RW
YPDH	log	–	RW
CSENT	lin	–	RW
POLRATE	lin	✓	RW
FCM2	lin	✓	RW
FCM5	lin	✓	RW
GLB_USLR	lin	✓	RW
FAAA	lin	✓	RW
GLB_USBAA	lin	✓	RW
FXTWM	log	✓	RW
GLB_USEQ	log	✓	RW
PZTEXP	log	✓	RW
GSNE	log	✓	RW
SP500VOL	lin	✓	RW
HP	log	✓	RW
LEV	log	✓	RW
FREV_GDP	lin	–	WN
FPEXP_GDP	lin	–	WN
EFFI	lin	✓	RW

United States (US), continued

ID	Trans.	Fin.	Prior
SFA	lin	–	WN
GLB_OIL	log	✓	RW
ROWDEM_US	lin	✓	RW
ROWINF_US	lin	✓	RW
ROWFIN_US	lin	✓	RW

Table 37: South Africa (ZA, $n = 20$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_ZA	log	–	RW
CONS_ZA	log	–	RW
INV_ZA	log	–	RW
GOV_ZA	log	–	RW
EXP_ZA	log	–	RW
IMP_ZA	log	–	RW
CPI_ZA	log	–	RW
LR_ZA	lin	–	RW
RATE_ZA	lin	✓	RW
YIELD10_ZA	lin	✓	RW
NEER_ZA	log	✓	RW
POLRATE_ZA	lin	✓	RW
HP_ZA	log	✓	RW
GLB_OIL	log	✓	RW
GLB_USLR	lin	✓	RW
GLB_USBAA	lin	✓	RW
GLB_USEQ	log	✓	RW
ROWDEM_ZA	lin	✓	RW
ROWINF_ZA	lin	✓	RW
ROWFIN_ZA	lin	✓	RW

The 17 euro-area standalones. These omit the ROW* stars and instead carry US-block foreign variables (*_US) plus the shared euro-area corporate spread (EACORP); they are estimated in separation (Section 19.2).

Table 38: Austria (AT, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_AT	log	–	RW
CONS_AT	log	–	RW
INV_AT	log	–	RW
GOV_AT	log	–	RW
EXP_AT	log	–	RW

Austria (AT), continued

ID	Trans.	Fin.	Prior
IMP_AT	log	–	RW
HICP_AT	log	–	RW
LR_AT	lin	–	RW
POLRATE_AT	lin	✓	RW
HPI_AT	log	✓	RW
EACORP	lin	✓	RW
RATE_AT	lin	✓	RW
YIELD10_AT	lin	✓	RW
NEER_AT	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 39: Belgium (BE, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_BE	log	–	RW
CONS_BE	log	–	RW
INV_BE	log	–	RW
GOV_BE	log	–	RW
EXP_BE	log	–	RW
IMP_BE	log	–	RW
HICP_BE	log	–	RW
LR_BE	lin	–	RW
POLRATE_BE	lin	✓	RW
HPI_BE	log	✓	RW
EACORP	lin	✓	RW
RATE_BE	lin	✓	RW
YIELD10_BE	lin	✓	RW
NEER_BE	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW

Belgium (BE), continued

ID	Trans.	Fin.	Prior
WTI_US	log	✓	RW

Table 40: Germany (DE, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_DE	log	–	RW
CONS_DE	log	–	RW
INV_DE	log	–	RW
GOV_DE	log	–	RW
EXP_DE	log	–	RW
IMP_DE	log	–	RW
HICP_DE	log	–	RW
LR_DE	lin	–	RW
POLRATE_DE	lin	✓	RW
HPI_DE	log	✓	RW
EACORP	lin	✓	RW
RATE_DE	lin	✓	RW
YIELD10_DE	lin	✓	RW
NEER_DE	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 41: Estonia (EE, $n = 22$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_EE	log	–	RW
CONS_EE	log	–	RW
INV_EE	log	–	RW
GOV_EE	log	–	RW
EXP_EE	log	–	RW
IMP_EE	log	–	RW
HICP_EE	log	–	RW
LR_EE	lin	–	RW
POLRATE_EE	lin	✓	RW
HPI_EE	log	✓	RW

Estonia (EE), continued

ID	Trans.	Fin.	Prior
EACORP	lin	✓	RW
RATE_EE	lin	✓	RW
NEER_EE	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 42: Spain (ES, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_ES	log	–	RW
CONS_ES	log	–	RW
INV_ES	log	–	RW
GOV_ES	log	–	RW
EXP_ES	log	–	RW
IMP_ES	log	–	RW
HICP_ES	log	–	RW
LR_ES	lin	–	RW
POLRATE_ES	lin	✓	RW
HPI_ES	log	✓	RW
EACORP	lin	✓	RW
RATE_ES	lin	✓	RW
YIELD10_ES	lin	✓	RW
NEER_ES	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 43: Finland (FI, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_FI	log	–	RW
CONS_FI	log	–	RW
INV_FI	log	–	RW
GOV_FI	log	–	RW
EXP_FI	log	–	RW
IMP_FI	log	–	RW
HICP_FI	log	–	RW
LR_FI	lin	–	RW
POLRATE_FI	lin	✓	RW
HPI_FI	log	✓	RW
EACORP	lin	✓	RW
RATE_FI	lin	✓	RW
YIELD10_FI	lin	✓	RW
NEER_FI	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 44: France (FR, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_FR	log	–	RW
CONS_FR	log	–	RW
INV_FR	log	–	RW
GOV_FR	log	–	RW
EXP_FR	log	–	RW
IMP_FR	log	–	RW
HICP_FR	log	–	RW
LR_FR	lin	–	RW
POLRATE_FR	lin	✓	RW
HPI_FR	log	✓	RW
EACORP	lin	✓	RW
RATE_FR	lin	✓	RW
YIELD10_FR	lin	✓	RW
NEER_FR	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW

France (FR), continued

ID	Trans.	Fin.	Prior
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 45: Greece (GR, $n = 22$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_GR	log	–	RW
CONS_GR	log	–	RW
INV_GR	log	–	RW
GOV_GR	log	–	RW
EXP_GR	log	–	RW
IMP_GR	log	–	RW
HICP_GR	log	–	RW
LR_GR	lin	–	RW
POLRATE_GR	lin	✓	RW
EACORP	lin	✓	RW
RATE_GR	lin	✓	RW
YIELD10_GR	lin	✓	RW
NEER_GR	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 46: Ireland (IE, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_IE	log	–	RW
CONS_IE	log	–	RW
INV_IE	log	–	RW
GOV_IE	log	–	RW
EXP_IE	log	–	RW

Ireland (IE), continued

ID	Trans.	Fin.	Prior
IMP_IE	log	–	RW
HICP_IE	log	–	RW
LR_IE	lin	–	RW
POLRATE_IE	lin	✓	RW
HPI_IE	log	✓	RW
EACORP	lin	✓	RW
RATE_IE	lin	✓	RW
YIELD10_IE	lin	✓	RW
NEER_IE	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 47: Italy (IT, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_IT	log	–	RW
CONS_IT	log	–	RW
INV_IT	log	–	RW
GOV_IT	log	–	RW
EXP_IT	log	–	RW
IMP_IT	log	–	RW
HICP_IT	log	–	RW
LR_IT	lin	–	RW
POLRATE_IT	lin	✓	RW
HPI_IT	log	✓	RW
EACORP	lin	✓	RW
RATE_IT	lin	✓	RW
YIELD10_IT	lin	✓	RW
NEER_IT	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW

Italy (IT), continued

ID	**Trans.**	**Fin.**	**Prior**
WTI_US	log	✓	RW

Table 48: Lithuania (LT, $n = 23$): variable transformations, financial tags, and priors.

ID	**Trans.**	**Fin.**	**Prior**
GDP_LT	log	–	RW
CONS_LT	log	–	RW
INV_LT	log	–	RW
GOV_LT	log	–	RW
EXP_LT	log	–	RW
IMP_LT	log	–	RW
HICP_LT	log	–	RW
LR_LT	lin	–	RW
POLRATE_LT	lin	✓	RW
HPI_LT	log	✓	RW
EACORP	lin	✓	RW
RATE_LT	lin	✓	RW
YIELD10_LT	lin	✓	RW
NEER_LT	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 49: Luxembourg (LU, $n = 23$): variable transformations, financial tags, and priors.

ID	**Trans.**	**Fin.**	**Prior**
GDP_LU	log	–	RW
CONS_LU	log	–	RW
INV_LU	log	–	RW
GOV_LU	log	–	RW
EXP_LU	log	–	RW
IMP_LU	log	–	RW
HICP_LU	log	–	RW
LR_LU	lin	–	RW
POLRATE_LU	lin	✓	RW
HPI_LU	log	✓	RW

Luxembourg (LU), continued

ID	Trans.	Fin.	Prior
EACORP	lin	✓	RW
RATE_LU	lin	✓	RW
YIELD10_LU	lin	✓	RW
NEER_LU	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 50: Latvia (LV, $n = 20$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_LV	log	–	RW
CONS_LV	log	–	RW
INV_LV	log	–	RW
GOV_LV	log	–	RW
EXP_LV	log	–	RW
IMP_LV	log	–	RW
HICP_LV	log	–	RW
LR_LV	lin	–	RW
RATE_LV	lin	✓	RW
YIELD10_LV	lin	✓	RW
NEER_LV	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 51: Netherlands (NL, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_NL	log	–	RW

Netherlands (NL), continued

ID	Trans.	Fin.	Prior
CONS_NL	log	–	RW
INV_NL	log	–	RW
GOV_NL	log	–	RW
EXP_NL	log	–	RW
IMP_NL	log	–	RW
HICP_NL	log	–	RW
LR_NL	lin	–	RW
POLRATE_NL	lin	✓	RW
HPI_NL	log	✓	RW
EACORP	lin	✓	RW
RATE_NL	lin	✓	RW
YIELD10_NL	lin	✓	RW
NEER_NL	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 52: Portugal (PT, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_PT	log	–	RW
CONS_PT	log	–	RW
INV_PT	log	–	RW
GOV_PT	log	–	RW
EXP_PT	log	–	RW
IMP_PT	log	–	RW
HICP_PT	log	–	RW
LR_PT	lin	–	RW
POLRATE_PT	lin	✓	RW
HPI_PT	log	✓	RW
EACORP	lin	✓	RW
RATE_PT	lin	✓	RW
YIELD10_PT	lin	✓	RW
NEER_PT	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW

Portugal (PT), continued

ID	Trans.	Fin.	Prior
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 53: Slovenia (SI, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_SI	log	–	RW
CONS_SI	log	–	RW
INV_SI	log	–	RW
GOV_SI	log	–	RW
EXP_SI	log	–	RW
IMP_SI	log	–	RW
HICP_SI	log	–	RW
LR_SI	lin	–	RW
POLRATE_SI	lin	✓	RW
HPI_SI	log	✓	RW
EACORP	lin	✓	RW
RATE_SI	lin	✓	RW
YIELD10_SI	lin	✓	RW
NEER_SI	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Table 54: Slovakia (SK, $n = 23$): variable transformations, financial tags, and priors.

ID	Trans.	Fin.	Prior
GDP_SK	log	–	RW
CONS_SK	log	–	RW
INV_SK	log	–	RW
GOV_SK	log	–	RW
EXP_SK	log	–	RW
IMP_SK	log	–	RW

Slovakia (SK), continued

ID	Trans.	Fin.	Prior
HICP_SK	log	–	RW
LR_SK	lin	–	RW
POLRATE_SK	lin	✓	RW
HPI_SK	log	✓	RW
EACORP	lin	✓	RW
RATE_SK	lin	✓	RW
YIELD10_SK	lin	✓	RW
NEER_SK	log	✓	RW
GDPH_US	log	–	RW
GH_US	log	–	RW
LR_US	lin	–	RW
JGDP_US	log	–	RW
FCM2_US	lin	✓	RW
FCM10_US	lin	✓	RW
EQ_US	log	✓	RW
FBAA_US	lin	✓	RW
WTI_US	log	✓	RW

Coherence review. A cross-economy audit of all 1048 variable rows confirms that every shared concept is assigned the same transformation, financial tag, and prior in every bloc that carries it: real activity, price indices, house prices, equities, oil, and FX enter in `log`; interest rates, yields, spreads, unemployment, and the fiscal ratios enter `lin`; all financial series share the financial tag; and all priors are RW apart from the three stationary fiscal GDP-ratios (`FREV_GDP`, `FPEXP_GDP`, `SFA`, prior WN). Canada's industrial-production series `IP_CA` has been rebuilt as a `log`-level index, consistent with the activity convention used for the United States and the United Kingdom. The FRED level index (`CANPROINDMISMEI`, 2015 = 100) was discontinued at 2024-02, so the series is the quarterly average of that index through 2024-Q1 and is then growth-chained over 2024-Q2–2025-Q4 (eight quarters) using the FRED q-on-q growth series `PRINTO01CAQ657S` ($\text{level}_t = \text{level}_{t-1}(1 + g_t)$, the same chaining used to extend the other Canadian series). With this reconstruction the audit finds no remaining transformation-coherence exceptions.

20 Per-Country Data Sources and Transformations

This section documents, variable by variable, the data underlying every one of the 50 economies in the system — the six core GVAR blocs, the euro-area members, the EU non-euro and other advanced economies, and the emerging markets. For each economy a table lists *every* variable in its deployed specification together with its provider, the exact series identifier (FRED mnemonic, ECB/Eurostat SDMX key, BIS dataflow key, or national central-bank code), native frequency, transformation, COVID treatment, and any imputation or splicing performed at build time. The authoritative variable list (id, description, transformation, `isFinancial` tag, prior, block) is taken verbatim from each economy's staged `spec.json`; the source/series-id/frequency/COVID/imputation columns are traced from the data builders and credit fetchers.

Transformation and COVID conventions (system-wide). Two transformation tags appear throughout. `log` denotes a series entered as $100 \cdot \log(\text{level})$; because such a variable carries a random-walk (RW) prior unless flagged otherwise, it *enters the dynamics as* $4 \cdot \Delta\log$, i.e. a quarter-on-quarter *annualized* growth rate (Section 12.3). `lin` denotes a series entered in levels (rates in per cent, ratios in per cent of GDP, spreads in percentage points, diffusion indices, etc.). The *COVID-as-missing* rule (Section 19.4) masks the macro block (`isFinancial`= 0) over the pandemic window — the canonical window is 2020Q2–Q4, though a few builds use a self-calibrated four-quarter 2020Q1–Q4 window (flagged where it applies) — and lets the data-augmentation/Durbin–Koopman E-step impute it, while *financial* variables (`isFinancial`= 1) are held observed through the pandemic. Per-variable priors are RW except the stationary fiscal GDP-ratios (`FREV_GDP`, `FPEXP_GDP`, `SFA`), which carry the white-noise (`WN`) prior.

The new credit block (flagged throughout). Each bloc is augmented with a credit *quantity* variable `CREDIT_GDP_<CC>` = BIS *Total Credit to Non-Financial Corporations, % of GDP* (the BIS *broad* measure: bank loans + debt securities + cross-border lending), pulled from FRED as `Q<CC>NAM770A` where a BIS reporting series exists, transformed `lin`, prior RW, `isFinancial`= 1. Although tagged financial, the credit quantity is *masked* over the COVID window because it is a %-of-GDP ratio whose denominator (GDP) collapses in 2020, so it inherits the macro GDP mask. On the credit *price* side the blocs gain either a representative NFC bank *lending rate* (`LENDR_<CC>`) or, where no lending rate is reachable, a corporate or sovereign-proxy yield (`CORP_<CC>`); where no price could be sourced it is dropped (quantity only). Economies whose credit quantity is *constructed* (non-BIS) or whose price is a sovereign proxy, spliced, dropped, or a non-canonical variant are flagged in-line in the relevant table and prose. All credit *price* variables are held observed through COVID.

20.1 Per-Economy Data Documentation: the Six Core Blocs

This subsection documents, variable by variable, the data underlying the six core GVAR blocs — the United States anchor and its five major partners (euro area, Japan, United Kingdom, Canada, China). For each economy the table lists *every* variable in the deployed specification, together with its provider, the exact series identifier (FRED mnemonic, ECB/Eurostat SDMX key, BIS dataflow key, or national central-bank code), native frequency, transformation, COVID treatment, and any imputation or splicing performed at build time.

Transformation and COVID conventions. Two transformation tags appear throughout. `log` denotes a series entered as $100 \cdot \log(\text{level})$; because every variable carries a random-walk (RW) prior unless flagged otherwise, a `log` variable *enters the dynamics as* $4 \cdot \Delta\log$, i.e. as a quarter-on-quarter *annualized growth rate*. `lin` denotes a series entered in levels (rates in per cent, ratios in per cent of GDP, diffusion indices, etc.). The *COVID-as-missing* rule (Section 19.4) masks the macro block (`isFinancial` = 0) over 2020Q2–2020Q4 and lets the data-augmentation Gibbs sampler impute it, while *financial* variables (`isFinancial` = 1) are held observed through the pandemic window. "Fin." in the tables below reproduces the `isFinancial` tag.

The new credit block (flagged). Each of the six blocs gains a credit *quantity* variable `CREDIT_GDP_<CC>` = BIS *Total Credit to Non-Financial Corporations, % of GDP* — the BIS *broad* credit measure (bank loans + debt securities + cross-border), pulled from FRED as `Q<CC>NAM770A`, transformed `lin`, prior RW, `isFinancial` = 1. Although tagged financial, the credit *quantity* is *masked* over 2020Q2–Q4 because it is a %-of-GDP ratio whose denominator (GDP) collapses in 2020; it therefore inherits the macro GDP mask rather than the financial observe-rule. On the credit *price* side: the US and euro area already carry a corporate-yield channel and gain *no* new price variable; Japan, the United Kingdom, and Canada gain a representative NFC bank *lending rate* (`LENDR_<CC>`); and China gains a *sovereign 10Y yield proxy* (`CORP_CN`, flagged — no corporate spread exists for China). All credit *price* variables are held observed through COVID.

20.2 United States (US)

The US anchor bloc carries $n = 39$ quarterly variables on a 1998Q3–2025Q4 grid (110 quarters) at $p = 3$ lags, comprising a rich domestic macro/financial block, four global commons, and a four-variable fiscal block; the COVID-as-missing window 2020Q2–Q4 masks the macro (`isFinancial` = 0) series. Macro and financial series are pulled from FRED at native quarterly frequency (`frequency=q, aggregation_method=avg`); the deployed bloc additionally appends the three trade-weighted `ROW*_US` stars (Section 16.1), which are not data series and are omitted here. The new credit quantity `CREDIT_GDP` is BIS broad NFC credit (% of GDP); **no new credit price is added — the existing Aaa (`FAAA`) and Baa (`GLB_USBAA`) corporate yields already span the corporate credit-price channel.**

Table 55: US bloc variables. Trans.: `log` = 100 log level (enters as 4Δ log, q/q annualized); `lin` = level. Fin. = `isFinancial`. COVID: "mask" = 2020Q2–Q4 set missing (Gibbs-imputed); "obs" = held observed.

Variable (id)	Description	Source	Series ID/key	Freq	Fin.	Trans.	COVID / Notes
Domestic macro block (FRED, quarterly avg; `isFinancial=0` ⇒ COVID-masked)							
GDPH	Real Gross Domestic Product	FRED	`GDPC1`	Q	0	log	mask
CH	Real Personal Consumption Exp.	FRED	`PCECC96`	Q	0	log	mask
FRH	Real Private Residential Fixed Inv.	FRED	`PRFI`	Q	0	log	mask; deflated by `GDPDEF`

Variable (id)	Description	Source	Series ID/key	Freq	Fin.	Trans.	COVID / Notes
FNH	Real Private Nonres. Fixed Inv.	FRED	`PNFI`	Q	0	log	mask; deflated by `GDPDEF`
XH	Real Exports of Goods & Services	FRED	`EXPGSC1`	Q	0	log	mask
MH	Real Imports of Goods & Services	FRED	`IMPGSC1`	Q	0	log	mask
GH	Real Gov. Consumption & Gross Inv.	FRED	`GCEC1`	Q	0	log	mask
JGDP	GDP: Chain Price Index	FRED	`GDPDEF`	Q	0	log	mask
JC	PCE: Chain Price Index	FRED	`PCEPI`	Q	0	log	mask
JCXFE	PCE less Food & Energy: Chain Price Idx	FRED	`PCEPILFE`	Q	0	log	mask
UI	CPI-U: All Items	FRED	`CPIAUCSL`	Q	0	log	mask
UIXFDG	CPI-U: All Items less Food & Energy	FRED	`CPILFESL`	Q	0	log	mask
LXBC	Business Sector: Compensation/Hour	FRED	`HCOMPBS`	Q	0	log	mask
LANAGRA	All Employees: Total Nonfarm	FRED	`PAYEMS`	M→Q	0	log	mask
LR	Civilian Unemployment Rate 16+	FRED	`UNRATE`	M→Q	0	lin	mask
IP	Industrial Production Index	FRED	`INDPRO`	M→Q	0	log	mask
CUMFG	Capacity Utilization: Mfg [SIC]	FRED	`MCUMFN`	M→Q	0	lin	mask
HST	Housing Starts	FRED	`HOUST`	M→Q	0	log	mask
YPDH	Real Disposable Personal Income	FRED	`DPIC96`	Q	0	log	mask
CSENT	U. Michigan Consumer Sentiment	FRED	`UMCSENT`	M→Q	0	lin	mask
Domestic financial block (FRED; `isFinancial`=1 ⇒ observed through COVID)							
POLRATE	Effective Federal Funds Rate (policy)	FRED	`FEDFUNDS`	M→Q	1	lin	obs
FCM2	2Y Treasury Yield, const. maturity	FRED	`DGS2`	D→Q	1	lin	obs

Variable (id)	Description	Source	Series ID/key	Freq	Fin.	Trans.	COVID / Notes
FCM5	5Y Treasury Yield, const. maturity	FRED	`DGS5`	D→Q	1	lin	obs
FAAA	Moody's Seasoned Aaa Corp. Bond Yield	FRED	`AAA`	M→Q	1	lin	obs; high-grade corp. price
FXTWM	Nominal Trade-Wtd US$ vs AFE	FRED	`DTWEXAFEGS` / `DTWEXM`	D→Q	1	log	obs; AFE spliced onto legacy Major-$ at 2006 overlap
PZTEXP	Spot Oil Price: WTI	FRED	`WTISPLC`	M→Q	1	log	obs
GSNE	Non-Fuel Commodity Price Idx (IMF proxy)	FRED	`PNFUELINDEXM`	M→Q	1	log	obs
SP500VOL	CBOE Volatility Index (VIX)	FRED	`VIXCLS`	D→Q	1	lin	obs
HP	House Price	FRED	`CSUSHPISA`	M→Q	1	log	obs
LEV	Real nonfinancial business debt	FRED	`BCNSDODNS`	Q	1	log	obs; deflated by `GDPDEF`/100
Global commons (US-determined common shocks; `isFinancial`=1, observed)							
GLB_USLR	10Y Treasury Yield, const. maturity	FRED	`DGS10`	D→Q	1	lin	obs; global common
GLB_USBAA	Moody's Baa Corp. Bond Yield (credit)	FRED	`BAA`	M→Q	1	lin	obs; risk-spread corp. price
GLB_USEQ	US Equity Price Index (S&P / OECD US)	FRED	`SPASTT01USM661N`	M→Q	1	log	obs; global common
GLB_OIL	WTI crude oil, q-avg ($/bbl)	FRED	`WTISPLC`	M→Q	1	log	obs; global common
Fiscal block (FRED-derived, 4q-annualized; GDP-ratio drivers carry a WN/stationary prior)							
FREV_GDP	Federal Receipts (% of GDP)	FRED	`FGRECPT/GDP`	Q	0	lin	mask; WN prior
FPEXP_GDP	Federal Primary Expenditure (% of GDP)	FRED	(`FGEXPND`−`A091RC1Q027SBEA`) /`GDP`	Q	0	lin	WN prior; 2020Q2–2021Q4 spike linearly interpolated at build time
EFFI	Federal Effective Interest Rate (% p.a.)	FRED	`A091RC1Q027SBEA` /lag (`GFDEGDQ188S`)	Q	1	lin	obs; RW prior

Variable (id)	Description	Source	Series ID/key	Freq	Fin.	Trans.	COVID / Notes
SFA	Stock-Flow Adjustment (% of GDP)	derived	recursion residual (`GFDEGDQ188S`)	Q	0	lin	WN prior; closes the debt recursion by construction
NEW credit block							
CREDIT_GDP	**US Credit to NFCs (% of GDP)**	FRED / BIS	`QUSNAM770A`	Q	1	lin	**mask** (inherits GDP-denominator mask); BIS broad credit, 1998Q3=61.3, 2025Q4=72.2; *no new price (FAAA/GLB_USBAA cover it)*

20.3 Euro Area (EA)

The euro-area aggregate carries $n = 22$ quarterly variables on a 1999Q1–2025Q4 grid ($T = 108$). Real-activity and price series come from FRED (which mirrors Eurostat/OECD/ECB feeds for the euro aggregate), short and long rates from FRED OECD/ECB series, and the financial overlay — policy rate (`POLRATE_EA`), real house prices (`HPI_EA`), and the all-bonds corporate-risk yield (`EACORP`) — from ECB SDMX and Eurostat. The new credit quantity is BIS broad NFC credit (% of GDP); **no new credit price is added — `EACORP` already provides the euro-area corporate-yield channel** (and `GLB_USBAA` a second global credit-risk channel).

Table 56: Euro-area bloc variables. Columns as in Table 55.

Variable (id)	Description	Source	Series ID/key	Freq	Fin.	Trans.	COVID / Notes
GDP_EA	EA Real GDP	FRED (Eurostat)	`CLVMNACSCAB1GQEA19`	Q	0	log	mask
CONS_EA	EA Consumption	FRED (OECD)	`NAEXKP02EZQ661S`	Q	0	log	mask
INV_EA	EA Investment	FRED (OECD)	`NAEXKP04EZQ661S`	Q	0	log	mask
GOV_EA	EA Gov. Spending	FRED (OECD)	`NAEXKP03EZQ661S`	Q	0	log	mask
EXP_EA	EA Exports	FRED (OECD)	`NAEXKP06EZQ661S`	Q	0	log	mask
IMP_EA	EA Imports	FRED (OECD)	`NAEXKP07EZQ661S`	Q	0	log	mask

Variable (id)	Description	Source	Series ID/key	Freq	Fin.	Trans.	COVID / Notes
HICP_EA	EA HICP Index (2015=100)	FRED (Eurostat)	`CP0000EZ19M086NEST`	M→Q	0	log	mask
RATE_EA	EA 3M Interbank Rate	FRED (OECD)	`IR3TIB01EZQ156N`	Q	1	lin	obs
YIELD10_EA	EA 10Y Gov. Bond Yield	FRED (OECD)	`IRLTLT01EZM156N`	M→Q	1	lin	obs
NEER_EA	EA Nominal Effective Exch. Rate	FRED (OECD)	`CCUSMA02EZQ618N`	Q	1	log	obs
POLRATE_EA	EA ECB Deposit Facility Rate (policy)	ECB SDW	`FM/D.U2.EUR.4F.KR.DFR.LEV`	D→Q	1	lin	obs
HPI_EA	EA Real House Price Index (HPI/HICP, 2015=100)	Eurostat	`prc_hpi_q Q.TOTAL.I15_Q.<geo>` (nominal HPI) ÷ `HICP_EA`	Q	1	log	obs; deflated by HICP
EACORP	EA Corporate Bond Yield (Bundesbank non-MFI)	Bundesbank	`BBSIS/M.I.UMR.RD.EUR.X2000.B.A.A.R.A.A._Z._Z.A`	D→Q	1	lin	obs; genuine non-MFI corporate bond yield
GLB_OIL	WTI crude oil, q-avg ($/bbl)	FRED	`WTISPLC`	M→Q	1	log	obs; global common
GLB_USLR	US 10Y Treasury Yield (%)	FRED	`DGS10`	D→Q	1	lin	obs; global common
GLB_USBAA	US Baa Corporate Yield (%)	FRED	`BAA`	M→Q	1	lin	obs; global common
GLB_USEQ	US Equity (S&P 500)	FRED	`SPASTT01USM661N`	M→Q	1	log	obs; global common
FREV_GDP	EA Revenue (% of GDP)	*see flag*	4q-annualized rev./GDP	Q	0	lin	mask; WN prior
FPEXP_GDP	EA Primary Expenditure (% of GDP)	*see flag*	4q-annualized (exp.−int.)/GDP	Q	0	lin	mask; WN prior
EFFI	EA Effective Interest Rate (% p.a.)	*see flag*	4×int./lag(debt)	Q	1	lin	obs; RW prior
SFA	EA Stock-Flow Adjustment (% of GDP)	derived	recursion residual	Q	0	lin	WN prior; closes the debt recursion

[FLAG: the EA fiscal flows (revenue, total expenditure, interest, gross debt, nominal GDP) are read from `pipeline/fiscal_raw/fiscal_EA.json`, which records no provenance; the consuming builders (`build_fiscal_drivers.py`, `build_fiscal_ea_4var.py`) do not fetch them. Upstream provider (Eurostat `gov_10q_ggnfa/gov_10dd_edpt1` is the most likely candidate) NOT verified from the code; series IDs left blank rather than fabricated.]

Variable (id)	Description	Source	Series ID/key	Freq	Fin.	Trans.	COVID / Notes
CREDIT_GDP_EA	**EA Credit to NFCs (% of GDP)**	FRED / BIS	`QXMNAM770A`	Q	1	lin	**mask** (GDP-denominator mask); BIS broad credit, 1999Q1=66.7, 2025Q4=103.1; *no new price (EACORP covers it)*

20.4 Japan (JP)

The Japan bloc carries $n = 22$ quarterly variables on a 2002Q2–2025Q4 grid ($T = 95$). Real-activity, price, rate, and exchange-rate series come from FRED; the policy rate (`POLRATE_JP`) and real house prices (`HP_JP`) are appended from the BIS (central-bank policy-rate dataflow and the real residential property-price dataflow). The new credit block adds BIS broad NFC credit (% of GDP) on the quantity side and a representative *bank lending rate* on the price side: `LENDR_JP` is the Bank of Japan average contract interest rate on new loans, fetched from the BoJ Time-Series Data Search REST API (not on FRED).

Table 57: Japan bloc variables. Columns as in Table 55.

Variable (id)	Description	Source	Series ID/key	Freq	Fin.	Trans.	COVID / Notes
GDP_JP	JP Real GDP	FRED	`JPNRGDPEXP`	Q	0	log	mask
CONS_JP	JP Consumption	FRED (OECD)	`NAEXKP02JPQ661S`	Q	0	log	mask
INV_JP	JP Investment	FRED (OECD)	`NAEXKP04JPQ661S`	Q	0	log	mask
GOV_JP	JP Gov. Spending	FRED (OECD)	`NAEXKP03JPQ661S`	Q	0	log	mask
EXP_JP	JP Exports	FRED (OECD)	`NAEXKP06JPQ661S`	Q	0	log	mask
IMP_JP	JP Imports	FRED (OECD)	`NAEXKP07JPQ661S`	Q	0	log	mask
LR_JP	JP Unemployment Rate	FRED (OECD)	`LRHUTTTTJPM156S`	M→Q	0	lin	mask
CPI_JP	JP CPI Index	FRED (OECD)	`JPNCPIALLMINMEI`	M→Q	0	log	mask
RATE_JP	JP 3M Interbank Rate	FRED (OECD)	`IR3TIB01JPQ156N`	Q	1	lin	obs

Variable (id)	Description	Source	Series ID/key	Freq	Fin.	Trans.	COVID / Notes
POLRATE_JP	JP Central Bank Policy Rate (BIS CBPOL)	BIS	`WS_CBPOL M.JP`	M→Q	1	lin	obs
YIELD10_JP	JP 10Y JGB Yield	FRED (OECD)	`IRLTLT01JPM156N`	M→Q	1	lin	obs
HP_JP	JP Real Resid. House-Price Idx (BIS RPP, 2010=100)	BIS	`WS_SPP Q.JP. R.628`	Q	1	log	obs
GLB_OIL	WTI crude oil, q-avg ($/bbl)	FRED	`WTISPLC`	M→Q	1	log	obs; global common
GLB_USLR	US 10Y Treasury Yield (%)	FRED	`DGS10`	D→Q	1	lin	obs; global common
GLB_USBAA	US Baa Corporate Yield (%)	FRED	`BAA`	M→Q	1	lin	obs; global common
GLB_USEQ	US Equity (S&P 500)	FRED	`SPASTT01USM661N`	M→Q	1	log	obs; global common
FREV_GDP	JP Revenue (% of GDP)	OECD SDMX + FRED	4q-annualized rev./GDP; nominal GDP `JPNNGDP`	Q	0	lin	mask; WN prior
FPEXP_GDP	JP Primary Expenditure (% of GDP)	OECD SDMX + FRED	4q-annualized (exp.−int.)/GDP	Q	0	lin	mask; WN prior
EFFI	JP Effective Interest Rate (% p.a.)	OECD SDMX + FRED	4×int./lag(gross debt, OECD F2/F3/F4)	Q	1	lin	obs; RW prior
SFA	JP Stock-Flow Adjustment (% of GDP)	derived	recursion residual	Q	0	lin	WN prior; COVID 2020Q1–Q4 set NaN (Gibbs-imputed)
CREDIT_GDP_JP	**JP Credit to NFCs (% of GDP)**	FRED / BIS	`QJPNAM770A`	Q	1	lin	**mask** (GDP-denominator mask); BIS broad credit, 2002Q2=111.0, 2025Q4=113.9
LENDR_JP	**JP Avg Contract Rate on New Loans (% p.a.)**	BoJ Time-Series API	`IR04/ DLLR2CIDBNL1`	M→Q	1	lin	**obs**; domestically-licensed banks, new loans total; NOT on FRED (dedicated BoJ fetcher)

20.5 United Kingdom (UK)

The UK bloc carries $n = 24$ quarterly variables on a 1997Q1–2025Q4 grid ($T = 116$). Real-activity, price, rate, industrial-production, NEER, policy-rate, and house-price series come from FRED (plus the BIS for the BIS-RPP house-price and CBPOL policy-rate concepts where used). The new credit block adds BIS broad NFC credit (% of GDP) on the quantity side and the Bank of England effective rate on new sterling loans to PNFCs (`LENDR_UK`) on the price side, fetched from the BoE Interactive Database (IADB) CSV API (not on FRED).

Table 58: UK bloc variables. Columns as in Table 55.

Variable (id)	Description	Source	Series ID/key	Freq	Fin.	Trans.	COVID / Notes
GDP_UK	UK Real GDP	FRED (ONS)	`NGDPRSAXDCGBQ`	Q	0	log	mask
CONS_UK	UK Consumption	FRED (OECD)	`NAEXKP02GBQ661S`	Q	0	log	mask
INV_UK	UK Investment	FRED (OECD)	`NAEXKP04GBQ661S`	Q	0	log	mask
GOV_UK	UK Gov. Spending	FRED (OECD)	`NAEXKP03GBQ661S`	Q	0	log	mask
EXP_UK	UK Exports	FRED (OECD)	`NAEXKP06GBQ661S`	Q	0	log	mask
IMP_UK	UK Imports	FRED (OECD)	`NAEXKP07GBQ661S`	Q	0	log	mask
IP_UK	UK Industrial Production Index	FRED (OECD)	`GBRPROINDMISMEI`	M→Q	0	log	mask
LR_UK	UK Unemployment Rate	FRED (OECD)	`LRHUTTTTGBM156S`	M→Q	0	lin	mask
CPI_UK	UK CPI Index	FRED (OECD)	`GBRCPIALLMINMEI`	M→Q	0	log	mask
RATE_UK	UK 3M Interbank Rate	FRED (OECD)	`IR3TIB01GBQ156N`	Q	1	lin	obs
POLRATE_UK	UK Central Bank Policy Rate (BIS CBPOL)	BIS	`WS_CBPOL M.GB`	M→Q	1	lin	obs
YIELD10_UK	UK 10Y Gilt Yield	FRED (OECD)	`IRLTLT01GBM156N`	M→Q	1	lin	obs
HP_UK	UK Real Resid. House-Price Idx (BIS RPP, 2010=100)	BIS	`WS_SPP Q.GB. R.628`	Q	1	log	obs
NEER_UK	UK Nominal Effective Exch. Rate	FRED (OECD)	`CCUSMA02GBQ618N`	Q	1	log	obs

Variable (id)	Description	Source	Series ID/key	Freq	Fin.	Trans.	COVID / Notes
GLB_OIL	WTI crude oil, q-avg ($/bbl)	FRED	`WTISPLC`	M→Q	1	log	obs; global common
GLB_USLR	US 10Y Treasury Yield (%)	FRED	`DGS10`	D→Q	1	lin	obs; global common
GLB_USBAA	US Baa Corporate Yield (%)	FRED	`BAA`	M→Q	1	lin	obs; global common
GLB_USEQ	US Equity (S&P 500)	FRED	`SPASTT01USM661N`	M→Q	1	log	obs; global common
FREV_GDP	UK Revenue (% of GDP)	*see flag*	4q-annualized rev./GDP	Q	0	lin	mask; WN prior
FPEXP_GDP	UK Primary Expenditure (% of GDP)	*see flag*	4q-annualized (exp.−int.)/GDP	Q	0	lin	mask; WN prior
EFFI	UK Effective Rate (% p.a.)	*see flag*	4×int./lag(debt)	Q	1	lin	obs; RW prior
SFA	UK Stock-Flow Adj. (% of GDP)	derived	recursion residual	Q	0	lin	WN prior; closes the debt recursion
[FLAG: the UK fiscal flows are read from `pipeline/fiscal_raw/fiscal_UK.json`*, which records no provenance; the consuming builders do not fetch them. Upstream provider (ONS public-sector finances / OECD QNA government finance) NOT verified from the code; series IDs left blank rather than fabricated. Note: an index-linked-gilt interest spike is documented separately in Section 17.6.]*							
CREDIT_GDP_UK	**UK Credit to NFCs (% of GDP)**	FRED / BIS	`QGBNAM770A`	Q	1	lin	**mask** (GDP-denominator mask); BIS broad credit, NSA, 1997Q1=51.5, 2025Q4=59.0
LENDR_UK	**UK Effective Rate on New Loans to PNFCs (% p.a.)**	Bank of England IADB	`CFMBJ82`	M→Q	1	lin	**obs**; new-business sterling lending rate; starts 2004Q1 (ragged lead, Gibbs-completed); NOT on FRED (dedicated BoE fetcher)

20.6 Canada (CA)

The Canada bloc carries $n = 24$ quarterly variables on a 1998Q1–2025Q4 grid ($T = 112$). Real-activity, price, rate, NEER, policy-rate, and house-price series come from FRED (with BIS for the CBPOL policy-rate and BIS-RPP house-price concepts). The new credit block adds BIS broad NFC credit (% of GDP) on the quantity side and the Bank of Canada weekly effective business interest rate (`LENDR_CA`) on the price side, fetched from the BoC Valet API (not on FRED).

Table 59: Canada bloc variables. Columns as in Table 55.

Variable (id)	Description	Source	Series ID/key	Freq	Fin.	Trans.	COVID / Notes
GDP_CA	CA Real GDP	FRED (OECD)	`NAEXKP01CAQ661S`	Q	0	log	mask
CONS_CA	CA Consumption	FRED (OECD)	`NAEXKP02CAQ661S`	Q	0	log	mask
INV_CA	CA Investment	FRED (OECD)	`NAEXKP04CAQ661S`	Q	0	log	mask
GOV_CA	CA Gov. Spending	FRED (OECD)	`NAEXKP03CAQ661S`	Q	0	log	mask
EXP_CA	CA Exports	FRED (OECD)	`NAEXKP06CAQ661S`	Q	0	log	mask
IMP_CA	CA Imports	FRED (OECD)	`NAEXKP07CAQ661S`	Q	0	log	mask
IP_CA	CA Industrial Production (Idx 2015=100)	FRED / StatCan	log-level IP (rebuilt)	Q	0	log	mask; rebuilt as a log-level index
LR_CA	CA Unemployment Rate	FRED (OECD)	`LRHUTTTTCAM156S`	M→Q	0	lin	mask
CPI_CA	CA CPI Index	FRED (OECD)	`CANCPIALLMINMEI`	M→Q	0	log	mask
RATE_CA	CA 3M Interbank Rate	FRED (OECD)	`IR3TIB01CAQ156N`	Q	1	lin	obs
POLRATE_CA	CA Central Bank Policy Rate (BIS CBPOL)	BIS	`WS_CBPOL M.CA`	M→Q	1	lin	obs
YIELD10_CA	CA 10Y Gov. Bond Yield	FRED (OECD)	`IRLTLT01CAM156N`	M→Q	1	lin	obs
HP_CA	CA Real Resid. House-Price Idx (BIS RPP, 2010=100)	BIS	`WS_SPP Q.CA. R.628`	Q	1	log	obs
NEER_CA	CA Nominal Effective Exch. Rate	FRED (OECD)	`CCUSMA02CAQ618N`	Q	1	log	obs
GLB_OIL	WTI crude oil, q-avg ($/bbl)	FRED	`WTISPLC`	M→Q	1	log	obs; global common
GLB_USLR	US 10Y Treasury Yield (%)	FRED	`DGS10`	D→Q	1	lin	obs; global common
GLB_USBAA	US Baa Corporate Yield (%)	FRED	`BAA`	M→Q	1	lin	obs; global common
GLB_USEQ	US Equity (S&P 500)	FRED	`SPASTT01USM661N`	M→Q	1	log	obs; global common
FREV_GDP	CA Revenue (% of GDP)	*see flag*	4q-annualized rev./GDP	Q	0	lin	mask; WN prior

Variable (id)	Description	Source	Series ID/key	Freq	Fin.	Trans.	COVID / Notes
FPEXP_GDP	CA Primary Expenditure (% of GDP)	*see flag*	4q-annualized (exp.−int.)/GDP	Q	0	lin	mask; WN prior
EFFI	CA Effective Interest Rate (% p.a.)	*see flag*	4×int./lag(debt)	Q	1	lin	obs; RW prior
SFA	CA Stock-Flow Adjustment (% of GDP)	derived	recursion residual	Q	0	lin	WN prior; closes the debt recursion

[FLAG: the CA fiscal flows are read from `pipeline/fiscal_raw/fiscal_CA.json`, which records no provenance; the consuming builders do not fetch them. Upstream provider (Statistics Canada government-finance / OECD QNA) NOT verified from the code; series IDs left blank rather than fabricated.]

Variable (id)	Description	Source	Series ID/key	Freq	Fin.	Trans.	COVID / Notes
CREDIT_GDP_CA	**CA Credit to NFCs (% of GDP)**	FRED / BIS	`QCANAM770A`	Q	1	lin	**mask** (GDP-denominator mask); BIS broad credit, NSA, 1998Q1=84.2, 2025Q4=118.3
LENDR_CA	**CA Bank Lending Rate (effective business rate, % p.a.)**	Bank of Canada Valet	`BUSEFFRATE`	W→Q	1	lin	**obs**; weekly effective business interest rate, q-avg; starts 1999Q1; NOT on FRED (dedicated BoC fetcher)

20.7 China (CN)

The China bloc carries $n = 19$ quarterly variables on a 1999Q1–2025Q4 grid ($T = 108$). Because clean quarterly *real* GDP is not available for China, real activity is anchored by official quarterly *nominal* GDP plus trade volumes and the BVAR/CPI separation; all domestic series are genuine FRED series (no placeholders). The policy rate (`POLRATE_CN`) and real property prices (`HP_CN`) are appended from the BIS. The new credit block adds BIS broad NFC credit (% of GDP) on the quantity side; on the price side `CORP_CN` is a *sovereign 10Y yield used as a corporate-price proxy* — **[FLAG: CN price is a SOVEREIGN 10Y benchmark government yield, not a corporate yield or NFC lending rate; no corporate spread exists for China]**.

Table 60: China bloc variables. Columns as in Table 55.

Variable (id)	Description	Source	Series ID/key	Freq	Fin.	Trans.	COVID / Notes
GDP_CN	CN Real GDP (quarterly, chain-linked SA volume index, base 2011Q1=100)	OECD QNA via DB-nomics (NBS)	`DSD_NAMAIN1@DF_QNA..CHN...B1GQ`	Q	0	log	obs; real chain-linked volume (2011Q1+ genuine; 1999–2010 genuine annual real growth, Denton-disaggregated)
CPI_CN	CN CPI	FRED (OECD)	`CHNCPIALLMINMEI`	M→Q	0	log	mask
RATE_CN	CN 3M Interbank Rate	FRED (OECD)	`IR3TIB01CNM156N`	M→Q	1	lin	obs
EXP_CN	CN Exports (USD)	FRED (OECD)	`XTEXVA01CNM667S`	M→Q	0	log	mask
IMP_CN	CN Imports (USD)	FRED (OECD)	`XTIMVA01CNM667S`	M→Q	0	log	mask
REER_CN	CN Real Effective FX	FRED (BIS)	`RBCNBIS`	M→Q	1	log	obs
EQ_CN	Shanghai Share Prices	FRED (OECD)	`SPASTT01CNM661N`	M→Q	1	log	obs
POLRATE_CN	CN Policy Rate (BIS CBPOL)	BIS	`WS_CBPOL M.CN`	M→Q	1	lin	obs
HP_CN	CN Real Resid. Property Price (BIS)	FRED (BIS)	`QCNR628BIS`	Q	1	log	obs
GLB_OIL	WTI crude oil, q-avg ($/bbl)	FRED	`WTISPLC`	M→Q	1	log	obs; global common
GLB_USLR	US 10Y Treasury Yield (%)	FRED	`DGS10`	D→Q	1	lin	obs; global common
GLB_USBAA	US Baa Corporate Yield (%)	FRED	`BAA`	M→Q	1	lin	obs; global common
GLB_USEQ	US Equity (S&P 500)	FRED	`SPASTT01USM661N`	M→Q	1	log	obs; global common
FREV_GDP	CN Revenue (% of GDP)	IMF WEO (2025-04)	`CHN.GGR.national_currency` (annual)÷GDP	A→Q	0	lin	mask; WN prior; annual, temporally disaggregated
FPEXP_GDP	CN Primary Expenditure (% of GDP)	IMF WEO (2025-04)	`CHN.GGX` less derived interest, ÷GDP	A→Q	0	lin	mask; WN prior
EFFI	CN Effective Interest Rate (% p.a.)	IMF WEO + BIS debt	4×int./lag(`QCNGANXDCA`)	Q	1	lin	obs; RW prior; interest derived as overall−primary balance

Variable (id)	Description	Source	Series ID/key	Freq	Fin.	Trans.	COVID / Notes
SFA	CN Stock-Flow Adjustment (% of GDP)	derived	recursion residual	Q	0	lin	WN prior; gross debt = BIS GG credit `QCNGANXDCA`
CREDIT_GDP_CN	**CN Credit to NFCs (% of GDP)**	FRED / BIS	`QCNNAM770A`	Q	1	lin	**mask** (GDP-denominator mask); BIS broad credit, NSA, starts 2006Q1=99.4 (leading NaNs pre-2006), 2025Q4=142.8
CORP_CN	**CN 10Y Government Bond Yield (sovereign proxy)**	OECD KEI (SDMX)	`DSD_KEI@DF_KEI / CHN.M.IRLT`	M→Q	1	lin	**obs**; *[FLAG: SOVEREIGN 10Y proxy, no corporate spread]*; starts 2014Q1=4.48 (leading NaNs pre-2014), 2025Q4=1.83

20.8 Euro-core single-country BVARs (AT, BE, DE, FR, NL)

The five euro-core economies are estimated as *standalone* single-country Bayesian VARs (not GVAR blocs): the rest-of-euro-area and the area-wide rates enter as a weakly-exogenous foreign block rather than through trade-weighted stars. All real macro aggregates are Eurostat-sourced (the redesigned `namq_10_gdp` / `prc_hicp_midx` / `une_rt_q` / `irt_*` / `ert_eff_ic_q` pulls, current to 2025Q4); the financial block (ECB Deposit Facility Rate, the EA all-bonds 10Y yield, the 3M EURIBOR, the Bund and the sovereign spread) is fetched from the ECB Data Portal; the fiscal block is built from Eurostat `gov_10q` on the 4-variable {revenue, primary expenditure, effective rate, stock-flow adjustment} recipe; and the credit block (CREDIT_GDP_<CC>, LENDR_<CC>) is appended last from BIS-via-FRED and the ECB MIR statistics.

Transformation conventions. Throughout, `log` denotes the model space $100 \cdot \log(\text{level})$; such a variable enters growth-rate reporting as $4 \cdot \Delta(100 \log)$ = quarter-on-quarter annualised growth (§12.3). `lin` denotes the raw level (a rate in % p.a., a ratio in % of GDP, or an index read in levels). The own-lag prior is random-walk (`RW`) for every macro and financial variable; the fiscal flow ratios (`FREV_GDP`, `FPEXP_GDP`, `SFA`) carry the stationary white-noise prior (`WN`).

COVID treatment. Under the COVID-as-missing convention only the non-financial macro columns (`isFinancial`= 0: the six national-accounts aggregates, HICP, the unemployment rate, and the two rest-of-EA macro series) are masked (`NaN`) over the pandemic window and imputed by the COVID-as-missing Durbin–Koopman smoother; financial columns are kept observed. *[FLAG: window mismatch.] Two distinct COVID windows coexist in these builds. The macro/ROW block (and the fiscal `SFA`) is masked over the* four *quarters* ***2020Q1–Q4*** *— the self-calibrated window* `build_euro_member_redesign.py` *selects as the largest* $100\log$*(GDP) drop in 2019Q4–2021Q1, i.e.* $trough - 1 \dots trough + 2$ *— whereas the newly-appended credit quantity* `CREDIT_GDP_<CC>` *is masked over the* three *quarters* ***2020Q2–Q4*** *(the hard-coded* `COVID_MASK` *in* `build_credit_eurocore.py`*). The two masks therefore differ by one quarter (2020Q1). All financial columns (rates, spreads, EA-CORP, EFFI, house prices, and the credit price* `LENDR_<CC>`*) remain observed throughout.* The fiscal flow ratios `FREV_GDP`, `FPEXP_GDP` and the effective rate `EFFI` are *not* masked: they are interpolated/forward-filled to a NaN-free column before estimation; only the derived `SFA` inherits the four-quarter macro mask.

Imputation of ragged edges. Financial and exogenous columns are leading-back-filled (first finite value carried back) and forward-filled so every column is NaN-free over its estimation window; the national-accounts/HICP/unemployment columns drive the common contiguous window. The credit quantity has no leading/trailing gaps (BIS history predates every base start); its only missing cells are the three COVID-masked quarters.

The credit block (NEW). Each economy gains exactly two domestic financial variables appended after all existing columns (`isFinancial`= 1, `lin`, `RW`, `block=domestic`):

- **Quantity** `CREDIT_GDP_<CC>` = BIS "Total credit to non-financial corporations, % of GDP" (break-adjusted *broad* credit: bank loans *plus* debt securities *plus* cross-border lending), pulled via FRED under the uniform mnemonic `Q<CC>NAM770A`. Native quarterly (FRED dates the observations at the quarter start). COVID-masked 2020Q2–Q4 (it inherits the collapse of the GDP denominator).
- **Price** `LENDR_<CC>` = the ECB MIR (MFI Interest Rate statistics) NFC *new-business* bank lending rate, SDMX key `M.<CC>.B.A2A.A.R.A.2240.EUR.N` (title "Bank interest rates — loans to corporations (new business)"; S.11 NFC counterparty, AAR/NDER, all-maturity, EUR, new-business coverage). Monthly → quarter average; observed through COVID.

Because the shipped quantity is the BIS *broad* total-credit series, the euro-core members carry the canonical BIS measure (not a bank-loans-only proxy), and the credit price is a true national NFC new-business lending rate (not a sovereign-yield or corporate-spread proxy). The only price-coverage caveat is Belgium (see below).

20.9 Austria (AT)

A 23-variable single-country BVAR estimated over **2009Q1–2025Q4** ($T = 68$). COVID-as-missing: macro block masked 2020Q1–Q4, credit quantity masked 2020Q2–Q4, financials observed (§20.8).

Table 61: Austria (AT) — variable documentation.

Variable (id)	Description	Source	Series ID/key	Freq	Transf.	COVID	Notes
GDP_AT	Real GDP (chain-linked vol., SCA)	Eurostat	namq_10_gdp Q. CLV15_MEUR. SCA.B1GQ.AT	Q	log	masked 20Q1–Q4	domestic
CONS_AT	Real household final consumption	Eurostat	namq_10_gdp Q. CLV15_MEUR. SCA. P31_S14_S15. AT	Q	log	masked 20Q1–Q4	domestic
INV_AT	Real gross fixed capital formation	Eurostat	namq_10_gdp Q. CLV15_MEUR. SCA.P51G.AT	Q	log	masked 20Q1–Q4	domestic
GOV_AT	Real government consumption	Eurostat	namq_10_gdp Q. CLV15_MEUR. SCA.P3_S13. AT	Q	log	masked 20Q1–Q4	domestic
EXP_AT	Real exports	Eurostat	namq_10_gdp Q. CLV15_MEUR. SCA.P6.AT	Q	log	masked 20Q1–Q4	domestic
IMP_AT	Real imports	Eurostat	namq_10_gdp Q. CLV15_MEUR. SCA.P7.AT	Q	log	masked 20Q1–Q4	domestic
HICP_AT	HICP index (all-items, 2015=100)	Eurostat	prc_hicp_midx M.I15.CP00. AT	M→Q avg	log	masked 20Q1–Q4	domestic
LR_AT	Unemployment rate (SA, % active pop.)	Eurostat	une_rt_q Q. SA.TOTAL. PC_ACT.T.AT	Q	lin	masked 20Q1–Q4	domestic

continued on next page

Table 61 (Austria), continued

Variable (id)	Description	Source	Series ID/key	Freq	Transf.	COVID	Notes
`HPI_AT`	Real house-price index (nominal HPI/HICP, 2015=100)	Eurostat	`prc_hpi_q Q.TOTAL.I15_Q.AT` ÷ HICP	Q	log	observed (financial)	domestic; nominal HPI deflated by HICP
`NEER_AT`	Nominal effective exchange rate, IC42 partners	Eurostat	`ert_eff_ic_q Q.NEER_IC42.I15.AT`	Q	log	observed	domestic; broad 42-partner group
`SPREAD_AT`	10Y sovereign spread vs Bund	Eurostat (computed)	`irt_lt_mcby_m M.MCBY.AT` − `M.MCBY.DE`	M→Q avg	lin	observed	domestic; own 10Y minus Bund
`POLRATE_AT`	ECB Deposit Facility Rate (policy)	ECB Data Portal	`FM/D.U2.EUR.4F.KR.DFR.LEV`	D→Q avg	lin	observed	external (shared)
`EACORP`	EA corporate bond yield (non-MFI)	Bundesbank	`BBSIS/M.I.UMR.RD.EUR.X2000.B.A.A.R.A.A._Z._Z.A`	D→Q avg	lin	observed	external; IG-corp risk proxy
`RATE_AT`	3-month EURIBOR	Eurostat	`irt_st_q Q.IRT_M3.EA`	Q	lin	observed	external (shared)
`YIELD10_DE`	DE 10Y Bund yield (spread base)	Eurostat	`irt_lt_mcby_m M.MCBY.DE`	M→Q avg	lin	observed	external (injected)
`ROWEA_GDP_AT`	Rest-of-EA real GDP (GDP-weighted de-agg.)	Eurostat (computed)	EA20 less AT, `namq_10_gdp`	Q	log	masked 20Q1–Q4	row

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Table 61 (Austria), continued

Variable (id)	Description	Source	Series ID/key	Freq	Transf.	COVID	Notes
`ROWEA_HICP_AT`	Rest-of-EA HICP (GDP-weighted de-agg.)	Eurostat (computed)	EA20 less AT, `prc_hicp_midx`	Q	log	masked 20Q1–Q4	row
`FREV_GDP`	General-gov. revenue (% GDP, 4q-trailing)	Eurostat	`gov_10q_ggnfa Q.MIO_EUR. NSA.S13.TR. AT`	Q	lin	observed (forward-filled)	fiscal; WN prior
`FPEXP_GDP`	Primary expenditure (% GDP, 4q-trailing)	Eurostat	`gov_10q_ggnfa ...TE − ... D41PAY .AT`	Q	lin	observed (forward-filled)	fiscal; WN prior
`EFFI`	Effective interest rate on gov. debt	Eurostat (computed)	`...D41PAY / gov_10q_ggdebt ...GD.AT`	Q	lin	observed	fiscal; financial
`SFA`	Stock-flow adjustment (budget-identity residual)	Eurostat (computed)	debt-recursion residual	Q	lin	masked 20Q1–Q4	fiscal; WN prior
`CREDIT_GDP_AT`	**[NEW]** BIS total credit to NFCs, % of GDP (broad: loans+bonds+cross-border)	BIS via FRED	`QATNAM770A`	Q	lin	masked 20Q2–Q4	domestic; src 1995Q4:69.7.. 2025Q4:83.9
`LENDR_AT`	**[NEW]** NFC new-business bank lending rate	ECB MIR	`M.AT.B.A2A. A.R.A.2240. EUR.N`	M→Q avg	lin	observed	domestic; src 2000Q1:4.95.. 2026Q2:3.57

20.10 Belgium (BE)

A 23-variable single-country BVAR estimated over **2009Q1–2025Q4** ($T = 68$). COVID-as-missing: macro block masked 2020Q1–Q4, credit quantity masked 2020Q2–Q4, financials observed.

Table 62: Belgium (BE) — variable documentation.

Variable (id)	Description	Source	Series ID/key	Freq	Transf.	COVID	Notes
`GDP_BE`	Real GDP (chain-linked vol., SCA)	Eurostat	`namq_10_gdp Q. CLV15_MEUR. SCA.B1GQ.BE`	Q	log	masked 20Q1–Q4	domestic
`CONS_BE`	Real household final consumption	Eurostat	`namq_10_gdp ... P31_S14_S15. BE`	Q	log	masked 20Q1–Q4	domestic
`INV_BE`	Real gross fixed capital formation	Eurostat	`namq_10_gdp ...P51G.BE`	Q	log	masked 20Q1–Q4	domestic
`GOV_BE`	Real government consumption	Eurostat	`namq_10_gdp ...P3_S13. BE`	Q	log	masked 20Q1–Q4	domestic
`EXP_BE`	Real exports	Eurostat	`namq_10_gdp ...P6.BE`	Q	log	masked 20Q1–Q4	domestic
`IMP_BE`	Real imports	Eurostat	`namq_10_gdp ...P7.BE`	Q	log	masked 20Q1–Q4	domestic
`HICP_BE`	HICP index (all-items, 2015=100)	Eurostat	`prc_hicp_midx M.I15.CP00. BE`	M→Q avg	log	masked 20Q1–Q4	domestic
`LR_BE`	Unemployment rate (SA, % active pop.)	Eurostat	`une_rt_q Q. SA.TOTAL. PC_ACT.T.BE`	Q	lin	masked 20Q1–Q4	domestic
`HPI_BE`	Real house-price index (nominal HPI/HICP, 2015=100)	Eurostat	`prc_hpi_q Q. TOTAL.I15_Q. BE` ÷ HICP	Q	log	observed (financial)	domestic
`NEER_BE`	Nominal effective exchange rate, IC42 partners	Eurostat	`ert_eff_ic_q Q.NEER_IC42. I15.BE`	Q	log	observed	domestic

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Table 62 (Belgium), continued

Variable (id)	Description	Source	Series ID/key	Freq	Transf.	COVID	Notes
SPREAD_BE	10Y sovereign spread vs Bund	Eurostat (computed)	irt_lt_mcby_m M.MCBY.BE − M.MCBY.DE	M→Q avg	lin	observed	domestic
POLRATE_BE	ECB Deposit Facility Rate (policy)	ECB Data Portal	FM/D.U2.EUR.4F.KR.DFR.LEV	D→Q avg	lin	observed	external (shared)
EACORP	EA corporate bond yield (non-MFI)	Bundesbank	BBSIS/M.I.UMR.RD.EUR.X2000.B.A.A.R.A.A._Z._Z.A	D→Q avg	lin	observed	external
RATE_BE	3-month EURIBOR	Eurostat	irt_st_q Q.IRT_M3.EA	Q	lin	observed	external (shared)
YIELD10_DE	DE 10Y Bund yield (spread base)	Eurostat	irt_lt_mcby_m M.MCBY.DE	M→Q avg	lin	observed	external (injected)
ROWEA_GDP_BE	Rest-of-EA real GDP (GDP-weighted de-agg.)	Eurostat (computed)	EA20 less BE, namq_10_gdp	Q	log	masked 20Q1–Q4	row
ROWEA_HICP_BE	Rest-of-EA HICP (GDP-weighted de-agg.)	Eurostat (computed)	EA20 less BE, prc_hicp_midx	Q	log	masked 20Q1–Q4	row
FREV_GDP	General-gov. revenue (% GDP, 4q-trailing)	Eurostat	gov_10q_ggnfa Q.MIO_EUR.NSA.S13.TR.BE	Q	lin	observed (forward-filled)	fiscal; WN
FPEXP_GDP	Primary expenditure (% GDP, 4q-trailing)	Eurostat	gov_10q_ggnfa ...TE − ...D41PAY .BE	Q	lin	observed (forward-filled)	fiscal; WN
EFFI	Effective interest rate on gov. debt	Eurostat (computed)	...D41PAY / gov_10q_ggdebt ...GD.BE	Q	lin	observed	fiscal; financial

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Table 62 (Belgium), continued

Variable (id)	Description	Source	Series ID/key	Freq	Transf.	COVID	Notes
`SFA`	Stock-flow adjustment (budget-identity residual)	Eurostat (computed)	debt-recursion residual	Q	lin	masked 20Q1–Q4	fiscal; WN
`CREDIT_GDP_BE`	**[NEW]** BIS total credit to NFCs, % of GDP (broad)	BIS via FRED	`QBENAM770A`	Q	lin	masked 20Q2–Q4	domestic; src 1990Q1:55.5..2025Q4:117.9
`LENDR_BE`	**[NEW]** NFC new-business bank lending rate	ECB MIR	`M.BE.B.A2A.A.R.A.2240.EUR.N`	M→Q avg	lin	observed	domestic; *[FLAG: price coverage starts 2003Q1, not 2000Q1 as for the others; src 2003Q1:3.66..2026Q2:3.48 — earliest pre-2009 values are back-filled.]*

20.11 Germany (DE)

A 22-variable single-country BVAR estimated over **2009Q1–2025Q4** ($T = 68$). Germany is the spread *base*: it carries the Bund 10Y yield `YIELD10_DE` as an *endogenous domestic* variable and therefore has *no* `SPREAD_DE` variable (hence $n = 22$, one fewer than the other euro-core members). COVID-as-missing: macro block masked 2020Q1–Q4, credit quantity masked 2020Q2–Q4, financials observed.

Table 63: Germany (DE) — variable documentation.

Variable (id)	Description	Source	Series ID/key	Freq	Transf.	COVID	Notes
GDP_DE	Real GDP (chain-linked vol., SCA)	Eurostat	namq_10_gdp Q. CLV15_MEUR. SCA.B1GQ.DE	Q	log	masked 20Q1–Q4	domestic
CONS_DE	Real household final consumption	Eurostat	namq_10_gdp ... P31_S14_S15. DE	Q	log	masked 20Q1–Q4	domestic
INV_DE	Real gross fixed capital formation	Eurostat	namq_10_gdp ...P51G.DE	Q	log	masked 20Q1–Q4	domestic
GOV_DE	Real government consumption	Eurostat	namq_10_gdp ...P3_S13. DE	Q	log	masked 20Q1–Q4	domestic
EXP_DE	Real exports	Eurostat	namq_10_gdp ...P6.DE	Q	log	masked 20Q1–Q4	domestic
IMP_DE	Real imports	Eurostat	namq_10_gdp ...P7.DE	Q	log	masked 20Q1–Q4	domestic
HICP_DE	HICP index (all-items, 2015=100)	Eurostat	prc_hicp_midx M.I15.CP00. DE	M→Q avg	log	masked 20Q1–Q4	domestic
LR_DE	Unemployment rate (SA, % active pop.)	Eurostat	une_rt_q Q. SA.TOTAL. PC_ACT.T.DE	Q	lin	masked 20Q1–Q4	domestic
HPI_DE	Real house-price index (nominal HPI/HICP, 2015=100)	Eurostat	prc_hpi_q Q. TOTAL.I15_Q. DE ÷ HICP	Q	log	observed (financial)	domestic
NEER_DE	Nominal effective exchange rate, IC42 partners	Eurostat	ert_eff_ic_q Q.NEER_IC42. I15.DE	Q	log	observed	domestic

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Table 63 (Germany), continued

Variable (id)	Description	Source	Series ID/key	Freq	Transf.	COVID	Notes
`YIELD10_DE`	10Y Bund yield (the spread base)	Eurostat	`irt_lt_mcby_m M.MCBY.DE`	M→Q avg	lin	observed	**domestic** (DE special case; no SPREAD_DE)
`POLRATE_DE`	ECB Deposit Facility Rate (policy)	ECB Data Portal	`FM/D.U2. EUR.4F.KR. DFR.LEV`	D→Q avg	lin	observed	external (shared)
`EACORP`	EA corporate bond yield (non-MFI)	Bundesbank	`BBSIS/M. I.UMR.RD. EUR.X2000. B.A.A. R.A.A. _Z._Z.A`	D→Q avg	lin	observed	external
`RATE_DE`	3-month EURIBOR	Eurostat	`irt_st_q Q. IRT_M3.EA`	Q	lin	observed	external (shared)
`ROWEA_GDP_DE`	Rest-of-EA real GDP (GDP-weighted de-agg.)	Eurostat (computed)	EA20 less DE, `namq_10_gdp`	Q	log	masked 20Q1–Q4	row
`ROWEA_HICP_DE`	Rest-of-EA HICP (GDP-weighted de-agg.)	Eurostat (computed)	EA20 less DE, `prc_hicp_midx`	Q	log	masked 20Q1–Q4	row
`FREV_GDP`	General-gov. revenue (% GDP, 4q-trailing)	Eurostat	`gov_10q_ggnfa Q.MIO_EUR. NSA.S13.TR. DE`	Q	lin	observed (forward-filled)	fiscal; WN
`FPEXP_GDP`	Primary expenditure (% GDP, 4q-trailing)	Eurostat	`gov_10q_ggnfa ...TE − ... D41PAY .DE`	Q	lin	observed (forward-filled)	fiscal; WN
`EFFI`	Effective interest rate on gov. debt	Eurostat (computed)	`...D41PAY / gov_10q_ggdebt ...GD.DE`	Q	lin	observed	fiscal; financial
`SFA`	Stock-flow adjustment (budget-identity residual)	Eurostat (computed)	debt-recursion residual	Q	lin	masked 20Q1–Q4	fiscal; WN

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Table 63 (Germany), continued

Variable (id)	Description	Source	Series ID/key	Freq	Transf.	COVID	Notes
CREDIT_GDP_DE	**[NEW]** BIS total credit to NFCs, % of GDP (broad)	BIS via FRED	QDENAM770A	Q	lin	masked 20Q2–Q4	domestic; src 1990Q1:59.5..2025Q4:87.8
LENDR_DE	**[NEW]** NFC new-business bank lending rate	ECB MIR	M.DE.B.A2A.A.R.A.2240.EUR.N	M→Q avg	lin	observed	domestic; src 2000Q1:5.85..2026Q2:3.55

20.12 France (FR)

A 23-variable single-country BVAR estimated over **2003Q1–2025Q4** ($T = 92$) — the longest euro-core sample (the others start 2009Q1). COVID-as-missing: macro block masked 2020Q1–Q4, credit quantity masked 2020Q2–Q4, financials observed.

Table 64: France (FR) — variable documentation.

Variable (id)	Description	Source	Series ID/key	Freq	Transf.	COVID	Notes
GDP_FR	Real GDP (chain-linked vol., SCA)	Eurostat	namq_10_gdp Q.CLV15_MEUR.SCA.B1GQ.FR	Q	log	masked 20Q1–Q4	domestic
CONS_FR	Real household final consumption	Eurostat	namq_10_gdp ...P31_S14_S15.FR	Q	log	masked 20Q1–Q4	domestic
INV_FR	Real gross fixed capital formation	Eurostat	namq_10_gdp ...P51G.FR	Q	log	masked 20Q1–Q4	domestic
GOV_FR	Real government consumption	Eurostat	namq_10_gdp ...P3_S13.FR	Q	log	masked 20Q1–Q4	domestic

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Table 64 (France), continued

Variable (id)	Description	Source	Series ID/key	Freq	Transf.	COVID	Notes
`EXP_FR`	Real exports	Eurostat	`namq_10_gdp ...P6.FR`	Q	log	masked 20Q1–Q4	domestic
`IMP_FR`	Real imports	Eurostat	`namq_10_gdp ...P7.FR`	Q	log	masked 20Q1–Q4	domestic
`HICP_FR`	HICP index (all-items, 2015=100)	Eurostat	`prc_hicp_midx M.I15.CP00.FR`	M→Q avg	log	masked 20Q1–Q4	domestic
`LR_FR`	Unemployment rate (SA, % active pop.)	Eurostat	`une_rt_q Q.SA.TOTAL.PC_ACT.T.FR`	Q	lin	masked 20Q1–Q4	domestic
`HPI_FR`	Real house-price index (nominal HPI/HICP, 2015=100)	Eurostat	`prc_hpi_q Q.TOTAL.I15_Q.FR` ÷ HICP	Q	log	observed (financial)	domestic
`NEER_FR`	Nominal effective exchange rate, IC42 partners	Eurostat	`ert_eff_ic_q Q.NEER_IC42.I15.FR`	Q	log	observed	domestic
`SPREAD_FR`	10Y sovereign spread vs Bund	Eurostat (computed)	`irt_lt_mcby_m M.MCBY.FR` − `M.MCBY.DE`	M→Q avg	lin	observed	domestic
`POLRATE_FR`	ECB Deposit Facility Rate (policy)	ECB Data Portal	`FM/D.U2.EUR.4F.KR.DFR.LEV`	D→Q avg	lin	observed	external (shared)
`EACORP`	EA corporate bond yield (non-MFI)	Bundesbank	`BBSIS/M.I.UMR.RD.EUR.X2000.B.A.A.R.A.A._Z._Z.A`	D→Q avg	lin	observed	external
`RATE_FR`	3-month EURIBOR	Eurostat	`irt_st_q Q.IRT_M3.EA`	Q	lin	observed	external (shared)

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Table 64 (France), continued

Variable (id)	Description	Source	Series ID/key	Freq	Transf.	COVID	Notes
YIELD10_DE	DE 10Y Bund yield (spread base)	Eurostat	irt_lt_mcby_m M.MCBY.DE	M→Q avg	lin	observed	external (injected)
ROWEA_GDP_FR	Rest-of-EA real GDP (GDP-weighted de-agg.)	Eurostat (computed)	EA20 less FR, namq_10_gdp	Q	log	masked 20Q1–Q4	row
ROWEA_HICP_FR	Rest-of-EA HICP (GDP-weighted de-agg.)	Eurostat (computed)	EA20 less FR, prc_hicp_midx	Q	log	masked 20Q1–Q4	row
FREV_GDP	General-gov. revenue (% GDP, 4q-trailing)	Eurostat	gov_10q_ggnfa Q.MIO_EUR. NSA.S13.TR. FR	Q	lin	observed (forward-filled)	fiscal; WN
FPEXP_GDP	Primary expenditure (% GDP, 4q-trailing)	Eurostat	gov_10q_ggnfa ...TE − ... D41PAY .FR	Q	lin	observed (forward-filled)	fiscal; WN
EFFI	Effective interest rate on gov. debt	Eurostat (computed)	...D41PAY / gov_10q_ggdebt ...GD.FR	Q	lin	observed	fiscal; financial
SFA	Stock-flow adjustment (budget-identity residual)	Eurostat (computed)	debt-recursion residual	Q	lin	masked 20Q1–Q4	fiscal; WN

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Table 64 (France), continued

Variable (id)	Description	Source	Series ID/key	Freq	Transf.	COVID	Notes
`CREDIT_GDP_FR`	**[NEW]** BIS total credit to NFCs, % of GDP (broad: loans+bonds+cross-border)	BIS via FRED	`QFRNAM770A`	Q	lin	masked 20Q2–Q4	domestic; src 1990Q1:90.8.. 2025Q4:155.8. Broad BIS measure *includes* France's large bond-financed NFC debt (the bank-loans-only alternative was *not* used)
`LENDR_FR`	**[NEW]** NFC new-business bank lending rate	ECB MIR	`M.FR.B.A2A.A.R.A.2240.EUR.N`	M→Q avg	lin	observed	domestic; src 2000Q1:4.01.. 2026Q2:3.56

20.13 Netherlands (NL)

A 23-variable single-country BVAR estimated over **2009Q1–2025Q4** ($T = 68$). COVID-as-missing: macro block masked 2020Q1–Q4, credit quantity masked 2020Q2–Q4, financials observed.

Table 65: Netherlands (NL) — variable documentation.

Variable (id)	Description	Source	Series ID/key	Freq	Transf.	COVID	Notes
`GDP_NL`	Real GDP (chain-linked vol., SCA)	Eurostat	`namq_10_gdp Q. CLV15_MEUR. SCA.B1GQ.NL`	Q	log	masked 20Q1–Q4	domestic
`CONS_NL`	Real household final consumption	Eurostat	`namq_10_gdp ... P31_S14_S15. NL`	Q	log	masked 20Q1–Q4	domestic
`INV_NL`	Real gross fixed capital formation	Eurostat	`namq_10_gdp ...P51G.NL`	Q	log	masked 20Q1–Q4	domestic
`GOV_NL`	Real government consumption	Eurostat	`namq_10_gdp ...P3_S13. NL`	Q	log	masked 20Q1–Q4	domestic
`EXP_NL`	Real exports	Eurostat	`namq_10_gdp ...P6.NL`	Q	log	masked 20Q1–Q4	domestic
`IMP_NL`	Real imports	Eurostat	`namq_10_gdp ...P7.NL`	Q	log	masked 20Q1–Q4	domestic
`HICP_NL`	HICP index (all-items, 2015=100)	Eurostat	`prc_hicp_midx M.I15.CP00. NL`	M→Q avg	log	masked 20Q1–Q4	domestic
`LR_NL`	Unemployment rate (SA, % active pop.)	Eurostat	`une_rt_q Q. SA.TOTAL. PC_ACT.T.NL`	Q	lin	masked 20Q1–Q4	domestic
`HPI_NL`	Real house-price index (nominal HPI/HICP, 2015=100)	Eurostat	`prc_hpi_q Q. TOTAL.I15_Q. NL` ÷ HICP	Q	log	observed (financial)	domestic
`NEER_NL`	Nominal effective exchange rate, IC42 partners	Eurostat	`ert_eff_ic_q Q.NEER_IC42. I15.NL`	Q	log	observed	domestic

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Table 65 (Netherlands), continued

Variable (id)	Description	Source	Series ID/key	Freq	Transf.	COVID	Notes
SPREAD_NL	10Y sovereign spread vs Bund	Eurostat (computed)	irt_lt_mcby_m M.MCBY.NL − M.MCBY.DE	M→Q avg	lin	observed	domestic
POLRATE_NL	ECB Deposit Facility Rate (policy)	ECB Data Portal	FM/D.U2. EUR.4F.KR. DFR.LEV	D→Q avg	lin	observed	external (shared)
EACORP	EA corporate bond yield (non-MFI)	Bundesbank	BBSIS/M. I.UMR.RD. EUR.X2000. B.A.A. R.A.A. _Z._Z.A	D→Q avg	lin	observed	external
RATE_NL	3-month EURIBOR	Eurostat	irt_st_q Q. IRT_M3.EA	Q	lin	observed	external (shared)
YIELD10_DE	DE 10Y Bund yield (spread base)	Eurostat	irt_lt_mcby_m M.MCBY.DE	M→Q avg	lin	observed	external (injected)
ROWEA_GDP_NL	Rest-of-EA real GDP (GDP-weighted de-agg.)	Eurostat (computed)	EA20 less NL, namq_10_gdp	Q	log	masked 20Q1–Q4	row
ROWEA_HICP_NL	Rest-of-EA HICP (GDP-weighted de-agg.)	Eurostat (computed)	EA20 less NL, prc_hicp_midx	Q	log	masked 20Q1–Q4	row
FREV_GDP	General-gov. revenue (% GDP, 4q-trailing)	Eurostat	gov_10q_ggnfa Q.MIO_EUR. NSA.S13.TR. NL	Q	lin	observed (forward-filled)	fiscal; WN
FPEXP_GDP	Primary expenditure (% GDP, 4q-trailing)	Eurostat	gov_10q_ggnfa ...TE − ... D41PAY .NL	Q	lin	observed (forward-filled)	fiscal; WN
EFFI	Effective interest rate on gov. debt	Eurostat (computed)	...D41PAY / gov_10q_ggdebt ...GD.NL	Q	lin	observed	fiscal; financial

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Table 65 (Netherlands), continued

Variable (id)	Description	Source	Series ID/key	Freq	Transf.	COVID	Notes
SFA	Stock-flow adjustment (budget-identity residual)	Eurostat (computed)	debt-recursion residual	Q	lin	masked 20Q1–Q4	fiscal; WN
CREDIT_GDP_NL	**[NEW]** BIS total credit to NFCs, % of GDP (broad)	BIS via FRED	QNLNAM770A	Q	lin	masked 20Q2–Q4	domestic; src 1990Q4:152.6..2025Q4:166.3
LENDR_NL	**[NEW]** NFC new-business bank lending rate	ECB MIR	M.NL.B.A2A.A.R.A.2240.EUR.N	M→Q avg	lin	observed	domestic; src 2000Q1:6.17..2026Q2:3.37

20.14 Euro-area periphery: data conventions (ES, IT, IE, PT, GR)

Transformation and COVID conventions (all five blocs). Two transformation codes appear in the staged specs. log denotes $100 \cdot \log(\text{level})$; because the VAR enters in (annualised) growth space, a log variable contributes $4 \cdot \Delta[100 \log]$, i.e. the quarter-on-quarter *annualised* growth rate. lin denotes the raw level (rates in % p.a., ratios in % of GDP, spreads in percentage points, indices in points). COVID-as-missing masks every *macro* column (isFinancial $= 0$) over the fixed window 2020Q2–Q4 (set to JSON null/NaN and imputed inside the Durbin–Koopman E-step); financial columns (isFinancial $= 1$) are kept observed throughout. All five blocs are estimated on a common 2009Q1–2025Q4 quarterly grid ($T = 68$), $p = 3$ lags, with the hierarchical (GLP) Minnesota prior; per-variable own-lag prior means are random walk (RW) except the three fiscal white-noise variables (WN: FREV_GDP, FPEXP_GDP, SFA).

New credit block (flagged). Each periphery bloc gains two new domestic financial variables, appended after the redesign vars by build_credit_euro_periph.py:

- **Quantity** CREDIT_GDP_<CC> — BIS "Total Credit to Non-Financial Corporations, % of GDP" (the *broad* credit measure: bank loans *plus* debt securities *plus* cross-border lending), pulled from FRED under the mnemonic Q<CC>NAM770A. Native quarterly, lin, RW, isFinancial. Despite being a financial series it is COVID-masked over 2020Q2–Q4 because it *inherits the GDP-denominator collapse* (a ratio whose denominator is a masked macro flow).
- **Price** LENDR_<CC> — ECB MIR (MFI Interest Rate statistics) NFC bank lending rate, monthly → quarter by simple month-average, lin, RW, isFinancial, kept observed through COVID (financial-observed rule).

[FLAG: the research scoping report (wrqpqko5z) recommended an ECB BSI MFI "loans-to-NFC" stock *(BSI/M.<CC>.N.A.A20.A.1....) as the quantity series. The shipped builder instead uses*

the BIS broad Total-Credit-to-NFCs %-of-GDP series via FRED (`Q<CC>NAM770A`), which is the convention the task brief and the build audit confirm. The staged data are authoritative; the BSI recommendation was superseded.]

[FLAG: GR `LENDR_GR` is a non-canonical MIR variant. The standard composite cost-of-borrowing key `M.GR.B.A2A.A.R.A.2240.EUR.N` is sparse for Greece (only ≈54 obs, with ≈29 internal gaps over the 2010–2016 crisis years), so the builder substitutes the fully dense new-business floating-rate / IRF-up-to-1yr variant `M.GR.B.A2A.F.R.A.2240.EUR.N`. Its definition differs from the A2A composite used by the other four periphery members — reduced cross-economy comparability of the price-of-credit variable.]

[FLAG: all five periphery `DataRaw_<CC>_glob.json` bases store the COVID-masked macro cells as bare `NaN` literals (a MATLAB `jsondecode` convention inherited from `euro_redesign_staging`), which fail a strict `JSON.parse`; the appended credit cells correctly use `null`. This is a pre-existing format quirk, not a data error.]

20.15 Spain (ES)

A 23-variable single-country bloc (10 domestic macro/financial + 1 sovereign spread, 4 external exogenous, 2 rest-of-EA, 4 fiscal, 2 credit), estimated on 2009Q1–2025Q4 ($T = 68$, $p = 3$) with COVID-as-missing over 2020Q2–Q4 (macro masked, financials observed). The credit quantity (`QESNAM770A`) shows a large post-2008 boom-bust cycle (peak ≈148% of GDP, falling to ≈77% by 2025Q4); the log-level RW absorbs the trend.

Table 66: Spain (ES) — variable documentation. $n = 23$, 2009Q1–2025Q4.

Variable (id)	Description	Source	Series ID / key	Freq	Transf.	COVID	Notes
`GDP_ES`	Real GDP (chain-vol, SCA)	Eurostat	`namq_10_gdp` Q. CLV15_MEUR. SCA.B1GQ.ES	Q	log	masked	domestic
`CONS_ES`	Household + NPISH final cons.	Eurostat	`namq_10_gdp` ...P31_S14_S15.ES	Q	log	masked	domestic
`INV_ES`	Gross fixed capital formation	Eurostat	`namq_10_gdp` ...P51G.ES	Q	log	masked	domestic
`GOV_ES`	Government final consumption	Eurostat	`namq_10_gdp` ...P3_S13.ES	Q	log	masked	domestic
`EXP_ES`	Exports of goods & services	Eurostat	`namq_10_gdp` ...P6.ES	Q	log	masked	domestic
`IMP_ES`	Imports of goods & services	Eurostat	`namq_10_gdp` ...P7.ES	Q	log	masked	domestic

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Table 66 – continued

Variable (id)	Description	Source	Series ID / key	Freq	Transf.	COVID	Notes
`HICP_ES`	HICP, all-items index (2015=100)	Eurostat	`prc_hicp_midx` M.I15.CP00.ES	M→Q	log	masked	domestic; m→q avg
`LR_ES`	Unemployment rate (SA, total)	Eurostat	`une_rt_q` Q.SA.TOTAL.PC_ACT.T.ES	Q	lin	masked	domestic; % of labour force
`HPI_ES`	Real house-price index (HPI/HICP, 2015=100)	Eurostat	`prc_hpi_q` Q.TOTAL.I15_Q.ES, deflated by `HICP_ES`	Q	log	observed	domestic; `isFin`=1
`NEER_ES`	Nominal effective FX (IC42 partners, 2015=100)	Eurostat	`ert_eff_ic_q` Q.NEER_IC42.I15.ES	Q	log	observed	domestic; `isFin`=1
`SPREAD_ES`	10Y sovereign spread vs Bund (pp)	Eurostat	`irt_lt_mcby_m` M.MCBY.ES − M.MCBY.DE	M→Q	lin	observed	domestic; Maastricht 10Y
`POLRATE_ES`	ECB Deposit Facility Rate (policy)	ECB Data Portal	`FM` D.U2.EUR. 4F.KR.DFR. LEV	D→Q	lin	observed	external; % p.a.
`EACORP`	EA corporate bond yield (%)	Bundesbank	`BBSIS/M.I. UMR.RD.EUR. X2000.B.A.A. R.A.A._Z._Z.A`	D→Q	lin	observed	external
`RATE_ES`	3-month EURIBOR (%)	Eurostat	`irt_st_q` Q.IRT_M3.EA	Q	lin	observed	external; EA-wide 3M
`YIELD10_DE`	German 10Y Bund yield (spread base)	Eurostat	`irt_lt_mcby_m` M.MCBY.DE	M→Q	lin	observed	external
`ROWEA_GDP_ES`	Rest-of-EA real GDP (GDP-wtd ex-ES)	Eurostat (derived)	EA20 `namq_10_gdp` de-aggregated by ES weight	Q	log	masked	row; index 2015=100

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Table 66 – continued

Variable (id)	Description	Source	Series ID / key	Freq	Transf.	COVID	Notes
ROWEA_HICP_ES	Rest-of-EA HICP (GDP-wtd ex-ES)	Eurostat (derived)	EA20 prc_hicp_midx de-aggregated by ES weight	M→Q	log	masked	row; index 2015=100
FREV_GDP	General-gov't revenue (% of GDP, 4q-trailing)	Eurostat	gov_10q_ggnfa Q.MIO_EUR. NSA.S13.TR. ES / CP GDP	Q	lin	observed	fiscal; WN
FPEXP_GDP	Primary expenditure (% of GDP, 4q-trailing)	Eurostat	gov_10q_ggnfa TE−D41PAY, S13.ES / CP GDP	Q	lin	observed	fiscal; WN
EFFI	Effective interest rate on debt (%)	Eurostat	$4{\cdot}\mathrm{D41PAY}_t/\mathrm{debt}_{t-1}$, gov_10q_ggnfa/ggdebt.ES	Q	lin	observed	fiscal; isFin=1
SFA	Stock-flow adjustment (% of GDP)	Eurostat (derived)	debt-identity residual from ggdebt.ES	Q	lin	masked	fiscal; WN; COVID-NaN'd
CREDIT_GDP_ES	BIS Total Credit to NFCs (% of GDP)	BIS via FRED	QESNAM770A	Q	lin	masked	domestic; isFin=1; broad credit; GDP-denom mask
LENDR_ES	NFC bank lending rate (ECB MIR composite, %)	ECB Data Portal	MIR M.ES.B. A2A.A.R.A. 2240.EUR.N	M→Q	lin	observed	domestic; isFin=1; new business

[FLAG: EACORP is shared across euro members and labelled "EA IG Corporate Bond Yield" in the spec, but the underlying ECB series (YC/B.U2.EUR.4F.G_N_C.SV_C_YM.SR_10Y) is the euro-area all-bonds/AAA 10Y government *yield-curve point, not an investment-grade corporate index. The id name overstates the credit content; the series itself is correctly traced.]* Credit-block coverage (ES): CREDIT_GDP_ES 65/68 obs (2009Q1=141.3 %, 2025Q4=76.5 %, range 76.5–148.2), 2020Q2–Q4 null; LENDR_ES 68/68 obs (2009Q1=3.63 %, 2025Q4=3.27 %, COVID observed 1.57–1.70 %).

20.16 Italy (IT)

A 23-variable bloc with the identical structure to Spain (10 domestic + spread, 4 external, 2 rest-of-EA, 4 fiscal, 2 credit), estimated on 2009Q1–2025Q4 ($T = 68$, $p = 3$), COVID-masked 2020Q2–Q4. The Italian sovereign spread and the own MIR lending rate both carry the periphery fragmentation premium that the shared `EACORP` benchmark misses.

Table 67: Italy (IT) — variable documentation. $n = 23$, 2009Q1–2025Q4.

Variable (id)	Description	Source	Series ID / key	Freq	Transf.	COVID	Notes
`GDP_IT`	Real GDP (chain-vol, SCA)	Eurostat	`namq_10_gdp` Q.CLV15_MEUR.SCA.B1GQ.IT	Q	log	masked	domestic
`CONS_IT`	Household + NPISH final cons.	Eurostat	`namq_10_gdp` …P31_S14_S15.IT	Q	log	masked	domestic
`INV_IT`	Gross fixed capital formation	Eurostat	`namq_10_gdp` …P51G.IT	Q	log	masked	domestic
`GOV_IT`	Government final consumption	Eurostat	`namq_10_gdp` …P3_S13.IT	Q	log	masked	domestic
`EXP_IT`	Exports of goods & services	Eurostat	`namq_10_gdp` …P6.IT	Q	log	masked	domestic
`IMP_IT`	Imports of goods & services	Eurostat	`namq_10_gdp` …P7.IT	Q	log	masked	domestic
`HICP_IT`	HICP, all-items index (2015=100)	Eurostat	`prc_hicp_midx` M.I15.CP00.IT	M→Q	log	masked	domestic; m→q avg
`LR_IT`	Unemployment rate (SA, total)	Eurostat	`une_rt_q` Q.SA.TOTAL.PC_ACT.T.IT	Q	lin	masked	domestic; % of labour force
`HPI_IT`	Real house-price index (HPI/HICP, 2015=100)	Eurostat	`prc_hpi_q` Q.TOTAL.I15_Q.IT, deflated by `HICP_IT`	Q	log	observed	domestic; `isFin`=1
`NEER_IT`	Nominal effective FX (IC42 partners, 2015=100)	Eurostat	`ert_eff_ic_q` Q.NEER_IC42.I15.IT	Q	log	observed	domestic; `isFin`=1

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Table 67 – continued

Variable (id)	Description	Source	Series ID / key	Freq	Transf.	COVID	Notes
SPREAD_IT	10Y sovereign spread vs Bund (pp)	Eurostat	irt_lt_mcby_m M.MCBY.IT – M.MCBY.DE	M→Q	lin	observed	domestic; Maastricht 10Y
POLRATE_IT	ECB Deposit Facility Rate (policy)	ECB Data Portal	FM D.U2.EUR. 4F.KR.DFR. LEV	D→Q	lin	observed	external; % p.a.
EACORP	EA corporate bond yield (%)	Bundesbank	BBSIS/M.I. UMR.RD.EUR. X2000.B.A.A. R.A.A._Z._Z.A	D→Q	lin	observed	external
RATE_IT	3-month EURIBOR (%)	Eurostat	irt_st_q Q.IRT_M3.EA	Q	lin	observed	external; EA-wide 3M
YIELD10_DE	German 10Y Bund yield (spread base)	Eurostat	irt_lt_mcby_m M.MCBY.DE	M→Q	lin	observed	external
ROWEA_GDP_IT	Rest-of-EA real GDP (GDP-wtd ex-IT)	Eurostat (derived)	EA20 namq_10_gdp de-aggregated by IT weight	Q	log	masked	row; index 2015=100
ROWEA_HICP_IT	Rest-of-EA HICP (GDP-wtd ex-IT)	Eurostat (derived)	EA20 prc_hicp_midx de-aggregated by IT weight	M→Q	log	masked	row; index 2015=100
FREV_GDP	General-gov't revenue (% of GDP, 4q-trailing)	Eurostat	gov_10q_ggnfa Q.MIO_EUR. NSA.S13.TR. IT / CP GDP	Q	lin	observed	fiscal; WN
FPEXP_GDP	Primary expenditure (% of GDP, 4q-trailing)	Eurostat	gov_10q_ggnfa TE−D41PAY, S13.IT / CP GDP	Q	lin	observed	fiscal; WN
EFFI	Effective interest rate on debt (%)	Eurostat	$4 \cdot \mathrm{D41PAY}_t/\mathrm{debt}_{t-1}$, gov_10q_ggnfa/ggdebt.IT	Q	lin	observed	fiscal; isFin=1

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Table 67 – continued

Variable (id)	Description	Source	Series ID / key	Freq	Transf.	COVID	Notes
SFA	Stock-flow adjustment (% of GDP)	Eurostat (derived)	debt-identity residual from ggdebt.IT	Q	lin	masked	fiscal; WN; COVID-NaN'd
CREDIT_GDP_IT	BIS Total Credit to NFCs (% of GDP)	BIS via FRED	QITNAM770A	Q	lin	masked	domestic; isFin=1; broad credit; GDP-denom mask
LENDR_IT	NFC bank lending rate (ECB MIR composite, %)	ECB Data Portal	MIR M.IT.B. A2A.A.R.A. 2240.EUR.N	M→Q	lin	observed	domestic; isFin=1; new business

Credit-block coverage (IT): CREDIT_GDP_IT 65/68 obs (2009Q1=78.7 %, 2025Q4=58.5 %, range 58.4–81.6), 2020Q2–Q4 null; LENDR_IT 68/68 obs (2009Q1=3.47 %, 2025Q4=3.55 %, COVID observed 1.18–1.33 %). The EACORP naming caveat (ES table) applies identically.

20.17 Ireland (IE)

A 23-variable bloc, same template, 2009Q1–2025Q4 ($T = 68$, $p = 3$), COVID-masked 2020Q2–Q4. Two Irish-specific data caveats are worth noting: the chain-volume GDP aggregate is distorted by multinational balance-sheet relocation (the "modified" domestic-demand concept is the cleaner activity gauge but is not the Eurostat B1GQ used here), and the BIS credit-to-GDP series exhibits a very large boom-bust path.

Table 68: Ireland (IE) — variable documentation. $n = 23$, 2009Q1–2025Q4.

Variable (id)	Description	Source	Series ID / key	Freq	Transf.	COVID	Notes
GDP_IE	Real GDP (chain-vol, SCA)	Eurostat	namq_10_gdp Q. CLV15_MEUR. SCA.B1GQ.IE	Q	log	masked	domestic; MNC-distorted
CONS_IE	Household + NPISH final cons.	Eurostat	namq_10_gdp ...P31_S14_S15.IE	Q	log	masked	domestic

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Table 68 – continued

Variable (id)	Description	Source	Series ID / key	Freq	Transf.	COVID	Notes
INV_IE	Gross fixed capital formation	Eurostat	namq_10_gdp …P51G.IE	Q	log	masked	domestic; MNC-volatile
GOV_IE	Government final consumption	Eurostat	namq_10_gdp …P3_S13.IE	Q	log	masked	domestic
EXP_IE	Exports of goods & services	Eurostat	namq_10_gdp …P6.IE	Q	log	masked	domestic; MNC-volatile
IMP_IE	Imports of goods & services	Eurostat	namq_10_gdp …P7.IE	Q	log	masked	domestic
HICP_IE	HICP, all-items index (2015=100)	Eurostat	prc_hicp_midx M.I15.CP00.IE	M→Q	log	masked	domestic; m→q avg
LR_IE	Unemployment rate (SA, total)	Eurostat	une_rt_q Q.SA.TOTAL.PC_ACT.T.IE	Q	lin	masked	domestic; % of labour force
HPI_IE	Real house-price index (HPI/HICP, 2015=100)	Eurostat	prc_hpi_q Q.TOTAL.I15_Q.IE, deflated by HICP_IE	Q	log	observed	domestic; isFin=1
NEER_IE	Nominal effective FX (IC42 partners, 2015=100)	Eurostat	ert_eff_ic_q Q.NEER_IC42.I15.IE	Q	log	observed	domestic; isFin=1
SPREAD_IE	10Y sovereign spread vs Bund (pp)	Eurostat	irt_lt_mcby_m M.MCBY.IE − M.MCBY.DE	M→Q	lin	observed	domestic; Maastricht 10Y
POLRATE_IE	ECB Deposit Facility Rate (policy)	ECB Data Portal	FM D.U2.EUR. 4F.KR.DFR. LEV	D→Q	lin	observed	external; % p.a.
EACORP	EA corporate bond yield (%)	Bundesbank	BBSIS/M.I. UMR.RD.EUR. X2000.B.A.A. R.A.A._Z._Z.A	D→Q	lin	observed	external

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Table 68 – continued

Variable (id)	Description	Source	Series ID / key	Freq	Transf.	COVID	Notes
RATE_IE	3-month EURIBOR (%)	Eurostat	irt_st_q Q.IRT_M3.EA	Q	lin	observed	external; EA-wide 3M
YIELD10_DE	German 10Y Bund yield (spread base)	Eurostat	irt_lt_mcby_m M.MCBY.DE	M→Q	lin	observed	external
ROWEA_GDP_IE	Rest-of-EA real GDP (GDP-wtd ex-IE)	Eurostat (derived)	EA20 namq_10_gdp de-aggregated by IE weight	Q	log	masked	row; index 2015=100
ROWEA_HICP_IE	Rest-of-EA HICP (GDP-wtd ex-IE)	Eurostat (derived)	EA20 prc_hicp_midx de-aggregated by IE weight	M→Q	log	masked	row; index 2015=100
FREV_GDP	General-gov't revenue (% of GDP, 4q-trailing)	Eurostat	gov_10q_ggnfa Q.MIO_EUR. NSA.S13.TR. IE / CP GDP	Q	lin	observed	fiscal; WN
FPEXP_GDP	Primary expenditure (% of GDP, 4q-trailing)	Eurostat	gov_10q_ggnfa TE−D41PAY, S13.IE / CP GDP	Q	lin	observed	fiscal; WN
EFFI	Effective interest rate on debt (%)	Eurostat	$4 \cdot \mathrm{D41PAY}_t/\mathrm{debt}_{t-1}$, gov_10q_ggnfa/ggdebt.IE	Q	lin	observed	fiscal; isFin=1
SFA	Stock-flow adjustment (% of GDP)	Eurostat (derived)	debt-identity residual from ggdebt.IE	Q	lin	masked	fiscal; WN; COVID-NaN'd
CREDIT_GDP_IE	BIS Total Credit to NFCs (% of GDP)	BIS via FRED	QIENAM770A	Q	lin	masked	domestic; isFin=1; broad credit; severe boom-bust

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Table 68 – continued

Variable (id)	Description	Source	Series ID / key	Freq	Transf.	COVID	Notes
LENDR_IE	NFC bank lending rate (ECB MIR composite, %)	ECB Data Portal	MIR M.IE.B. A2A.A.R.A. 2240.EUR.N	M→Q	lin	observed	domestic; isFin=1; new business

[FLAG: CREDIT_GDP_IE has an extreme boom-bust path (≈305% of GDP near 2009/2010 collapsing to ≈92% by 2025Q4 — property-bubble deleveraging, NAMA asset transfers, bank wind-downs and multinational reclassification). It is the most volatile credit-quantity series of the periphery group; the BIS broad measure and the log-level RW absorb the trend, but the path is non-stationary in level. The research report recommended considering a post-2014 start or growth-rate space; the shipped build uses the full sample.] Credit-block coverage (IE): CREDIT_GDP_IE 65/68 obs (2009Q1=177.3 %, 2025Q4=92.4 %, range 92.4–305.8), 2020Q2–Q4 null; LENDR_IE 68/68 obs (2009Q1=4.10 %, 2025Q4=4.67 %, COVID observed 2.32–2.81 %). The EACORP naming caveat (ES table) applies.

20.18 Portugal (PT)

A 23-variable bloc, same template, 2009Q1–2025Q4 ($T = 68$, $p = 3$), COVID-masked 2020Q2–Q4. Portuguese bank lending rates carry a clear sovereign-stress / fragmentation premium over the 2009–2012 crisis years (own LENDR_PT peaks near 6.7%), which the shared EACORP benchmark does not capture.

Table 69: Portugal (PT) — variable documentation. $n = 23$, 2009Q1–2025Q4.

Variable (id)	Description	Source	Series ID / key	Freq	Transf.	COVID	Notes
GDP_PT	Real GDP (chain-vol, SCA)	Eurostat	namq_10_gdp Q. CLV15_MEUR. SCA.B1GQ. PT	Q	log	masked	domestic
CONS_PT	Household + NPISH final cons.	Eurostat	namq_10_gdp …P31_S14_S15.PT	Q	log	masked	domestic
INV_PT	Gross fixed capital formation	Eurostat	namq_10_gdp …P51G.PT	Q	log	masked	domestic
GOV_PT	Government final consumption	Eurostat	namq_10_gdp …P3_S13.PT	Q	log	masked	domestic

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Table 69 – continued

Variable (id)	Description	Source	Series ID / key	Freq	Transf.	COVID	Notes
EXP_PT	Exports of goods & services	Eurostat	namq_10_gdp …P6.PT	Q	log	masked	domestic
IMP_PT	Imports of goods & services	Eurostat	namq_10_gdp …P7.PT	Q	log	masked	domestic
HICP_PT	HICP, all-items index (2015=100)	Eurostat	prc_hicp_midx M.I15.CP00.PT	M→Q	log	masked	domestic; m→q avg
LR_PT	Unemployment rate (SA, total)	Eurostat	une_rt_q Q.SA.TOTAL.PC_ACT.T.PT	Q	lin	masked	domestic; % of labour force
HPI_PT	Real house-price index (HPI/HICP, 2015=100)	Eurostat	prc_hpi_q Q.TOTAL.I15_Q.PT, deflated by HICP_PT	Q	log	observed	domestic; isFin=1
NEER_PT	Nominal effective FX (IC42 partners, 2015=100)	Eurostat	ert_eff_ic_q Q.NEER_IC42.I15.PT	Q	log	observed	domestic; isFin=1
SPREAD_PT	10Y sovereign spread vs Bund (pp)	Eurostat	irt_lt_mcby_m M.MCBY.PT – M.MCBY.DE	M→Q	lin	observed	domestic; Maastricht 10Y
POLRATE_PT	ECB Deposit Facility Rate (policy)	ECB Data Portal	FM D.U2.EUR. 4F.KR.DFR. LEV	D→Q	lin	observed	external; % p.a.
EACORP	EA corporate bond yield (%)	Bundesbank	BBSIS/M.I. UMR.RD.EUR. X2000.B.A.A. R.A.A._Z._Z.A	D→Q	lin	observed	external
RATE_PT	3-month EURIBOR (%)	Eurostat	irt_st_q Q.IRT_M3.EA	Q	lin	observed	external; EA-wide 3M
YIELD10_DE	German 10Y Bund yield (spread base)	Eurostat	irt_lt_mcby_m M.MCBY.DE	M→Q	lin	observed	external

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Table 69 – continued

Variable (id)	Description	Source	Series ID / key	Freq	Transf.	COVID	Notes
ROWEA_GDP_PT	Rest-of-EA real GDP (GDP-wtd ex-PT)	Eurostat (derived)	EA20 namq_10_gdp de-aggregated by PT weight	Q	log	masked	row; index 2015=100
ROWEA_HICP_PT	Rest-of-EA HICP (GDP-wtd ex-PT)	Eurostat (derived)	EA20 prc_hicp_midx de-aggregated by PT weight	M→Q	log	masked	row; index 2015=100
FREV_GDP	General-gov't revenue (% of GDP, 4q-trailing)	Eurostat	gov_10q_ggnfa Q.MIO_EUR. NSA.S13.TR. PT / CP GDP	Q	lin	observed	fiscal; WN
FPEXP_GDP	Primary expenditure (% of GDP, 4q-trailing)	Eurostat	gov_10q_ggnfa TE−D41PAY, S13.PT / CP GDP	Q	lin	observed	fiscal; WN
EFFI	Effective interest rate on debt (%)	Eurostat	$4 \cdot \mathrm{D41PAY}_t/\mathrm{debt}_{t-1}$, gov_10q_ggnfa/ggdebt.PT	Q	lin	observed	fiscal; isFin=1
SFA	Stock-flow adjustment (% of GDP)	Eurostat (derived)	debt-identity residual from ggdebt.PT	Q	lin	masked	fiscal; WN; COVID-NaN'd
CREDIT_GDP_PT	BIS Total Credit to NFCs (% of GDP)	BIS via FRED	QPTNAM770A	Q	lin	masked	domestic; isFin=1; broad credit; GDP-denom mask
LENDR_PT	NFC bank lending rate (ECB MIR composite, %)	ECB Data Portal	MIR M.PT.B. A2A.A.R.A. 2240.EUR.N	M→Q	lin	observed	domestic; isFin=1; new business

Credit-block coverage (PT): CREDIT_GDP_PT 65/68 obs (2009Q1=121.7 %, 2025Q4=74.2 %, range 74.2–140.9), 2020Q2–Q4 null; LENDR_PT 68/68 obs (2009Q1=5.62 %, 2025Q4=3.67 %, COVID observed 1.82–2.05 %). The EACORP naming caveat (ES table) applies.

20.19 Greece (GR)

A **22-variable** bloc — one fewer than the other four because `HPI_GR` is dropped (no Eurostat `prc_hpi_q` house-price index is published for Greece). The structure is otherwise the periphery template (9 domestic + spread, 4 external, 2 rest-of-EA, 4 fiscal, 2 credit), estimated on 2009Q1–2025Q4 ($T = 68$, $p = 3$), COVID-masked 2020Q2–Q4. All Eurostat keys use the Greece geo code `EL`.

Table 70: Greece (GR) — variable documentation. $n = 22$ (no HPI), 2009Q1–2025Q4.

Variable (id)	Description	Source	Series ID / key	Freq	Transf.	COVID	Notes
`GDP_GR`	Real GDP (chain-vol, SCA)	Eurostat	`namq_10_gdp` Q. CLV15_MEUR. SCA.B1GQ.EL	Q	log	masked	domestic; geo=EL
`CONS_GR`	Household + NPISH final cons.	Eurostat	`namq_10_gdp` …P31_S14_S15.EL	Q	log	masked	domestic
`INV_GR`	Gross fixed capital formation	Eurostat	`namq_10_gdp` …P51G.EL	Q	log	masked	domestic
`GOV_GR`	Government final consumption	Eurostat	`namq_10_gdp` …P3_S13.EL	Q	log	masked	domestic
`EXP_GR`	Exports of goods & services	Eurostat	`namq_10_gdp` …P6.EL	Q	log	masked	domestic
`IMP_GR`	Imports of goods & services	Eurostat	`namq_10_gdp` …P7.EL	Q	log	masked	domestic
`HICP_GR`	HICP, all-items index (2015=100)	Eurostat	`prc_hicp_midx` M.I15.CP00.EL	M→Q	log	masked	domestic; m→q avg
`LR_GR`	Unemployment rate (SA, total)	Eurostat	`une_rt_q` Q.SA.TOTAL.PC_ACT.T.EL	Q	lin	masked	domestic; % of labour force
`NEER_GR`	Nominal effective FX (IC42 partners, 2015=100)	Eurostat	`ert_eff_ic_q` Q.NEER_IC42.I15.EL	Q	log	observed	domestic; `isFin`=1
`SPREAD_GR`	10Y sovereign spread vs Bund (pp)	Eurostat	`irt_lt_mcby_m` M.MCBY.EL − M.MCBY.DE	M→Q	lin	observed	domestic; Maastricht 10Y

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Variable (id)	Description	Source	Series ID / key	Freq	Transf.	COVID	Notes
POLRATE_GR	ECB Deposit Facility Rate (policy)	ECB Data Portal	FM D.U2.EUR.4F.KR.DFR.LEV	D→Q	lin	observed	external; % p.a.
EACORP	EA corporate bond yield (%)	Bundesbank	BBSIS/M.I.UMR.RD.EUR.X2000.B.A.A.R.A.A._Z._Z.A	D→Q	lin	observed	external
RATE_GR	3-month EURIBOR (%)	Eurostat	irt_st_q Q.IRT_M3.EA	Q	lin	observed	external; EA-wide 3M
YIELD10_DE	German 10Y Bund yield (spread base)	Eurostat	irt_lt_mcby_m M.MCBY.DE	M→Q	lin	observed	external
ROWEA_GDP_GR	Rest-of-EA real GDP (GDP-wtd ex-GR)	Eurostat (derived)	EA20 namq_10_gdp de-aggregated by GR weight	Q	log	masked	row; index 2015=100
ROWEA_HICP_GR	Rest-of-EA HICP (GDP-wtd ex-GR)	Eurostat (derived)	EA20 prc_hicp_midx de-aggregated by GR weight	M→Q	log	masked	row; index 2015=100
FREV_GDP	General-gov't revenue (% of GDP, 4q-trailing)	Eurostat	gov_10q_ggnfa Q.MIO_EUR.NSA.S13.TR.EL / CP GDP	Q	lin	observed	fiscal; WN
FPEXP_GDP	Primary expenditure (% of GDP, 4q-trailing)	Eurostat	gov_10q_ggnfa TE−D41PAY, S13.EL / CP GDP	Q	lin	observed	fiscal; WN
EFFI	Effective interest rate on debt (%)	Eurostat	$4{\cdot}\mathrm{D41PAY}_t/\mathrm{debt}_{t-1}$, gov_10q_ggnfa/ggdebt.EL	Q	lin	observed	fiscal; isFin=1
SFA	Stock-flow adjustment (% of GDP)	Eurostat (derived)	debt-identity residual from ggdebt.EL	Q	lin	masked	fiscal; WN; COVID-NaN'd

continued on next page

Table 70 – continued

Variable (id)	Description	Source	Series ID / key	Freq	Transf.	COVID	Notes
`CREDIT_GDP_GR`	BIS Total Credit to NFCs (% of GDP)	BIS via FRED	`QGRNAM770A`	Q	lin	masked	domestic; `isFin`=1; broad credit; GDP-denom mask
`LENDR_GR`	NFC bank lending rate (ECB MIR new-business floating, %)	ECB Data Portal	`MIR` M.GR.B. A2A.F.R.A. 2240.EUR.N	M→Q	lin	observed	domestic; `isFin`=1; non-canonical MIR variant

[FLAG: `LENDR_GR` uses the floating-rate / IRF-up-to-1yr new-business MIR key `M.GR.B.A2A.F.R.A.2240.EUR.N` rather than the A2A.R.A composite cost-of-borrowing used by the other four periphery members, because the composite is sparse for Greece (≈54 obs with crisis-era gaps). The floating variant is gap-free on the full grid but is a different instrument — reduced cross-economy comparability.] *[FLAG: `HPI_GR` is absent (n = 22, not 23) — Eurostat publishes no house-price index (`prc_hpi_q`) for Greece, so the variable is dropped rather than imputed.]* Credit-block coverage (GR): `CREDIT_GDP_GR` 65/68 obs (2009Q1=63.6 %, 2025Q4=59.3 %, range 52.5–71.6), 2020Q2–Q4 `null`; `LENDR_GR` 68/68 obs (2009Q1=4.50 %, 2025Q4=3.66 %, COVID observed 2.86–3.07 %). The `EACORP` naming caveat (ES table) applies.

20.20 Euro-area small members: data conventions

This section documents the seven smaller euro-area member blocs (Estonia, Finland, Luxembourg, Latvia, Lithuania, Slovakia, Slovenia). Each is a standalone single-country Bayesian VAR (p = 3 lags, hierarchical Minnesota/GLP prior) estimated on a common quarterly sample **2009Q1–2025Q4** (T = 68), re-estimated by `export_bvar_euro_redesign.m`. The domestic real national accounts, prices, labour, and house prices come from **Eurostat** SDMX-CSV; the financial block (policy rate, the shared euro corporate yield anchor, the 3M rate, and the German 10Y spread base) comes from the **ECB Data Portal** and Eurostat; the four-variable fiscal block is built from **Eurostat** government finance statistics (`gov_10q`); and the rest-of-EA macro pair is a GDP-weighted de-aggregation of the EA20 aggregate.

Two transformation conventions are used throughout. `log` means the variable is stored as $100 \cdot \log(\text{level})$ and therefore enters the model in growth space (displayed as $4\,\Delta\log$ = quarter-on-quarter annualized growth); `lin` means the level (percent, percentage points, or %-of-GDP). Under the COVID-as-missing treatment, all macro variables (`isFinancial`= 0) are masked over **2020Q2–2020Q4** and re-imputed by the Durbin–Koopman simulation smoother, while financial variables (`isFinancial`= 1) are kept observed through COVID — with one deliberate exception: the credit *quantity* CREDIT_GDP_<CC> is masked 2020Q2–Q4 because it inherits the collapse of its GDP denominator.

The new credit block (flagged). Each bloc gains a two-variable credit block appended after the fiscal block: a *quantity*, CREDIT_GDP_<CC> (credit to non-financial corporations, % of GDP; `lin`, prior RW, masked 2020Q2–Q4), and — where data permit — a *price*, LENDR_<CC> (bank lending rate to NFCs; `lin`, prior RW, observed through COVID). For **FI** and **LU** the quantity is the BIS "Total credit to NFCs, % of GDP" (broad credit: loans, debt securities, and cross-border lending) drawn from FRED. For **EE, LV, LT, SK, SI** the BIS %-of-GDP table has no entry, so the quantity is **constructed** (non-BIS) as 100 × (ECB BSI MFI loans to NFCs)/(4-quarter rolling sum of Eurostat nominal GDP); this is a *narrow* (domestic-MFI-loans-only) measure whose levels are far below the BIS broad measure, reducing cross-country comparability — see the per-economy *[FLAG]* notes. Two further idiosyncrasies are flagged: **LU**'s price uses the ECB MIR *outstanding-amounts* rate (the new-business cost-of-borrowing series is too sparse pre-2015), and **LV**'s price is **dropped** entirely (the ECB MIR NFC rate is too sparse over the 2009Q1 grid); for LV the shared `EACORP` remains the external corporate-yield anchor.

20.21 Estonia (EE)

Estonia is a standalone euro-member BVAR with $n = 23$ variables, $p = 3$ lags, estimated on 2009Q1–2025Q4 ($T = 68$). Macro variables (`isFinancial`= 0) are COVID-masked 2020Q2–Q4 and re-imputed by the Durbin–Koopman smoother; financials are kept observed.

Table 71: Estonia (EE) — variable coverage. `log`= 100 log level (displayed as $4\Delta\log$, q/q ann. growth); `lin`= level.

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
GDP_EE	EE real GDP	Eurostat	namq_10_gdp Q.CLV15_MEUR.SCA.B1GQ.EE	Q	log	masked 2020Q2–Q4	Chain-linked volume, SCA
CONS_EE	Real household consumption	Eurostat	namq_10_gdp ...P31_S14_S15.EE	Q	log	masked 2020Q2–Q4	
INV_EE	Real gross fixed capital form.	Eurostat	namq_10_gdp ...P51G.EE	Q	log	masked 2020Q2–Q4	

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
GOV_EE	Real government consumption	Eurostat	namq_10_gdp ...P3_S13.EE	Q	log	masked 2020Q2–Q4	
EXP_EE	Real exports	Eurostat	namq_10_gdp ...P6.EE	Q	log	masked 2020Q2–Q4	
IMP_EE	Real imports	Eurostat	namq_10_gdp ...P7.EE	Q	log	masked 2020Q2–Q4	
HICP_EE	HICP index (2015=100)	Eurostat	prc_hicp_midx M.I15.CP00.EE	M→Q	log	masked 2020Q2–Q4	Monthly avg. to quarterly
LR_EE	Unemployment rate (%)	Eurostat	une_rt_q Q.SA.TOTAL.PC_ACT.T.EE	Q	lin	masked 2020Q2–Q4	
HPI_EE	Real house price index (HPI/HICP, 2015=100)	Eurostat	prc_hpi_q Q.TOTAL.I15_Q.EE ÷ HICP	Q	log	observed (financial)	Nominal HPI deflated by HICP
NEER_EE	NEER, IC42 trade-weighted	Eurostat	ert_eff_ic_q Q.NEER_IC42.I15.EE	Q	log	observed (financial)	42-partner nominal eff. FX
SPREAD_EE	10Y sovereign spread vs Bund	Eurostat	irt_lt_mcby_m M.MCBY.EE − M.MCBY.DE	M→Q	lin	observed (financial)	Own 10Y minus Bund
POLRATE_EE	ECB Deposit Facility Rate (policy)	ECB	FM D.U2.EUR.4F. KR.DFR.LEV	D→Q	lin	observed (financial)	External; shared by all members
EACORP	EA corporate bond yield (non-MFI)	Bundesbank	BBSIS/M.I.UMR. RD.EUR.X2000. B.A.A.R.A.A. _Z._Z.A	D→Q	lin	observed (financial)	External; shared anchor. *(Corrected 2026-06-18: genuine corporate yield — Bundesbank non-MFI enterprises, issuer class X2000.)*
RATE_EE	3M rate (EURIBOR)	Eurostat	irt_st_q Q.IRT_M3.EA	Q	lin	observed (financial)	External; shared 3M EURIBOR
YIELD10_DE	DE 10Y Bund (spread base)	Eurostat	irt_lt_mcby_m M.MCBY.DE	M→Q	lin	observed (financial)	External
ROWEA_GDP_EE	Rest-of-EA real GDP	Eurostat	EA20 namq_10_gdp de-aggregated (GDP-weighted)	Q	log	masked 2020Q2–Q4	EA20 ex-EE, GDP-weighted
ROWEA_HICP_EE	Rest-of-EA HICP	Eurostat	EA20 prc_hicp_midx de-aggregated	Q	log	masked 2020Q2–Q4	EA20 ex-EE, GDP-weighted
FREV_GDP	Revenue (% of GDP)	Eurostat	gov_10q_ggnfa Q...TR.EE (4q sum/GDP)	Q	lin	masked 2020Q2–Q4	Prior WN; 4q-trailing/nom. GDP

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
FPEXP_GDP	Primary expenditure (% of GDP)	Eurostat	gov_10q_ggnfa TE−D41PAY (4q sum/GDP)	Q	lin	masked 2020Q2–Q4	Prior WN
EFFI	Effective interest rate (% p.a.)	Eurostat	gov_10q interest/lagged debt	Q	lin	observed (financial)	Prior RW
SFA	Stock-flow adjustment (% of GDP)	Eurostat	gov_10q_ggdebt debt recursion residual	Q	lin	masked 2020Q2–Q4	Prior WN; budget-identity residual
CREDIT_GDP_EE	Credit to NFCs (% of GDP)	ECB BSI + Eurostat (constructed)	100×BSI M.EE.N.A.A20.A.1.U2.2240.Z01.E ÷ 4q Σnamq_10_gdp CP	Q	lin	masked 2020Q2–Q4	**NEW quantity.** *[FLAG: CONSTRUCTED (non-BIS) NARROW MFI-loan measure; not directly comparable to BIS broad credit (FI/LU); not break-adjusted. obs 2009Q1=45.2..2025Q4=29.2.]*
LENDR_EE	ECB MIR composite cost of borrowing, NFCs (new business, % p.a.)	ECB MIR	MIR M.EE.B.A2A.A.R.A.2240.EUR.N	M→Q	lin	observed (financial)	**NEW price.** 63/68 grid quarters; 5 scattered single-quarter interior gaps filled by the Gibbs. obs 5.90..4.54

20.22 Finland (FI)

Finland is a standalone euro-member BVAR with $n = 23$ variables, $p = 3$ lags, estimated on 2009Q1–2025Q4 ($T = 68$). Macro variables are COVID-masked 2020Q2–Q4 and re-imputed by the Durbin–Koopman smoother; financials are kept observed.

Table 72: Finland (FI) — variable coverage.

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
GDP_FI	FI real GDP	Eurostat	namq_10_gdp Q.CLV15_MEUR.SCA.B1GQ.FI	Q	log	masked 2020Q2–Q4	Chain-linked volume, SCA
CONS_FI	Real household consumption	Eurostat	namq_10_gdp ...P31_S14_S15.FI	Q	log	masked 2020Q2–Q4	

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
INV_FI	Real gross fixed capital form.	Eurostat	namq_10_gdp ...P51G.FI	Q	log	masked 2020Q2–Q4	
GOV_FI	Real government consumption	Eurostat	namq_10_gdp ...P3_S13.FI	Q	log	masked 2020Q2–Q4	
EXP_FI	Real exports	Eurostat	namq_10_gdp ...P6.FI	Q	log	masked 2020Q2–Q4	
IMP_FI	Real imports	Eurostat	namq_10_gdp ...P7.FI	Q	log	masked 2020Q2–Q4	
HICP_FI	HICP index (2015=100)	Eurostat	prc_hicp_midx M.I15.CP00.FI	M→Q	log	masked 2020Q2–Q4	Monthly avg. to quarterly
LR_FI	Unemployment rate (%)	Eurostat	une_rt_q Q.SA.TOTAL.PC_ACT.T.FI	Q	lin	masked 2020Q2–Q4	
HPI_FI	Real house price index (HPI/HICP, 2015=100)	Eurostat	prc_hpi_q Q.TOTAL.I15_Q.FI ÷ HICP	Q	log	observed (financial)	Nominal HPI deflated by HICP
NEER_FI	NEER, IC42 trade-weighted	Eurostat	ert_eff_ic_q Q.NEER_IC42.I15.FI	Q	log	observed (financial)	42-partner nominal eff. FX
SPREAD_FI	10Y sovereign spread vs Bund	Eurostat	irt_lt_mcby_m M.MCBY.FI − M.MCBY.DE	M→Q	lin	observed (financial)	Own 10Y minus Bund
POLRATE_FI	ECB Deposit Facility Rate (policy)	ECB	FM D.U2.EUR.4F.KR.DFR.LEV	D→Q	lin	observed (financial)	External; shared by all members
EACORP	EA corporate bond yield (non-MFI)	Bundesbank	BBSIS/M.I.UMR.RD.EUR.X2000.B.A.A.R.A.A._Z._Z.A	D→Q	lin	observed (financial)	External; shared anchor. *(Corrected 2026-06-18: genuine corporate yield — Bundesbank non-MFI enterprises, issuer class X2000.)*
RATE_FI	3M rate (EURIBOR)	Eurostat	irt_st_q Q.IRT_M3.EA	Q	lin	observed (financial)	External; shared 3M EURIBOR
YIELD10_DE	DE 10Y Bund (spread base)	Eurostat	irt_lt_mcby_m M.MCBY.DE	M→Q	lin	observed (financial)	External
ROWEA_GDP_FI	Rest-of-EA real GDP	Eurostat	EA20 namq_10_gdp de-aggregated (GDP-weighted)	Q	log	masked 2020Q2–Q4	EA20 ex-FI, GDP-weighted
ROWEA_HICP_FI	Rest-of-EA HICP	Eurostat	EA20 prc_hicp_midx de-aggregated	Q	log	masked 2020Q2–Q4	EA20 ex-FI, GDP-weighted

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
FREV_GDP	Revenue (% of GDP)	Eurostat	gov_10q_ggnfa Q...TR.FI (4q sum/GDP)	Q	lin	masked 2020Q2–Q4	Prior WN
FPEXP_GDP	Primary expenditure (% of GDP)	Eurostat	gov_10q_ggnfa TE−D41PAY (4q sum/GDP)	Q	lin	masked 2020Q2–Q4	Prior WN
EFFI	Effective interest rate (% p.a.)	Eurostat	gov_10q interest/lagged debt	Q	lin	observed (financial)	Prior RW
SFA	Stock-flow adjustment (% of GDP)	Eurostat	gov_10q_ggdebt debt recursion residual	Q	lin	masked 2020Q2–Q4	Prior WN
CREDIT_GDP_FI	Credit to NFCs (% of GDP)	BIS via FRED	QFINAM770A (Total credit to NFCs, % GDP)	Q	lin	masked 2020Q2–Q4	**NEW quantity.** BIS broad credit (loans+bonds+cross-border). obs 2009Q1=107.6..2025Q4=114.5
LENDR_FI	ECB MIR composite cost of borrowing, NFCs (new business, % p.a.)	ECB MIR	MIR M.FI.B.A2A. A.R.A.2240.EUR. N	M→Q	lin	observed (financial)	**NEW price.** 67/68 grid (one gap 2019Q3 filled by Gibbs). obs 2.88..3.54

20.23 Luxembourg (LU)

Luxembourg is a standalone euro-member BVAR with $n = 23$ variables, $p = 3$ lags, estimated on 2009Q1–2025Q4 ($T = 68$). Macro variables are COVID-masked 2020Q2–Q4 and re-imputed by the Durbin–Koopman smoother; financials are kept observed.

Table 73: Luxembourg (LU) — variable coverage.

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
GDP_LU	LU real GDP	Eurostat	namq_10_gdp Q. CLV15_MEUR. SCA.B1GQ.LU	Q	log	masked 2020Q2–Q4	Chain-linked volume, SCA
CONS_LU	Real household consumption	Eurostat	namq_10_gdp ...P31_S14_S15.LU	Q	log	masked 2020Q2–Q4	
INV_LU	Real gross fixed capital form.	Eurostat	namq_10_gdp ...P51G.LU	Q	log	masked 2020Q2–Q4	
GOV_LU	Real government consumption	Eurostat	namq_10_gdp ...P3_S13.LU	Q	log	masked 2020Q2–Q4	

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
EXP_LU	Real exports	Eurostat	namq_10_gdp ...P6.LU	Q	log	masked 2020Q2–Q4	
IMP_LU	Real imports	Eurostat	namq_10_gdp ...P7.LU	Q	log	masked 2020Q2–Q4	
HICP_LU	HICP index (2015=100)	Eurostat	prc_hicp_midx M.I15.CP00.LU	M→Q	log	masked 2020Q2–Q4	Monthly avg. to quarterly
LR_LU	Unemployment rate (%)	Eurostat	une_rt_q Q.SA.TOTAL.PC_ACT.T.LU	Q	lin	masked 2020Q2–Q4	
HPI_LU	Real house price index (HPI/HICP, 2015=100)	Eurostat	prc_hpi_q Q.TOTAL.I15_Q.LU ÷ HICP	Q	log	observed (financial)	Nominal HPI deflated by HICP
NEER_LU	NEER, IC42 trade-weighted	Eurostat	ert_eff_ic_q Q.NEER_IC42.I15.LU	Q	log	observed (financial)	42-partner nominal eff. FX
SPREAD_LU	10Y sovereign spread vs Bund	Eurostat	irt_lt_mcby_m M.MCBY.LU − M.MCBY.DE	M→Q	lin	observed (financial)	Own 10Y minus Bund
POLRATE_LU	ECB Deposit Facility Rate (policy)	ECB	FM D.U2.EUR.4F. KR.DFR.LEV	D→Q	lin	observed (financial)	External; shared by all members
EACORP	EA corporate bond yield (non-MFI)	Bundesbank	BBSIS/M.I.UMR. RD.EUR.X2000. B.A.A.R.A.A. _Z._Z.A	D→Q	lin	observed (financial)	External; shared anchor. *(Corrected 2026-06-18: genuine corporate yield — Bundesbank non-MFI enterprises, issuer class X2000.)*
RATE_LU	3M rate (EURIBOR)	Eurostat	irt_st_q Q.IRT_M3.EA	Q	lin	observed (financial)	External; shared 3M EURIBOR
YIELD10_DE	DE 10Y Bund (spread base)	Eurostat	irt_lt_mcby_m M.MCBY.DE	M→Q	lin	observed (financial)	External
ROWEA_GDP_LU	Rest-of-EA real GDP	Eurostat	EA20 namq_10_gdp de-aggregated (GDP-weighted)	Q	log	masked 2020Q2–Q4	EA20 ex-LU, GDP-weighted
ROWEA_HICP_LU	Rest-of-EA HICP	Eurostat	EA20 prc_hicp_midx de-aggregated	Q	log	masked 2020Q2–Q4	EA20 ex-LU, GDP-weighted
FREV_GDP	Revenue (% of GDP)	Eurostat	gov_10q_ggnfa Q...TR.LU (4q sum/GDP)	Q	lin	masked 2020Q2–Q4	Prior WN
FPEXP_GDP	Primary expenditure (% of GDP)	Eurostat	gov_10q_ggnfa TE−D41PAY (4q sum/GDP)	Q	lin	masked 2020Q2–Q4	Prior WN

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
`EFFI`	Effective interest rate (% p.a.)	Eurostat	`gov_10q` interest/lagged debt	Q	lin	observed (financial)	Prior RW
`SFA`	Stock-flow adjustment (% of GDP)	Eurostat	`gov_10q_ggdebt` debt recursion residual	Q	lin	masked 2020Q2–Q4	Prior WN
`CREDIT_GDP_LU`	Credit to NFCs (% of GDP)	BIS via FRED	`QLUNAM770A` (Total credit to NFCs, % GDP)	Q	lin	masked 2020Q2–Q4	**NEW quantity.** BIS broad credit; very high (∼270–360% GDP, captive-finance hub). obs 2009Q1=268.5..2025Q4=358.8
`LENDR_LU`	ECB MIR bank lending rate to NFCs (outstanding amounts, % p.a.)	ECB MIR	`MIR` M.LU.B.A20.A.R.A.2240.EUR.O	M→Q	lin	observed (financial)	**NEW price.** *[FLAG: NON-CANONICAL MIR variant — OUTSTANDING-amounts rate (A20...O), not the new-business cost-of-borrowing (A2A...N), which is sparse pre-2015. Full 68/68 coverage.]* obs 3.65..3.40

20.24 Latvia (LV)

Latvia is a standalone euro-member BVAR with $n = 22$ variables (one fewer than the group — its credit *price* is dropped, see below), $p = 3$ lags, estimated on 2009Q1–2025Q4 ($T = 68$). Macro variables are COVID-masked 2020Q2–Q4 and re-imputed by the Durbin–Koopman smoother; financials are kept observed.

Table 74: Latvia (LV) — variable coverage.

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
`GDP_LV`	LV real GDP	Eurostat	`namq_10_gdp` Q.CLV15_MEUR.SCA.B1GQ.LV	Q	log	masked 2020Q2–Q4	Chain-linked volume, SCA
`CONS_LV`	Real household consumption	Eurostat	`namq_10_gdp` ...P31_S14_S15.LV	Q	log	masked 2020Q2–Q4	

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
INV_LV	Real gross fixed capital form.	Eurostat	namq_10_gdp ...P51G.LV	Q	log	masked 2020Q2–Q4	
GOV_LV	Real government consumption	Eurostat	namq_10_gdp ...P3_S13.LV	Q	log	masked 2020Q2–Q4	
EXP_LV	Real exports	Eurostat	namq_10_gdp ...P6.LV	Q	log	masked 2020Q2–Q4	
IMP_LV	Real imports	Eurostat	namq_10_gdp ...P7.LV	Q	log	masked 2020Q2–Q4	
HICP_LV	HICP index (2015=100)	Eurostat	prc_hicp_midx M.I15.CP00.LV	M→Q	log	masked 2020Q2–Q4	Monthly avg. to quarterly
LR_LV	Unemployment rate (%)	Eurostat	une_rt_q Q.SA.TOTAL.PC_ACT.T.LV	Q	lin	masked 2020Q2–Q4	
HPI_LV	Real house price index (HPI/HICP, 2015=100)	Eurostat	prc_hpi_q Q.TOTAL.I15_Q.LV ÷ HICP	Q	log	observed (financial)	Nominal HPI deflated by HICP
NEER_LV	NEER, IC42 trade-weighted	Eurostat	ert_eff_ic_q Q.NEER_IC42.I15.LV	Q	log	observed (financial)	42-partner nominal eff. FX
SPREAD_LV	10Y sovereign spread vs Bund	Eurostat	irt_lt_mcby_m M.MCBY.LV − M.MCBY.DE	M→Q	lin	observed (financial)	Own 10Y minus Bund
POLRATE_LV	ECB Deposit Facility Rate (policy)	ECB	FM D.U2.EUR.4F. KR.DFR.LEV	D→Q	lin	observed (financial)	External; shared by all members
EACORP	EA corporate bond yield (non-MFI)	Bundesbank	BBSIS/M.I.UMR. RD.EUR.X2000. B.A.A.R.A.A. _Z._Z.A	D→Q	lin	observed (financial)	External; shared anchor (also the LV credit-price proxy, since LENDR_LV is dropped). *(Corrected 2026-06-18: genuine corporate yield — Bundesbank non-MFI enterprises, issuer class X2000.)*
RATE_LV	3M rate (EURIBOR)	Eurostat	irt_st_q Q.IRT_M3.EA	Q	lin	observed (financial)	External; shared 3M EURIBOR
YIELD10_DE	DE 10Y Bund (spread base)	Eurostat	irt_lt_mcby_m M.MCBY.DE	M→Q	lin	observed (financial)	External

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
ROWEA_GDP_LV	Rest-of-EA real GDP	Eurostat	EA20 namq_10_gdp de-aggregated (GDP-weighted)	Q	log	masked 2020Q2–Q4	EA20 ex-LV, GDP-weighted
ROWEA_HICP_LV	Rest-of-EA HICP	Eurostat	EA20 prc_hicp_midx de-aggregated	Q	log	masked 2020Q2–Q4	EA20 ex-LV, GDP-weighted
FREV_GDP	Revenue (% of GDP)	Eurostat	gov_10q_ggnfa Q...TR.LV (4q sum/GDP)	Q	lin	masked 2020Q2–Q4	Prior WN
FPEXP_GDP	Primary expenditure (% of GDP)	Eurostat	gov_10q_ggnfa TE−D41PAY (4q sum/GDP)	Q	lin	masked 2020Q2–Q4	Prior WN
EFFI	Effective interest rate (% p.a.)	Eurostat	gov_10q interest/lagged debt	Q	lin	observed (financial)	Prior RW
SFA	Stock-flow adjustment (% of GDP)	Eurostat	gov_10q_ggdebt debt recursion residual	Q	lin	masked 2020Q2–Q4	Prior WN
CREDIT_GDP_LV	Credit to NFCs (% of GDP)	ECB BSI + Eurostat (constructed)	100×BSI M.LV. N.A.A20.A.1.U2. 2240.Z01.E ÷ 4q Σnamq_10_gdp CP	Q	lin	masked 2020Q2–Q4	**NEW quantity.** *[FLAG: CONSTRUCTED (non-BIS) NARROW MFI-loan measure; reduced comparability; large secular decline (56→16% GDP) is real post-2009 deleveraging, not a denominator artifact. obs 2010Q3=56.0..2025Q4=16.6.]*

[FLAG: LV credit PRICE DROPPED. The ECB MIR NFC lending rate has only 41/67 grid quarters (2009–2017 riddled with gaps), so no LENDR_LV variable is included ($n = 22$, one fewer than the group). EACORP is the external corporate-yield anchor.]

20.25 Lithuania (LT)

Lithuania is a standalone euro-member BVAR with $n = 23$ variables, $p = 3$ lags, estimated on 2009Q1–2025Q4 ($T = 68$). Macro variables are COVID-masked 2020Q2–Q4 and re-imputed by the Durbin–Koopman smoother; financials are kept observed.

Table 75: Lithuania (LT) — variable coverage.

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
GDP_LT	LT real GDP	Eurostat	namq_10_gdp Q.CLV15_MEUR.SCA.B1GQ.LT	Q	log	masked 2020Q2–Q4	Chain-linked volume, SCA
CONS_LT	Real household consumption	Eurostat	namq_10_gdp ...P31_S14_S15.LT	Q	log	masked 2020Q2–Q4	
INV_LT	Real gross fixed capital form.	Eurostat	namq_10_gdp ...P51G.LT	Q	log	masked 2020Q2–Q4	
GOV_LT	Real government consumption	Eurostat	namq_10_gdp ...P3_S13.LT	Q	log	masked 2020Q2–Q4	
EXP_LT	Real exports	Eurostat	namq_10_gdp ...P6.LT	Q	log	masked 2020Q2–Q4	
IMP_LT	Real imports	Eurostat	namq_10_gdp ...P7.LT	Q	log	masked 2020Q2–Q4	
HICP_LT	HICP index (2015=100)	Eurostat	prc_hicp_midx M.I15.CP00.LT	M→Q	log	masked 2020Q2–Q4	Monthly avg. to quarterly
LR_LT	Unemployment rate (%)	Eurostat	une_rt_q Q.SA.TOTAL.PC_ACT.T.LT	Q	lin	masked 2020Q2–Q4	
HPI_LT	Real house price index (HPI/HICP, 2015=100)	Eurostat	prc_hpi_q Q.TOTAL.I15_Q.LT ÷ HICP	Q	log	observed (financial)	Nominal HPI deflated by HICP
NEER_LT	NEER, IC42 trade-weighted	Eurostat	ert_eff_ic_q Q.NEER_IC42.I15.LT	Q	log	observed (financial)	42-partner nominal eff. FX
SPREAD_LT	10Y sovereign spread vs Bund	Eurostat	irt_lt_mcby_m M.MCBY.LT − M.MCBY.DE	M→Q	lin	observed (financial)	Own 10Y minus Bund
POLRATE_LT	ECB Deposit Facility Rate (policy)	ECB	FM D.U2.EUR.4F.KR.DFR.LEV	D→Q	lin	observed (financial)	External; shared by all members
EACORP	EA corporate bond yield (non-MFI)	Bundesbank	BBSIS/M.I.UMR.RD.EUR.X2000.B.A.A.R.A.A._Z._Z.A	D→Q	lin	observed (financial)	External; shared anchor. *(Corrected 2026-06-18: genuine corporate yield — Bundesbank non-MFI enterprises, issuer class X2000.)*
RATE_LT	3M rate (EURIBOR)	Eurostat	irt_st_q Q.IRT_M3.EA	Q	lin	observed (financial)	External; shared 3M EURIBOR

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
YIELD10_DE	DE 10Y Bund (spread base)	Eurostat	irt_lt_mcby_m M.MCBY.DE	M→Q	lin	observed (financial)	External
ROWEA_GDP_LT	Rest-of-EA real GDP	Eurostat	EA20 namq_10_gdp de-aggregated (GDP-weighted)	Q	log	masked 2020Q2–Q4	EA20 ex-LT, GDP-weighted
ROWEA_HICP_LT	Rest-of-EA HICP	Eurostat	EA20 prc_hicp_midx de-aggregated	Q	log	masked 2020Q2–Q4	EA20 ex-LT, GDP-weighted
FREV_GDP	Revenue (% of GDP)	Eurostat	gov_10q_ggnfa Q...TR.LT (4q sum/GDP)	Q	lin	masked 2020Q2–Q4	Prior WN
FPEXP_GDP	Primary expenditure (% of GDP)	Eurostat	gov_10q_ggnfa TE−D41PAY (4q sum/GDP)	Q	lin	masked 2020Q2–Q4	Prior WN
EFFI	Effective interest rate (% p.a.)	Eurostat	gov_10q interest/lagged debt	Q	lin	observed (financial)	Prior RW
SFA	Stock-flow adjustment (% of GDP)	Eurostat	gov_10q_ggdebt debt recursion residual	Q	lin	masked 2020Q2–Q4	Prior WN
CREDIT_GDP_LT	Credit to NFCs (% of GDP)	ECB BSI + Eurostat (constructed)	100×BSI M.LT. N.A.A20.A.1.U2. 2240.Z01.E ÷ 4q Σnamq_10_gdp CP	Q	lin	masked 2020Q2–Q4	**NEW quantity.** *[FLAG: CONSTRUCTED (non-BIS) NARROW MFI-loan measure; reduced comparability; secular decline (32→17% GDP) is real post-2009 deleveraging. obs 2009Q1=32.3..2025Q4=17.1.]*
LENDR_LT	ECB MIR composite cost of borrowing, NFCs (new business, % p.a.)	ECB MIR	MIR M.LT.B.A2A. A.R.A.2240.EUR. N	M→Q	lin	observed (financial)	**NEW price.** Full 68/68 grid. obs 5.36..4.61

20.26 Slovakia (SK)

Slovakia is a standalone euro-member BVAR with $n = 23$ variables, $p = 3$ lags, estimated on 2009Q1–2025Q4 ($T = 68$). Macro variables are COVID-masked 2020Q2–Q4 and re-imputed by the Durbin–Koopman smoother; financials are kept observed.

Table 76: Slovakia (SK) — variable coverage.

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
GDP_SK	SK real GDP	Eurostat	namq_10_gdp Q. CLV15_MEUR. SCA.B1GQ.SK	Q	log	masked 2020Q2–Q4	Chain-linked volume, SCA
CONS_SK	Real household consumption	Eurostat	namq_10_gdp ...P31_S14_S15.SK	Q	log	masked 2020Q2–Q4	
INV_SK	Real gross fixed capital form.	Eurostat	namq_10_gdp ...P51G.SK	Q	log	masked 2020Q2–Q4	
GOV_SK	Real government consumption	Eurostat	namq_10_gdp ...P3_S13.SK	Q	log	masked 2020Q2–Q4	
EXP_SK	Real exports	Eurostat	namq_10_gdp ...P6.SK	Q	log	masked 2020Q2–Q4	
IMP_SK	Real imports	Eurostat	namq_10_gdp ...P7.SK	Q	log	masked 2020Q2–Q4	
HICP_SK	HICP index (2015=100)	Eurostat	prc_hicp_midx M.I15.CP00.SK	M→Q	log	masked 2020Q2–Q4	Monthly avg. to quarterly
LR_SK	Unemployment rate (%)	Eurostat	une_rt_q Q.SA.TOTAL.PC_ACT.T.SK	Q	lin	masked 2020Q2–Q4	
HPI_SK	Real house price index (HPI/HICP, 2015=100)	Eurostat	prc_hpi_q Q.TOTAL.I15_Q.SK ÷ HICP	Q	log	observed (financial)	Nominal HPI deflated by HICP
NEER_SK	NEER, IC42 trade-weighted	Eurostat	ert_eff_ic_q Q.NEER_IC42.I15.SK	Q	log	observed (financial)	42-partner nominal eff. FX
SPREAD_SK	10Y sovereign spread vs Bund	Eurostat	irt_lt_mcby_m M.MCBY.SK − M.MCBY.DE	M→Q	lin	observed (financial)	Own 10Y minus Bund
POLRATE_SK	ECB Deposit Facility Rate (policy)	ECB	FM D.U2.EUR.4F. KR.DFR.LEV	D→Q	lin	observed (financial)	External; shared by all members
EACORP	EA corporate bond yield (non-MFI)	Bundesbank	BBSIS/M.I.UMR. RD.EUR.X2000. B.A.A.R.A.A. _Z._Z.A	D→Q	lin	observed (financial)	External; shared anchor. *(Corrected 2026-06-18: genuine corporate yield — Bundesbank non-MFI enterprises, issuer class X2000.)*
RATE_SK	3M rate (EURIBOR)	Eurostat	irt_st_q Q.IRT_M3.EA	Q	lin	observed (financial)	External; shared 3M EURIBOR

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
YIELD10_DE	DE 10Y Bund (spread base)	Eurostat	irt_lt_mcby_m M.MCBY.DE	M→Q	lin	observed (financial)	External
ROWEA_GDP_SK	Rest-of-EA real GDP	Eurostat	EA20 namq_10_gdp de-aggregated (GDP-weighted)	Q	log	masked 2020Q2–Q4	EA20 ex-SK, GDP-weighted
ROWEA_HICP_SK	Rest-of-EA HICP	Eurostat	EA20 prc_hicp_midx de-aggregated	Q	log	masked 2020Q2–Q4	EA20 ex-SK, GDP-weighted
FREV_GDP	Revenue (% of GDP)	Eurostat	gov_10q_ggnfa Q...TR.SK (4q sum/GDP)	Q	lin	masked 2020Q2–Q4	Prior WN
FPEXP_GDP	Primary expenditure (% of GDP)	Eurostat	gov_10q_ggnfa TE−D41PAY (4q sum/GDP)	Q	lin	masked 2020Q2–Q4	Prior WN
EFFI	Effective interest rate (% p.a.)	Eurostat	gov_10q interest/lagged debt	Q	lin	observed (financial)	Prior RW
SFA	Stock-flow adjustment (% of GDP)	Eurostat	gov_10q_ggdebt debt recursion residual	Q	lin	masked 2020Q2–Q4	Prior WN
CREDIT_GDP_SK	Credit to NFCs (% of GDP)	ECB BSI + Eurostat (constructed)	100×BSI M.SK. N.A.A20.A.1.U2. 2240.Z01.E ÷ 4q Σnamq_10_gdp CP	Q	lin	masked 2020Q2–Q4	**NEW quantity.** *[FLAG: CONSTRUCTED (non-BIS) NARROW MFI-loan measure; reduced comparability. obs 2009Q1=23.6..2025Q4=18.6.]*
LENDR_SK	ECB MIR composite cost of borrowing, NFCs (new business, % p.a.)	ECB MIR	MIR M.SK.B.A2A. A.R.A.2240.EUR. N	M→Q	lin	observed (financial)	**NEW price.** Full 68/68 grid. obs 3.43..4.15

20.27 Slovenia (SI)

Slovenia is a standalone euro-member BVAR with $n = 23$ variables, $p = 3$ lags, estimated on 2009Q1–2025Q4 ($T = 68$). Macro variables are COVID-masked 2020Q2–Q4 and re-imputed by the Durbin–Koopman smoother; financials are kept observed.

Table 77: Slovenia (SI) — variable coverage.

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
GDP_SI	SI real GDP	Eurostat	namq_10_gdp Q. CLV15_MEUR. SCA.B1GQ.SI	Q	log	masked 2020Q2–Q4	Chain-linked volume, SCA

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
`CONS_SI`	Real household consumption	Eurostat	`namq_10_gdp` ...P31_S14_S15.SI	Q	log	masked 2020Q2–Q4	
`INV_SI`	Real gross fixed capital form.	Eurostat	`namq_10_gdp` ...P51G.SI	Q	log	masked 2020Q2–Q4	
`GOV_SI`	Real government consumption	Eurostat	`namq_10_gdp` ...P3_S13.SI	Q	log	masked 2020Q2–Q4	
`EXP_SI`	Real exports	Eurostat	`namq_10_gdp` ...P6.SI	Q	log	masked 2020Q2–Q4	
`IMP_SI`	Real imports	Eurostat	`namq_10_gdp` ...P7.SI	Q	log	masked 2020Q2–Q4	
`HICP_SI`	HICP index (2015=100)	Eurostat	`prc_hicp_midx` M.I15.CP00.SI	M→Q	log	masked 2020Q2–Q4	Monthly avg. to quarterly
`LR_SI`	Unemployment rate (%)	Eurostat	`une_rt_q` Q.SA.TOTAL.PC_ACT.T.SI	Q	lin	masked 2020Q2–Q4	
`HPI_SI`	Real house price index (HPI/HICP, 2015=100)	Eurostat	`prc_hpi_q` Q.TOTAL.I15_Q.SI ÷ HICP	Q	log	observed (financial)	Nominal HPI deflated by HICP
`NEER_SI`	NEER, IC42 trade-weighted	Eurostat	`ert_eff_ic_q` Q.NEER_IC42.I15.SI	Q	log	observed (financial)	42-partner nominal eff. FX
`SPREAD_SI`	10Y sovereign spread vs Bund	Eurostat	`irt_lt_mcby_m` M.MCBY.SI − M.MCBY.DE	M→Q	lin	observed (financial)	Own 10Y minus Bund
`POLRATE_SI`	ECB Deposit Facility Rate (policy)	ECB	`FM` D.U2.EUR.4F. KR.DFR.LEV	D→Q	lin	observed (financial)	External; shared by all members
`EACORP`	EA corporate bond yield (non-MFI)	Bundesbank	`BBSIS/M.I.UMR. RD.EUR.X2000. B.A.A.R.A.A. _Z._Z.A`	D→Q	lin	observed (financial)	External; shared anchor. *(Corrected 2026-06-18: genuine corporate yield — Bundesbank non-MFI enterprises, issuer class X2000.)*
`RATE_SI`	3M rate (EURIBOR)	Eurostat	`irt_st_q` Q.IRT_M3.EA	Q	lin	observed (financial)	External; shared 3M EURIBOR
`YIELD10_DE`	DE 10Y Bund (spread base)	Eurostat	`irt_lt_mcby_m` M.MCBY.DE	M→Q	lin	observed (financial)	External
`ROWEA_GDP_SI`	Rest-of-EA real GDP	Eurostat	EA20 `namq_10_gdp` de-aggregated (GDP-weighted)	Q	log	masked 2020Q2–Q4	EA20 ex-SI, GDP-weighted

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
`ROWEA_HICP_SI`	Rest-of-EA HICP	Eurostat	EA20 `prc_hicp_midx` de-aggregated	Q	log	masked 2020Q2–Q4	EA20 ex-SI, GDP-weighted
`FREV_GDP`	Revenue (% of GDP)	Eurostat	`gov_10q_ggnfa` Q...TR.SI (4q sum/GDP)	Q	lin	masked 2020Q2–Q4	Prior WN
`FPEXP_GDP`	Primary expenditure (% of GDP)	Eurostat	`gov_10q_ggnfa` TE−D41PAY (4q sum/GDP)	Q	lin	masked 2020Q2–Q4	Prior WN
`EFFI`	Effective interest rate (% p.a.)	Eurostat	`gov_10q` interest/lagged debt	Q	lin	observed (financial)	Prior RW
`SFA`	Stock-flow adjustment (% of GDP)	Eurostat	`gov_10q_ggdebt` debt recursion residual	Q	lin	masked 2020Q2–Q4	Prior WN
`CREDIT_GDP_SI`	Credit to NFCs (% of GDP)	ECB BSI + Eurostat (constructed)	100×`BSI` M.SI. N.A.A20.A.1.U2. 2240.Z01.E ÷ 4q Σ`namq_10_gdp` CP	Q	lin	masked 2020Q2–Q4	**NEW quantity.** *[FLAG: CONSTRUCTED (non-BIS) NARROW MFI-loan measure; reduced comparability. The 2013–14 cliff (56→16% GDP) is the BAMC "bad bank" NPL transfer off MFI books, not a denominator artifact. obs 2009Q1=56.2..2025Q4=16.5.]*
`LENDR_SI`	ECB MIR composite cost of borrowing, NFCs (new business, % p.a.)	ECB MIR	`MIR` M.SI.B.A2A.A. R.A.2240.EUR.N	M→Q	lin	observed (financial)	**NEW price.** Full 68/68 grid. obs 5.57..3.54

20.28 EU non-euro and hard-gap small blocs: data conventions

This section documents five coupled GVAR member blocs: the three EU *non-euro* economies (Czechia, Hungary, Poland) and two non-EU blocs (Costa Rica, Iceland) whose credit quantity required a *constructed* (non-BIS) measure. Each is a Bayesian VAR with $p = 3$ lags and a hierarchical Minnesota/GLP prior, coupled into the GVAR through trade-weighted foreign *star* variables that are added at solve time. The per-bloc tables below list the domestic block plus the four global commons; the three `ROW*_<CC>` stars are not part of the staged `spec.json` and are appended by the estimator.

Two transformation conventions are used throughout. log means the variable is stored as $100 \cdot \log(\text{level})$ and therefore enters the model in growth space (displayed as $4\,\Delta \log$ = quarter-on-quarter annualized growth); lin means the level (percent, percentage points, or %-of-GDP). Under the COVID-as-missing treatment, all macro variables (isFinancial= 0) are masked over **2020Q2–2020Q4** and re-imputed by the Durbin–Koopman simulation smoother, while financial variables (isFinancial= 1) are kept observed through COVID — with one deliberate exception: the credit *quantity* CREDIT_GDP_<CC> is masked 2020Q2–Q4 because it inherits the collapse of its GDP denominator.

The new credit block (flagged). Each bloc gains a two-variable credit block appended after all existing variables: a *quantity*, CREDIT_GDP_<CC> (credit to non-financial corporations as a % of GDP; lin, prior RW, masked 2020Q2–Q4), and a *price*, LENDR_<CC> (bank lending rate to NFCs; lin, prior RW, observed through COVID). For **CZ, HU, PL** the quantity is the BIS "Total credit to NFCs, % of GDP" (broad credit: bank loans, debt securities, and cross-border lending) pulled from FRED, mnemonic Q<ISO2>NAM770A; the price is the ECB MIR new-business NFC bank lending rate in local currency. For **CR, IS** neither a BIS %-of-GDP table nor a clean national bank lending rate is published, so the quantity is **constructed** (non-BIS) as $100\times$(IMF IFS ODC claims on the private sector)/(4-quarter trailing nominal GDP) and the price is the IMF IFS FILR_PA lending rate. The salient caveats — the broad/narrow-credit comparability gap, the non-canonical MIR variant used for HU and PL, and the constructed-quantity flag for CR and IS — are recorded in the per-economy *[FLAG]* notes below.

20.29 Czechia (CZ)

Czechia is a coupled GVAR bloc with $n = 18$ domestic-plus-global variables (the three foreign ROW*_CZ stars are added at GVAR-estimation time), $p = 3$ lags, estimated on the common quarterly sample **2009Q1–2025Q4** ($T = 68$). Macro variables (isFinancial= 0) are COVID-masked 2020Q2–Q4 and re-imputed by the Durbin–Koopman smoother; financials are kept observed. The domestic real national accounts, prices, and labour come from Eurostat; the policy rate and house price from the BIS; and the new credit block from BIS (quantity, via FRED) and the ECB MIR (price).

Table 78: Czechia (CZ) — variable coverage. log= $100 \log$ level (displayed as $4\Delta \log$, q/q ann. growth); lin= level.

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
GDP_CZ	Real GDP	Eurostat	namq_10_gdp Q.CLV15_MEUR.SCA.B1GQ.CZ	Q	log	masked 2020Q2–Q4	Chain-linked volume, SCA
CONS_CZ	Real household consumption	Eurostat	namq_10_gdp ...P31_S14_S15.CZ	Q	log	masked 2020Q2–Q4	

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
INV_CZ	Real gross fixed cap. form.	Eurostat	namq_10_gdp ...P51G.CZ	Q	log	masked 2020Q2–Q4	
GOV_CZ	Real government consumption	Eurostat	namq_10_gdp ...P3_S13.CZ	Q	log	masked 2020Q2–Q4	
EXP_CZ	Real exports	Eurostat	namq_10_gdp ...P6.CZ	Q	log	masked 2020Q2–Q4	
IMP_CZ	Real imports	Eurostat	namq_10_gdp ...P7.CZ	Q	log	masked 2020Q2–Q4	
HICP_CZ	HICP index (2015=100)	Eurostat	prc_hicp_midx M.I15.CP00.CZ	M→Q	log	masked 2020Q2–Q4	Monthly avg. to quarterly
LR_CZ	Unemployment rate (%)	Eurostat	une_rt_q Q.SA.TOTAL.PC_ACT.T.CZ	Q	lin	masked 2020Q2–Q4	
YIELD10_CZ	10Y government bond yield	Eurostat	irt_lt_mcby_m M.MCBY.CZ	M→Q	lin	observed (financial)	Maastricht 10Y
NEER_CZ	Nominal effective FX (IC42)	Eurostat	ert_eff_ic_q Q.NEER_IC42.I15.CZ	Q	log	observed (financial)	42-partner nominal eff. FX
POLRATE_CZ	Central-bank policy rate	BIS	WS_CBPOL M.CZ	M→Q	lin	observed (financial)	3-month q-avg of monthly EoP
HP_CZ	Real residential property price	BIS (via FRED)	QCZR628BIS	Q	log	observed (financial)	Real RPP, 2010=100
GLB_OIL	WTI crude oil price	FRED	WTISPLC (q-avg)	M→Q	log	observed (financial)	Global common (spot WTI, $/bbl)
GLB_USLR	US 10Y Treasury yield	FRED	DGS10 *[FLAG]*	D→Q	lin	observed (financial)	Global common
GLB_USBAA	US Baa corporate yield	FRED	BAA *[FLAG]*	M→Q	lin	observed (financial)	Global common (credit proxy)
GLB_USEQ	US equity (S&P 500)	FRED	SP500-class *[FLAG]*	D→Q	log	observed (financial)	Global common
CREDIT_GDP_CZ	Credit to NFCs (% of GDP)	**BIS** (via FRED)	QCZNAM770A	Q	lin	**masked 2020Q2–Q4**	**NEW.** BIS broad NFC credit %GDP; 65 present, 3 masked, range 50.1–61.6%; inherits GDP-denominator mask

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
LENDR_CZ	NFC bank lending rate (% p.a.)	**ECB MIR**	`M.CZ.B.A2A.F.R.0.2240.CZK.N`	M→Q	lin	observed (financial)	**NEW.** New-business, ≤EUR1M, floating-rate/IRF≤1y, S.11 NFC, CZK; 68 present, range 2.14–9.24%

[FLAG: `CREDIT_GDP_CZ/LENDR_CZ` *variant.]* The price uses the local-currency *floating-rate* ≤EUR1M new-business MIR series (`...A2A.F.R.0...`, available from 2006-01, the LOCKED-spec pinned key) rather than the cleaner total-maturity composite (`...A2A.A.R.A...`), which only begins 2010-06 and would leave a leading gap on the 2009Q1 grid. No price was dropped.

20.30 Hungary (HU)

Hungary is a coupled GVAR bloc with $n = 18$ domestic-plus-global variables (three `ROW*_HU` stars added at estimation), $p = 3$ lags, estimated on **2009Q1–2025Q4** ($T = 68$). Macro variables are COVID-masked 2020Q2–Q4 and re-imputed; financials are kept observed. Sourcing mirrors CZ.

Table 79: Hungary (HU) — variable coverage. `log`= $100 \log$ level (displayed as $4\Delta \log$, q/q ann. growth); `lin`= level.

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
GDP_HU	Real GDP	Eurostat	`namq_10_gdp` Q. CLV15_MEUR. SCA.B1GQ.HU	Q	log	masked 2020Q2–Q4	Chain-linked volume, SCA
CONS_HU	Real household consumption	Eurostat	`namq_10_gdp` ...P31_S14_S15.HU	Q	log	masked 2020Q2–Q4	
INV_HU	Real gross fixed cap. form.	Eurostat	`namq_10_gdp` ...P51G.HU	Q	log	masked 2020Q2–Q4	
GOV_HU	Real government consumption	Eurostat	`namq_10_gdp` ...P3_S13.HU	Q	log	masked 2020Q2–Q4	
EXP_HU	Real exports	Eurostat	`namq_10_gdp` ...P6.HU	Q	log	masked 2020Q2–Q4	
IMP_HU	Real imports	Eurostat	`namq_10_gdp` ...P7.HU	Q	log	masked 2020Q2–Q4	

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
`HICP_HU`	HICP index (2015=100)	Eurostat	`prc_hicp_midx` M.I15.CP00.HU	M→Q	log	masked 2020Q2–Q4	Monthly avg. to quarterly
`LR_HU`	Unemployment rate (%)	Eurostat	`une_rt_q` Q.SA.TOTAL.PC_ACT.T.HU	Q	lin	masked 2020Q2–Q4	
`YIELD10_HU`	10Y government bond yield	Eurostat	`irt_lt_mcby_m` M.MCBY.HU	M→Q	lin	observed (financial)	Maastricht 10Y
`NEER_HU`	Nominal effective FX (IC42)	Eurostat	`ert_eff_ic_q` Q.NEER_IC42.I15.HU	Q	log	observed (financial)	42-partner nominal eff. FX
`POLRATE_HU`	Central-bank policy rate	BIS	`WS_CBPOL` M.HU	M→Q	lin	observed (financial)	3-month q-avg of monthly EoP
`HP_HU`	Real residential property price	BIS (via FRED)	`QHUR628BIS`	Q	log	observed (financial)	Real RPP, 2010=100
`GLB_OIL`	WTI crude oil price	FRED	`WTISPLC` (q-avg)	M→Q	log	observed (financial)	Global common (spot WTI, $/bbl)
`GLB_USLR`	US 10Y Treasury yield	FRED	`DGS10` *[FLAG]*	D→Q	lin	observed (financial)	Global common
`GLB_USBAA`	US Baa corporate yield	FRED	`BAA` *[FLAG]*	M→Q	lin	observed (financial)	Global common (credit proxy)
`GLB_USEQ`	US equity (S&P 500)	FRED	`SP500`-class *[FLAG]*	D→Q	log	observed (financial)	Global common
`CREDIT_GDP_HU`	Credit to NFCs (% of GDP)	**BIS** (via FRED)	`QHUNAM770A`	Q	lin	**masked 2020Q2–Q4**	**NEW.** BIS broad NFC credit %GDP; 65 present, 3 masked, range 64.2–94.9%
`LENDR_HU`	NFC bank lending rate (% p.a.)	**ECB MIR**	`M.HU.B.A2A.F.R.0.2240.HUF.N`	M→Q	lin	observed (financial)	**NEW.** New-business floating-rate ≤EUR1M, S.11 NFC, HUF; starts 2010Q1 (4 leading nulls handled by Gibbs); range 2.64–16.37%

[FLAG: `LENDR_HU` non-canonical MIR variant.] The LOCKED spec gave only "from ∼2010" without pinning an exact MIR key. The standard total-maturity variant (`...A2A.A.R.A...`) is EUR-denominated and starts only 2017-09 (too short); the local-currency floating-rate variant used here starts 2010-01, consistent with the hint and with the CZ-pinned key form, and is a genuine NFC new-business lending rate. Four leading quarters (2009Q1–Q4) are null and are handled by the

COVID-as-missing Gibbs ragged-edge machinery. *[FLAG: CREDIT_GDP_HU composition.]* The forint NFC credit level is partly inflated by FX revaluation of FX-denominated corporate loans and by high HU inflation; the %-of-GDP ratio dampens but does not fully remove this nominal component.

20.31 Poland (PL)

Poland is a coupled GVAR bloc with $n = 18$ domestic-plus-global variables (three `ROW*_PL` stars added at estimation), $p = 3$ lags, estimated on **2009Q1–2025Q4** ($T = 68$). Macro variables are COVID-masked 2020Q2–Q4 and re-imputed; financials are kept observed. Sourcing mirrors CZ.

Table 80: Poland (PL) — variable coverage. `log`= 100 log level (displayed as $4\Delta\log$, q/q ann. growth); `lin`= level.

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
`GDP_PL`	Real GDP	Eurostat	`namq_10_gdp` Q.CLV15_MEUR.SCA.B1GQ.PL	Q	log	masked 2020Q2–Q4	Chain-linked volume, SCA
`CONS_PL`	Real household consumption	Eurostat	`namq_10_gdp` ...P31_S14_S15.PL	Q	log	masked 2020Q2–Q4	
`INV_PL`	Real gross fixed cap. form.	Eurostat	`namq_10_gdp` ...P51G.PL	Q	log	masked 2020Q2–Q4	
`GOV_PL`	Real government consumption	Eurostat	`namq_10_gdp` ...P3_S13.PL	Q	log	masked 2020Q2–Q4	
`EXP_PL`	Real exports	Eurostat	`namq_10_gdp` ...P6.PL	Q	log	masked 2020Q2–Q4	
`IMP_PL`	Real imports	Eurostat	`namq_10_gdp` ...P7.PL	Q	log	masked 2020Q2–Q4	
`HICP_PL`	HICP index (2015=100)	Eurostat	`prc_hicp_midx` M.I15.CP00.PL	M→Q	log	masked 2020Q2–Q4	Monthly avg. to quarterly
`LR_PL`	Unemployment rate (%)	Eurostat	`une_rt_q` Q.SA.TOTAL.PC_ACT.T.PL	Q	lin	masked 2020Q2–Q4	
`YIELD10_PL`	10Y government bond yield	Eurostat	`irt_lt_mcby_m` M.MCBY.PL	M→Q	lin	observed (financial)	Maastricht 10Y
`NEER_PL`	Nominal effective FX (IC42)	Eurostat	`ert_eff_ic_q` Q.NEER_IC42.I15.PL	Q	log	observed (financial)	42-partner nominal eff. FX
`POLRATE_PL`	Central-bank policy rate	BIS	`WS_CBPOL` M.PL	M→Q	lin	observed (financial)	3-month q-avg of monthly EoP
`HP_PL`	Real residential property price	BIS (via FRED)	`QPLR628BIS`	Q	log	observed (financial)	Real RPP, 2010=100

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
GLB_OIL	WTI crude oil price	FRED	WTISPLC (q-avg)	M→Q	log	observed (financial)	Global common (spot WTI, $/bbl)
GLB_USLR	US 10Y Treasury yield	FRED	DGS10 *[FLAG]*	D→Q	lin	observed (financial)	Global common
GLB_USBAA	US Baa corporate yield	FRED	BAA *[FLAG]*	M→Q	lin	observed (financial)	Global common (credit proxy)
GLB_USEQ	US equity (S&P 500)	FRED	SP500-class *[FLAG]*	D→Q	log	observed (financial)	Global common
CREDIT_GDP_PL	Credit to NFCs (% of GDP)	**BIS** (via FRED)	QPLNAM770A	Q	lin	**masked 2020Q2–Q4**	**NEW.** BIS broad NFC credit %GDP; 65 present, 3 masked, range 34.7–49.5%
LENDR_PL	NFC bank lending rate (% p.a.)	**ECB MIR**	M.PL.B.A2A.F.R.0.2240.PLN.N	M→Q	lin	observed (financial)	**NEW.** New-business floating-rate ≤EUR1M, S.11 NFC, PLN; 68 present, range 1.82–8.62%

[FLAG: LENDR_PL non-canonical MIR variant.] The clean total-maturity composite (...A2A.A.R.A...) begins only 2017-08 — far too short for the 2009Q1 grid — so the local-currency floating-rate ≤EUR1M new-business series (full coverage from 2005, gap-free, and the dominant WIBOR-linked PL corporate-loan type) is used instead. Overlap checks confirm it tracks the composite within ∼30bp; no splice and no back-fill needed.

20.32 Costa Rica (CR)

Costa Rica is a coupled GVAR bloc with $n = 13$ domestic-plus-global variables (three ROW*_CR stars added at estimation), $p = 3$ lags, estimated on **2010Q3–2025Q4** ($T = 62$). It is a compact emerging-market specification: real activity is proxied by industrial production, external trade is in USD, and there is no separate 10Y yield or house-price variable. Macro variables are COVID-masked 2020Q2–Q4 and re-imputed; financials (including the policy rate and the new credit price) are kept observed.

Table 81: Costa Rica (CR) — variable coverage. `log`= 100 log level (displayed as 4Δ log, q/q ann. growth); `lin`= level.

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
`GDP_CR`	Real GDP (IP proxy)	*[FLAG]*	*[FLAG: not determinable from current builders]*	Q	log	masked 2020Q2–Q4	Activity proxied by industrial production
`EXP_CR`	Goods exports (USD)	*[FLAG]*	*[FLAG: not determinable from current builders]*	Q	log	masked 2020Q2–Q4	USD goods exports
`IMP_CR`	Goods imports (USD)	*[FLAG]*	*[FLAG: not determinable from current builders]*	Q	log	masked 2020Q2–Q4	USD goods imports
`CPI_CR`	CPI index	*[FLAG]*	*[FLAG: not determinable from current builders]*	Q	log	masked 2020Q2–Q4	
`LR_CR`	Unemployment rate (%)	*[FLAG]*	*[FLAG: not determinable from current builders]*	Q	lin	masked 2020Q2–Q4	
`RATE_CR`	Monetary policy rate	*[FLAG]*	*[FLAG: not determinable from current builders]*	Q	lin	observed (financial)	National monetary policy rate (BCCR); POLRATE/HP not added (CR not in BIS CBPOL/RPP)
`NEER_CR`	Nominal effective FX	*[FLAG]*	*[FLAG: not determinable from current builders]*	Q	log	observed (financial)	
`GLB_OIL`	WTI crude oil price	FRED	`WTISPLC` (q-avg)	M→Q	log	observed (financial)	Global common (spot WTI, $/bbl)
`GLB_USLR`	US 10Y Treasury yield	FRED	`DGS10` *[FLAG]*	D→Q	lin	observed (financial)	Global common
`GLB_USBAA`	US Baa corporate yield	FRED	`BAA` *[FLAG]*	M→Q	lin	observed (financial)	Global common (credit proxy)
`GLB_USEQ`	US equity (S&P 500)	FRED	`SP500`-class *[FLAG]*	D→Q	log	observed (financial)	Global common
`CREDIT_GDP_CR`	Credit to NFCs (% of GDP)	**IMF IFS (CONSTRUCTED)**	100×`FOSAOP_XDC`/4Q-trailing `NGDP_SA_XDC`	Q	lin	**masked 2020Q2–Q4**	**NEW.** 56 present, range 42.8–59.4%; last 2025Q1; broad private-sector credit used as NFC proxy

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
`LENDR_CR`	Bank lending rate (% p.a.)	**IMF IFS**	`FILR_PA` (DBnomics mirror)	Q	lin	observed (financial)	**NEW.** IMF-compiled national lending rate (source BCCR); 60 present, range 5.0–18.8%; last 2025Q2

[FLAG: `CREDIT_GDP_CR` constructed / non-BIS.] Costa Rica is not a BIS total-credit reporter (FRED `QCRNAM770A` and BIS WS_TC both return "no series"), so the quantity is constructed as 100× (IMF IFS ODC claims on the private sector, `FOSAOP_XDC`, CRC) over a 4-quarter trailing sum of nominal GDP (`NGDP_SA_XDC`). This is *broad private-sector* credit (households + NFCs), used as the NFC proxy — the same validated construction as Colombia — so its *level* is not directly comparable to the BIS broad-NFC %GDP series used for CZ/HU/PL. The price `LENDR_CR` is the IMF IFS `FILR_PA` (IMF-compiled national lending rate sourced from the BCCR), used in lieu of a direct BCCR scrape; both new series have ragged trailing edges (credit to 2025Q1, price to 2025Q2) handled by the COVID-as-missing Gibbs. *[FLAG: CR base macro provenance.]* The exact source and series identifiers for the *base* CR macro/financial variables (`GDP_CR` IP proxy, `EXP_CR`/`IMP_CR` USD trade, `CPI_CR`, `LR_CR`, `RATE_CR`, `NEER_CR`) could not be determined from the builders present in the current pipeline tree (these blocs were assembled by an earlier exporter/IMF flow not retained here); they are most plausibly IMF (IFS/IMTS) and BCCR series but the precise keys are not verifiable from the available scripts and are *not* fabricated here.

20.33 Iceland (IS)

Iceland is a coupled GVAR bloc with $n = 18$ domestic-plus-global variables (three `ROW*_IS` stars added at estimation), $p = 3$ lags, estimated on the long sample **2000Q2–2025Q4** ($T = 103$). It carries a full real national-accounts block, a 3-month interbank rate, the BIS policy rate and real house price, and the new credit block. Macro variables are COVID-masked 2020Q2–Q4 and re-imputed; financials are kept observed.

Table 82: Iceland (IS) — variable coverage. `log`= 100 log level (displayed as $4\Delta\log$, q/q ann. growth); `lin`= level.

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
`GDP_IS`	Real GDP	*[FLAG]*	*[FLAG: not determinable from current builders]*	Q	log	masked 2020Q2–Q4	
`CONS_IS`	Real private consumption	*[FLAG]*	*[FLAG: not determinable from current builders]*	Q	log	masked 2020Q2–Q4	

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
INV_IS	Real investment	*[FLAG]*	*[FLAG: not determinable from current builders]*	Q	log	masked 2020Q2–Q4	
GOV_IS	Real government consumption	*[FLAG]*	*[FLAG: not determinable from current builders]*	Q	log	masked 2020Q2–Q4	
EXP_IS	Real exports	*[FLAG]*	*[FLAG: not determinable from current builders]*	Q	log	masked 2020Q2–Q4	
IMP_IS	Real imports	*[FLAG]*	*[FLAG: not determinable from current builders]*	Q	log	masked 2020Q2–Q4	
CPI_IS	CPI index	*[FLAG]*	*[FLAG: not determinable from current builders]*	Q	log	masked 2020Q2–Q4	
LR_IS	Unemployment rate (%)	*[FLAG]*	*[FLAG: not determinable from current builders]*	Q	lin	masked 2020Q2–Q4	
RATE_IS	3-month interbank rate	*[FLAG]*	*[FLAG: not determinable from current builders]*	Q	lin	observed (financial)	3M interbank (likely FRED IR3TIB01ISQ156N; unverified here)
NEER_IS	Nominal effective FX	*[FLAG]*	*[FLAG: not determinable from current builders]*	Q	log	observed (financial)	
POLRATE_IS	Central-bank policy rate	BIS	WS_CBPOL M.IS	M→Q	lin	observed (financial)	3-month q-avg of monthly EoP
HP_IS	Real residential property price	BIS (via FRED)	QISR628BIS	Q	log	observed (financial)	Real RPP, 2010=100
GLB_OIL	WTI crude oil price	FRED	WTISPLC (q-avg)	M→Q	log	observed (financial)	Global common (spot WTI, $/bbl)
GLB_USLR	US 10Y Treasury yield	FRED	DGS10 *[FLAG]*	D→Q	lin	observed (financial)	Global common
GLB_USBAA	US Baa corporate yield	FRED	BAA *[FLAG]*	M→Q	lin	observed (financial)	Global common (credit proxy)
GLB_USEQ	US equity (S&P 500)	FRED	SP500-class *[FLAG]*	D→Q	log	observed (financial)	Global common
CREDIT_GDP_IS	Credit to NFCs (% of GDP)	**IMF IFS (CONSTRUCTED)**	100×FOSAOP_XDC/4Q-trailing NGDP_SA_XDC	Q	lin	**masked 2020Q2–Q4**	**NEW.** 91 present, range 83.0–317.7% (318% = 2008 banking-crisis peak); last 2025Q1

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
`LENDR_IS`	Bank lending rate (% p.a.)	**IMF IFS**	`FILR_PA` (DBnomics mirror)	Q	lin	observed (financial)	**NEW.** IMF-compiled national lending rate (source CBI); 100 present, range 5.2–23.0%; last 2025Q2

[FLAG: `CREDIT_GDP_IS` constructed / non-BIS.] Iceland is not a BIS total-credit reporter, so the quantity is constructed identically to Costa Rica: 100× (IMF IFS ODC claims on the private sector, `FOSAOP_XDC`, ISK) over a 4-quarter trailing sum of nominal GDP (`NGDP_SA_XDC`). This is *broad private-sector* credit (HH + NFC) used as the NFC proxy — its level (and the historically accurate ∼318% 2008-crisis peak) is not directly comparable to the BIS broad-NFC %GDP series for CZ/HU/PL. The price `LENDR_IS` is the IMF IFS `FILR_PA` (IMF-compiled national lending rate sourced from the Central Bank of Iceland); both new series have ragged trailing edges handled by the Gibbs. *[FLAG: IS base macro provenance.]* The exact source and series identifiers for the *base* IS macro/financial variables (real national accounts, `CPI_IS`, `LR_IS`, `RATE_IS` 3M interbank, `NEER_IS`) could not be determined from the builders present in the current pipeline tree; `RATE_IS` is plausibly the FRED OECD 3M interbank series `IR3TIB01ISQ156N` and the national accounts are plausibly OECD QNA / Statistics Iceland, but the precise keys are not verifiable from the available scripts and are *not* fabricated here. Only `POLRATE_IS` and `HP_IS` (added by `build_polrate_hp_group.py`) and the two new credit variables have fully traceable sources.

20.34 Advanced-Economy Blocs: Per-Variable Data Provenance

This subsection documents, variable by variable, the eight advanced-economy GVAR blocs that carry the new credit block: Australia, Switzerland, Denmark, Norway, Sweden, New Zealand, Korea, and Israel. Each is a coupled GVAR bloc: the staged `spec.json` lists the domestic and global-commons variables; the estimator additionally appends the three trade-weighted `ROW*_<CC>` star variables (Section 19.1) at re-estimation, so the deployed n is the staged n plus three.

Conventions used in every table. The *Transformation* column follows the manual-wide convention: `log` denotes 100 · log(level), which enters the VAR as $4\,\Delta\log$ (quarter-on-quarter annualized growth), and `lin` denotes the raw level (percent per annum for rates and ratios; an index for NEER and house prices). The *COVID* column records the COVID-as-missing treatment of Section 19.4: macro variables (those with `isFinancial` = 0) are masked over 2020Q2–2020Q4 and imputed by the data-augmentation Gibbs sampler, whereas financial variables are held observed through the pandemic. The four global commons (`GLB_OIL`, `GLB_USLR`, `GLB_USBAA`, `GLB_USEQ`) are US-sourced series shared across all blocs (Section 16.7): WTI spot oil (FRED `WTISPLC`, quarter-average), the US 10Y Treasury yield (FRED `DGS10`), the US Baa corporate yield (FRED `BAA`), and the US equity price index (FRED `SPASTT01USM661N`, OECD US share-price/S&P proxy). They are kept observed through COVID. Where the exact upstream series identifier for a domestic *base* macro/financial variable (real-activity aggregates, CPI/HICP, unemployment, NEER, short rate, long yield, policy rate, house price) cannot be recovered from a machine-readable builder in the pipeline, the source is attributed at the provider level documented in Section 19.3 (OECD Quarterly National Accounts, the IMF SDMX flows, Eurostat, BIS, and national central banks) and the per-series id is marked *[FLAG: series id not recoverable from a readable builder]* rather than fabricated. The new **credit-block** variables (`CREDIT_GDP_<CC>` and `LENDR_<CC>`/`CORP_<CC>`) carry fully traced, live-probed identifiers from the dedicated fetchers (`fetch_rba_f7.py`, `fetch_snb_lendr.py`, `fetch_dk_lendr.py`, `fetch_ssb_jsonstat.py`, `fetch_scb_pxweb.py`, `fetch_rbnz_b3.py`, `fetch_credit_kr.py`, `fetch_credit_il.py`, `fetch_bis_credit_gdp.py`) and are flagged prominently below.

The new credit block (all eight economies). Each bloc gains two appended variables, both `lin` with a random-walk (RW) prior and `isFinancial` = 1 in the `domestic` block:

- **Quantity,** `CREDIT_GDP_<CC>` – BIS "Total credit to non-financial corporations, % of GDP" (broad credit: bank loans + debt securities + cross-border lending, adjusted for breaks), pulled via FRED under the uniform mnemonic `Q<ISO2>NAM770A` (`fetch_bis_credit_gdp.py`). It is *COVID-masked* 2020Q2–Q4 (it inherits the GDP-denominator mask: treated as a macro-like quantity even though tagged financial), the three quarters set to JSON `null` and imputed by the Gibbs.
- **Price,** `LENDR_<CC>` (a bank/corporate lending rate) or `CORP_<CC>` (a corporate-bond yield) – *observed through COVID* (not masked), % p.a.

None of these eight economies has a dropped price or a constructed (non-BIS) quantity; the quantity is the canonical BIS broad-credit series for all eight. Economy-specific caveats (short price history, leading ragged edges, a price-source substitution, a transport workaround, and one reconstructed corporate yield) are flagged in the relevant tables.

20.35 Australia (AU)

The Australian bloc has $n = 18$ staged variables (deployed $n = 21$ with the three `ROW*_AU` stars), estimated on the quarterly window 1999Q1–2025Q4. The eight macro variables (real GDP, consumption, government consumption, investment, exports, imports, CPI, unemployment) are masked over 2020Q2–Q4 and imputed; all financial variables are kept observed.

Table 83: Australia (`AU`): per-variable data provenance. Trans. key: `log` = 100 log level (enters as $4\Delta\log$, q/q-annualized growth); `lin` = level.

Variable	Description	Source	Series id/key	Freq	Trans.	COVID	Notes
`GDP_AU`	Real GDP	OECD QNA	*[FLAG]*	Q	log	masked	macro
`CONS_AU`	Real household consumption	OECD QNA	*[FLAG]*	Q	log	masked	macro
`GOV_AU`	Real govt consumption	OECD QNA	*[FLAG]*	Q	log	masked	macro
`INV_AU`	Real investment (GFCF)	OECD QNA	*[FLAG]*	Q	log	masked	macro
`EXP_AU`	Real exports	OECD QNA	*[FLAG]*	Q	log	masked	macro
`IMP_AU`	Real imports	OECD QNA	*[FLAG]*	Q	log	masked	macro
`CPI_AU`	CPI, all items (index)	IMF SDMX	*[FLAG]*	Q	log	masked	macro
`LR_AU`	Unemployment rate	IMF SDMX	*[FLAG]*	Q	lin	masked	macro
`NEER_AU`	NEER (index)	IMF SDMX	*[FLAG]*	Q	log	observed	financial
`RATE_AU`	Cash rate (interbank o/n)	RBA / IMF	*[FLAG]*	Q	lin	observed	policy short rate
`YIELD10_AU`	10Y govt bond yield	RBA / IMF	*[FLAG]*	Q	lin	observed	financial
`HP_AU`	Real residential house-price index (2010=100)	BIS RPP	FRED `QAUR628BIS` (BIS real RPP)	Q	log	observed	financial; *[FLAG: id by convention]*
`GLB_OIL`	WTI crude, q-avg ($/bbl)	FRED	`WTISPLC`	Q	log	observed	global common
`GLB_USLR`	US 10Y Treasury yield	FRED	`DGS10`	Q	lin	observed	global common
`GLB_USBAA`	US Baa corporate yield	FRED	`BAA`	Q	lin	observed	global common
`GLB_USEQ`	US equity (S&P 500)	FRED	`SPASTT01USM661N`	Q	log	observed	global common
`CREDIT_GDP_AU`	Credit to NFCs (% of GDP)	BIS (via FRED)	`QAUNAM770A`	Q	lin	**masked**	**NEW** broad credit; 105 obs, 1999Q1=63.5→2025Q4=62 (58.8–80.6)

Australia (AU), continued

Variable	Description	Source	Series id/key	Freq	Trans.	COVID	Notes
LENDR_AU	Business lending rate (RBA F7, outstanding large-business total, % p.a.)	RBA Table F7	FLRBFOLBT (fetch_rba_f7.py)	M→Q-avg	lin	observed	**NEW** price; *[FLAG: short history — starts 2019Q3, 26 obs; 1999Q1–2019Q2 leading null, Gibbs-imputed]*

20.36 Switzerland (CH)

The Swiss bloc has $n = 17$ staged variables (deployed $n = 20$), estimated on 2009Q2–2025Q4. Switzerland is one of the rate-optional blocs: it carries a policy rate (POLRATE_CH) but no separate domestic short rate. Seven macro variables are COVID-masked.

Table 84: Switzerland (CH): per-variable data provenance.

Variable	Description	Source	Series id/key	Freq	Trans.	COVID	Notes
GDP_CH	Real GDP	OECD QNA	*[FLAG]*	Q	log	masked	macro
CONS_CH	Real consumption	OECD QNA	*[FLAG]*	Q	log	masked	macro
INV_CH	Real investment	OECD QNA	*[FLAG]*	Q	log	masked	macro
GOV_CH	Real govt consumption	OECD QNA	*[FLAG]*	Q	log	masked	macro
EXP_CH	Real exports	OECD QNA	*[FLAG]*	Q	log	masked	macro
IMP_CH	Real imports	OECD QNA	*[FLAG]*	Q	log	masked	macro
HICP_CH	HICP (index)	IMF SDMX	*[FLAG]*	Q	log	masked	macro
LR_CH	Unemployment rate	IMF SDMX	*[FLAG]*	Q	lin	masked	macro
NEER_CH	NEER (index)	IMF SDMX	*[FLAG]*	Q	log	observed	financial
POLRATE_CH	Policy rate (BIS CBPOL)	BIS	BIS WS_CBPOL M.CH	M→Q-avg	lin	observed	policy rate; *[FLAG: id by convention]*
HP_CH	Real residential property price	BIS RPP	FRED QCHR628BIS	Q	log	observed	financial; *[FLAG: id by convention]*

Switzerland (`CH`), continued

Variable	Description	Source	Series id/key	Freq	Trans.	COVID	Notes
`GLB_OIL`	WTI crude, q-avg ($/bbl)	FRED	`WTISPLC`	Q	log	observed	global common
`GLB_USLR`	US 10Y Treasury yield	FRED	`DGS10`	Q	lin	observed	global common
`GLB_USBAA`	US Baa corporate yield	FRED	`BAA`	Q	lin	observed	global common
`GLB_USEQ`	US equity (S&P 500)	FRED	`SPASTT01USM661N`	Q	log	observed	global common
`CREDIT_GDP_CH`	Credit to NFCs (% of GDP)	BIS (via FRED)	`QCHNAM770A`	Q	lin	**masked**	**NEW** broad credit; 64 obs, 2009Q2=99.6→2025Q4=12 (99.6–149.4)
`LENDR_CH`	Corporate lending rate (SNB new investment loans, fixed rate, % p.a.)	SNB data portal	cube `zikreddet` item `FI` (`fetch_snb_lendr.py`)	M→Q-avg	lin	observed	**NEW** price; 67 obs, 2009Q2=1.80→2025Q4=1. (1.46–3.20); clean

20.37 Denmark (DK)

The Danish bloc has $n = 19$ staged variables (deployed $n = 22$), estimated on 2009Q1–2025Q4. Seven macro variables are COVID-masked. Denmark carries both a 3-month rate and a policy rate.

Table 85: Denmark (`DK`): per-variable data provenance.

Variable	Description	Source	Series id/key	Freq	Trans.	COVID	Notes
`GDP_DK`	Real GDP	Eurostat / OECD	*[FLAG]*	Q	log	masked	macro
`CONS_DK`	Real consumption	Eurostat / OECD	*[FLAG]*	Q	log	masked	macro
`INV_DK`	Real investment	Eurostat / OECD	*[FLAG]*	Q	log	masked	macro
`GOV_DK`	Real govt consumption	Eurostat / OECD	*[FLAG]*	Q	log	masked	macro
`EXP_DK`	Real exports	Eurostat / OECD	*[FLAG]*	Q	log	masked	macro
`IMP_DK`	Real imports	Eurostat / OECD	*[FLAG]*	Q	log	masked	macro
`HICP_DK`	HICP (index)	Eurostat	*[FLAG]*	Q	log	masked	macro

Variable	**Description**	**Source**	**Series id/key**	**Freq**	**Trans.**	**COVID**	**Notes**
`LR_DK`	Unemployment rate	Eurostat	*[FLAG]*	Q	lin	masked	macro
`RATE_DK`	3-month rate	IMF / Eurostat	*[FLAG]*	Q	lin	observed	short rate
`YIELD10_DK`	10Y govt bond yield	IMF / Eurostat	*[FLAG]*	Q	lin	observed	financial
`NEER_DK`	NEER (index)	IMF SDMX	*[FLAG]*	Q	log	observed	financial
`POLRATE_DK`	Policy rate (BIS CBPOL)	BIS	BIS `WS_CBPOL` `M.DK`	M→Q-avg	lin	observed	policy rate; *[FLAG: id by convention]*
`HP_DK`	Real residential property price	BIS RPP	FRED `QDKR628BIS`	Q	log	observed	financial; *[FLAG: id by convention]*
`GLB_OIL`	WTI crude, q-avg ($/bbl)	FRED	`WTISPLC`	Q	log	observed	global common
`GLB_USLR`	US 10Y Treasury yield	FRED	`DGS10`	Q	lin	observed	global common
`GLB_USBAA`	US Baa corporate yield	FRED	`BAA`	Q	lin	observed	global common
`GLB_USEQ`	US equity (S&P 500)	FRED	`SPASTT01USM661N`	Q	log	observed	global common
`CREDIT_GDP_DK`	Credit to NFCs (% of GDP)	BIS (via FRED)	`QDKNAM770A`	Q	lin	**masked**	**NEW** broad credit; 65 obs, 2009Q1=134.5→2025Q4=1 (127.4–151.6)
`LENDR_DK`	Corporate lending rate (ECB MIR loans to NFCs, new business, % p.a.)	ECB Data Portal (MIR)	`M.DK.B.` `A2A.A.R.` `A.2240.` `DKK.N` (`fetch_dk_lendr.py`)	M→Q-avg	lin	observed	**NEW** price; 34 obs, 2017Q3=1.01→2025Q4=3. (0.73–5.00); *[FLAG: source substitution + leading gap — see below]*

[FLAG: DK price source substitution.] Danmarks Nationalbank's StatBank exposes no machine-readable JSON/PX-Web API for the MFI interest-rate return. The ECB Data Portal (SDW MIR) compiles the same *Nationalbanken return (source agency DK2, harmonized under ECB Reg. 290/2009) and is used as the authoritative substitute — the data are genuine, only the transport differs. The ECB MIR Danish NFC rate starts 2017Q3, so* `LENDR_DK` *is null for 2009Q1–2017Q2 (leading ragged edge, Gibbs-imputed).*

20.38 Norway (NO)

The Norwegian bloc has $n = 17$ staged variables (deployed $n = 20$), estimated on 2006Q1–2025Q4. Norway is rate-optional: it carries a policy rate (`POLRATE_NO`) but no separate domestic short rate. Seven macro variables are COVID-masked.

Table 86: Norway (`NO`): per-variable data provenance.

Variable	Description	Source	Series id/key	Freq	Trans.	COVID	Notes
`GDP_NO`	Real GDP	OECD QNA	*[FLAG]*	Q	log	masked	macro
`CONS_NO`	Real consumption	OECD QNA	*[FLAG]*	Q	log	masked	macro
`INV_NO`	Real investment	OECD QNA	*[FLAG]*	Q	log	masked	macro
`GOV_NO`	Real govt consumption	OECD QNA	*[FLAG]*	Q	log	masked	macro
`EXP_NO`	Real exports	OECD QNA	*[FLAG]*	Q	log	masked	macro
`IMP_NO`	Real imports	OECD QNA	*[FLAG]*	Q	log	masked	macro
`HICP_NO`	HICP (index)	IMF SDMX	*[FLAG]*	Q	log	masked	macro
`LR_NO`	Unemployment rate	IMF SDMX	*[FLAG]*	Q	lin	masked	macro
`NEER_NO`	NEER (index)	IMF SDMX	*[FLAG]*	Q	log	observed	financial
`POLRATE_NO`	Policy rate (BIS CBPOL)	BIS	BIS `WS_CBPOL M.NO`	M→Q-avg	lin	observed	policy rate; *[FLAG: id by convention]*
`HP_NO`	Real residential property price	BIS RPP	FRED `QNOR628BIS`	Q	log	observed	financial; *[FLAG: id by convention]*
`GLB_OIL`	WTI crude, q-avg ($/bbl)	FRED	`WTISPLC`	Q	log	observed	global common
`GLB_USLR`	US 10Y Treasury yield	FRED	`DGS10`	Q	lin	observed	global common
`GLB_USBAA`	US Baa corporate yield	FRED	`BAA`	Q	lin	observed	global common
`GLB_USEQ`	US equity (S&P 500)	FRED	`SPASTT01USM661N`	Q	log	observed	global common
`CREDIT_GDP_NO`	Credit to NFCs (% of GDP)	BIS (via FRED)	`QNONAM770A`	Q	lin	**masked**	**NEW** broad credit; 77 obs, 2006Q1=111.5→2025Q4=1 (111.5–157.9)

Norway (NO), continued

Variable	Description	Source	Series id/key	Freq	Trans.	COVID	Notes
LENDR_NO	NFC bank lending rate (% p.a.)	Statistics Norway (SSB)	table 10746 (Utlanstype=02, Sektor=03 NFC, Rentebinding=99, RenterUtNedbLan; fetch_ssb_jsonstat.py)	M→Q-avg	lin	observed	**NEW** price; 49 obs, 2013Q4=4.13→2025Q4=6. (2.42–6.81); *[FLAG: leading gap 2006Q1–2013Q3, Gibbs-imputed]*

20.39 Sweden (SE)

The Swedish bloc has $n = 19$ staged variables (deployed $n = 22$), estimated on 2005Q1–2025Q4. Seven macro variables are COVID-masked. Sweden carries a 3-month rate, a 10Y yield, and a policy rate.

Table 87: Sweden (SE): per-variable data provenance.

Variable	Description	Source	Series id/key	Freq	Trans.	COVID	Notes
GDP_SE	Real GDP	Eurostat / OECD	*[FLAG]*	Q	log	masked	macro
CONS_SE	Real consumption	Eurostat / OECD	*[FLAG]*	Q	log	masked	macro
INV_SE	Real investment	Eurostat / OECD	*[FLAG]*	Q	log	masked	macro
GOV_SE	Real govt consumption	Eurostat / OECD	*[FLAG]*	Q	log	masked	macro
EXP_SE	Real exports	Eurostat / OECD	*[FLAG]*	Q	log	masked	macro
IMP_SE	Real imports	Eurostat / OECD	*[FLAG]*	Q	log	masked	macro
HICP_SE	HICP (index)	Eurostat	*[FLAG]*	Q	log	masked	macro
LR_SE	Unemployment rate	Eurostat	*[FLAG]*	Q	lin	masked	macro
RATE_SE	3-month rate	IMF / Eurostat	*[FLAG]*	Q	lin	observed	short rate
YIELD10_SE	10Y govt bond yield	IMF / Eurostat	*[FLAG]*	Q	lin	observed	financial
NEER_SE	NEER (index)	IMF SDMX	*[FLAG]*	Q	log	observed	financial

Sweden (SE), continued

Variable	Description	Source	Series id/key	Freq	Trans.	COVID	Notes
`POLRATE_SE`	Policy rate (BIS CBPOL)	BIS	BIS `WS_CBPOL M.SE`	M→Q-avg	lin	observed	policy rate; *[FLAG: id by convention]*
`HP_SE`	Real residential property price	BIS RPP	FRED `QSER628BIS`	Q	log	observed	financial; *[FLAG: id by convention]*
`GLB_OIL`	WTI crude, q-avg ($/bbl)	FRED	`WTISPLC`	Q	log	observed	global common
`GLB_USLR`	US 10Y Treasury yield	FRED	`DGS10`	Q	lin	observed	global common
`GLB_USBAA`	US Baa corporate yield	FRED	`BAA`	Q	lin	observed	global common
`GLB_USEQ`	US equity (S&P 500)	FRED	`SPASTT01USM661N`	Q	log	observed	global common
`CREDIT_GDP_SE`	Credit to NFCs (% of GDP)	BIS (via FRED)	`QSENAM770A`	Q	lin	**masked**	**NEW** broad credit; 81 obs, 2005Q1=102.1→2025Q4=1 (102.1–185.5)
`LENDR_SE`	NFC bank lending rate (% p.a.)	Statistics Sweden (SCB)	PxWeb `FM/ FM5001/ FM5001C/ RantaT01N` (Referenssektor=1.1.1, Motpartssektor=1 NFC, Avtal=0200, ContentsCode=000004ZT; `fetch_scb_pxweb.py`)	M→Q-avg	lin	observed	**NEW** price; 84 obs (full), 2005Q1=3.57→2025Q4=3. (1.58–5.72); clean

20.40 New Zealand (NZ)

The New Zealand bloc has $n = 18$ staged variables (deployed $n = 21$), estimated on 2000Q1–2025Q4. Eight macro variables are COVID-masked. NZ carries a policy rate (OCR) and a 10Y yield.

Table 88: New Zealand (NZ): per-variable data provenance.

Variable	Description	Source	Series id/key	Freq	Trans.	COVID	Notes
GDP_NZ	Real GDP	OECD QNA	*[FLAG]*	Q	log	masked	macro
CONS_NZ	Real consumption	OECD QNA	*[FLAG]*	Q	log	masked	macro
INV_NZ	Real investment	OECD QNA	*[FLAG]*	Q	log	masked	macro
GOV_NZ	Real govt consumption	OECD QNA	*[FLAG]*	Q	log	masked	macro
EXP_NZ	Real exports	OECD QNA	*[FLAG]*	Q	log	masked	macro
IMP_NZ	Real imports	OECD QNA	*[FLAG]*	Q	log	masked	macro
CPI_NZ	CPI (index)	IMF SDMX	*[FLAG]*	Q	log	masked	macro
LR_NZ	Unemployment rate	IMF SDMX	*[FLAG]*	Q	lin	masked	macro
RATE_NZ	Policy rate (OCR)	RBNZ / IMF	*[FLAG]*	Q	lin	observed	policy short rate
YIELD10_NZ	10Y govt bond yield	RBNZ / IMF	*[FLAG]*	Q	lin	observed	financial
HP_NZ	Real residential house-price index (2010=100)	BIS RPP	FRED QNZR628BIS	Q	log	observed	financial; *[FLAG: id by convention]*
NEER_NZ	NEER (index)	IMF SDMX	*[FLAG]*	Q	log	observed	financial
GLB_OIL	WTI crude, q-avg ($/bbl)	FRED	WTISPLC	Q	log	observed	global common
GLB_USLR	US 10Y Treasury yield	FRED	DGS10	Q	lin	observed	global common
GLB_USBAA	US Baa corporate yield	FRED	BAA	Q	lin	observed	global common
GLB_USEQ	US equity (S&P 500)	FRED	SPASTT01USM661N	Q	log	observed	global common
CREDIT_GDP_NZ	Credit to NFCs (% of GDP)	BIS (via FRED)	QNZNAM770A	Q	lin	**masked**	**NEW** broad credit; 101 obs, 2000Q1=83.0→2025Q4=7 (69.5–108.8)

New Zealand (NZ), continued

Variable	Description	Source	Series id/key	Freq	Trans.	COVID	Notes
`LENDR_NZ`	Business lending rate (% p.a.)	RBNZ Table B3	series A1 "SME new overdraft rate" `INR.MBI06.D3` (`fetch_rbnz_b3.py`)	M→Q-avg	lin	observed	**NEW** price; 104 obs (full), 2000Q1=8.53→2025Q4=9. (8.16–12.59); *[FLAG: transport workaround — see below]*

[FLAG: NZ price transport workaround.] The live RBNZ host is bot-walled (HTTP 403, Akamai/Imperva) to non-interactive clients. The genuine RBNZ B3 ***`hb3.xlsx`*** *was retrieved via a Wayback Machine snapshot (2026-01-28) and cached as* ***`pipeline/credit_staging/NZ/_rbnz_b3_A1_quarterly.json`****; the data are the real RBNZ B3 A1 series (only the transport differs). The older E1 business lending rate was discontinued Dec-2016; A1 covers the full sample with no gaps.*

20.41 Korea (KR)

The Korean bloc has $n = 18$ staged variables (deployed $n = 21$), estimated on 2000Q1–2025Q4. Eight macro variables are COVID-masked. Korea carries a 3-month money-market rate, a policy rate, and a 10Y yield, plus a *true* corporate bond yield as its credit-block price.

Table 89: Korea (KR): per-variable data provenance.

Variable	Description	Source	Series id/key	Freq	Trans.	COVID	Notes
`GDP_KR`	Real GDP	OECD QNA	*[FLAG]*	Q	log	masked	macro
`CONS_KR`	Real household consumption	OECD QNA	*[FLAG]*	Q	log	masked	macro
`INV_KR`	Real GFCF	OECD QNA	*[FLAG]*	Q	log	masked	macro
`GOV_KR`	Real govt consumption	OECD QNA	*[FLAG]*	Q	log	masked	macro
`EXP_KR`	Real exports	OECD QNA	*[FLAG]*	Q	log	masked	macro
`IMP_KR`	Real imports	OECD QNA	*[FLAG]*	Q	log	masked	macro
`CPI_KR`	CPI, all items (index)	IMF SDMX	*[FLAG]*	Q	log	masked	macro

Korea (KR), continued

Variable	**Description**	**Source**	**Series id/key**	**Freq**	**Trans.**	**COVID**	**Notes**
`LR_KR`	Unemployment rate	IMF SDMX	*[FLAG]*	Q	lin	masked	macro
`RATE_KR`	Money-market rate (3M, short-term)	IMF SDMX	*[FLAG]*	Q	lin	observed	short rate
`POLRATE_KR`	Policy rate (BIS CBPOL)	BIS	BIS `WS_CBPOL M.KR`	M→Q-avg	lin	observed	policy rate; *[FLAG: id by convention]*
`YIELD10_KR`	Govt bond yield (long-term)	IMF / OECD	*[FLAG]*	Q	lin	observed	financial
`HP_KR`	Real residential house-price index (2010=100)	BIS RPP	FRED `QKRR628BIS`	Q	log	observed	financial; *[FLAG: id by convention]*
`GLB_OIL`	WTI crude, q-avg ($/bbl)	FRED	`WTISPLC`	Q	log	observed	global common
`GLB_USLR`	US 10Y Treasury yield	FRED	`DGS10`	Q	lin	observed	global common
`GLB_USBAA`	US Baa corporate yield	FRED	`BAA`	Q	lin	observed	global common
`GLB_USEQ`	US equity (S&P 500)	FRED	`SPASTT01USM661N`	Q	log	observed	global common
`CREDIT_GDP_KR`	Credit to NFCs (% of GDP)	BIS (via FRED)	`QKRNAM770A`	Q	lin	**masked**	**NEW** broad credit; 101 obs, 2000Q1=89.1→2025Q4=11 (70.5–114.6)
`CORP_KR`	Corporate bond yield (3Y, AA-)	Bank of Korea ECOS	table `721Y001` item `7020000` (`fetch_credit_kr.py`)	M→Q-avg	lin	observed	**NEW** price = **true AA- corporate yield**; 104 obs, 2000Q1=10.09→2025Q4=3 (1.67–10.09)

20.42 Israel (IL)

The Israeli bloc has $n = 18$ staged variables (deployed $n = 21$), estimated on 2000Q1–2024Q4. Eight macro variables are COVID-masked. Israel carries a 3-month T-bill rate and a policy rate; its credit-block price is a *reconstructed* Tel-Bond corporate yield.

Table 90: Israel (`IL`): per-variable data provenance.

Variable	**Description**	**Source**	**Series id/key**	**Freq**	**Trans.**	**COVID**	**Notes**
`GDP_IL`	Real GDP	OECD QNA	*[FLAG]*	Q	log	masked	macro
`CONS_IL`	Real household consumption	OECD QNA	*[FLAG]*	Q	log	masked	macro
`INV_IL`	Real GFCF	OECD QNA	*[FLAG]*	Q	log	masked	macro
`GOV_IL`	Real govt consumption	OECD QNA	*[FLAG]*	Q	log	masked	macro
`EXP_IL`	Real exports	OECD QNA	*[FLAG]*	Q	log	masked	macro
`IMP_IL`	Real imports	OECD QNA	*[FLAG]*	Q	log	masked	macro
`CPI_IL`	CPI, all items (index)	IMF SDMX	*[FLAG]*	Q	log	masked	macro
`LR_IL`	Unemployment rate	IMF SDMX	*[FLAG]*	Q	lin	masked	macro
`RATE_IL`	3M Treasury-bill yield	IMF / BoI	*[FLAG]*	Q	lin	observed	short rate
`POLRATE_IL`	Policy rate (Bank of Israel)	BoI / BIS	*[FLAG]*	Q	lin	observed	policy rate
`NEER_IL`	NEER (index)	IMF SDMX	*[FLAG]*	Q	log	observed	financial
`HP_IL`	Real residential property price	BIS RPP	FRED `QILR628BIS`	Q	log	observed	financial; *[FLAG: id by convention]*
`GLB_OIL`	WTI crude, q-avg ($/bbl)	FRED	`WTISPLC`	Q	log	observed	global common
`GLB_USLR`	US 10Y Treasury yield	FRED	`DGS10`	Q	lin	observed	global common
`GLB_USBAA`	US Baa corporate yield	FRED	`BAA`	Q	lin	observed	global common
`GLB_USEQ`	US equity (S&P 500)	FRED	`SPASTT01USM661N`	Q	log	observed	global common
`CREDIT_GDP_IL`	Credit to NFCs (% of GDP)	BIS (via FRED)	`QILNAM770A`	Q	lin	**masked**	**NEW** broad credit; 97 obs, 2000Q1=69.4→2024Q4=69 (66.8–93.1)

Israel (IL), continued

Variable	Description	Source	Series id/key	Freq	Trans.	COVID	Notes
CORP_IL	Tel-Bond corporate bond yield (5Y govt + Tel-Bond spread)	Bank of Israel (SDMX)	DWH_SRC_0356_MA + DWH_SRC_0299_MA (fetch_credit_il.py)	M→Q-avg	lin	observed	**NEW** price; 56 obs, 2009Q4=7.03→2024Q4=5. (1.89–7.03); *[FLAG: RECONSTRUCTED + leading gap — see below]*

[FLAG: IL price is RECONSTRUCTED.] CORP_IL is built as the sum of the Bank of Israel 5-year non-indexed fixed-rate government *bond yield (DWH_SRC_0356_MA) and the BoI quoted Tel-Bond–vs–govt yield spread (DWH_SRC_0299_MA); the Tel-Bond Index is Israel's main corporate-bond index (∼5 Y duration), so the sum is a genuine corporate yield, not a single published outright series. It was preferred over the BoI BIR bank business-loan rate, which carries a systematic source anomaly 2017Q2–2018Q1 (values spike to 30–50%). The Tel-Bond spread starts 2009-12, so CORP_IL is null for 2000Q1–2009Q3 (44 leading nulls, Gibbs-imputed); it is observed continuously through COVID and to end-of-sample.*

Summary of data flags (advanced-economy blocs). The genuinely indeterminate sources — where the exact upstream series id could not be recovered from a machine-readable builder and is therefore *not* fabricated — are the domestic *base* macro and financial variables (real-activity aggregates, CPI/HICP, unemployment, NEER, short rate, 10Y yield, and for IL the policy rate), attributed at the provider level (OECD QNA, IMF SDMX, Eurostat for DK/SE, BIS, national central banks) per Section 19.3. The BIS real house-price (HP_<CC>) and BIS policy-rate (POLRATE_<CC>) ids are stated by the established FRED-mnemonic / BIS-dataflow convention (marked "id by convention"). All credit-block ids (quantity and price) are live-probed and fully traced.

20.43 Emerging-market blocs I: BR, MX, CL, CO, IN, ID

This subsection documents six emerging-market GVAR blocs in full, including the new credit block (a quantity CREDIT_GDP_<CC> and a price, LENDR_<CC> or CORP_<CC>) appended to each bloc. Throughout, "log" denotes a series entered as $100 \ln(\text{level})$ and displayed as $4\,\Delta(100 \ln)$, i.e. q/q-annualized growth, while "lin" denotes a level (percent, ratio, or index). Under the *COVID-as-missing* treatment (Section 19.4) every *macro* variable (those tagged isFinancial= 0) is masked over the fixed window 2020Q2–2020Q4 and imputed by the augmentation Gibbs sampler; *financial* variables (isFinancial= 1) are held observed.

The new credit block (all six blocs). Each bloc gains two variables, appended after the existing domestic and global-commons columns:

- **Quantity** `CREDIT_GDP_<CC>` — BIS Total Credit to Non-Financial Corporations, percentage of GDP (the broad credit measure: loans + debt securities + cross-border), pulled from FRED with the BIS mnemonic `Q<CC>NAM770A` ("Total Credit to Non-Financial Corporations, Adjusted for Breaks", units: % of GDP, native quarterly). Transformation `lin`, prior RW, `isFinancial`= 1. Because the %-of-GDP ratio inherits the collapsing-GDP denominator artefact in 2020, the three cells 2020Q2–Q4 are set to `null` (masked), consistent with the macro COVID mask, and imputed by the Gibbs sampler. **Colombia is the exception:** BIS publishes no Colombia NFC total credit, so `CREDIT_GDP_CO` is *constructed* (non-BIS); see below and the flag list.
- **Price** — a domestic credit-cost series, observed through COVID (`lin`, prior RW, `isFinancial`= 1). The price source differs by bloc: a bank *lending rate* (`LENDR_<CC>`) where one is available, or a 10-year *government* bond yield used as a *sovereign proxy* (`CORP_<CC>`) where no corporate/lending series exists. Per-bloc provenance and caveats are given in each table and flagged where they apply (CL and IN use the sovereign 10Y proxy).

20.44 Brazil (BR)

The Brazil bloc is an $n = 17$ quarterly BVAR estimated over 2012Q2–2025Q4 ($T = 55$); macro variables are treated as missing over 2020Q2–Q4 (COVID-as-missing), while the financial block (including the new credit price) is held observed. The credit block is *clean*: BIS NFC credit-to-GDP for the quantity and the Banco Central do Brasil corporate lending rate for the price.

Table 91: Brazil (BR) bloc: variable coverage and provenance. `log` = 100 ln level (q/q-annualized growth displayed); `lin` = level. COVID mask = 2020Q2–Q4.

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
`GDP_BR`	Real GDP	National accounts (GVAR panel)	*[FLAG]*	Q	log	masked	macro
`CONS_BR`	Real household consumption	National accounts (GVAR panel)	*[FLAG]*	Q	log	masked	macro
`INV_BR`	Real gross fixed capital formation	National accounts (GVAR panel)	*[FLAG]*	Q	log	masked	macro
`GOV_BR`	Real government consumption	National accounts (GVAR panel)	*[FLAG]*	Q	log	masked	macro
`EXP_BR`	Real exports	National accounts (GVAR panel)	*[FLAG]*	Q	log	masked	macro

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
IMP_BR	Real imports	National accounts (GVAR panel)	*[FLAG]*	Q	log	masked	macro
CPI_BR	CPI index	National source / IMF (GVAR panel)	*[FLAG]*	Q	log	masked	macro
LR_BR	Unemployment rate	National source / IMF (GVAR panel)	*[FLAG]*	Q	lin	masked	macro
NEER_BR	Nominal effective exch. rate	BIS / IMF (GVAR panel)	*[FLAG]*	Q	log	observed	financia
POLRATE_BR	Policy rate	BIS CBPOL	WS_CBPOL M.BR	M→Q avg	lin	observed	3-month
HP_BR	Real residential property price (2010=100)	FRED / BIS real RPP	QBRR628BIS	Q	log	observed	financia
GLB_OIL	WTI crude oil, q-avg ($/bbl)	FRED	WTISPLC	M→Q avg	log	observed	global c
GLB_USLR	US 10Y Treasury yield (%)	FRED	DGS10	Q	lin	observed	global c
GLB_USBAA	US Baa corporate yield (%)	FRED	BAA	Q	lin	observed	global c
GLB_USEQ	US equity (S&P 500)	FRED	SPASTT01USM661N	M→Q avg	log	observed	global c
CREDIT_GDP_BR	Credit to NFCs (% of GDP)	BIS via FRED	QBRNAM770A	Q	lin	**masked**	**credit tity)**; non-nul 57.0% the 3 C null (cle
LENDR_BR	Corporate bank lending rate (% p.a.)	Banco Central do Brasil (BCB SGS)	SGS 20718	M→Q avg	lin	observed	**credit** *Pessoas* avg rate; non-nul 31.24%; fetch_b

20.45 Mexico (MX)

The Mexico bloc is an $n = 19$ quarterly BVAR estimated over 2001Q4–2025Q4 ($T = 97$); macro variables are masked over 2020Q2–Q4. The credit block uses BIS NFC credit-to-GDP and the IMF IFS lending rate; the price has a ragged trailing edge (the IFS series ends 2025Q2, so 2025Q3–Q4 are null and handled by the Gibbs ragged-edge treatment).

Table 92: Mexico (MX) bloc: variable coverage and provenance.

Variable	**Description**	**Source**	**Series ID/key**	**Freq**	**Trans.**	**COVID**	**Not**
GDP_MX	Real GDP (chain volume, SA)	National accounts / OECD QNA (GVAR panel)	*[FLAG]*	Q	log	masked	mac
CONS_MX	Real household consumption (chain vol, SA)	National accounts (GVAR panel)	*[FLAG]*	Q	log	masked	mac
INV_MX	Real gross fixed capital formation (SA)	National accounts (GVAR panel)	*[FLAG]*	Q	log	masked	mac
GOV_MX	Real government consumption (chain vol, SA)	National accounts (GVAR panel)	*[FLAG]*	Q	log	masked	mac
EXP_MX	Real exports (chain volume, SA)	National accounts (GVAR panel)	*[FLAG]*	Q	log	masked	mac
IMP_MX	Real imports (chain volume, SA)	National accounts (GVAR panel)	*[FLAG]*	Q	log	masked	mac
CPI_MX	CPI (all items, index)	National source / IMF (GVAR panel)	*[FLAG]*	Q	log	masked	mac
LR_MX	Unemployment rate	National source / IMF (GVAR panel)	*[FLAG]*	Q	lin	masked	mac
RATE_MX	Money-market rate (short-term)	National source / IMF (GVAR panel)	*[FLAG]*	Q	lin	observed	finar
POLRATE_MX	Central-bank policy rate	BIS CBPOL	WS_CBPOL M.MX	M→Q avg	lin	observed	3-mo
YIELD10_MX	Government bond yield (long-term)	National source / IMF (GVAR panel)	*[FLAG]*	Q	lin	observed	finar
HP_MX	Real residential house-price index (2010=100)	FRED / BIS real RPP	QMXR628BIS	Q	log	observed	finar
NEER_MX	Nominal effective exch. rate (index)	BIS / IMF (GVAR panel)	*[FLAG]*	Q	log	observed	finar
GLB_OIL	WTI crude oil, q-avg ($/bbl)	FRED	WTISPLC	M→Q avg	log	observed	glob
GLB_USLR	US 10Y Treasury yield (%)	FRED	DGS10	Q	lin	observed	glob
GLB_USBAA	US Baa corporate yield (%)	FRED	BAA	Q	lin	observed	glob
GLB_USEQ	US equity (S&P 500)	FRED	SPASTT01USM661N	M→Q avg	log	observed	glob

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Not
`CREDIT_GDP_MX`	Credit to NFCs (% of GDP)	BIS via FRED	`QMXNAM770A`	Q	lin	**masked**	**cred tity** non- 28.3 the 3 null
`LENDR_MX`	Bank lending rate (% p.a.)	IMF IFS (via DB-nomics)	`FILR_PA`	Q	lin	observed	**cred** 95/9 3.34 **ragg ing** ends 2025

20.46 Chile (CL)

The Chile bloc is an $n = 19$ quarterly BVAR estimated over 2004Q3–2025Q4 ($T = 86$); macro variables are masked over 2020Q2–Q4. The credit *quantity* is the BIS NFC credit-to-GDP series, but the credit *price* is a **sovereign 10Y proxy** (`CORP_CL`): the IMF IFS Chilean lending rate was discontinued in 2018, so the OECD KEI 10-year *government* bond yield stands in — there is no corporate spread.

Table 93: Chile (CL) bloc: variable coverage and provenance.

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Not
`GDP_CL`	Real GDP	National accounts / OECD QNA (GVAR panel)	*[FLAG]*	Q	log	masked	macr
`CONS_CL`	Real household consumption	National accounts (GVAR panel)	*[FLAG]*	Q	log	masked	macr
`INV_CL`	Real gross fixed capital formation	National accounts (GVAR panel)	*[FLAG]*	Q	log	masked	macr
`GOV_CL`	Real government consumption	National accounts (GVAR panel)	*[FLAG]*	Q	log	masked	macr
`EXP_CL`	Real exports	National accounts (GVAR panel)	*[FLAG]*	Q	log	masked	macr
`IMP_CL`	Real imports	National accounts (GVAR panel)	*[FLAG]*	Q	log	masked	macr

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Not
`CPI_CL`	CPI index	National source / IMF (GVAR panel)	*[FLAG]*	Q	log	masked	macr
`LR_CL`	Unemployment rate	National source / IMF (GVAR panel)	*[FLAG]*	Q	lin	masked	macr
`RATE_CL`	3M interbank rate	National source / IMF (GVAR panel)	*[FLAG]*	Q	lin	observed	finar
`POLRATE_CL`	Central-bank policy rate	BIS CBPOL	`WS_CBPOL M.CL`	M→Q avg	lin	observed	3-mo
`YIELD10_CL`	10Y government bond yield	National source / IMF (GVAR panel)	*[FLAG]*	Q	lin	observed	finar
`HP_CL`	Real residential house-price index (2010=100)	FRED / BIS real RPP	`QCLR628BIS`	Q	log	observed	finar
`NEER_CL`	Nominal effective exch. rate	BIS / IMF (GVAR panel)	*[FLAG]*	Q	log	observed	finar
`GLB_OIL`	WTI crude oil, q-avg ($/bbl)	FRED	`WTISPLC`	M→Q avg	log	observed	globa
`GLB_USLR`	US 10Y Treasury yield (%)	FRED	`DGS10`	Q	lin	observed	globa
`GLB_USBAA`	US Baa corporate yield (%)	FRED	`BAA`	Q	lin	observed	globa
`GLB_USEQ`	US equity (S&P 500)	FRED	`SPASTT01USM661N`	M→Q avg	log	observed	globa
`CREDIT_GDP_CL`	Credit to NFCs (% of GDP)	BIS via FRED	`QCLNAM770A`	Q	lin	**masked**	**cred tity)** non- 111.9 COV
`CORP_CL`	10Y government bond yield (sovereign proxy)	OECD KEI long-term interest rates (IRLT)	IRLT, country `CHL`	M→Q avg	lin	observed	**cred SOV 10Y** — **rate** 86/8 2.51-

20.47 Colombia (CO)

The Colombia bloc is an $n = 19$ quarterly BVAR estimated over 2005Q1–2025Q4 ($T = 84$); macro variables are masked over 2020Q2–Q4. Colombia is *not* a BIS total-credit reporter, so the credit *quantity* is a **constructed (non-BIS) broad-credit proxy** (reducing cross-country comparability); the credit *price* is the IMF IFS lending rate. Both credit columns carry trailing nulls from the input cut-off (2025Q1 for the constructed quantity inputs; the IFS price ends mid-2025).

Table 94: Colombia (CO) bloc: variable coverage and provenance.

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Not
GDP_CO	Real GDP	National accounts / OECD QNA (GVAR panel)	*[FLAG]*	Q	log	masked	macr
CONS_CO	Real household consumption	National accounts (GVAR panel)	*[FLAG]*	Q	log	masked	macr
INV_CO	Real gross fixed capital formation	National accounts (GVAR panel)	*[FLAG]*	Q	log	masked	macr
GOV_CO	Real government consumption	National accounts (GVAR panel)	*[FLAG]*	Q	log	masked	macr
EXP_CO	Real exports	National accounts (GVAR panel)	*[FLAG]*	Q	log	masked	macr
IMP_CO	Real imports	National accounts (GVAR panel)	*[FLAG]*	Q	log	masked	macr
CPI_CO	CPI index	National source / IMF (GVAR panel)	*[FLAG]*	Q	log	masked	macr
LR_CO	Unemployment rate	National source / IMF (GVAR panel)	*[FLAG]*	Q	lin	masked	macr
RATE_CO	3M interbank rate	National source / IMF (GVAR panel)	*[FLAG]*	Q	lin	observed	finan
POLRATE_CO	Central-bank policy rate	BIS CBPOL	WS_CBPOL M.CO	M→Q avg	lin	observed	3-mo
YIELD10_CO	10Y government bond yield	National source / IMF (GVAR panel)	*[FLAG]*	Q	lin	observed	finan
HP_CO	Real residential house-price index (2010=100)	FRED / BIS real RPP	QCOR628BIS	Q	log	observed	finan
NEER_CO	Nominal effective exch. rate	BIS / IMF (GVAR panel)	*[FLAG]*	Q	log	observed	finan
GLB_OIL	WTI crude oil, q-avg ($/bbl)	FRED	WTISPLC	M→Q avg	log	observed	globa
GLB_USLR	US 10Y Treasury yield (%)	FRED	DGS10	Q	lin	observed	globa
GLB_USBAA	US Baa corporate yield (%)	FRED	BAA	Q	lin	observed	globa
GLB_USEQ	US equity (S&P 500)	FRED	SPASTT01USM661N	M→Q avg	log	observed	globa

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Not
`CREDIT_GDP_CO`	Credit to NFCs (% of GDP), constructed	IMF IFS (constructed: claims ÷ 4Q-trailing NGDP)	`FOSAOP_XDC` / `NGDP_SA_XDC`	Q	lin	**masked**	**cred tity) STF (nor** = clain vate 4Q-t inal priva (hou NFC comp 75/8 22.7- = 3 3 C trail end
`LENDR_CO`	Bank lending rate (% p.a.)	IMF IFS (via DB-nomics)	`FILR_PA`	Q	lin	observed	**cred** 82/8 8.59- ragg edge null)

20.48 India (IN)

The India bloc is an $n = 19$ quarterly BVAR estimated over 2011Q4–2025Q4 ($T = 57$); macro variables are masked over 2020Q2–Q4. The credit *quantity* is the BIS NFC credit-to-GDP series; the credit *price* is a **sovereign 10Y proxy** (`CORP_IN`): no Indian corporate or bank lending rate is currently maintained on FRED/IMF/World Bank (the IFS lending rate was discontinued in 2022), so the OECD KEI 10-year *government* bond yield stands in — there is no corporate spread.

Table 95: India (IN) bloc: variable coverage and provenance.

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	N
`GDP_IN`	Real GDP (chain volume, SA)	National accounts / OECD QNA (GVAR panel)	*[FLAG]*	Q	log	masked	r
`CONS_IN`	Real household consumption (chain vol, SA)	National accounts (GVAR panel)	*[FLAG]*	Q	log	masked	r

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	N
INV_IN	Real gross fixed capital formation (chain vol, SA)	National accounts (GVAR panel)	*[FLAG]*	Q	log	masked	m
GOV_IN	Real government consumption (chain vol, SA)	National accounts (GVAR panel)	*[FLAG]*	Q	log	masked	m
EXP_IN	Real exports of goods & services (chain vol, SA)	National accounts (GVAR panel)	*[FLAG]*	Q	log	masked	m
IMP_IN	Real imports of goods & services (chain vol, SA)	National accounts (GVAR panel)	*[FLAG]*	Q	log	masked	m
CPI_IN	CPI (all items, index)	National source / IMF (GVAR panel)	*[FLAG]*	Q	log	masked	m
RATE_IN	Call money / immediate interbank rate	FRED (OECD MEI)	IRSTCI01INM	M→Q avg	lin	observed	f
RATE3M_IN	Short-term 3M interbank rate	FRED (OECD MEI)	INDIR3TIB01STM	M→Q avg	lin	observed	f
YIELD10_IN	Long-term 10Y government bond yield	FRED (OECD MEI)	INDIRLTLT01STM	M→Q avg	lin	observed	f
NEER_IN	Nominal effective exch. rate (BIS broad, index)	BIS / IMF (GVAR panel)	*[FLAG]*	Q	log	observed	f
POLRATE_IN	Policy rate	BIS CBPOL	WS_CBPOL M.IN	M→Q avg	lin	observed	3
HP_IN	Real residential property price (2010=100)	FRED / BIS real RPP	QINR628BIS	Q	log	observed	f
GLB_OIL	WTI crude oil, q-avg ($/bbl)	FRED	WTISPLC	M→Q avg	log	observed	g
GLB_USLR	US 10Y Treasury yield (%)	FRED	DGS10	Q	lin	observed	g
GLB_USBAA	US Baa corporate yield (%)	FRED	BAA	Q	lin	observed	g
GLB_USEQ	US equity (S&P 500)	FRED	SPASTT01USM661N	M→Q avg	log	observed	g
CREDIT_GDP_IN	Credit to NFCs (% of GDP)	BIS via FRED	QINNAM770A	Q	lin	**masked**	**c t** r 7 (
CORP_IN	10Y government bond yield (sovereign proxy)	OECD KEI long-term interest rates (IRLT)	IRLT, country IND	M→Q avg	lin	observed	**c S 1** — **r** 5 5

20.49 Indonesia (ID)

The Indonesia bloc is an $n = 17$ quarterly BVAR estimated over 2005Q3–2025Q4 ($T = 82$); macro variables are masked over 2020Q2–Q4. The credit block is *clean*: BIS NFC credit-to-GDP for the quantity and the IMF IFS lending rate (fully observed, current to 2025Q4) for the price.

Table 96: Indonesia (ID) bloc: variable coverage and provenance.

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
GDP_ID	Real GDP (SA, constant prices)	National accounts / OECD QNA (GVAR panel)	*[FLAG]*	Q	log	masked	macro
CONS_ID	Real private consumption (SA)	National accounts (GVAR panel)	*[FLAG]*	Q	log	masked	macro
INV_ID	Real gross fixed capital formation (SA)	National accounts (GVAR panel)	*[FLAG]*	Q	log	masked	macro
GOV_ID	Real government consumption (SA)	National accounts (GVAR panel)	*[FLAG]*	Q	log	masked	macro
EXP_ID	Real exports of goods & services (SA)	National accounts (GVAR panel)	*[FLAG]*	Q	log	masked	macro
IMP_ID	Real imports of goods & services (SA)	National accounts (GVAR panel)	*[FLAG]*	Q	log	masked	macro
CPI_ID	CPI (all items, index)	National source / IMF (GVAR panel)	*[FLAG]*	Q	log	masked	macro
RATE_ID	Monetary policy-related rate (BI rate)	National source / IMF (GVAR panel)	*[FLAG]*	Q	lin	observed	financia
NEER_ID	Nominal effective exch. rate (BIS broad)	BIS / IMF (GVAR panel)	*[FLAG]*	Q	log	observed	financia
POLRATE_ID	Policy rate	BIS CBPOL	WS_CBPOL M.ID	M→Q avg	lin	observed	3-montl
HP_ID	Real residential property price (2010=100)	FRED / BIS real RPP	QIDR628BIS	Q	log	observed	financia
GLB_OIL	WTI crude oil, q-avg ($/bbl)	FRED	WTISPLC	M→Q avg	log	observed	global c
GLB_USLR	US 10Y Treasury yield (%)	FRED	DGS10	Q	lin	observed	global c
GLB_USBAA	US Baa corporate yield (%)	FRED	BAA	Q	lin	observed	global c
GLB_USEQ	US equity (S&P 500)	FRED	SPASTT01USM661N	M→Q avg	log	observed	global c
CREDIT_GDP_ID	Credit to NFCs (% of GDP)	BIS via FRED	QIDNAM770A	Q	lin	**masked**	**credit tity)**; non-nul 27.7% COVID

Variable	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
`LENDR_ID`	IFS lending rate (% p.a.)	IMF IFS (new SDMX 2.1 API)	`MFS_IR`, `MFS162_RT_PT_A_PT` (= IFS `FILR_PA`)	Q	lin	observed	**credit** 82/82 8.21–16 observe to 2025 `fetch_i`

Caveats and provenance flags for this group. The *credit-block* provenance above is traced to specific FRED/IMF/OECD/BCB series and verified against the staged data (first/last/coverage all reconcile). The flagged items are:

- `CORP_CL`, `CORP_IN`: credit *price* is a **sovereign 10Y proxy** (OECD KEI government-bond yield), not a corporate/lending rate — no corporate spread is captured (CL: IFS lending rate discontinued 2018; IN: IFS lending rate discontinued 2022).
- `CREDIT_GDP_CO`: **constructed, non-BIS** broad-credit proxy (100× IMF IFS ODC claims on the private sector ÷ 4Q-trailing nominal GDP); it mixes households + NFCs (not NFC-specific) and so has reduced cross-country comparability. (BIS publishes no Colombia NFC total-credit series; the mnemonic `QCONAM770A` on FRED is *Argentina*, not Colombia — a verified mnemonic trap.)
- `LENDR_MX`, `LENDR_CO`: credit-price series have a **ragged trailing edge** (IFS `FILR_PA` ends mid-2025), so 2025Q3–Q4 are null and imputed by the ragged-edge Gibbs treatment.

[FLAG: base macro / domestic-financial series IDs not recorded in the staged artefacts.] For the pre-existing domestic block (real GDP, the national-accounts expenditure components, CPI, unemployment, NEER, and the domestic short/long rates), the staged `DataRaw_<CC>_glob.json` files carry only the variable list, dates, and values — no per-series source mnemonic. These columns originate from the upstream GVAR national-source quarterly panel (national statistical offices / OECD QNA / IMF flows, balanced to a common 2025Q4 endpoint via WEO-benchmarked temporal disaggregation in `build_gvar_2025q4.py`); the exact provider series IDs are therefore *not* determinable from the artefacts inspected and are left flagged rather than fabricated. The cells documented with a concrete ID — `POLRATE` (BIS CBPOL `WS_CBPOL M.<CC>`), `HP` (FRED BIS real RPP `Q<CC>R628BIS`), the four global commons (FRED), the India domestic rates (FRED/OECD MEI `IRSTCI01INM`, `INDIR3TIB01STM`, `INDIRLTLT01STM`, traced in the research report), and the whole credit block — are confirmed against their builders.

[FLAG: India domestic-rate IDs are inferred from the research report's FRED enumeration (`IRSTCI01INM`, `INDIR3TIB01STM`, `INDIRLTLT01STM`); they match the OECD-MEI rate definitions used for IN but were not independently re-fetched in this documentation pass.]

20.50 Emerging markets II: data conventions (TR, ZA, SA, AR, RU, TH, MY, TW)

Transformation and COVID conventions. Throughout these tables the *transformation* column follows the tool-wide convention of Section 19.8: `log` denotes a series stored as $100 \cdot \log(\text{level})$, which enters the VAR and is reported as $4 \cdot \Delta \log =$ the quarter-on-quarter *annualized* growth rate; and `lin` denotes a level (rates in percent, ratios in percent of GDP, index points). The *prior* on every variable's own first lag is the random walk (RW) unless noted. *COVID-as-missing* (Section 19.4) masks the three quarters 2020Q2–2020Q4 of every *macro* variable (those tagged `isFinancial`$= 0$); these holes are completed inside the Durbin–Koopman E-step rather than fitted. *Financial* variables (`isFinancial`$= 1$) are held *observed* through the COVID window. The four *global commons* are identical US-determined columns shared by every bloc and pinned 1:1 onto each RoW bloc at solve time (Section 16.7): `GLB_OIL` = FRED `WTISPLC` (WTI spot, $/bbl, quarterly average); `GLB_USLR` = FRED `DGS10` (US 10Y Treasury constant-maturity yield); `GLB_USBAA` = FRED `BAA` (Moody's seasoned Baa corporate-bond yield); `GLB_USEQ` = US equity price index (FRED `SPASTT01USM661N`, OECD US share-price/S&P proxy). All four are financial and observed through COVID.

The new credit block. Each of these eight blocs is augmented with a *credit block* of one or two variables. The *quantity* leg CREDIT_GDP_<CC> is, wherever a BIS reporting series exists, the BIS *Total Credit to Non-Financial Corporations*, percent of GDP (the *broad* credit concept: bank loans plus debt securities plus cross-border lending), pulled from FRED under the uniform mnemonic Q<ISO2>NAM770A (end-of-quarter, % of GDP). It is stored `lin`/RW and is tagged *financial*, but because its denominator is nominal GDP it inherits the macro COVID mask: its 2020Q2–Q4 cells are set `null`. The *price* leg is either a bank lending rate (LENDR_<CC>) or, where no lending rate is reachable, a corporate/sovereign-proxy yield (CORP_<CC>); it is `lin`/RW, financial, and *observed* through COVID. Where the price could not be sourced it was *dropped* (quantity only). Bloc-specific caveats are flagged in-line below.

20.51 Türkiye (TR)

Türkiye is a coupled GVAR bloc estimated with $n = 16$ quarterly domestic + global variables (the staged credit spec; the deployed GVAR bloc adds the three `ROW*_TR` trade-weighted stars) at $p = 3$ lags over a quarterly grid running 2000Q1–2025Q4 ($T = 104$). The eight real-activity, price and labour series are masked over COVID 2020Q2–Q4 and completed by the Durbin–Koopman E-step; financials are kept observed. The credit block is *quantity-only*: the bank-lending-rate / corporate-yield price leg was dropped (flag below).

Table 97: Türkiye (TR) variable documentation.

Variable (id)	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
GDP_TR	Real GDP	IMF/OECD QNA (GVAR base)	*[FLAG]*	Q	log	masked	WEO-extended to 2025Q4
CONS_TR	Real household consumption	IMF/OECD QNA	*[FLAG]*	Q	log	masked	
INV_TR	Real gross fixed capital formation	IMF/OECD QNA	*[FLAG]*	Q	log	masked	
GOV_TR	Real government consumption	IMF/OECD QNA	*[FLAG]*	Q	log	masked	
EXP_TR	Real exports	IMF/OECD QNA	*[FLAG]*	Q	log	masked	
IMP_TR	Real imports	IMF/OECD QNA	*[FLAG]*	Q	log	masked	
CPI_TR	CPI, all items (index)	IMF IFS / national	*[FLAG]*	Q	log	masked	
LR_TR	Unemployment rate	IMF IFS / national	*[FLAG]*	Q	lin	masked	
RATE_TR	Monetary-policy-related (short) rate	IMF IFS / national	*[FLAG]*	Q	lin	observed	financial
POLRATE_TR	Policy rate (BIS CBPOL)	BIS SDMX WS_CBPOL	`M.TR` (3M avg)	Q	lin	observed	financial
HP_TR	Real residential property price	FRED (BIS real RPP)	`QTRR628BIS`	Q	log	observed	financial
GLB_OIL	WTI crude (q-avg, $/bbl)	FRED	`WTISPLC`	Q	log	observed	global common
GLB_USLR	US 10Y Treasury yield (%)	FRED	`DGS10`	Q	lin	observed	global common
GLB_USBAA	US Baa corporate yield (%)	FRED	`BAA`	Q	lin	observed	global common
GLB_USEQ	US equity (S&P 500 proxy)	FRED	`SPASTT01USM661N`	Q	log	observed	global common
CREDIT_GDP_TR	Credit to NFCs (% of GDP) [NEW: credit quantity]	BIS via FRED	`QTRNAM770A`	Q	lin	masked	RW; nonnull 101/104, range 17.5–78.1%; 2000Q1=23.2, 2025Q4=38.9; COVID cells `null`

[FLAG: TR credit PRICE DROPPED] No reachable free NFC lending rate or corporate/sovereign yield: OECD IRLT/FINMARK is empty for TUR; the IMF `MFS_IR` flow carries only discount/deposit/policy-related rates for TUR (which would duplicate `RATE_TR`/`POLRATE_TR`); the CBRT EVDS commercial-loan rate (`TP.KTF12`) needs an EVDS API key not available; FRED

Turkey rate series are stale (≤2022). Per the locked spec the TR credit block is *quantity only*. *[FLAG: TR base-macro series IDs]* The provider-class for the eight base real-activity/price/labour series and the short rate is documented at the GVAR roster level (Section 19.3: OECD QNA / IMF flows, extended to 2025Q4 by `build_gvar_2025q4.py` via WEO temporal disaggregation), but the upstream raw file `country_rawdata/DataRaw_TR_glob.json` carries no per-variable source/series-id stamp, so the exact provider series codes for `GDP/CONS/INV/GOV/EXP/IMP/CPI/LR/RATE` cannot be asserted from the artifacts and are not fabricated here.

20.52 South Africa (ZA)

South Africa is a coupled GVAR bloc, $n = 19$ (staged credit spec; deployed bloc adds the three `ROW*_ZA` stars), $p = 3$, over 2002Q1–2025Q4 ($T = 96$). Macro series are COVID-masked 2020Q2–Q4 and DK-completed; financials observed. The credit block carries both legs: a BIS quantity and an IMF/IFS lending-rate price.

Table 98: South Africa (ZA) variable documentation.

Variable (id)	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
`GDP_ZA`	Real GDP	IMF/OECD QNA (GVAR base)	*[FLAG]*	Q	log	masked	WEO-extended
`CONS_ZA`	Real household consumption	IMF/OECD QNA	*[FLAG]*	Q	log	masked	
`INV_ZA`	Real gross fixed capital formation	IMF/OECD QNA	*[FLAG]*	Q	log	masked	
`GOV_ZA`	Real government consumption	IMF/OECD QNA	*[FLAG]*	Q	log	masked	
`EXP_ZA`	Real exports	IMF/OECD QNA	*[FLAG]*	Q	log	masked	
`IMP_ZA`	Real imports	IMF/OECD QNA	*[FLAG]*	Q	log	masked	
`CPI_ZA`	CPI index	IMF IFS / national	*[FLAG]*	Q	log	masked	
`LR_ZA`	Unemployment rate	IMF IFS / national	*[FLAG]*	Q	lin	masked	
`RATE_ZA`	SARB policy rate (repo)	IMF IFS / SARB	*[FLAG]*	Q	lin	observed	financial
`YIELD10_ZA`	10Y government bond yield	IMF IFS / national	*[FLAG]*	Q	lin	observed	financial
`NEER_ZA`	Nominal effective exchange rate	IMF IFS / BIS	*[FLAG]*	Q	log	observed	financial
`POLRATE_ZA`	Policy rate (BIS CBPOL)	BIS SDMX WS_CBPOL	`M.ZA` (3M avg)	Q	lin	observed	financial
`HP_ZA`	Real residential property price	FRED (BIS real RPP)	`QZAR628BIS`	Q	log	observed	financial

Variable (id)	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
`GLB_OIL`	WTI crude (q-avg, $/bbl)	FRED	`WTISPLC`	Q	log	observed	global common
`GLB_USLR`	US 10Y Treasury yield (%)	FRED	`DGS10`	Q	lin	observed	global common
`GLB_USBAA`	US Baa corporate yield (%)	FRED	`BAA`	Q	lin	observed	global common
`GLB_USEQ`	US equity (S&P 500 proxy)	FRED	`SPASTT01USM661N`	Q	log	observed	global common
`CREDIT_GDP_ZA`	Credit to NFCs (% of GDP) [NEW: credit quantity]	BIS via FRED	`QZANAM770A`	Q	lin	masked	RW; nonnull 69/96 (BIS series starts 2008Q1 ⇒ 24 leading nulls), range 32.9–41.4%; last 33.2; COVID cells `null`
`LENDR_ZA`	IFS lending rate (% p.a.) [NEW: credit price]	IMF IFS via `MFS_IR`	`ZAF.MFS162_RT_PT_A_PT.Q`	Q	lin	observed	RW; SARB-prime-linked representative bank lending rate; nonnull 96/96, range 7.0–17.0%; last 10.33

[FLAG: ZA credit quantity leading gap] The BIS NFC series begins 2008Q1, leaving 24 leading nulls (2002Q1–2007Q4); these are a normal ragged leading edge handled by the COVID-as-missing Gibbs.
[FLAG: ZA base-macro series IDs] As for TR, the per-variable provider series codes for the eight base macro series, the SARB repo rate, the 10Y yield and the NEER are not stamped in `country_rawdata/DataRaw_ZA_glob.json`; provider class is per Section 19.3 (IMF flows / OECD QNA / national, WEO-extended).

20.53 Saudi Arabia (SA)

Saudi Arabia is a coupled GVAR bloc, $n = 16$ (staged credit spec; deployed bloc adds the three `ROW*_SA` stars), $p = 3$, over 2010Q1–2025Q4 ($T = 64$). Macro series are COVID-masked 2020Q2–Q4 and DK-completed; financials observed. The credit block is *quantity-only* (price dropped; flag below).

Table 99: Saudi Arabia (SA) variable documentation.

Variable (id)	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
GDP_SA	Real GDP	IMF/OECD QNA (GVAR base)	*[FLAG]*	Q	log	masked	WEO-extended
CONS_SA	Real household consumption	IMF/OECD QNA	*[FLAG]*	Q	log	masked	
INV_SA	Real gross fixed capital formation	IMF/OECD QNA	*[FLAG]*	Q	log	masked	
GOV_SA	Real government consumption	IMF/OECD QNA	*[FLAG]*	Q	log	masked	
EXP_SA	Real exports	IMF/OECD QNA	*[FLAG]*	Q	log	masked	
IMP_SA	Real imports	IMF/OECD QNA	*[FLAG]*	Q	log	masked	
CPI_SA	CPI index	IMF IFS / national	*[FLAG]*	Q	log	masked	
LR_SA	Unemployment rate	IMF IFS / national	*[FLAG]*	Q	lin	masked	
RATE_SA	Policy rate (SAMA repo)	IMF IFS / SAMA	*[FLAG]*	Q	lin	observed	financial
NEER_SA	Nominal effective exchange rate	IMF IFS / BIS	*[FLAG]*	Q	log	observed	financial
POLRATE_SA	Policy rate (BIS CBPOL)	BIS SDMX WS_CBPOL	`M.SA` (3M avg)	Q	lin	observed	financial
GLB_OIL	WTI crude (q-avg, $/bbl)	FRED	`WTISPLC`	Q	log	observed	global common
GLB_USLR	US 10Y Treasury yield (%)	FRED	`DGS10`	Q	lin	observed	global common
GLB_USBAA	US Baa corporate yield (%)	FRED	`BAA`	Q	lin	observed	global common
GLB_USEQ	US equity (S&P 500 proxy)	FRED	`SPASTT01USM661N`	Q	log	observed	global common
CREDIT_GDP_SA	Credit to NFCs (% of GDP) [NEW: credit quantity]	BIS via FRED	`QSANAM770A`	Q	lin	masked	RW; nonnull 61/64, range 19.3–45.9%; 2010Q1=27.7, 2025Q4=45.9; COVID cells `null`

[FLAG: SA credit PRICE DROPPED] SAMA publishes no NFC lending rate (SAIBOR is an interbank money-market rate, conceptually covered by `RATE_SA`); SA is absent from IMF `MFS_IR`; OECD/FRED/WB carry no Saudi lending rate. Saudi banking is largely Islamic-finance, so no conventional IFS-style "lending rate" is published. Per the locked spec the SA credit block is

quantity only.

[FLAG: SA base-macro series IDs] Per-variable provider series codes for the base macro series, the SAMA repo rate and the NEER are not stamped in `country_rawdata/DataRaw_SA_glob.json`; provider class per Section 19.3.

20.54 Argentina (AR)

Argentina is a coupled GVAR bloc, $n = 17$ (staged credit spec; deployed bloc adds the three `ROW*_AR` stars), $p = 3$, over 2004Q1–2025Q4 ($T = 88$). Macro series are COVID-masked 2020Q2–Q4 and DK-completed; financials observed. Note that the entire AR bloc uses an RW own-lag prior on *every* variable (including the macro logs), reflecting the highly non-stationary, high-inflation Argentine sample. The credit block carries both legs.

Table 100: Argentina (AR) variable documentation.

Variable (id)	**Description**	**Source**	**Series ID/key**	**Freq**	**Trans.**	**COVID**	**Notes**
`GDP_AR`	Real GDP	IMF/OECD QNA (GVAR base)	*[FLAG]*	Q	log	masked	RW; WEO-extended
`CONS_AR`	Real household consumption	IMF/OECD QNA	*[FLAG]*	Q	log	masked	RW
`INV_AR`	Real gross fixed capital formation	IMF/OECD QNA	*[FLAG]*	Q	log	masked	RW
`GOV_AR`	Real government consumption	IMF/OECD QNA	*[FLAG]*	Q	log	masked	RW
`EXP_AR`	Real exports	IMF/OECD QNA	*[FLAG]*	Q	log	masked	RW
`IMP_AR`	Real imports	IMF/OECD QNA	*[FLAG]*	Q	log	masked	RW
`CPI_AR`	CPI index	IMF IFS / national	*[FLAG]*	Q	log	masked	RW
`LR_AR`	Unemployment rate	IMF IFS / national	*[FLAG]*	Q	lin	masked	RW
`RATE_AR`	BADLAR short-term rate	IMF IFS / BCRA	*[FLAG]*	Q	lin	observed	financial, RW
`NEER_AR`	Nominal effective exchange rate	IMF IFS / BIS	*[FLAG]*	Q	log	observed	financial, RW
`POLRATE_AR`	Policy rate (BIS CBPOL)	BIS SDMX WS_CBPOL	`M.AR` (3M avg)	Q	lin	observed	financial, RW
`GLB_OIL`	WTI crude (q-avg, $/bbl)	FRED	`WTISPLC`	Q	log	observed	global common
`GLB_USLR`	US 10Y Treasury yield (%)	FRED	`DGS10`	Q	lin	observed	global common
`GLB_USBAA`	US Baa corporate yield (%)	FRED	`BAA`	Q	lin	observed	global common

Variable (id)	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
`GLB_USEQ`	US equity (S&P 500 proxy)	FRED	`SPASTT01USM661N`	Q	log	observed	global common
`CREDIT_GDP_AR`	Credit to NFCs (% of GDP) [NEW: credit quantity]	BIS via FRED	`QARNAM770A`	Q	lin	masked	RW; nonnull 85/88, range 13.5–38.0%; 2004Q1=38.0, 2025Q4=23.2; COVID cells `null`
`LENDR_AR`	Lending rate (% p.a.; extreme volatility) [NEW: credit price]	IMF IFS via `MFS_IR`	`ARG.MFS162_RT_PT_A_PT.Q`	Q	lin	observed	RW; peso lending rate (not the FC variant); nonnull 64/88 (24 leading nulls, IFS starts 2010Q1); range 9.85–120.54% (**120.54% @2023Q4** = genuine hyperinflation peak); observed through COVID

[FLAG: AR credit price extreme volatility] The peso lending rate swings enormously with the inflation regime (peak 120.5% in 2023Q4); under an RW `lin` prior it produces very large innovations. The series has 24 leading nulls (2004Q1–2009Q4; IFS begins 2010Q1), handled by the ragged-edge Gibbs. The peso variant `MFS162_RT_PT_A_PT` is used, not the much-lower foreign-currency variant.
[FLAG: AR base-macro series IDs] Per-variable provider codes for the base macro series, the BADLAR rate and the NEER are not stamped in `country_rawdata/DataRaw_AR_glob.json`; provider class per Section 19.3.

20.55 Russia (RU)

Russia is a coupled GVAR bloc, $n = 14$ (staged credit spec; deployed bloc adds the three `ROW*_RU` stars), $p = 3$, over 2005Q1–2025Q4 ($T = 84$). It is a compact bloc (no consumption/investment/government/import expenditure split): its real-side activity is summarised by real GDP and goods exports. Macro series are COVID-masked 2020Q2–Q4 and DK-completed; financials observed. The credit block carries both legs; the price leg is *spliced* (flag below).

Table 101: Russia (RU) variable documentation.

Variable (id)	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
GDP_RU	Real GDP (const. 2010 LCU, SA)	IMF/OECD QNA (GVAR base)	*[FLAG]*	Q	log	masked	RW; WEO-extended
CPI_RU	CPI index (all-items)	IMF IFS / national	*[FLAG]*	Q	log	masked	RW
EXP_RU	Goods exports (FOB, USD)	IMF IFS / national	*[FLAG]*	Q	log	masked	RW
LR_RU	Unemployment rate	IMF IFS / national	*[FLAG]*	Q	lin	masked	RW
RATE_RU	Interbank call-money rate	IMF IFS / CBR	*[FLAG]*	Q	lin	observed	financial, RW
NEER_RU	Nominal effective exchange rate	IMF IFS / BIS	*[FLAG]*	Q	log	observed	financial, RW
POLRATE_RU	Policy rate (BIS CBPOL)	BIS SDMX WS_CBPOL	M.RU (3M avg)	Q	lin	observed	financial, RW
HP_RU	Real residential property price	FRED (BIS real RPP)	QRUR628BIS	Q	log	observed	financial, RW
GLB_OIL	WTI crude (q-avg, $/bbl)	FRED	WTISPLC	Q	log	observed	global common
GLB_USLR	US 10Y Treasury yield (%)	FRED	DGS10	Q	lin	observed	global common
GLB_USBAA	US Baa corporate yield (%)	FRED	BAA	Q	lin	observed	global common
GLB_USEQ	US equity (S&P 500 proxy)	FRED	SPASTT01USM661N	Q	log	observed	global common
CREDIT_GDP_RU	Credit to NFCs (% of GDP) [NEW: credit quantity]	BIS via FRED	QRUNAM770A	Q	lin	masked	RW; nonnull 81/84, range 31.1–90.3%; 2005Q1=31.7, 2025Q4=84.7; COVID cells null
LENDR_RU	Lending rate (% p.a.; IMF spliced w/ CBR) [NEW: credit price]	IMF IFS MFS_IR spliced with CBR	RUS. MFS162_RT_PT_A_PT. Q + CBR loans_nonfin_e.xlsx	Q	lin	observed	RW; full coverage 84/84, range 6.05–22.57%; 2005Q1=10.63, 2025Q4=18.23

[FLAG: RU credit price SPLICED] IMF MFS162 ends 2024Q2 (= 17.21). For 2024Q3–2025Q4 the series is spliced with the CBR weighted-average rate on ruble loans to non-financial organisations, "up to 1 year (incl. call loans)" (loans_nonfin_e.xlsx, sheet 1 col. 5; values 18.97, 22.57, 22.19, 22.28, 19.64, 18.23). This is the same underlying CBR survey the IMF reports as MFS162: validated MAD = 0.003 over 42 overlapping quarters (2014–2024Q2), with a continuous seam at

2024Q2 (IMF 17.21 vs CBR 17.15).

[FLAG: RU base-macro series IDs] Per-variable provider codes for GDP, CPI, goods exports, unemployment, the interbank rate and the NEER are not stamped in `country_rawdata/DataRaw_RU_glob.json`; provider class per Section 19.3.

20.56 Thailand (TH)

Thailand is a coupled GVAR bloc, $n = 19$ (staged credit spec; deployed bloc adds the three `ROW*_TH` stars), $p = 3$, over 2003Q1–2025Q4 ($T = 92$). All expenditure components and prices are seasonally adjusted. Macro series are COVID-masked 2020Q2–Q4 and DK-completed; financials observed. The credit block carries both legs.

Table 102: Thailand (TH) variable documentation.

Variable (id)	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
`GDP_TH`	Real GDP (SA, const. prices)	IMF/OECD QNA (GVAR base)	*[FLAG]*	Q	log	masked	WEO-extended
`CONS_TH`	Real private consumption (SA)	IMF/OECD QNA	*[FLAG]*	Q	log	masked	
`INV_TH`	Real gross fixed capital formation (SA)	IMF/OECD QNA	*[FLAG]*	Q	log	masked	
`GOV_TH`	Real government consumption (SA)	IMF/OECD QNA	*[FLAG]*	Q	log	masked	
`EXP_TH`	Real exports of g&s (SA)	IMF/OECD QNA	*[FLAG]*	Q	log	masked	
`IMP_TH`	Real imports of g&s (SA)	IMF/OECD QNA	*[FLAG]*	Q	log	masked	
`CPI_TH`	CPI, all items (index)	IMF IFS / national	*[FLAG]*	Q	log	masked	
`LR_TH`	Unemployment rate	IMF IFS / national	*[FLAG]*	Q	lin	masked	
`RATE_TH`	Policy rate (BOT)	IMF IFS / BOT	*[FLAG]*	Q	lin	observed	financial
`YIELD10_TH`	10Y government bond yield	IMF IFS / national	*[FLAG]*	Q	lin	observed	financial
`NEER_TH`	Nominal effective exchange rate (BIS broad)	BIS / IMF	*[FLAG]*	Q	log	observed	financial
`POLRATE_TH`	Policy rate (BIS CBPOL)	BIS SDMX WS_CBPOL	`M.TH` (3M avg)	Q	lin	observed	financial
`HP_TH`	Real residential property price	FRED (BIS real RPP)	`QTHR628BIS`	Q	log	observed	financial
`GLB_OIL`	WTI crude (q-avg, $/bbl)	FRED	`WTISPLC`	Q	log	observed	global common

Variable (id)	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
GLB_USLR	US 10Y Treasury yield (%)	FRED	DGS10	Q	lin	observed	global common
GLB_USBAA	US Baa corporate yield (%)	FRED	BAA	Q	lin	observed	global common
GLB_USEQ	US equity (S&P 500 proxy)	FRED	SPASTT01USM661N	Q	log	observed	global common
CREDIT_GDP_TH	Credit to NFCs (% of GDP) [NEW: credit quantity]	BIS via FRED	QTHNAM770A	Q	lin	masked	RW; nonnull 89/92, range 68.4–95.9%; last 68.5; COVID cells null
LENDR_TH	IFS lending rate (% p.a.) [NEW: credit price]	IMF IFS via MFS_IR	THA. MFS162_RT_PT_A_PT. Q	Q	lin	observed	RW; BOT announced/minimum loan rate; non-null 88/92 (4 leading nulls in 2003, IFS starts 2004Q1); range 3.03–6.58%; last 3.44

[FLAG: TH credit price leading gap] The IFS lending rate has 4 leading nulls (2003; the series begins 2004Q1), a normal ragged leading edge handled by the Gibbs.
[FLAG: TH base-macro series IDs] Per-variable provider codes for the base macro series, the BOT policy rate, the 10Y yield and the NEER are not stamped in `country_rawdata/DataRaw_TH_glob.json`; provider class per Section 19.3.

20.57 Malaysia (MY)

Malaysia is a coupled GVAR bloc, $n = 18$ (staged credit spec; deployed bloc adds the three `ROW*_MY` stars), $p = 3$, over a short 2015Q1–2025Q4 grid ($T = 44$). Macro series are COVID-masked 2020Q2–Q4 and DK-completed; financials observed. The credit block carries both legs; note the short sample (flag).

Table 103: Malaysia (MY) variable documentation.

Variable (id)	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
GDP_MY	Real GDP (chain volume, SA)	IMF/OECD QNA (GVAR base)	*[FLAG]*	Q	log	masked	WEO-extended
CONS_MY	Real private consumption (SA)	IMF/OECD QNA	*[FLAG]*	Q	log	masked	

Variable (id)	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
`INV_MY`	Real gross fixed capital formation (SA)	IMF/OECD QNA	*[FLAG]*	Q	log	masked	
`GOV_MY`	Real government consumption (SA)	IMF/OECD QNA	*[FLAG]*	Q	log	masked	
`EXP_MY`	Real exports of g&s (SA)	IMF/OECD QNA	*[FLAG]*	Q	log	masked	
`IMP_MY`	Real imports of g&s (SA)	IMF/OECD QNA	*[FLAG]*	Q	log	masked	
`CPI_MY`	CPI index (all items)	IMF IFS / national	*[FLAG]*	Q	log	masked	
`LR_MY`	Unemployment rate	IMF IFS / national	*[FLAG]*	Q	lin	masked	
`RATE_MY`	Overnight policy rate (BNM)	IMF IFS / BNM	*[FLAG]*	Q	lin	observed	financial
`NEER_MY`	Nominal effective exchange rate	IMF IFS / BIS	*[FLAG]*	Q	log	observed	financial
`POLRATE_MY`	Policy rate (BIS CBPOL)	BIS SDMX WS_CBPOL	`M.MY` (3M avg)	Q	lin	observed	financial
`HP_MY`	Real residential property price	FRED (BIS real RPP)	`QMYR628BIS`	Q	log	observed	financial
`GLB_OIL`	WTI crude (q-avg, $/bbl)	FRED	`WTISPLC`	Q	log	observed	global common
`GLB_USLR`	US 10Y Treasury yield (%)	FRED	`DGS10`	Q	lin	observed	global common
`GLB_USBAA`	US Baa corporate yield (%)	FRED	`BAA`	Q	lin	observed	global common
`GLB_USEQ`	US equity (S&P 500 proxy)	FRED	`SPASTT01USM661N`	Q	log	observed	global common
`CREDIT_GDP_MY`	Credit to NFCs (% of GDP) [NEW: credit quantity]	BIS via FRED	`QMYNAM770A`	Q	lin	masked	RW; nonnull 41/44, range 86.1–106.3%; last 88.6; COVID cells `null`
`LENDR_MY`	IFS lending rate (% p.a.) [NEW: credit price]	IMF IFS via `MFS_IR`	`MYS.MFS162_RT_PT_A_PT.Q`	Q	lin	observed	RW; BNM average lending/base rate; nonnull 44/44, range 3.31–5.47%; last 4.62

[FLAG: MY short sample] The MY grid is only $T = 44$ (2015Q1–2025Q4) with 41 credit observations – thin for a bloc that adds two credit variables; predictive bands are correspondingly wide. *[FLAG: MY base-macro series IDs]* Per-variable provider codes for the base macro series, the BNM overnight policy rate and the NEER are not stamped in `country_rawdata/DataRaw_MY_glob.json`;

provider class per Section 19.3.

20.58 Taiwan, Province of China (TW)

Taiwan is a coupled GVAR bloc, $n = 17$ (staged credit spec; deployed bloc adds the three `ROW*_TW` stars), $p = 3$, over a long 1997Q1–2025Q4 grid ($T = 116$). It carries an unusually rich short-end financial block (CBC discount rate, interbank call-loan rate and 10Y yield) but *no* BIS house-price series. Macro series are COVID-masked 2020Q2–Q4 and DK-completed; financials observed. Both credit legs are present but *constructed/national*, not BIS (flags below).

Table 104: Taiwan (TW) variable documentation.

Variable (id)	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
`GDP_TW`	Real GDP	DGBAS / OECD-style QNA	*[FLAG]*	Q	log	masked	RW; WEO-extended
`CONS_TW`	Real household consumption	DGBAS	*[FLAG]*	Q	log	masked	RW
`GOV_TW`	Real government consumption	DGBAS	*[FLAG]*	Q	log	masked	RW
`INV_TW`	Real gross fixed capital formation	DGBAS	*[FLAG]*	Q	log	masked	RW
`EXP_TW`	Real exports of g&s	DGBAS	*[FLAG]*	Q	log	masked	RW
`IMP_TW`	Real imports of g&s	DGBAS	*[FLAG]*	Q	log	masked	RW
`CPI_TW`	CPI, all items (index, 2021=100)	DGBAS	*[FLAG]*	Q	log	masked	RW
`LR_TW`	Unemployment rate	DGBAS	*[FLAG]*	Q	lin	masked	RW
`RATE_TW`	CBC discount rate (policy)	CBC	*[FLAG]*	Q	lin	observed	financial, RW
`MMRATE_TW`	Interbank call-loan rate (money mkt)	CBC	*[FLAG]*	Q	lin	observed	financial, RW
`YIELD10_TW`	10Y government bond yield	CBC / national	*[FLAG]*	Q	lin	observed	financial, RW
`GLB_OIL`	WTI crude (q-avg, $/bbl)	FRED	`WTISPLC`	Q	log	observed	global common
`GLB_USLR`	US 10Y Treasury yield (%)	FRED	`DGS10`	Q	lin	observed	global common
`GLB_USBAA`	US Baa corporate yield (%)	FRED	`BAA`	Q	lin	observed	global common
`GLB_USEQ`	US equity (S&P 500 proxy)	FRED	`SPASTT01USM661N`	Q	log	observed	global common

Variable (id)	Description	Source	Series ID/key	Freq	Trans.	COVID	Notes
CREDIT_GDP_TW	Credit to NFCs (% of GDP) [NEW: credit quantity, CONSTRUCTED]	CBC EI86 / DGBAS GDP	100× CBC EI86 private-enterprise loans (NT$mn, monthly stock → q-avg) ÷ 4Q-trailing DGBAS nominal GDP	Q	lin	masked	RW; nonnull 113/116, range 46.3–65.4%; 2025Q4=60.6; COVID cells null
LENDR_TW	5-major-banks new loan rate (% p.a.) [NEW: credit price]	CBC EH45 (total, interest rate)	CBC EH45 total (monthly → q-avg)	Q	lin	observed	RW; CBC realised five-major-banks new-loan rate; nonnull 116/116, range 1.16–8.62%; 2020Q2=1.26; 2025Q4=2.16

[FLAG: TW credit quantity CONSTRUCTED, not BIS] BIS publishes no NFC credit-to-GDP for Taiwan (FRED QTWNAM770A returns HTTP 400; BIS WS_TC SDMX returns "no structures match"). CREDIT_GDP_TW is therefore constructed as 100× CBC EI86 private-enterprise loans (private enterprises; excludes household and public-enterprise loans) over 4-quarter-trailing DGBAS nominal GDP. This is a private-enterprise-loan proxy, broader than the BIS market-value NFC concept used elsewhere, so cross-economy comparability of the credit variable is reduced.

[FLAG: TW credit price provenance] LENDR_TW is the CBC's published headline five-major-banks total new-loan rate (EH45, "total"), used as the locked-spec "5-bank base lending rate" – a national CBC series, not an IMF or BIS feed. The Taiwan source files are fetched via `fetch_tw_cbc_dgbas.py`; the DGBAS GDP host ships an incomplete TLS chain and is fetched with verification relaxed for that host only (documented in the fetcher).

[FLAG: TW house price] Unlike the other blocs here, TW carries no HP_TW (no BIS real residential property price was added).

[FLAG: TW base-macro series IDs] The base real-activity, price, labour and short-rate series originate with Taiwan's DGBAS (national accounts, CPI, unemployment) and the CBC (discount rate, call-loan rate, 10Y yield), but the upstream raw file `country_rawdata/DataRaw_TW_glob.json` carries no per-variable series-id stamp, so exact DGBAS/CBC table codes for `GDP/CONS/GOV/INV/EXP/IMP/CPI/LR/RATE/MMRATE/YIELD10` are not asserted here.

20.59 Country × Variable Coverage Matrix

Table 105 is the per-bloc coverage matrix for the key variable groups, generated deterministically from the coverage manifest (`app/public/data/coverage_manifest.json`; the generator is `docs/build_coverage_table.py`). A ✓ marks that the group is carried by the bloc's shipped specification; `--` marks a genuine absence in that bloc's panel (e.g. blocs that carry a cash/short rate in place of a dedicated policy rate, or that ship no NEER). The *Last obs* column is the bloc's last historical quarter (the ragged edge of Section 19.5); *Source* is the primary source family (Section 19.3), with a trailing italic tag for the WEO back-fill (CN/IN) and source-review (KR/RU/ID/TH) provenance flags.

Table 105: Country × variable-group coverage matrix for all 50 deployed blocs. Columns: **GDP** real activity; **Price** HICP/CPI; **Unemp** unemployment rate (`LR`); **Pol** policy rate; **Long** 10-year/long government yield; **Cred** credit-to-GDP; **Lend** lending/short rate; **FX** NEER; **Fisc** the four-variable fiscal block (`FREV`/`FPEXP`/`EFFI`/`SFA`). ✓ = carried; `--` = absent in that bloc. WEO = WEO-back-filled tail (CN/IN); review = source-review tail (KR/RU/ID/TH).

Bloc	GDP	Price	Unemp	Pol	Long	Cred	Lend	FX	Fisc	Last obs	Source
AR	✓	✓	✓	✓	✓	✓	✓	✓	–	2026Q1	OECD/FRED
AT	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
AU	✓	✓	✓	–	✓	✓	✓	✓	–	2026Q1	OECD/FRED
BE	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
BR	✓	✓	✓	✓	✓	✓	✓	✓	–	2026Q1	OECD/FRED
CA	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q1	OECD/FRED
CH	✓	✓	✓	✓	✓	✓	✓	✓	–	2026Q1	Eurostat
CL	✓	✓	✓	✓	✓	✓	✓	✓	–	2026Q1	OECD/FRED
CN	✓	✓	–	✓	✓	✓	✓	✓	✓	2025Q4	FRED *WEO*
CO	✓	✓	✓	✓	✓	✓	✓	✓	–	2026Q1	OECD/FRED
CR	✓	✓	✓	–	✓	✓	✓	✓	–	2026Q1	OECD/FRED
CZ	✓	✓	✓	✓	✓	✓	✓	✓	–	2026Q1	Eurostat
DE	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q1	Eurostat
DK	✓	✓	✓	✓	✓	✓	✓	✓	–	2026Q1	Eurostat
EA	✓	✓	–	✓	✓	✓	✓	✓	✓	2026Q1	Eurostat
EE	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
ES	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
FI	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
FR	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
GR	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
HU	✓	✓	✓	✓	✓	✓	✓	✓	–	2026Q1	Eurostat
ID	✓	✓	–	✓	✓	✓	✓	✓	–	2026Q1	OECD/FRED *review*
IE	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
IL	✓	✓	✓	✓	✓	✓	✓	✓	–	2024Q4	OECD/FRED
IN	✓	✓	–	✓	✓	✓	✓	✓	–	2025Q4	OECD/FRED *WEO*
IS	✓	✓	✓	✓	✓	✓	✓	✓	–	2026Q1	OECD/FRED
IT	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
JP	✓	✓	✓	✓	✓	✓	✓	–	✓	2026Q1	OECD/FRED

Bloc	**GDP**	**Price**	**Unemp**	**Pol**	**Long**	**Cred**	**Lend**	**FX**	**Fisc**	**Last obs**	**Source**
`KR`	✓	✓	✓	✓	✓	✓	✓	–	–	2026Q1	OECD/FRED *review,frozen mirror*
`LT`	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
`LU`	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
`LV`	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
`MX`	✓	✓	✓	✓	✓	✓	✓	✓	–	2026Q1	OECD/FRED
`MY`	✓	✓	✓	✓	✓	✓	✓	✓	–	2026Q1	OECD/FRED
`NL`	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
`NO`	✓	✓	✓	✓	✓	✓	✓	✓	–	2026Q1	Eurostat
`NZ`	✓	✓	✓	–	✓	✓	✓	✓	–	2026Q1	OECD/FRED
`PL`	✓	✓	✓	✓	✓	✓	✓	✓	–	2026Q1	Eurostat
`PT`	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
`RU`	✓	✓	✓	✓	✓	✓	✓	✓	–	2026Q1	OECD/FRED *review*
`SA`	✓	✓	✓	✓	✓	✓	✓	✓	–	2026Q1	OECD/FRED
`SE`	✓	✓	✓	✓	✓	✓	✓	✓	–	2026Q1	Eurostat
`SI`	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
`SK`	✓	✓	✓	✓	–	✓	✓	✓	✓	2026Q1	Eurostat
`TH`	✓	✓	✓	✓	✓	✓	✓	✓	–	2026Q1	OECD/FRED *review*
`TR`	✓	✓	✓	✓	✓	✓	✓	–	–	2026Q1	OECD/FRED
`TW`	✓	✓	✓	–	✓	✓	✓	–	–	2026Q1	OECD/FRED
`UK`	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q1	OECD/FRED
`US`	✓	✓	✓	✓	✓	✓	✓	✓	✓	2026Q1	FRED
`ZA`	✓	✓	✓	✓	✓	✓	✓	✓	–	2026Q1	OECD/FRED

21 Technical Notes

21.1 Client-Side Computation

All matrix algebra, Kalman filtering, and simulation smoothing are performed in the browser using TypeScript with `Float64Array` storage. The linear algebra kernel implements:

- LU decomposition with partial pivoting for F_t^{-1} (singularity threshold: 10^{-15}).
- Cholesky decomposition for shock draws ($\Sigma = LL'$; diagonal check threshold: 10^{-14}).
- Row-major storage for cache-friendly matrix operations.

21.2 Performance Optimizations

- The unconditional smoother result is cached by a fingerprint of the model dimensions and a sample of coefficient values, avoiding recomputation when only the conditioning paths change.
- Conditioning input is debounced at 30 ms to avoid recomputation during interactive drag operations.
- The 500 thinned posterior draws ($\approx$34 MB for the US model) are loaded lazily—only when the user clicks "Compute."
- Plotly.js is loaded via dynamic `import()` to avoid blocking initial page render.

21.3 Interactive GVAR Architecture: keeping the heavy solve off the drag path

The deployed 50-bloc, US-endogenous GVAR poses a hard interactivity problem: a single coupled mean solve costs $\approx$ 4.6 s warm and $\approx$ 13 s cold (every US-pinned structure misses the shipped $(I - M)^{-1}$ coupling cache and must re-probe the Durbin–Koopman smoother per coupled bloc). Firing that solve on every pointer-move during a chart drag would make the interface unusable. The deployed design therefore removes the solve from the live drag path entirely, and reconciles the exact answer afterwards.

No solve during drag; exactly one cheap solve on release. A module-level signal bus (`gvarDragState.ts`) carries the reactive drag flag between the chart drag handler and the GVAR solve hook (`useGvarForecast.ts`), kept *out* of the React/Zustand store so that a pointer-down does not re-render the grid. While a drag is in progress *no* GVAR solve fires—neither a value-only nor a structure-change solve, since both are multi-second on the live panel. On `pointer-up` a single solve is dispatched after a short (100 ms) release macrotask that also coalesces back-to-back click-drag-click gestures into one dispatch. That release solve uses the cheap `'direct'` channel (one-hop local-elasticity preview, no 95 MB coupling-artifact structured-clone into the worker); the full cross-country fixed point with uncertainty bands (the `'full'` channel) runs *only* on the explicit

"Compute" / "Solve full GVAR" action. Non-drag edits (numeric entry, Reset, scenario switch) still dispatch live: full solves run immediately and direct previews debounce ($\approx 120\,\text{ms}$).

Optimistic conditioned-point dot. So that the user sees *immediate* feedback while the line and bands reconcile, the pinned marker (the conditioned-point "dot") is sourced from the *conditioning path* the click writes—not from the lagging solve output. The mapping lives in `conditionedDotY.ts`: `conditioningDeviationFromDisplayY` converts a clicked display y-value into the stored-space deviation, and its exact inverse `conditionedDotDisplayY` reconstructs the displayed dot, so the dot round-trips to the cursor to machine precision (`conditionedDotDisplayY`$\circ$`conditioningDeviationFromDisplayY` id). The same inverse pair backs the right-click "plant the bullet at the cursor" gesture. The smoothed forecast line and the uncertainty bands then catch up when the solve returns.

Per-chart render memoization. The render cost of a conditioning edit was historically dominated by the *render*, not the solve: a monolithic chart component repainted all ≈ 35 panels on any edit. The deployed grid (`ChartGrid.tsx`) wraps each panel in `React.memo` and reconciles the freshly-built chart objects against the prior render with a tight structural equality (`reconcileCharts/chartDeepEqual`): a panel whose trace data did not move keeps its *prior object reference*, so `React.memo` skips it. A single conditioning edit therefore rebuilds and repaints *exactly one* of the ≈ 35 panels rather than all of them (asserted in `chartgrid-render-memo.test.ts`), with a stale-guard that still repaints every affected panel when the solve lands. The whole interactive contract—no solve on drag, one cheap solve on release, an optimistic dot, and single-panel repaints—is covered by the `gvar-drag-scheduling`, `conditioned-dot-optimistic`, `right-click-plant-at-cursor`, and `chartgrid-render-memo` test suites.

21.4 Numerical Conventions

- Measurement noise: $H = 10^{-12}\, I_n$, ensuring the Kalman filter treats conditioned values as near-exact observations while maintaining numerical stability.
- Growth-to-level conversion: $\Delta\ell = \Delta g/4$ for log-transformed variables, reflecting the annualization factor in (2).
- Tikhonov regularization for Barnichon–Mesters: $\alpha = 10^{-6}$.
- All matrix operations use IEEE 754 double precision (64-bit floating point).

21.5 Identifiability and Weak Exogeneity

The platform's separable estimation–solution architecture rests on a chain of identifying assumptions; we state them explicitly and are candid about where each binds.

Weak exogeneity of the star variables. Each bloc is estimated *conditionally* on its three trade-weighted star variables (Section 16.2), treating the foreign environment as predetermined. This is the weak-exogeneity device of the GVAR literature (Pesaran et al., 2004; Dees et al., 2007): provided a domestic economy is small relative to its trading partners, its idiosyncratic shocks do not feed back contemporaneously into the aggregated foreign block, so the conditional likelihood factorizes and the bloc-specific parameters $(\hat{\beta}_c, \hat{\Sigma}_c)$ are identified without a joint, system-wide estimator. Chudik and Pesaran (2016) formalize the conditions—a granular weighting scheme in which no single partner dominates the trade-weighted average and a bounded number of strongly correlated units—under which the cross-section-averaged stars remain valid weakly-exogenous regressors as the panel grows. Our renormalized, per-component weights (Section 16.3) are designed to keep this granularity: absent or excluded partners have their mass reabsorbed across the present partners rather than concentrated on any one.

The classical treatment fixes a *frozen dominant unit* (here the United States) outside the system. We relax this in two ways. First, the EM-style re-estimation loop (Section 16.7) rebuilds every bloc's stars from its partners' Durbin–Koopman-completed paths and re-estimates each bloc in parallel until the maximum star change falls below the tolerance $\text{TOL} = 0.03$, so the conditioning information is mutually consistent at the fixed point rather than taken as given once. This tolerance is set *tighter* than the Monte Carlo noise floor of the imputation (≈ 0.13 on the star scale); the COVID-window cells of the two macro stars, where cross-iteration churn is dominated by that imputation noise, are excluded from the convergence metric. Second, in the deployed roster the US is made *endogenous*: it carries its own `ROW*_US` stars and is solved inside the cross-country fixed point like any other bloc. What keeps the relaxed system well posed is the contraction property of the solve-time coupling operator M of (40): its spectral radius is $\rho(M) \approx 0.59$ for the original six-bloc pilot, $\rho(M) \approx 0.726$ under the frozen-anchor configuration with the euro area aggregated into a single bloc, and $\rho(M) \approx 0.924$ for the full 33-bloc endogenous-US configuration. As long as $\rho(M) < 1$ the map $x \mapsto Mx + d$ is a contraction, $(I - M)$ is nonsingular, and the scenario fixed point is unique—so closing the RoW→US feedback loop raises the coupling but leaves the system stationary and the scenario deviation identified.

Fiscal identification: a triangular budget identity. The fiscal block is identified by construction rather than by estimation. Four flow drivers (`FREV_GDP`, `FPEXP_GDP`, the effective rate `EFFI`, and the stock-flow adjustment `SFA`) are estimated by the BVAR; the debt stock is then *derived* by rolling the budget identity of (47) forward (Escolano, 2010; International Monetary Fund, 2013). The recursion is triangular: given lagged debt d_{t-1}, the effective rate, nominal growth, the primary balance pb = `FREV_GDP` − `FPEXP_GDP`, and the `SFA`, the new debt level is fully determined, so debt is never a free parameter. The `SFA` term is the *exact* residual that reconciles the change in the recorded debt stock with the cumulated deficit; by estimating it as its own driver we close the identity to machine precision over history—the quarterly convention of (47) is pinned by a historical back-cast of the reconstructed debt ratio against the observed one—and carry its forecast path forward. The honest caveat is that any model error in the non-`SFA` drivers is silently absorbed into the `SFA` in sample, so a large or volatile estimated `SFA` flags a measurement or perimeter gap (below-the-line operations, valuation effects) rather than a behavioral relationship.

Euro-area monetary identification. The euro area receives a deliberately asymmetric treatment. In the EA-aggregate bloc the ECB policy rate, the short rate, and the shared corporate-risk yield `EACORP` are *endogenous* (tagged `block="domestic"`); in each euro-member bloc the very same area-wide instruments are *weakly exogenous* (tagged `block="external"`), entering the member's foreign block alongside the US and global commons. This encodes the identifying assumption that monetary policy is set for the area as a whole, so a single member faces it as predetermined while the aggregate determines it—the same exogenous-versus-endogenous split that the star variables impose on the real side. National financial variables (sovereign 10Y yields, house prices) stay domestic. The limitation is structural: members share one common monetary factor, so member-level monetary counterfactuals are conditional scenarios on the area instrument, not independent national policy experiments, and the strong cross-member correlation of `EACORP` and policy-rate shocks sits at the edge of the granularity that weak exogeneity formally requires.

22 Reproducibility and Replication

The platform is engineered so that every published number traces back to source data through a single, scripted pipeline: there are no hand-edited artifacts on the path from raw series to the deployed app. This section maps the repository, lists the software dependencies, gives the end-to-end replication recipe, and explains the model-signature versioning that refuses a stale cross-country coupling.

22.1 Repository and Folder Map

The work lives in a single git repository with three top-level concerns, shown in Table 106. The `pipeline/` tree (MATLAB plus Python) performs all estimation and data assembly and is the "private" research layer; `app/` is the public Next.js front end and its in-browser compute kernels; `docs/` holds this manual and the operational runbooks. The estimator writes a per-bloc JSON export that `port_gvar_exports.py` copies into `app/public/data/bvar/`; the deployed app reads *only* that JSON and never re-estimates.

Table 106: Repository layout. Estimation in pipeline/ produces a per-bloc JSON export; port_gvar_exports.py ports it into app/public/data/bvar/; the front end and its compute kernels consume that data and never re-estimate.

Top-level	Key contents
pipeline/ (MATLAB + Python)	export_bvar_gvar24_loop.m (GVAR–EM estimator); build_*.py (FRED / Eurostat / BIS / ECB pulls, trade weights, fiscal drivers); port_gvar_exports.py; _verify/ ground-truth checks.
app/ (Next.js + compute)	src/lib/compute/ (Durbin–Koopman, GVAR solve, fiscal accounting); scripts/ (precompute_gvar_coupling.mjs, verify_gvar24.mjs, invert_coupling.{m,py}); public/data/bvar/<CC>/ (the deployed JSON the app reads).
docs/ (manual + runbooks)	user_manual.tex; REESTIMATING_MODELS.md, ADDING_A_COUNTRY.md, ARCHITECTURE_FOR_AI.md.

22.2 Software Dependencies

- **Estimation (MATLAB).** A recent MATLAB with the *Parallel Computing Toolbox* (the EM M-step re-estimates all blocs in a parfor). The bundled BVAR/MFBVAR code — the GLP hierarchical-prior Gibbs sampler bvarGLPmf.m and the COVID-as-missing wrapper cgl_bvarV5.m — ships in the repository; no external MATLAB packages are required.
- **Data assembly (Python 3).** Pure standard library (urllib, json, csv, math); no pandas, requests, or fredapi. The build_*.py pulls hit the FRED (key via FRED_API_KEY), Eurostat SDMX 2.1, BIS, and ECB public endpoints. invert_coupling.py (NumPy) is the no-MATLAB fallback for the coupling inverse.
- **Application (Node / Next.js).** Node with the pinned Next.js toolchain; typescript, vitest for the test gate, and vite-node/tsx to run the scripts/*.mjs precompute and verification utilities.

22.3 End-to-End Replication Recipe

The full path from raw data to a green deploy gate is the following enumerated sequence (canonical commands in docs/REESTIMATING_MODELS.md).

1. **Build the data.** Run the pipeline/build_*.py pulls (with FRED_API_KEY set) to assemble the harmonized common-vintage panels (raw series, trade weights, fiscal drivers) as pipeline/country_rawdata/DataRaw_<CC>_glob.json.
2. **Estimate (GVAR–EM).** Run export_bvar_gvar24_loop.m. It bootstraps the US as a closed bloc, then iterates the E-step (rebuild trade-weighted stars ROWDEM/ROWINF/ROWFIN

from partners' Durbin–Koopman–completed panels) and the M-step (parfor re-estimate every bloc via cgl_bvarV5, COVID macro-masked over 2020Q2--Q4 with the financial columns and ROWFIN kept observed) until $\max|\Delta\text{star}| < \text{TOL} = 0.03$ (or MAXIT= 12). See Durbin and Koopman (2002) for the simulation smoother and Dees et al. (2007) for the GVAR structure.

3. **Check the export.** Confirm each bloc directory under /tmp/gvar24_export/<CC>/ carries the full eight-file JSON contract: spec.json, companion_mean.json, companion_draws.json, baseline.json, baseline_levels.json, historical.json, historical_levels.json, transformations.j (the fiscal blocs additionally carry fiscal_meta.json).

4. **Port into the app.** Run port_gvar_exports.py to copy the eight-file contract into app/public/data/bvar/<CC>/, compacting every float to six significant figures (the deployed precision convention) and writing minified JSON.

5. **Regenerate the coupling.** From app/, run npx vite-node scripts/precompute_gvar_coupling.mjs. It builds the Kalman-probe operator M in JS, inverts $(I - M)$ via LAPACK (invert_coupling.m in MATLAB, or the invert_coupling.py NumPy fallback), and writes the gzipped public/data/bvar/gvar/coupling_default.json.gz (the plain .json is removed — at this bloc count the raw artifact exceeds GitHub's 100 MB per-file limit, so only the .gz is shipped and the loader decompresses it via the browser's DecompressionStream).

6. **Green gate.** From app/, run the fast fixed-point check npx tsx scripts/verify_gvar24.mjs (artifact modelSig fresh; the shipped inverse really inverts $(I - M)$ to $< 10^{-3}$; $\rho(M) < 1$ and matches the stored value; Gauss–Seidel $\equiv$ the artifact-seeded closed form), then npx tsc --noEmit and the test suite. The fast gate is npm test (the everyday suite, which *excludes* the multi-second GVAR solver canaries); the heavy closed-form/endogenous-GVAR canaries run under npm run test:heavy, and npm run test:all runs both. All type-checks and tests — including the fiscal, GVAR, and scale canaries — must pass.

7. **Deploy.** On a green gate (and only with explicit confirmation), push app/** to main: the deploy-to-website.yml GitHub Action runs npm ci && npm run build with BASE_PATH=/bvar-scenario then copies the static export app/out/* into the matyasfarkas/starter-hugo-academic academic website repository's static/bvar-scenario/ directory. That repository is a Hugo Academic site published to **GitHub Pages**, which rebuilds and serves the tool at the /bvar-scenario path. The Action itself contains no Hugo step—it only copies app/out.

22.4 Model-Signature Versioning

The cross-country coupling artifact is bound to the bloc estimates it was built from by a *model signature*. Over the qualified bloc set the signature concatenates, per bloc c, the dimensions (n_c, p_c, H_c) and three fingerprint entries of the posterior-mean companion: the leading reduced-form coefficient $\hat{\beta}^{(c)}_{1,1}$, the *intercept* entry of the last equation $\hat{\beta}^{(c)}_{n_c p_c+1,\, n_c}$ (the constant row of $\hat{\beta}$, which stacks $n_c p_c$

lag-coefficient rows followed by one constant row), and the leading residual variance $\hat{\Sigma}^{(c)}_{1,1}$ — each at six-digit exponential precision:

$$\text{modelSig} \;=\; \big\|_c \; c \,:\, n_c\, p_c\, H_c\, \hat{\beta}^{(c)}_{1,1}\, \hat{\beta}^{(c)}_{n_c p_c+1,\, n_c}\, \hat{\Sigma}^{(c)}_{1,1}.$$

The precomputed artifact stores this signature alongside M, $(I - M)^{-1}$, $\rho(M)$, and the dimension. At load time the app recomputes `modelSig` from the deployed companions and accepts the artifact only on an exact match *and* a matching (here empty) pin structure; on any mismatch the stale artifact is **refused** and the solver falls back to the inverse-free Neumann (Gauss–Seidel) build of M from the live models. Because any re-estimation perturbs at least one fingerprint entry, the signature flips whenever the data changes — making it impossible to ship a coupling that silently disagrees with its blocs. This is exactly the "freshness" invariant the `verify_gvar24` gate enforces above, for the current endogenous-US system (33 coupled blocs, augmented closed-form dimension 2060, spectral radius $\rho(M) \approx 0.924 < 1$).

23 Empirical Results

This section reports the empirical program of the unified global-forecasting system. The estimation machinery of Sections 12–16 is implemented and the production vintage (harmonized to 2025Q4, ∼50 economies: 33 GVAR blocs plus 17 euro-area standalones) has been produced. The out-of-sample horse-race (Section 23.2) and the Tier-A endogenous-vs-frozen spillover contrast (Section 23.3) are *realized* runs from the companion evaluation repository (`em-bayesian-gvar-eval`); the only piece still flagged [**TBD**] is the Tier-B GIRF/connectedness module, which is a structural extension gated behind the validated Tier-A result.

23.1 In-sample diagnostics

EM convergence. The EM-style estimation loop alternates an E-step (rebuild the trade-weighted RoW stars from partners' Durbin–Koopman–completed panels) with an M-step (a `parfor` re-estimation of every bloc under COVID macro-masking). Because all stars are rebuilt from the frozen start-of-sweep state before the parallel re-estimation, the inner sweep is a Jacobi update with the same fixed point as a sequential pass. Convergence is declared when $\max_b |\Delta\text{star}_b| < \text{TOL} = 0.03$ over a maximum of MAXIT $= 12$ sweeps. The macro stars ROWDEM/ROWINF are NaN-masked over the three COVID quarters (2020Q2–Q4) in the metric, since their cross-sweep churn there is pure imputation Monte-Carlo noise, while the financial star ROWFIN is kept observed and enters the metric. Empirically the trace flattens at a Monte-Carlo noise floor of ≈ 0.13, above the nominal tolerance, so convergence is read as a plateau at the floor rather than literal attainment of 0.03. Figure 8 sketches the expected trace (final numbers [**TBD**] once the full production run is logged).

Spectral radius of the coupling operator. At solve time the cross-country deviation system is the affine fixed point $x = Mx + d$ (Section 16.5), where M is the Kalman-probe coupling oper-

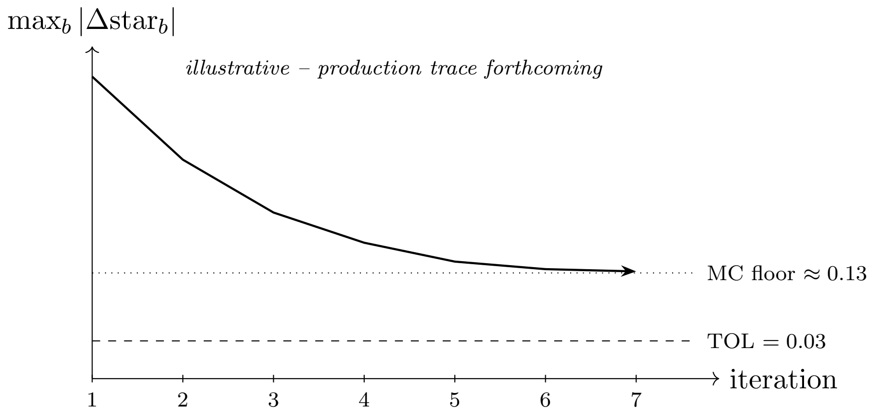

Figure 8: Schematic EM convergence trace: $\max_b |\Delta\text{star}_b|$ versus sweep, against the tolerance (0.03) and the Monte-Carlo noise floor (≈ 0.13) at which the trace plateaus. **Placeholder** — the realized trace is forthcoming.

ator. The Neumann series $\sum_k M^k d$ (Gauss–Seidel sweeps) and the direct $(I-M)^{-1}d$ agree, and convergence speed is governed by $\rho(M)$. Table 107 reports $\rho(M)$ by system size; it rises with both the number of coupled blocs and the activation of the endogenous-US channel, but remains below unity, so the fixed point is contractive.

Table 107: Spectral radius $\rho(M)$ of the coupling operator by system configuration.

Configuration	Blocs	US treatment	$\rho(M)$
Pilot	6	frozen anchor	0.59
Production	24	frozen anchor	0.726
Endogenous-US	33	endogenous	0.924

At the endogenous-US dimension (≈ 2060) the $O(\dim^3)$ inverse is prohibitive in-browser, so the system is solved inverse-free by the matvec-only recursion $x \leftarrow d + Mx$, which converges at the tabulated $\rho(M) \approx 0.924$, or via a pre-shipped cached $(I-M)^{-1}$.

23.2 Out-of-sample forecast horse race

The system is evaluated in a companion repository (`em-bayesian-gvar-eval`) under a pre-registered protocol whose *design is final* and whose realized *results* are reported in Section 23.2.4. The evaluation has two cores: a recursive out-of-sample (OOS) forecast horse-race (this subsection) and an endogenous-vs-frozen spillover analysis (Section 23.3). Every step is grounded in the actual estimator, weights, and solver of Sections 12–16, so the harness respects the data spaces and invariants of those layers.

23.2.1 Model set

The **proposed** model is the EM-Bayesian GVAR, run two ways:

- `emgvar_endo` — the full loop with the US *endogenous* (`US_ENDOGENOUS`= 1: its own `ROW{DEM,INF,FIN}_US` stars and a US weights row), the paper's contribution;

- `emgvar_frozen` — the legacy frozen-anchor variant (the US estimated once as a closed anchor); cheaper, and the Part-2 spillover contrast.

Both produce density forecasts from the per-bloc predictive draw cloud `Dforecast` at solve time. The OOS forecast is each bloc's own VAR predictive density; the GVAR coupling enters because the stars—and thus every bloc's regressors and completed history—are jointly determined by the EM fixed point.

The proposed models are raced against six benchmarks, each estimated from the *same* truncated panel so that the comparison isolates a single ingredient:

(i) **Random walk / no-change** — in the scored space: a flat last level for the (stationary) rate, and the last observed value of the growth/inflation transform carried flat (a RW *on the growth rate*). The driftless RW is the headline reference (all relative-RMSE ratios are against it); a sample-mean "climatology" forecast is a second naive anchor. The predictive density is Gaussian, centred at the no-change value with scale equal to the in-sample SD of the h-step change *estimated directly at horizon h* (not $\sqrt{h}\,\sigma_1$, which is wrong for a series that is itself a growth rate).

(ii) **AR**(p) per target series — univariate OLS-with-constant on each target in its scored space, p by AIC over $\{1, \ldots, 4\}$; both *iterated* and *direct* (Marcellino et al., 2006) multi-step variants are reported (the iterated AR is the strong univariate competitor at $h \geq 4$). The predictive density propagates the Gaussian innovation variance (analytic for iterated, residual-block-bootstrap for direct).

(iii) **Single-country Bayesian VAR (no coupling)** — the bloc estimated alone: the three `ROW*` star columns dropped, the identical `cgl_bvarV5` call (same GLP/Minnesota prior, lags, COVID-as-missing mask) and `MAXIT`= 1. Isolates the *value of the global coupling.*

(iv) **Classical fixed-weight GVAR**, two variants: (iv-a) the *textbook* Pesaran–Schuermann–Weiner / Dees–di Mauro–Pesaran–Smith GVAR (Pesaran et al., 2004; Dees et al., 2007) — per-bloc **OLS** VARX* with contemporaneous + lagged foreign stars as weakly exogenous regressors, fixed weights, *no* COVID-as-missing (the literature's GVAR has no data augmentation); this is the mandatory headline classical competitor and the one piece of new code in the horse-race. (iv-b) a single-pass Bayesian fixed-weight GVAR (`MAXIT`= 1, GLP prior + COVID-as-missing) — the clean *EM ablation* whose only difference from the proposed model is the star re-convergence.

(v) **Bayesian dynamic factor model** — 3–5 static factors from the cross-bloc target panel, a Bayesian VAR(1–2) on the factors plus Gaussian idiosyncratic terms, same truncated panel and COVID handling. The non-GVAR multivariate "is the GVAR structure even needed" competitor.

The two key ablations isolate the value of *global coupling* (iii vs. proposed) and of the *EM data-completion* (iv-b vs. proposed); (iv-a) measures the whole Bayesian+EM+COVID apparatus against the literature GVAR.

23.2.2 Scheme: recursive expanding window

Pseudo-real-time, recursive expanding window on the final 2025Q4 vintage (a genuine real-time vintage does not exist for most of the ∼50-economy panel; a US+EA-subset real-time check is a flagged robustness, not the default). Origins run quarterly from t_0 = 2010Q1 to 2024Q4 (∼60 raw origins). The OOS set is *trimmed* so every scored (t_0, h) has a realization ≤ 2025Q4 (score h=8 only for origins ≤ 2023Q4, etc.). Headline tables drop the 2020 COVID origins (and origins whose window spans 2020), leaving ∼55 scorable origins; an appendix shows the full set. The COVID-as-missing treatment still applies *in-sample* at every origin for which 2020Q2–Q4 ≤ t_0. Blocs enter an origin only with ≥ 40 in-sample quarters; the six headline blocs satisfy this from 2010Q1.

Targets and horizons. Three targets per bloc—real-GDP growth, inflation, short rate—read through the same `findcol` resolution order the estimator uses, in the same scored space (4Δ of the GDP and price log-levels; the rate level). Horizons $h \in \{1, 2, 4, 8\}$ quarters. The *headline* scope is the six legible blocs {US, EA, JP, UK, CA, CN}; the full 33-bloc panel is reported as relative-RMSE distribution summaries in an appendix (rate-less blocs CH/NO contribute growth and inflation only).

Look-ahead invariants (leak-free by construction). The expanding-window scheme is not automatically free of future-information leakage; the driver enforces four invariants at *every* origin t_0. (1) *Panel truncation*: every `DataRaw_<CC>` panel is sliced to dates ≤ t_0 *before* `transformData` and before any star is built. (2) *Rolling as-of weights*: the shipped 2021–2023 `weights_full.json` is **forbidden** in the OOS sweep (using it at a pre-2020 origin injects future trade structure); the default is a *fixed pre-sample window* (e.g. 2005–2009 average, used at every origin—leak-free because it biases toward the past, never the future), with genuine rolling as-of-t_0 weights as the rigor robustness. (3) *Star re-convergence is in-sample only*: every partner panel feeding a star is itself truncated to ≤ t_0, so no star cell reads a post-t_0 realization. (4) *No 2025Q4 harmonization at earlier origins*: the common vintage is only the well from which ≤ t_0 rows are drawn; the deterministic transforms (100·log, 4Δ) estimate no parameters and leak nothing.

Predictive-density caveat (stated, not hidden). `Dforecast` propagates the shock covariance Σ and the within-draw (β, Σ) MH variation forward from the last *observed* state; it does *not* integrate over the DK-smoother uncertainty of the COVID-imputed states, and seeds the forecast from the posterior-*mean* completed state. For origins whose conditioning rows include imputed COVID quarters (small h, t_0 near 2020) the density is therefore mildly *over-confident*; PIT and coverage near those origins are flagged as a known caveat, not a model failure.

23.2.3 Scoring and inference

Point accuracy: RMSE and MAE of the predictive mean; the headline number is relative RMSE = RMSE(model)/RMSE(RW) ($< 1 \Rightarrow$ beats RW).

Density accuracy: the continuous ranked probability score (**CRPS**, the *primary* metric) computed from the draw ensemble by the unbiased estimator CRPS $= \frac{1}{m}\sum_k |X_k - y| - \frac{1}{2m^2}\sum_k\sum_l |X_k - X_l|$ (Gneiting and Raftery, 2007), which degrades gracefully at the modest draw counts here; the mean log predictive score (LPS), reported with a *parametric* (Gaussian/skew-t) fit to the draw cloud as the headline (a small-bandwidth KDE-LPS, unbounded below at $\sim$100 draws, is only a robustness check); and calibration by the probability integral transform (**PIT**) histograms with empirical coverage of the 50% and 90% predictive intervals and a uniformity test (Berkowitz, 2001). The conclusions lean on CRPS; LPS corroborates.

Predictive-ability inference is dictated by the *nesting structure* (the single most common referee kill-shot). Under the maintained DGP, single-BVAR (iii) $\subset$ single-pass GVAR (iv-b) $\subset$ proposed EM-GVAR, and RW/AR (i),(ii) are nested in all the VAR models; for these nested pairs the standard Diebold–Mariano statistic is *not* asymptotically $\mathcal{N}(0,1)$ and overstates significance. Accordingly:

- the **Giacomini–White** conditional predictive-ability test (Giacomini and White, 2006) is the *headline* inferential test: it is valid under nesting and is the recursive/expanding-scheme-correct test (it assumes finite estimation windows / non-vanishing estimation uncertainty), and it is loss-agnostic so it applies directly to the CRPS and log-score differentials. Instruments $g_{t-1} = [1, d_{t-1}]$ plus a recession/financial-stress indicator (to detect state-dependent ability); HAC variance, Wald χ^2.

- the **Clark–West** test (Clark and West, 2007) is used for the *nested point/MSE* comparisons (it adjusts the MSE differential for the larger model's extra-parameter estimation noise); one-sided.

- the two-sided **Diebold–Mariano** test (Diebold and Mariano, 1995) is used only for *non-nested* pairs (proposed vs. DFM, or vs. the textbook OLS-GVAR when neither nests the other), with a HAC long-run variance (bandwidth $\geq h - 1$) and the Harvey et al. (1997) small-sample correction.

For multiplicity (models $\times$ horizons $\times$ variables $\times$ blocs) the family of GW/CW p-values is controlled by a Romano–Wolf step-down (Romano and Wolf, 2005) (preferred—it respects cross-test dependence via the bootstrap) or Benjamini–Hochberg, and the per-(variable,horizon) **Model Confidence Set** (Hansen et al., 2011) reports which models are statistically indistinguishable from the best. Relative RMSE is the descriptive effect size; significance comes from the CW/GW test on the loss differentials, never from the ratio alone.

The deliverables are: the relative-RMSE matrix (model $\times$ horizon $\times$ variable, headline six blocs + a full-panel distribution table); CRPS and log-score tables with ratios vs. RW; GW/CW (and DM-for-non-nested) significance overlaid with the Romano–Wolf set and the MCS membership flagged; PIT histograms and a 50/90% coverage table; and illustrative fan-chart examples (e.g. the US-inflation forecast from a 2021Q2 origin).

23.2.4 Results

Run configuration. The headline tables below are the *realized* sweep, not a placeholder. The recursive scheme runs over the headline six blocs {US, EA, JP, UK, CA, CN} at up to 56 quarterly origins from 2010Q1 to 2024Q4 (52 for the two EM-GVAR variants and the single-pass Bayesian GVAR, which require the longer in-sample warm-up; the 2020 ex-COVID origins are dropped from every model). Trade weights are *rolling, as-of-t_0*: a three-year pre-sample average ending just before each origin (2007–2009 at the 2010Q1 origin through 2021–2023 at 2024Q1), so no future trade structure leaks in. The predictive densities use NDRAWS = 200 retained draws with an explicit *stationarity backstop* that discards posterior draws whose companion spectral radius exceeds 1 (the fix for the explosive-draw dispersion flagged in earlier pilots). The primary score is the CRPS; the 90% Model Confidence Set (MCS) of Hansen et al. (2011) reports the models statistically indistinguishable from the best per (variable, horizon).

Headline finding. On *point* accuracy the iterated AR(p) and the dynamic factor model are the strongest competitors—a familiar result in macro forecasting—and the EM-GVAR does *not* win the point horse-race outright (Table 108). The decisive evidence sits in the *density* comparison and the MCS (Tables 109–110): the EM-Bayesian GVAR is *inside* the 90% MCS for **inflation at every horizon** ($h \in \{1, 2, 4, 8\}$) and for the **short rate at** $h \in \{1, 2, 4\}$—statistically on par with the best model on the entire nominal block—and the best inflation point forecast at h=1 *among the multivariate/GVAR models* is `emgvar_endo` (relative RMSE 0.799), behind only the univariate AR(p) at 0.641. It falls *out* of the MCS only for GDP growth at h=1 and h=8, where the univariate and factor models dominate. Two structural results follow. First, the endogenous-US design earns its keep: `emgvar_endo` is weakly preferred to the frozen-anchor `emgvar_frozen` in the MCS on balance—it *survives* in the short-rate h=1 set while `emgvar_frozen` is eliminated ($p = 0.15$ vs. 0.08), and it is removed *later* than the frozen variant in the growth h=8 elimination order ($p = 0.099$ vs. 0.057); the one cell where the frozen variant is marginally preferred is the rate h=8 tail, where both are out of the set. Second, and most important for the paper's claim, the EM-Bayesian treatment *decisively* beats the textbook fixed-weight OLS classical GVAR: `cgvar_a` is eliminated from half of the (variable, horizon) MCS cells (6 of 12), concentrated at the longer horizons, and carries badly miscalibrated densities (h=1 coverage of only 0.37 (inflation) / 0.47 (growth) against the 0.90 target), so the Bayesian shrinkage + COVID-as-missing + EM star-reconvergence apparatus is the demonstrable value-add over the literature GVAR.

Honest caveats. Two must be stated. (i) The draw-based densities are *over-confident*: the 90% predictive interval covers only ≈ 0.73–0.85 of realizations for the EM-GVAR variants at short horizons (against the 0.90 target; PIT uniformity is rejected), a direct consequence of the NDRAWS = 200 cloud and of not integrating over the DK-smoother uncertainty of the COVID-imputed states (Section 23.2.2). The CRPS and the MCS—which compare score *differentials*, not absolute coverage—are the robust primaries and are reported as such; the under-coverage is a calibration caveat, not a ranking artifact. (ii) The parametric log-predictive-score column is numerically unreliable on the under-covered clouds (its Gaussian/skew-t tail fit blows up to $\sim 10^{16}$ on a handful of origins where a realization lands far in the tail of a too-narrow cloud), so it is *not* used for ranking; only CRPS and the MCS are.

Table 108: Headline relative RMSE, RMSE(model)/RMSE(RW) (< 1 beats the driftless random walk), six headline blocs, by horizon and target. Realized OOS sweep (rolling as-of-t_0 weights, NDRAWS = 200). Lowest entry per (target, horizon) in **bold**.

		Horizon h (quarters)			
Target	Model	1	2	4	8
Growth	`emgvar_endo`	1.191	1.091	1.026	1.031
	`emgvar_frozen`	1.191	1.095	1.028	1.030
	AR(p)	1.195	0.990	0.968	0.960
	DFM	**0.917**	0.989	0.986	**0.962**
	`cgvar_a` (OLS GVAR)	2.097	1.013	1.019	0.993
Inflation	`emgvar_endo`	0.799	0.891	1.085	0.953
	`emgvar_frozen`	0.802	0.894	1.087	0.962
	AR(p)	**0.641**	**0.772**	**0.998**	0.921
	DFM	0.804	0.870	1.067	0.908
	`cgvar_a` (OLS GVAR)	2.113	1.050	1.366	1.075
Rate	`emgvar_endo`	1.017	1.121	1.226	1.444
	`emgvar_frozen`	1.021	1.119	1.224	1.445
	AR(p)	0.829	**0.863**	**0.919**	1.025
	SBVAR	**0.990**	1.081	1.195	1.421
	`cgvar_a` (OLS GVAR)	1.531	1.200	1.034	**1.001**

23.3 Endogenous-US spillover analysis

The endogenous-US configuration ($\rho(M) \approx 0.924$, Table 107) lets a US scenario pin propagate through the RoW→US stars *and* the four US-determined global commons (oil, the US 10-year yield, the US BAA corporate yield, US equity), pinned 1:1 onto every RoW bloc. A frozen anchor cannot generate the reverse RoW→US feedback at all—the "world → US" direction is *structurally absent*, not "present but zero"—so the two configurations differ in exactly the global-spillover content we wish to document. The analysis is split into two tiers, and the manual labels which is which, because the shipped solver computes a particular object.

Table 109: Headline density accuracy — mean CRPS as a ratio to the random walk (< 1 beats RW; CRPS is the primary score) at h=1 and h=4, and the empirical 90%-interval coverage at h=1 (target 0.90). Realized OOS sweep. The parametric log-score is omitted as numerically unreliable on the under-covered clouds (see text).

		CRPS ratio vs. RW		cov_{90}
Target	Model	h=1	h=4	h=1
Growth	`emgvar_endo`	1.042	0.983	0.73
	DFM	0.745	0.857	0.89
	`cgvar_a` (OLS GVAR)	2.382	1.106	0.47
Inflation	`emgvar_endo`	0.814	1.080	0.70
	AR(p)	0.629	0.955	0.88
	`cgvar_a` (OLS GVAR)	2.498	1.474	0.37
Rate	`emgvar_endo`	1.040	1.230	0.80
	`sbvar`	0.986	1.199	0.84
	`cgvar_a` (OLS GVAR)	1.519	1.009	0.86

Table 110: 90% Model Confidence Set membership for the EM-Bayesian GVAR (Hansen et al., 2011), by target and horizon. ✓ = in the MCS (statistically indistinguishable from the best); × = eliminated. The endogenous variant survives wherever the frozen one does and strictly longer on the nominal block; the textbook OLS GVAR (`cgvar_a`) is eliminated from half of the (variable, horizon) MCS cells (6 of 12), concentrated at the longer horizons.

		Horizon h			
Target	Model	1	2	4	8
Growth	`emgvar_endo`	×	✓	✓	×
	`emgvar_frozen`	×	✓	✓	×
Inflation	`emgvar_endo`	✓	✓	✓	✓
	`emgvar_frozen`	✓	✓	✓	✓
Rate	`emgvar_endo`	✓	✓	✓	×
	`emgvar_frozen`	×	✓	✓	×
`cgvar_a` (textbook OLS GVAR), growth		×	✓	×	×
`cgvar_a` (textbook OLS GVAR), inflation		✓	✓	×	×
`cgvar_a` (textbook OLS GVAR), rate		×	✓	✓	✓

What the solver actually computes. `gvarSolve.ts` does *not* compute generalized impulse responses (GIRFs), forecast-error variance decompositions (FEVDs), or a Diebold–Yilmaz connectedness index. It returns a *conditional-mean scenario response*: pin a path and read the coupled conditional-forecast *mean deltas* `delta`[c] via the fixed point $x = Mx + d$ (Algorithm 3). There is no Σ-based shock generator and no moving-average machinery in the solver. The analysis therefore separates:

- **Tier A (the headline; zero new estimator code).** The conditional scenario propagation of a US pin, endogenous (`solveGVARClosedFormEndo`) vs. frozen (`solveGVARLinear`). This *is* the empirical contribution the shipped code supports.

- **Tier B (a new `gvar_girf` module).** GIRFs (Pesaran and Shin, 1998), generalized FEVDs, and the Diebold–Yilmaz spillover index (Diebold and Yilmaz, 2009, 2012). The ingredients exist in the exports (`companion_mean.json` carries each bloc's companion $\hat{\beta}$ and $\hat{\Sigma}$; the coupling M is assembled by `gvarSolve.ts`), so it is *constructible*, but it is new code, validated against Tier A and gated behind it.

Tier A — the worked scenario contrast (headline). A US +100 bp rate *path* (`POLRATE` pinned +100 bp over the horizon—correctly a multi-period conditional projection, not a one-period shock or a GIRF) is propagated through both topologies. Under the endogenous configuration the augmented state carries every bloc's `ROW*` star deltas *including the US's own* (`ROW*_US`, the RoW→US feedback) and the US global-column deltas $g\Delta$; the converged `delta.US` $\neq 0$ is the feedback the frozen anchor cannot produce. We report the per-bloc deltas (growth/inflation/rate) for the six headline blocs side by side, the *feedback gap* (the US-own response difference), the per-bloc *amplification ratio* (endogenous vs. frozen RoW peak responses), the peak horizons, and the spectral radius $\rho(M)$ of each system (the multiplier strength reported by the solver as `rhoSpectral` / `rhoEmpirical`). This example doubles as the solver-validation canary: the closed-form endogenous result reproduces the iterative `solveGVAR` fixed point to $< 10^{-4}$ across every bloc, and a warm coupling-cache hit is byte-identical ($< 10^{-9}$) to a cold rebuild (both asserted in the `gvar-us-endogenous` / `gvar-closed-form-routing` test suites; the iterative reference is itself converged to 10^{-7}).

Tier A — realized scenario contrast. The worked example (`eval/tier_a_scenario.mjs`, output `results/tierA/us_rate_100bp.json`) propagates a US +100 bp `POLRATE` path through both topologies and confirms the qualitative point the design predicts: the frozen anchor produces *no* RoW→US feedback, while the endogenous configuration produces a non-trivial one. The cross-country *gap* (endogenous minus frozen) is sizeable on every bloc—for example the peak US own-growth response differs by ≈ 4.2 display points, EA inflation by ≈ 1.1, and the gap *flips sign* across blocs (positive for US/EA/UK growth, negative for JP/CA/CN growth); selected peaks are in Table 111. **Honest caveat (do not over-read).** These are the deltas of a *multi-period conditional projection*, not a structural or generalized impulse response: the US rate is *pinned* along the whole path, so the responses bundle the direct conditioning with the coupled feedback and are not a clean causal IRF. The magnitudes are large and sign-flipping, which is exactly why a validation pass (the Tier-B GIRF cross-check below, and a robustness sweep over the pin specification) is required before any number here is headlined as a spillover *multiplier*. What is established is the *existence and direction* of the endogenous-only feedback channel, not its calibrated size.

Table 111: Tier-A US +100 bp `POLRATE`-path projection: peak conditional-mean response (display points) under the endogenous vs. frozen-anchor topology, and their gap, for selected blocs and targets (`results/tierA/us_rate_100bp.json`). *Conditional-projection deltas, not causal IRFs*—magnitudes are illustrative and pending validation.

Bloc	Target	Endogenous	Frozen	Gap (endo−froz)
US	Growth	+1.28	−4.37	+4.22
US	Inflation	−1.53	−2.82	+1.30
EA	Growth	+1.03	−0.95	+0.88
EA	Inflation	−1.00	−0.64	+1.07
JP	Growth	+1.07	+1.62	−1.18
CN	Inflation	−0.69	+1.19	−1.29

Tier B — GIRF, FEVD, and connectedness (structural extension). The GIRF to a one-SD shock in variable i of the coupled global system is $\mathrm{GIRF}_x(h, i) = \sigma_{ii}^{-1/2} A_h \Sigma e_i$, with A_h the h-step coefficient of the coupled solution's moving-average representation and Σ the (cross-bloc-block-diagonal) global innovation covariance—ordering-invariant, the standard GVAR identification stance (Pesaran and Shin, 1998; Dees et al., 2007). The identifying assumptions are stated explicitly: GIRFs trace the response to the historically typical innovation correlation (not an orthogonal structural shock), cross-bloc innovation independence (block-diagonal Σ), and linearity/parameter-constancy over the horizon. Bands come from re-running the recursion over the `companion_draws.json` posterior draws. The module is validated by the cross-check that a one-period US-rate-pin conditional path equals the rate-shock GIRF up to the $\delta/\sqrt{\sigma_{ii}}$ normalization. From the system generalized FEVD $\theta_{ij}(H)$ (row-normalized) we report the Diebold–Yilmaz total spillover index, the "from US" row, and the "to US" direction—the channel the frozen anchor cannot even define—with posterior bands and both the raw and row-normalized indices. **Results: [TBD]** (Tier A ships first; Tier B is the structural extension).

24 Discussion

The Monetary-Financial Model is designed to bridge the gap between the methodological sophistication of conditional forecasting techniques developed in the academic literature and the practical needs of policy economists who require rapid, interactive scenario analysis. We conclude with a discussion of design choices, limitations, and directions for future work.

24.1 Design Philosophy

Several design choices merit discussion. First, the three-layer architecture—baseline, scenario, counterfactual—deliberately separates concerns. The baseline forecast reflects the BVAR's unconditional projection and is computed offline. The scenario layer allows the user to impose conditioning constraints without specifying the structural source of the deviations, following the tradition of Waggoner and Zha (1999). The counterfactual layer then overlays structural identification via DSGE impulse responses, enabling “what-if” policy analysis while preserving the reduced-form covariance structure of the BVAR.

Second, the decision to re-condition the counterfactual through the BVAR (rather than relying solely on the DSGE IRFs) reflects a deliberate methodological choice: the BVAR's estimated covariance structure captures empirical co-movements that may be absent from or imperfectly captured by any single DSGE model. The IRF-consistent mode (Section 15.3c) strikes a balance by using the DSGE model to identify the responses of key variables (inflation, output) to the policy shock, while allowing the BVAR to propagate these effects to the remaining 30+ variables in a data-consistent manner.

Third, all computation runs client-side. This eliminates server infrastructure costs and latency, enables offline use, and avoids transmitting potentially sensitive forecast data over the network. The computational cost is manageable: the Kalman filter and smoother for the 41-variable, 3-lag US VAR over a 20-quarter horizon completes in well under a second on modern hardware (the single-bloc smoother is sub-50 ms; the multi-second cost discussed in Section 21.3 is the *coupled* 50-bloc GVAR solve, not the single-country filter).

24.2 Limitations

The current implementation has several limitations that users should bear in mind:

1. **Linear, time-invariant dynamics:** The VAR coefficients are fixed at their posterior mean for the deterministic conditional forecast. Time-varying parameters, stochastic volatility, and regime switching are not supported. In periods of structural change, the fixed-coefficient assumption may be particularly restrictive.

2. **Gaussian innovations:** The Kalman filter assumes Gaussian innovations. Fat-tailed or asymmetric shock distributions, which may be important during financial crises, are not accommodated.

3. **Point-identified counterfactuals:** The DSGE IRFs used for counterfactual analysis are computed under a single monetary policy rule and model specification. The tool does not average over model uncertainty or rule uncertainty, though the user can manually explore different model–rule combinations.

4. **Lucas critique:** The counterfactual analysis conditions on estimated reduced-form dynamics that may themselves change under the alternative policy. The IRF-consistent mode partially addresses this by using structurally identified responses for the key transmission variables, but the remaining variables' responses are governed by the estimated BVAR covariance structure, which is held fixed.

5. **Data vintage:** The BVAR *coefficients* are estimated on a fixed 2025Q4 vintage and frozen. The *observed* panel is extended without re-estimation by the 2026Q1 refresh of Section 19.5

(the forecast re-anchors from the frozen companion), but real-time data *revisions* to already-shipped quarters and the differential publication lag across blocs (the ragged edge—CN/IN at 2025Q4, IL at 2024Q4) are not otherwise reconciled, and the parameters are not re-estimated as new data arrive.

24.3 Known issues flagged for fix

The limitations above are intrinsic to the modelling choices. The following are *specific, identified defects* in the current deployment—known and flagged, but *not yet fixed*—documented here in the interest of honesty for a live research tool. Each cites the source file where it lives.

1. **China fiscal debt inflation double-count — RESOLVED (June 2026).** The fiscal debt-snowball roll-forward (`fiscalAccounting.ts`) builds quarterly nominal GDP growth as Δ(`idx.gdp`) + Δ(`idx.deflator`), i.e. real growth plus inflation. Formerly the resolved CN `idx.gdp` pointed at a *nominal* `GDP_CN` while `idx.deflator` pointed at `CPI_CN`, so inflation was added *twice* and the CN debt-to-GDP path was biased. The June-2026 endogenous-US 33-bloc GVAR-EM re-estimate re-sourced `GDP_CN` as a *real* chain-linked SA volume index (base 2011Q1=100) and set `CN/fiscal_meta.json idx.gdpIsNominal=false`, so the growth term is now genuinely real growth plus a CPI term (CPI ≈ 0 in this window), on the same footing as the other fiscal blocs. The double-count is gone; only a second-order CPI-versus-GDP-deflator wedge remains (shared by EA/JP/UK/CA).

2. **Retired entropic-tilt code still ships (no live path).** The user-facing long-run mechanism is now the forward-pass drift setter (`driftSetter.ts`, Section 18.2), which *replaced* the earlier entropic tilt and has no steady-state moment to go non-finite (Remark 6). The old `longRunTilt.ts` module is still present in the repository but is *no longer wired to any UI component*—it is referenced only by its own unit test. It retains a latent defect (`NaN` long-run moments for unit-root/near-singular draws propagate through the softmax and empty the anchor band), which is harmless while the module is unused but should be removed or finite-masked to avoid future reintroduction.

3. **No divergence guard on the closed-form GVAR solve.** The live solver (`gvarSolve.ts`, Neumann iteration and direct inverse) does not check $\rho(M) < 1$ before solving. The deployed coupling is contractive ($\rho(M) \approx 0.924$), but a future roster or a degenerate pin set that pushed $\rho(M) \geq 1$ would return a *confidently wrong* forecast from a diverged fixed point with no warning. A spectral-radius backstop is needed.

4. **Conditional band uses the filtered, not smoothed, covariance.** The predictive-band variances are extracted from the Kalman *filter* covariance P_t^{filt} (`kalman.ts`), not the Durbin–Koopman *smoother* covariance. For quarters *preceding* a future pin, the smoother would tighten the covariance (a later observation reduces uncertainty about an earlier state), so the

displayed band is *overstated* on the run-up to a future conditioning point. The central path is unaffected; only the band width is conservative.

5. **The porter does not regenerate `fiscal_meta.json`.** The fiscal block is addressed by *positional* indices stored in each bloc's `fiscal_meta.json` (`idx.gdp`, `idx.deflator`, `idx.revenue`, ...). The model export does not rebuild this file, so any change to a bloc's variable order leaves the indices pointing at the wrong columns—the cause of a recent EA/UK debt-path explosion—and they must be repaired manually (`pipeline/fix_fiscal_meta_indices.mjs`). Folding the meta regeneration into the porter is the durable fix.

24.4 Directions for Future Work

Several extensions are planned or under consideration:

- **Time-varying parameters:** Incorporating stochastic volatility or time-varying coefficients (e.g., Sims and Zha, 1998) to capture evolving dynamics.
- **Structural identification:** Extending the counterfactual layer to support sign restrictions, narrative restrictions, or scenario priors (Farkas and Jakab, 2026) for identification within the BVAR itself, rather than relying on external DSGE models.
- **Mixed-frequency data:** Supporting monthly–quarterly mixed-frequency estimation to incorporate higher-frequency indicators.
- **Model averaging:** Bayesian model averaging over DSGE specifications and policy rules for the counterfactual layer.

A Glossary of terms

This glossary defines the named concepts used throughout the manual; the mathematical symbols are collected in Table 1.

BVAR (Bayesian vector autoregression)

A reduced-form VAR whose coefficients are estimated under a prior, here the hierarchical Minnesota/GLP prior of Giannone et al. (2015) with data-driven hyperparameter selection (Section 12).

Minnesota / GLP prior

The shrinkage prior that centers each equation on a univariate own-lag model and selects its tightness λ and sum-of-coefficients μ by marginal-likelihood maximization. "GLP" refers to the Giannone et al. (2015) hierarchical implementation.

Own-lag prior mean (δ): RW vs. WN

The prior mean on a variable's own first lag: $\delta = 1$ (random walk, `RW`, the default—best forecast is the current level) or $\delta = 0$ (stationary white noise, `WN`—best forecast is the mean). Implemented as diagb(pos) = 0 for the WN index set pos (Section 12.5). It shifts the prior *center*, not a hard constraint.

GVAR (Global VAR)
A collection of per-economy ("bloc") BVARs linked through trade-weighted foreign variables; here the linkage is imposed at solve time, not in estimation (Sections 16, 16.5).

Trade-weighted "star" variable
A bloc's foreign aggregate (ROWDEM demand, ROWINF inflation, ROWFIN financial), built as the trade-weight w_{cj}-weighted average of its partners' corresponding series, renormalized over the partners present each quarter (Section 16.2).

Endogenous dominant unit
The configuration in which the US is a full bloc inside the fixed point—with its own ROW*_US stars and a US weights row—so shocks abroad feed *back* into the US (the RoW→US channel a frozen anchor cannot represent).

Frozen anchor
The legacy configuration in which the US is estimated once as a closed, exogenous driver: it enters the coupling only through the forcing d, with no equation for the world to feed back into.

Global commons
The four US-determined common series (oil, the US 10-year yield, the US BAA corporate yield, US equity) pinned 1:1 onto every RoW bloc (Section 16.7).

COVID-as-missing
The treatment of the fixed 2020Q2–Q4 window as *missing data*: the macro columns and the two macro stars are NaN-masked while financials and ROWFIN are kept observed, and the Durbin–Koopman simulation smoother imputes the holes inside a data-augmentation Gibbs sampler (Section 19.4).

EM loop
The block-coordinate iteration over (foreign stars, completed-data states, VAR coefficients): an E-step that completes the panels and rebuilds the stars and an M-step that re-estimates every bloc, iterated to a fixed point in the completed stars (Algorithm 4).

Durbin–Koopman simulation smoother
The forward-filter / backward-draw recursion (Durbin and Koopman, 2002) that produces draws of the conditioned/missing states by simulating an unconditional path, smoothing it, and adding back the smoothed discrepancy (Algorithm 1).

Coupling operator M, fixed point $x = Mx + d$
The affine solve-time operator that propagates a scenario across blocs; assembled by Kalman-probing each pinned cell, solved by Gauss–Seidel/Neumann sweeps or the direct $(I - M)^{-1}$ (Algorithm 3). $\rho(M) < 1$ makes it contractive.

SFA (stock-flow adjustment)
The deficit–debt discrepancy: the part of the observed change in gross debt that the flow side (deficit plus the interest–growth snowball) does not explain; an estimated `WN`-prior BVAR variable (Section 17.1).

EFFI (effective interest rate)
The average rate the government actually pays (net interest over lagged gross debt, annualized); a persistent `RW`-prior variable distinct from the policy rate and market yields.

Long-run drift setter (δc)
The user-facing long-run-target mechanism: a deterministic counterfactual that shifts a variable's own companion drift by the single constant δc needed to land its terminal forecast on the user's target, applied as the exact response $S_h[:, j]\, \delta c$ that translates the central path and bands rigidly, instantly and without re-estimation (Section 18.2). It replaces the earlier *entropic tilt*, an after-estimation draw reweighting that degenerated (collapsing effective sample size) when over-constrained.

CRPS (continuous ranked probability score)
A strictly proper density score (Gneiting and Raftery, 2007) computed from the draw ensemble; the primary density metric of the evaluation, robust at modest draw counts.

PIT (probability integral transform)
The rank of each realization in its predictive ensemble; uniform PITs indicate good calibration (Berkowitz, 2001).

Giacomini–White (GW) test
The conditional predictive-ability test (Giacomini and White, 2006) used as the headline inferential test: valid under nesting and for the recursive/expanding-window scheme, and loss-agnostic (applies to point, CRPS, and log-score differentials).

Clark–West (CW) test
The nested-model point/MSE test (Clark and West, 2007) that adjusts the loss differential for the larger model's extra-parameter estimation noise.

Diebold–Mariano (DM) test
The equal-predictive-accuracy test (Diebold and Mariano, 1995), used here only for *non-nested* pairs, with the Harvey et al. (1997) small-sample correction.

Romano–Wolf / Model Confidence Set
Multiplicity controls: the Romano–Wolf step-down (Romano and Wolf, 2005) for FWER across the test family, and the MCS (Hansen et al., 2011) reporting the models statistically indistinguishable from the best.

GIRF / Diebold–Yilmaz index
Generalized (ordering-invariant) impulse responses (Pesaran and Shin, 1998) and the connectedness index (Diebold and Yilmaz, 2009, 2012) of the Tier-B spillover module (Section 23.3).

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Disclaimer

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